

TECHNICAL SUPPORT DOCUMENT: ENERGY EFFICIENCY PROGRAM FOR COMMERCIAL AND INDUSTRIAL EQUIPMENT:

AIR CLEANERS

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CHAPTER 1. INTRODUCTION

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CHAPTER 1. INTRODUCTION

1.1 PURPOSE OF THE DOCUMENT

This technical support document (TSD) is a stand-alone report that provides the technical analyses and results supporting the information presented in the direct final rule for air cleaners.

1.2 OVERVIEW OF APPLIANCE STANDARDS

Title III, Part B of the Energy Policy and Conservation Act of 1975 (EPCA or the Act), Pub. L. 94-163 (42 U.S.C. 6291–6309, as codified), established the Energy Conservation Program for Consumer Products Other Than Automobiles. In addition to specifying a list of covered residential products and commercial equipment, EPCA contains provisions that enable the Secretary of Energy to classify additional types of consumer products as covered products. (42 U.S.C. 6292(a)(20)) In a final determination of coverage published in the Federal Register on July 15, 2022 (July 2022 Final Determination), DOE classified air cleaners as covered consumer products under EPCA. 87 FR 42297.

1.3 PROCESS FOR SETTING ENERGY CONSERVATION STANDARDS

Pursuant to EPCA, any new or amended standard for a covered product must be designed to achieve the maximum improvement in energy efficiency that is technologically feasible and economically justified. (42 U.S.C. 6295(o)(2)(A) and (3)(B)) Furthermore, DOE may not adopt any standard that would not result in the significant conservation of energy. (42 U.S.C. 6295(o)(3)) Moreover, DOE may not prescribe a standard: (1) for certain products, including MREFs, if no test procedure has been established for the product, or (2) if DOE determines by rule that the new or amended standard is not technologically feasible or economically justified. (42 U.S.C. 6295(o)(3)(A)–(B)) In deciding whether a new or amended standard is economically justified, DOE must determine whether the benefits of the standard exceed its burdens. (42 U.S.C. 6295(o)(2)(B)(i)) DOE must make this determination after considering, to the greatest extent practicable, the following seven factors (42 U.S.C. 6295 (o)(2)(B)(i)):

- 1) the economic impact of the standard on the manufacturers and consumers of the affected products;
- 2) the savings in operating costs throughout the estimated average life of the product compared to any increases in the initial cost or maintenance expense;
- 3) the total projected amount of energy savings likely to result directly from the imposition of the standard;
- 4) any lessening of the utility or the performance of the products likely to result from the imposition of the standard;
- 5) the impact of any lessening of competition, as determined in writing by the Attorney General, that is likely to result from the imposition of the standard;

- 6) the need for national energy conservation; and
- 7) other factors the Secretary considers relevant.

Other statutory requirements are set forth in 42 U.S.C. 6295(o)(1)–(2)(A), (2)(B)(ii)–(iii), and (3)–(4).

DOE considers stakeholder participation to be a very important part of the process for setting energy conservation standards. Through formal public notifications (*i.e.*, *Federal Register* notices), DOE actively encourages the participation and interaction of all stakeholders during the comment period in each stage of a rulemaking. Beginning with the framework document, or request for information (RFI), and during subsequent comment periods, interactions among stakeholders provide a balanced discussion of the information that is required for a potential standards rulemaking.

Before DOE determines whether to adopt a proposed energy conservation standard, it must first solicit comments on the proposed standard. (42 U.S.C. 6295(m)(2)(B)) Any new or amended standard must be designed to achieve significant additional conservation of energy and be technologically feasible and economically justified. (42 U.S.C. 6295(o)(2)(A)) To determine whether economic justification exists, DOE must review comments on the proposal and determine that the benefits of the proposed standard exceed its burdens to the greatest extent practicable, weighing the seven factors listed above. (42 U.S.C. 6295 (o)(2)(B)(i))

After the publication of the framework document or RFI, the energy conservation standards rulemaking process involves three additional formal public notices, which DOE publishes in the *Federal Register*. The first of these notices is a notification of a webinar and availability of preliminary technical support document (preliminary analysis), which is designed to publicly vet the models and tools used in the preliminary rulemaking and to facilitate public participation before the notice of proposed rulemaking (NOPR) stage. The second notice is the NOPR, which presents a discussion of comments received in response to the preliminary analyses and analytical tools; analyses of the impacts of potential new or amended energy conservation standards on consumers, manufacturers, and the Nation; DOE’s weighting of these impacts of new or amended energy conservation standards; and the proposed energy conservation standards for the product. The third notice is the final rule, which presents a discussion of the comments received in response to the NOPR; the revised analyses; DOE’s weighting of these impacts; the new or amended energy conservation standards DOE is adopting for the product; and the compliance dates of the new or amended energy conservation standards.

As stated, DOE determined in the July 2022 Final Determination that coverage of air cleaners is necessary or appropriate to carry out the purposes of EPCA; the average U.S. household energy use for air cleaners is likely to exceed 100 kWh/yr; and thus, air cleaners qualify as a “covered product” under EPCA. 87 FR 42297.

On January 25, 2022, DOE published an RFI, seeking comments on potential test procedure and energy conservation standards for air cleaners. 87 FR 3702. In the January 2022

RFI, DOE requested information to aid in the development of the technical and economic analyses to support energy conservation standards for air cleaners, should they be warranted. 87 FR 3702, 3705.

On August 23, 2022, DOE received comments submitted jointly by groups representing manufacturers, energy and environmental advocates, and consumer groups (the Joint Stakeholders^a), titled “Joint Statement of Joint Stakeholder Proposal On Recommended Energy Conservation Standards And Test Procedure For Consumer Room Air Cleaners” (the Joint Proposal^b), which recommended specific energy conservation standards for certain air cleaners that, in the commenters’ view, would satisfy the EPCA requirements in 42 U.S.C. 6295(o).

The Joint Proposal submitted by the Joint Stakeholders urged DOE to publish final rules adopting the consumer room air cleaner test procedure and standards and compliance dates contained in the Joint Proposal, as soon as possible, but not later than December 31, 2022. The Joint Proposal also recommended that DOE adopt Association of Home Appliance Manufacturers (AHAM) AC-7-2022, “Energy Test Method for Consumer Room Air Cleaners,” as the DOE test procedure. In regards to energy conservation standards, the Joint Proposal specified two-tiered Tier 1 and Tier 2 standard levels, as shown in Table 1.3.1, for conventional room air cleaners with proposed compliance dates of December 31, 2023, and December 31, 2025, respectively. The recommended standards are based on the PM_{2.5} clean air delivery rate (CADR) of the air cleaner in cubic feet per minute (cfm), and consist of integrated energy factor (IEF) of the air cleaner, expressed in PM_{2.5} CADR per watt (PM_{2.5} CADR/W)

^a The Joint Stakeholders include the Association of Home Appliance Manufacturers (AHAM), Appliance Standards Awareness Project (ASAP), American Council for an Energy-Efficient Economy (ACEEE), Consumer Federation of America (CFA), Natural Resources Defense Council (NRDC), the New York State Energy Research and Development Authority (NYSERDA), and the Pacific Gas and Electric Company (PG&E). AHAM is representing the companies who manufacture consumer room air cleaners and are members of the Portable Appliance Division (DOE has included names of all manufacturers listed in the footnote on page 1 of the Joint Proposal and the signatories listed on pp. 13–14): 3M Co.; Access Business Group, LLC; ACCO Brands Corporation; Air King, Air King Ventilation Products; Airgle Corporation; Alticor, Inc.; Beijing Smartmi Electronic Technology Co., Ltd.; BISSELL Inc.; Blueair Inc.; BSH Home Appliances Corporation; De’Longhi America, Inc.; Dyson Limited; Essick Air Products; Fellowes Inc.; Field Controls; Foxconn Technology Group; GE Appliances, a Haier company; Gree Electric Appliances Inc.; Groupe SEB; Guardian Technologies, LLC; Haier Smart Home Co., Ltd.; Helen of Troy-Health & Home; iRobot; Lasko Products, Inc.; Molekule Inc.; Newell Brands Inc.; Oransi LLC; Phillips Domestic Appliances NA Corporation; SharkNinja Operating, LLC; Sharp Electronics Corporation; Sharp Electronics of Canada Ltd.; Sunbeam Products, Inc.; Trovac Industries Ltd; Vornado Air LLC; Whirlpool Corporation; Winix Inc.; and Zojirushi America Corporation.

^b DOE Docket No. EERE-2021-BT-STD-0035-0016. The Joint Petition may be accessed at: www.regulations.gov/comment/EERE-2021-BT-STD-0035-0016

Table 1.3.1 Tier 1 and Tier 2 Standards Proposed by the Joint Stakeholders in the Joint Proposal

Product Description	IEF ($PM_{2.5}$ CADR/W) Tier 1*	IEF ($PM_{2.5}$ CADR/W) Tier 2**
$10 \leq PM_{2.5} \text{ CADR} < 100$	1.69	1.89
$100 \leq PM_{2.5} \text{ CADR} < 150$	1.90	2.39
$PM_{2.5} \text{ CADR} \geq 150$	2.01	2.91

* Tier 1 standards would have an effective date of December 31, 2023.

** Tier 2 standards would have an effective date of December 31, 2025.

The Joint Stakeholders stated that the Tier 1 and Tier 2 standards are estimated to save 1.9 quads of fuel-fuel-cycle energy nationally over 30 years of sales.

EPCA allows DOE to issue a direct final rule establishing an energy conservation standard on receipt of a statement submitted jointly by interested persons that are fairly representative of relevant points of view (including representatives of manufacturers of covered products, States, and efficiency advocates) that contains recommendations with respect to an energy conservation standard that are in accordance with the provisions of 42 U.S.C. 6295(o). 42 U.S.C. 6295(p)(4) DOE has concluded that the recommendations of the Joint Petition satisfy the requirements of 42 U.S.C. 6295(o). In reaching this determination, DOE took into consideration the fact that the Joint Stakeholders consist of representatives of manufacturers of the covered product at issue, a state corporation, and efficiency advocates -- all of which are groups specifically identified by Congress as relevant parties to any consensus recommendation. (42 U.S.C. 6295(p)(4)(A))

DOE also evaluated whether the recommendation satisfies 42 U.S.C. 6295(o) by conducting an analysis to evaluate whether the potential energy conservation standards under consideration achieve the maximum improvement in energy efficiency that is technologically feasible and economically justified and result in significant energy conservation. The evaluation is the same comprehensive approach that DOE typically conducts whenever it considers potential energy conservation standards for a given type of product or equipment. Upon review, the Secretary determined that the Joint Proposal comports with the standard-setting criteria set forth under 42 U.S.C. 6295(p)(4)(A).

Therefore, DOE is publishing this direct final rule concurrently with a NPR that proposes identical energy conservation standards. If DOE receives adverse comment and determines that such comment may provide a reasonable basis for withdrawing the direct final rule, this final rule will be withdrawn and DOE will proceed with the proposed rule.

As part of the information gathering and sharing process, DOE organized and held interviews with manufacturers of the products considered in this rulemaking as part of the engineering analysis. DOE had four objectives for these interviews: (1) solicit manufacturer feedback on the draft inputs to the engineering analysis; (2) solicit feedback on topics related to the RFI; (3) provide an opportunity, early in the rulemaking process, to express manufacturers' concerns to DOE; and (4) foster cooperation between manufacturers and DOE. DOE incorporated the information gathered during the engineering interviews with manufacturers into its analysis.

On March 6, 2023, DOE published a final rule establishing a new test procedure at 10 CFR part 430, subpart B, appendix FF for air cleaners that references AHAM AC-7-2022 and includes methods to (1) measure the performance of the covered product and (2) use the measured results to calculate an IEF to represent the energy efficiency of air cleaners. 88 FR 14014.

DOE considered the Joint Stakeholders' recommendations in this direct final rule analysis. The standards established in the direct final rule are consistent with those recommended in the Joint Proposal.

This direct final rule TSD outlines the analytical methods used in this rulemaking and the results of the analyses performed.

1.4 STRUCTURE OF THE DOCUMENT

The TSD consists of 17 chapters and supporting appendices.

Chapter 1	Introduction: Describes the purpose of the TSD and outlines the structure of the document.
Chapter 2	Analytical Framework: Describes the general rulemaking process.
Chapter 3	Market and Technology Assessment: Characterizes the market for the considered products and the technologies available for increasing efficiency.
Chapter 4	Screening Analysis: Identifies all the design options that improve efficiency of the considered products, and determines which technology options are viable for consideration in the engineering analysis.
Chapter 5	Engineering Analysis: Describes the methods used for developing the relationship between increased efficiency and increased manufacturing cost and presents results of the analysis.
Chapter 6	Markups Analysis: Describes the methods used for establishing markups for converting manufacturer selling price to consumer purchase price and presents results of the analysis.
Chapter 7	Energy Use Analysis: Describes the sources and methods used for generating energy-use estimates for the considered products as a function of potential standard levels and presents results of the analysis.
Chapter 8	Life-Cycle Cost (LCC) and Payback Period Analysis (PBP): Describes the methods used for analyzing the economic effects of new efficiency standards on individual consumers and users of the products with respect to LCC savings and PBP of higher efficiency products and presents results of the analysis.

Chapter 9	Shipments Analysis: Describes the methods used for forecasting shipments with and without new efficiency standards and presents results of the analysis.
Chapter 10	National Impact Analysis: Describes the methods used for estimating the impacts of potential standards on national energy consumption and national economic benefit to consumers and presents results of the analysis.
Chapter 11	Consumer Subgroup Analysis: Describes the methods used for analyzing the effects of potential standards on different subgroups of consumers compared to all consumers and presents results of the analysis.
Chapter 12	Manufacturer Impact Analysis: Describes the methods used for analyzing the effects of potential standards on the finances and profitability of air cleaner manufacturers and presents results of the analysis.
Chapter 13	Emissions Impact Analysis: Describes the methods used for analyzing the impact of potential standards on national emissions of sulfur dioxide, nitrogen oxides, and mercury—as well as on carbon dioxide and other greenhouse gas emissions, and presents results of the analysis.
Chapter 14	Monetization of Emissions Reduction Benefits: Describes the methods used for estimating monetary benefits likely to result from reduced emissions expected to result from potential standards and presents results of the analysis.
Chapter 15	Utility Impact Analysis: Describes the methods used for analyzing key impacts of potential standards on electric utilities and presents results of the analysis.
Chapter 16	Employment Impact Analysis: Describes the methods used for analyzing the impact of potential standards on national employment and presents results of the analysis.
Chapter 17	Regulatory Impact Analysis: Describes the methods used for analyzing the impact of non-regulatory alternatives to energy conservation standards compared to potential new standards and presents results of the analysis.
Appendix 8A	User Instructions for the Life-Cycle Cost Analysis Spreadsheet
Appendix 8B	Uncertainty and Variability in the Life-Cycle Cost and Payback Period Analysis

Appendix 8C	Lifetime Distributions
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CHAPTER 2. ANALYTICAL FRAMEWORK

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CHAPTER 2. ANALYTICAL FRAMEWORK

2.1 INTRODUCTION

Section 6295(o)(2)(A) of the Energy Policy and Conservation Act (EPCA),^a Public Law 94-163, 42 U.S.C. 6291 *et seq.* requires the U.S. Department of Energy (DOE) to establish energy conservation standards that achieve the maximum improvement in energy efficiency of consumer products that is technologically feasible and economically justified. (42 U.S.C. 6295(o)(2)(A)) This chapter describes the general analytical framework that DOE uses in developing such standards, and in particular, energy conservation standards for air cleaners. The analytical framework is a description of the methodology, the analytical tools, and the relationships among the various analyses that are part of this rulemaking. The methodology that addresses the statutory requirement for economic justification, for example, includes analyses of life-cycle cost; economic impact on manufacturers and users; national benefits; effects, if any, on utility companies; and impacts from any lessening in competition among manufacturers.

Figure 2.1.1 summarizes the analytical components of the standards-setting process. The focus of this figure is the center column, identified as “Analyses.” The column labeled “Key Inputs” lists the types of data and information required for each analysis. Some key inputs come from public databases; DOE collects other inputs from interested parties or other knowledgeable experts within the field. The column labeled “Key Outputs” shows analytical results that feed directly into the standards-setting process. The figure shows how the analyses fit into the rulemaking process and how they relate to one another. Arrows connecting analyses show the types of information that feed from one analysis to another.

^a All references to EPCA in this document refer to the statute as amended through the Energy Act of 2020, Public Law 116-260 (Dec. 27, 2020), which reflect the last statutory amendments that impact Parts A and A-1 of EPCA.

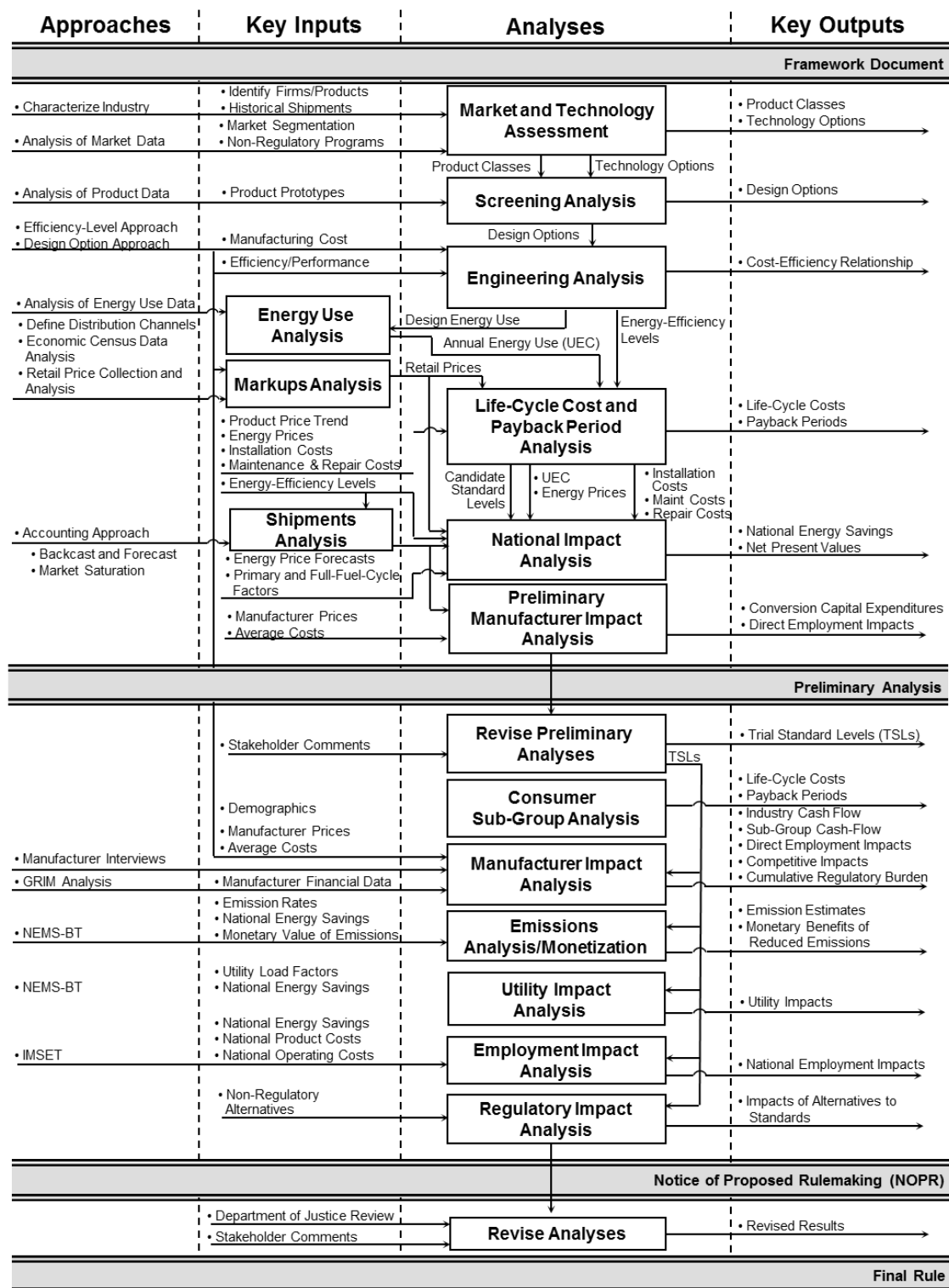


Figure 2.1.1 Flow Diagram of Analyses for the Rulemaking Process

The analyses performed as part of the direct final rule and reported in this technical support document (TSD) are listed below.

- A market and technology assessment to characterize the relevant equipment markets and technology options, including prototype designs.
- A screening analysis to review each technology option and determine if it is technologically feasible; is practical to manufacture, install, and service; would adversely affect product utility or product availability; would have adverse effects on health and safety; or would utilize proprietary technology that represents a unique pathway to achieving a given efficiency level.
- An engineering analysis to develop cost-efficiency relationships that show the manufacturer's cost of achieving increased energy efficiency.
- A markups analysis to develop markups throughout the distribution channel(s), relating the manufacturer selling price (MSP) to the cost to the consumer.
- An energy use analysis to determine the annual energy use of the considered equipment with and without standards for a representative group of users.
- A life-cycle cost (LCC) and payback period (PBP) analysis to calculate the savings in operating costs at the consumer level throughout the life of the considered equipment compared with any increase in the installed cost for the equipment likely to result directly from imposition of a standard.
- A shipments analysis to project equipment shipments, which then are used to calculate the national impacts of standards on energy, net present value (NPV), and future manufacturer cash flows.
- A national impact analysis (NIA) to assess the aggregate impacts at the national level of energy conservation standards for the equipment, as measured by the NPV of total consumer economic impacts and the national energy savings (NES).
- An LCC subgroup analysis to evaluate variations in consumer characteristics that might cause a standard to disproportionately affect particular consumer subpopulations.
- A manufacturer impact analysis (MIA) to estimate the financial impact of standards on manufacturers and to calculate impacts on costs, shipments, competition, direct employment, and manufacturing capacity.
- An emissions impacts analysis to assess the impacts of energy conservation standards on the environment.
- An emissions monetization to assign values to the benefits associated with emissions reductions.
- A utility impact analysis to estimate the effects of standards on electric utilities.

- An employment impact analysis to assess the aggregate impacts on national employment.
- A regulatory impact analysis to examine major alternatives to new energy conservation standards that potentially could achieve substantially the same regulatory goal at a lower cost.

DOE has not previously conducted an energy conservation standards rulemaking for air cleaners. For this rulemaking, DOE began the process by publishing a notice of proposed determination (NOPD) in the *Federal Register* on September 16, 2021 (the “September 2021 NOPD”) that sought comment on its proposal to classify air cleaners as covered products. 86 FR 51629. The September 2021 NOPD is available at www.regulations.gov/document/EERE-2021-BT-DET-0022-0001.

DOE then issued a request for information (RFI) in the *Federal Register* on January 25, 2022 (the “January 2022 RFI”) that sought input from interested parties regarding the development and evaluation of potential new test procedures and energy conservation standards for air cleaners, and whether such standards would result in significant energy savings, be technologically feasible, and economically justified for air cleaners. 87 FR 3702. The January 2022 RFI is available at www.regulations.gov/document/EERE-2021-BT-STD-0035-0001.

On July 15, 2022, DOE published a notice of final determination that coverage of air cleaners is necessary or appropriate to carry out the purposes of EPCA; the average U.S. household energy use for air cleaners is likely to exceed 100 kilowatt hours per year (kWh/yr); and thus, air cleaners qualify as a “covered product” under EPCA. 87 FR 42297. The July 2022 final determination is available at www.regulations.gov/document/EERE-2021-BT-DET-0022-0015.

The following sections provide a general description of the different analytical components of the direct final rule analysis. DOE has used the most reliable data available at the time of each analysis in this rulemaking. All data will be available for public review.

2.2 MARKET AND TECHNOLOGY ASSESSMENT

The market and technology assessment characterizes the relevant product markets and existing technology options, including working prototype designs, for the considered products.

2.2.1 Market Assessment

When DOE begins an energy conservation standards rulemaking, it develops information that provides an overall picture of the market for the products considered, including the nature of the products, the industry structure, and market characteristics for the products. This activity consists of both quantitative and qualitative efforts based primarily on publicly available information. The subjects addressed in the market assessment for air cleaners include manufacturers, trade associations, and the quantities and types of products sold and offered for sale. DOE examined both large and small and foreign and domestic manufacturers. Finally, DOE

reviewed other energy efficiency programs from utilities, individual States, and other organizations.

DOE reviewed relevant literature and interviewed manufacturers to develop an overall picture of the air cleaner industry in the United States. Industry publications and trade journals, government agencies, and trade organizations provided the bulk of the information, including (1) manufacturers and small businesses; (2) shipments; (3) detailed product information; and (4) industry trends. The analysis developed as part of the market and technology assessment is described in chapter 3 of this direct final rule TSD.

2.2.2 Technology Assessment

DOE typically uses information relating to existing and past technology options and prototype designs as inputs to determine what technologies manufacturers may use to attain higher performance levels. In consultation with interested parties, DOE develops a list of technologies for consideration. Initially, these technologies encompass all those it believes are technologically feasible.

DOE developed its list of technologically feasible design options for air cleaners through consultation with manufacturers, and from trade publications and technical papers. Because many options for improving product efficiency are available in existing units, product literature and direct examination provided additional information.

Chapter 3 of this direct final rule TSD includes the detailed list of all technology options identified for potential efficiency improvements in air cleaners.

2.3 SCREENING ANALYSIS

The screening analysis examines various technologies as to whether they: (1) are technologically feasible; (2) are practicable to manufacture, install, and service; (3) have an adverse impact on product utility or availability; (4) have adverse impacts on health and safety; and (5) utilize proprietary technology that represents a unique pathway to achieve a given efficiency level. DOE developed an initial list of efficiency-enhancement options from the technologies identified as technologically feasible in the technology assessment. Then DOE, in consultation with interested parties, reviewed the list to determine if these options are practicable to manufacture, install, and service; would adversely affect product utility or availability; would have adverse impacts on health and safety; or utilize proprietary technologies. In the engineering analysis, DOE further considered efficiency enhancement options that it did not screen out in the screening analysis. Chapter 4 of this direct final rule TSD contains details on the screening analysis for air cleaners.

2.4 ENGINEERING ANALYSIS

The purpose of the engineering analysis is to establish the relationship between the efficiency and cost of air cleaners. There are two elements to consider in the engineering analysis; the selection of efficiency levels to analyze (*i.e.*, the “efficiency analysis”) and the

determination of product cost at each efficiency level (*i.e.*, the “cost analysis”). In determining the performance of higher-efficiency products/equipment, DOE considers technologies and design option combinations not eliminated by the screening analysis. For each product class, DOE estimates the baseline cost, as well as the incremental cost for the product at efficiency levels above the baseline. The output of the engineering analysis is a set of cost-efficiency “curves” that are used in downstream analyses (*i.e.*, the LCC and PBP analyses and the NIA). Chapter 5 of this direct final rule TSD discusses the product classes DOE analyzed, the representative baseline units, the incremental efficiency levels, the methodology DOE used to develop the manufacturer production costs (MPCs), the cost-efficiency curves, and the impact of efficiency improvements on the considered products.

DOE typically uses one of two approaches to develop efficiency levels for the engineering analysis: (1) relying on observed efficiency levels in the market (*i.e.*, the efficiency-level approach), or (2) determining the incremental efficiency improvements associated with incorporating specific design options to a baseline model (*i.e.*, the design-option approach). Using the efficiency-level approach, the efficiency levels established for the analysis are determined based on the market distribution of existing products (in other words, based on the range of efficiencies and efficiency level “clusters” that already exist on the market). Using the design option approach, the efficiency levels established for the analysis are determined through detailed engineering calculations and/or computer simulations of the efficiency improvements from implementing specific design options that have been identified in the technology assessment. DOE may also rely on a combination of these two approaches. For example, the efficiency-level approach (based on actual products on the market) may be extended using the design option approach to interpolate to define “gap fill” levels (to bridge large gaps between other identified efficiency levels) and/or to extrapolate to the max-tech level (particularly in cases where the max-tech level exceeds the maximum efficiency level currently available on the market).

For this direct final rule, DOE primarily used the efficiency-level approach. DOE also used the design-option approach, testing and physically disassembling commercially available products to fill gaps where data was not available from the efficiency-level approach. DOE then considered the design options manufacturers would likely rely on to improve product efficiencies. From this information, DOE estimated the cost and efficiency impacts of incorporating specific design options at each efficiency level.

Chapter 5 of this direct final rule TSD describes the methodology and results of the analysis used to derive the cost-efficiency relationships.

2.5 MARKUPS ANALYSIS

DOE analyzed product markups to convert the MPC estimated in the engineering analysis to consumer prices, which then were used in the LCC and PBP. DOE calculated an industry-average manufacturer markup to convert MPC to MSP. DOE calculated distribution chain markups for baseline products (baseline markups) and for more efficient products (incremental markups). An incremental markup relates the change in the MSP of higher-

efficiency models (the incremental cost increase) to the change in the retailer or distributor sales price.

To develop markups, DOE identified how the products are distributed from the manufacturer to the consumer. After establishing appropriate distribution channels, DOE used data from sources that included the financial filings of manufacturers and distributors to determine how prices are marked up as the products pass from the manufacturer to the consumer.

Chapter 6 of this direct final rule TSD provides details on DOE's development of markups for air cleaners.

2.6 ENERGY USE ANALYSIS

The energy use characterization, which assesses the energy savings potential from higher efficiency levels, provides the basis for the energy savings values used in the LCC and subsequent analyses. Studies show that measurements of field energy use often vary considerably from the rated usage as determined by the DOE test procedure. To determine the field energy use by products that would meet possible energy efficiency standards, DOE used weather data from the National Oceanic and Atmospheric Administration in conjunction with housing and building data from the *Residential Energy Consumption Survey (RECS) 2020*¹ and *Commercial Building Energy Consumption Survey (CBECS) 2018*.²

Although *RECS 2020* and *CBECS 2018* are the most recent national residential household and commercial building surveys available, neither of them contains information about air cleaner ownership or usage. In the absence of such data, DOE included all household samples from *RECS 2020*, and all non-vacant and operating building samples from *CBECS 2018* for energy use calculations. The weighting indicates the number of residential households or commercial buildings in the United States that a particular sample represents. DOE adjusted the EIA's assigned sample weights to reflect geographic and principal building activity distributions of ownership of air cleaners for residential and commercial applications respectively based on the Evergreen Economics market research report.³

Chapter 7 of this direct final rule TSD provides more detail about DOE's approach for characterizing energy use of air cleaners.

2.7 LIFE-CYCLE COST AND PAYBACK PERIOD ANALYSIS

In determining whether an energy efficiency standard is economically justified, DOE considers the economic impacts of potential standards on consumers. The effects of new or amended standards on individual consumers usually include a reduction in operating cost and an increase in purchase price. DOE used the following two metrics to measure consumer impacts.

- LCC, the total consumer cost of a product, generally throughout the life of the product. The LCC calculation includes total installed cost (MSP, distribution markups, sales tax, and installation costs); operating costs (energy, repair, and maintenance costs); product

lifetime; and a discount rate. Future operating costs are discounted to the time of purchase and summed over the lifetime of the product.

- PBP, a measure of the amount of time it takes consumers to recover the assumed higher purchase price of more energy efficient products through reduced operating costs. Inputs to the calculation of PBP include the installed cost to the consumer and first-year operating costs.

In determining the LCC and PBP, DOE used data regarding engineering performance, energy use, and markups. DOE generated LCC and PBP results using a simulation approach in which certain key inputs to the analysis consist of probability distributions rather than single-point values. That analytical technique produces outcomes that also can be expressed as probability distributions. As a result, the analysis produces a range of LCC and PBP results, which enables DOE to identify the fraction of consumers achieving LCC savings or incurring net cost at each considered efficiency level.

Chapter 8 of this direct final rule TSD describes the LCC and PBP analysis.

2.8 SHIPMENTS ANALYSIS

DOE projected future shipments of air cleaners based on an analysis of key market drivers. Projections of shipments are needed to calculate the potential effects of standards on national energy use, NPV, and future manufacturer cash flows. DOE generated shipments projections for each product class. The projections estimate the total number of air cleaners shipped each year during the analysis period: 2024-2057 for trial standard level (TSL) 3 and 2028-2057 for the other TSLs. To create the projections, DOE combined current-year shipments with results of a shipments model that incorporates key market drivers for air cleaners. Chapter 9 of this direct final rule TSD provides additional details on the shipments analysis.

2.9 NATIONAL IMPACT ANALYSIS

The NIA assesses the aggregate impacts at the national level of potential energy conservation standards for each of the considered products, as measured by the NPV of total consumer economic impacts and the NES. DOE determined the NPV and NES for the efficiency levels considered for each of the product classes analyzed. To make the analysis more accessible and transparent to all interested parties, DOE prepared a Microsoft Excel spreadsheet model to forecast national energy consumption and the national consumer economic costs and savings resulting from new standards. The spreadsheet model uses typical values as inputs (as opposed to probability distributions). To assess the effect of input uncertainty on NES and NPV results, DOE may conduct sensitivity analyses by running scenarios on specific input variables.

2.9.1 National Energy Savings

The inputs for determining the NES for each product analyzed are: (1) annual energy consumption per unit, (2) shipments, (3) product stock, (4) national energy consumption, and (5)

site-to-power plant energy and full-fuel-cycle (FFC) conversion factors. DOE calculated the national energy consumption by multiplying the number of units, or stock, of each product (by vintage, or age) by the unit energy consumption (also by vintage). DOE calculated annual NES based on the difference in unit energy consumption between the no-new standards case (without new efficiency standards) and for each higher efficiency standard. DOE estimated energy consumption and savings based on site energy, and converted the electricity consumption and savings to FFC energy. Cumulative energy savings are the sum of the NES for each year.

2.9.2 Net Present Value of Consumer Benefits

The inputs for determining NPV of the total costs and benefits experienced by consumers of the considered appliances are: (1) total annual installed cost, (2) total annual savings in operating costs, (3) a discount factor, (4) present value of costs, and (5) present value of savings. DOE calculated net savings each year as the difference between the no-new-standards case and each standards case in total savings in operating costs and total increases in installed costs. DOE calculated savings over the life of each product. NPV is the difference between the present value of operating cost savings and the present value of total installed costs.

DOE calculated increases in total installed costs as the product of the difference in total installed cost between the no-new standards case and standards case (*i.e.*, once the standards take effect). DOE expressed savings in operating costs as decreases associated with the lower energy consumption of products bought in the standards case compared to the no-new-standards efficiency case. Total savings in operating costs are the product of savings per unit and the number of units of each vintage that survive in a given year.

To calculate the NPV, DOE used a discount factor based on real discount rates of 3 percent and 7 percent to discount future costs and savings to present values. Chapter 10 of this direct final rule TSD provides additional details on the NIA.

2.10 CONSUMER SUBGROUP ANALYSIS

The consumer subgroup analysis evaluates impacts on any identifiable groups or consumers who may be disproportionately affected by a national energy conservation standard. DOE analyzed the LCCs and PBPs for consumers that fall into any identifiable groups. Chapter 11 of this direct final rule TSD provides additional details on the consumer subgroup analysis.

2.11 MANUFACTURER IMPACT ANALYSIS

The purpose of the MIA is to identify and quantify the impacts of any new or amended energy conservation standards on manufacturers. The MIA has both quantitative and qualitative aspects, and includes the analyses of projected industry cash flows, the industry net present value (INPV), conversion costs, and direct employment. Additionally, the MIA seeks to describe how new or amended energy conservation standards might affect manufacturing capacity and competition, as well as how standards contribute to overall regulatory burden. Finally, the MIA seeks to identify any disproportionate impacts on manufacturer subgroups, including small

business manufacturers. The Department analyzes the impact of standards on manufacturers with substantial input from manufacturers and other interested parties.

DOE conducts the MIA in three phases and further tailors its analytical framework based on the comments it receives. In phase I, DOE creates an industry profile to characterize the industry and to identify important issues that require consideration. In phase II, DOE prepares an industry cash-flow model and determined what information to discuss with manufacturers during manufacturer interviews. In phase III, DOE interviews manufacturers and assesses the impacts of potential standards both quantitatively and qualitatively. DOE calculates industry and subgroup cash flow and INPV using the Government Regulatory Impact Model (GRIM). DOE then assesses impacts on competition, manufacturing capacity, direct employment, and regulatory burden based on manufacturer interview feedback. Chapter 12 of this direct final rule TSD provides additional details on the MIA.

2.12 EMISSIONS IMPACT ANALYSIS

In the emissions analysis, DOE estimates the reduction in power sector combustion emissions of carbon dioxide (CO₂), nitrogen oxides (NO_x), sulfur dioxide (SO₂), mercury (Hg), methane (CH₄) and nitrous oxide (N₂O) from potential energy conservation standards for the considered products. In addition, DOE estimates emissions impacts in production activities (extracting, processing, and transporting fuels) that provide the energy inputs to power plants and for site combustion. Together, these emissions account for the FFC.

The emissions analysis consists of two components. The first component estimates the effect of potential energy conservation standards on power sector and site (where applicable) combustion emissions of CO₂, NO_x, SO₂, and Hg. The second component estimates the impacts of potential standards on emissions of two additional greenhouse gases, CH₄ and N₂O, as well as the reductions to emissions of all species due to “upstream” activities in the fuel production chain. These upstream activities comprise extraction, processing, and transporting fuels to the site of combustion. The upstream emissions include emissions from fuel combustion during extraction, processing, and transportation of fuel, and “fugitive” emissions (direct leakage to the atmosphere) of CH₄ and CO₂.

The analysis of power sector emissions uses marginal emissions factors that are derived from data in *AEO*. The *AEO* incorporates the projected impacts of existing air quality regulations on emissions. *AEO* generally represents current legislation and environmental regulations, including recent government actions, for which implementing regulations were available as of the time of its preparation.

The methodology is described in more detail in chapter 13 of this direct final rule TSD.

2.13 MONETIZATION OF EMISSIONS REDUCTION BENEFITS

As part of its assessment of energy conservation standards, DOE considered the estimated monetary benefits likely to result from the reduced emissions of CO₂, CH₄, N₂O, SO₂,

and NO_x that are expected to result from each of the potential standard levels considered in the direct final rule phase of the rulemaking. To monetize the benefits of reducing greenhouse gas emissions this analysis uses the interim estimates presented in the *Technical Support Document: Social Cost of Carbon, Methane, and Nitrous Oxide Interim Estimates Under Executive Order 13990* published in February 2021 by the Interagency Working Group on the Social Cost of Greenhouse Gases (IWG).

To estimate the monetary value of reduced NO_x and SO₂ emissions from electricity generation attributable to the standard levels it considers, DOE uses benefit-per-ton estimates derived from analysis conducted by the U.S. Environmental Protection Agency (EPA). For NO_x and SO₂ emissions from combustion at the site of product use, DOE uses another set of benefit-per-ton estimates published by the EPA.

For more detail on the monetization of emissions analysis, see chapter 14 of this direct final rule TSD.

2.14 UTILITY IMPACT ANALYSIS

In the utility impact analysis, DOE analyzed the changes in electric installed capacity and electricity generation that are projected for each considered trial standard level. For electric utilities, the analysis is based on output of the DOE/EIA's National Energy Modeling System (NEMS). NEMS is a public domain, multi-sectored, partial equilibrium model of the U.S. energy sector. Each year, DOE/EIA uses NEMS to produce an energy forecast for the United States, the *AEO*. The EIA publishes a reference case, which incorporates all existing energy-related policies at the time of publication, and a variety of side cases which analyze the impact of different policies, energy price and market trends. DOE used a methodology based on results published for the *AEO 2022* reference case and a set of side cases that implemented a variety of efficiency-related policies. Further detail is provided in chapter 15 of this direct final rule TSD.

2.15 EMPLOYMENT IMPACT ANALYSIS

Energy conservation standards can affect employment both directly and indirectly. Direct employment impacts are changes in the number of employees at the plants that produce the covered products. DOE evaluated direct employment impacts in the MIA. Indirect employment impacts may result from expenditures shifting between goods (the substitution effect) and changes in income and overall expenditure levels (the income effect) that occur due to standards. DOE defines indirect employment impacts from standards as net jobs eliminated or created in the general economy as a result of increased spending driven by increased product prices and reduced spending on energy.

Indirect employment impacts were investigated in the employment impact analysis using the Pacific Northwest National Laboratory's "Impact of Sector Energy Technologies" (ImSET) model.⁴ The ImSET model was developed for DOE's Office of Planning, Budget, and Analysis to estimate the employment and income effects of energy-saving technologies in buildings, industry, and transportation. Compared with simple economic multiplier approaches, ImSET

allows for more complete and automated analysis of the economic impacts of energy conservation investments. Further detail is provided in chapter 16 of this direct final rule TSD.

2.16 REGULATORY IMPACT ANALYSIS

DOE prepared a regulatory impact analysis (RIA). DOE identified major alternatives to standards that represent feasible policy options to reduce the energy consumption of air cleaners. DOE evaluated each alternative in terms of its ability to achieve significant energy savings at a reasonable cost, and compared the effectiveness of each alternative to the effectiveness of the adopted standard. DOE recognized that voluntary or other non-regulatory efforts by manufacturers, utilities, and other interested parties can substantially affect energy efficiency or reduce energy consumption. DOE based its assessment on the recorded impacts of any such initiatives to date, but also considered information presented by interested parties regarding the impacts current initiatives may have in the future. Further detail on the RIA is provided in chapter 17 of this direct final rule TSD

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CHAPTER 3. MARKET AND TECHNOLOGY ASSESSMENT

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CHAPTER 3. MARKET AND TECHNOLOGY ASSESSMENT

3.1 INTRODUCTION

Air cleaners are a consumer product intended to improve indoor air quality. Based on a survey conducted by Evergreen Economics for Pacific Gas & Electric, and submitted by the California Investor Owned Utilities as part of their comment in response to DOE's initial request for information, the primary reasons for consumers to utilize air cleaners are to combat allergies and reduce germs.¹ Additionally, for consumers living in areas that are prone to wildfires, smoke reduction is a very important factor to purchase air cleaners.

This chapter provides an assessment of the air cleaners industry in the United States. The U.S. Department of Energy (DOE) developed the market and technology assessment presented in this chapter primarily from publicly available information. This assessment is helpful in identifying the major manufacturers and their product characteristics, which form the basis for the engineering and the life-cycle cost (LCC) analyses. Present and past industry structure and industry financial information help DOE in the process of conducting the manufacturer impact analysis.

3.2 PRODUCT DEFINITIONS

Title 10 of the Code of Federal Regulations (CFR) part 430.2 includes the following definitions pertaining to air cleaners:

Air cleaner means a product for improving indoor air quality, other than a central air conditioner, room air conditioner, portable air conditioner, dehumidifier, or furnace, that is an electrically-powered, self-contained, mechanically encased assembly that contains means to remove, destroy, or deactivate particulates, VOC [volatile organic compounds], and/or microorganisms from the air. It excludes products that operate solely by means of ultraviolet light without a fan for air circulation.

Conventional room air cleaner means an air cleaner that (1) is a portable or wall mounted (fixed) unit that plugs into an electrical outlet; (2) operates with a fan for air circulation; and, (3) contains means to remove, destroy, and/or deactivate particulates. The term portable is as defined in section 2.1.3.1 of AHAM AC-7-2022^a and fixed is as defined in section 2.1.3.2 of AHAM AC-7-2022.

As discussed elsewhere in this technical support document (TSD), DOE is establishing new energy conservation standards for air cleaners, consisting of standards for conventional

^a AHAM AC-7-2022, "Energy Test Method for Consumer Room Air Cleaners." AHAM AC-7-2022 may be purchased at:
<https://www.aham.org/ItemDetail?iProductCode=30014&Category=PADSTD&WebsiteKey=69a0a5fb-295a-4894-acd0-5785f146b899>

room air cleaners. Accordingly, DOE conducted its analyses for conventional room air cleaners as defined above.

3.3 PRODUCT CLASSES

When evaluating and establishing energy conservation standards, DOE may establish separate standards for a group of covered products (*i.e.*, establish a separate product class) if DOE determines that separate standards are justified based on the type of energy used, or if DOE determines that a product’s capacity or other performance-related feature justifies a different standard. (42 U.S.C. 6295(q)(1)(A) and (B)) In making a determination whether a performance related feature justifies a different standard, DOE must consider factors such as the utility to the consumer of the feature and other factors DOE determines are appropriate. (42 U.S.C. 6295(q)(1))

Air cleaners are a newly covered product; as such, DOE currently does not specify any energy conservation standards or associated product classes for air cleaners, and more specifically, conventional room air cleaners, which are in the scope of this rulemaking. First, to determine if separate product classes are warranted for air cleaners, DOE reviewed the U.S. Environmental Protection Agency’s (EPA’s) ENERGY STAR® Version 2.0 Room Air Cleaners Specification, Rev. May 2022, (ENERGY STAR V. 2.0), which specifies three separate “bins” based on capacity as defined by the smoke clean air delivery rate (CADR), expressed in cubic feet per minute (cfm). The ENERGY STAR V. 2.0 capacity bins are shown in Table 3.3.1.

Table 3.3.1 ENERGY STAR V. 2.0 Smoke CADR Bins

Smoke CADR Bins
$30 \leq \text{Smoke CADR} < 100$
$100 \leq \text{Smoke CADR} < 150$
$\text{Smoke CADR} \geq 150$

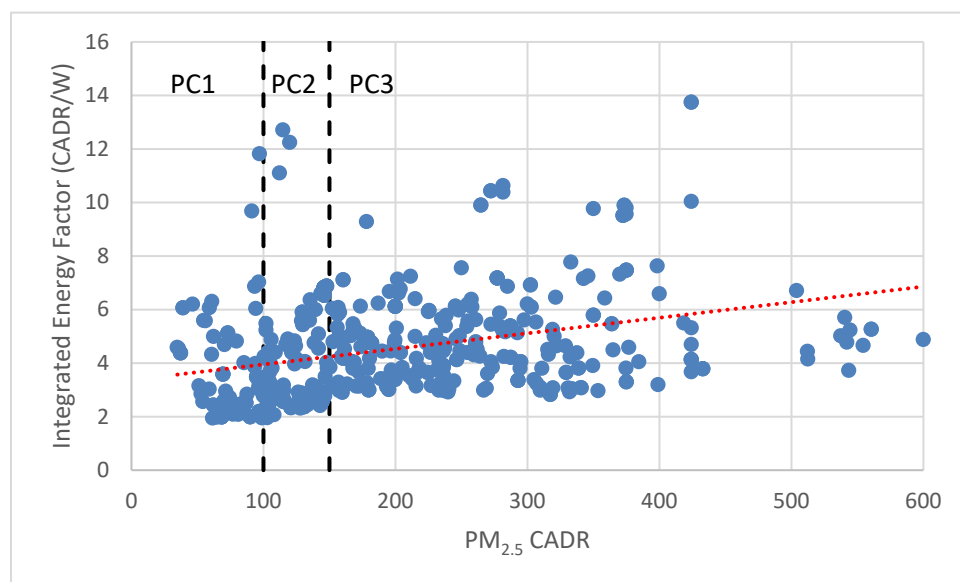
As discussed in section 3.4, DOE’s newly established air cleaners test procedure establishes the integrated energy factor (IEF) metric in terms of PM_{2.5} CADR, which is the geometric mean of smoke CADR and dust CADR and is also expressed in cfm. As part of the test procedure rulemaking, DOE determined that smoke CADR and PM_{2.5} CADR for a given air cleaner unit are very similar. Accordingly, given that the test procedure established a metric utilizing PM_{2.5} CADR, DOE determined that the appropriate capacity metric for establishing air cleaner product classes is PM_{2.5} CADR.

Next, to determine whether separate product classes based on capacity are justified and, if justified, the appropriate capacity range for each product class, DOE evaluated the ENERGY STAR product database² for conventional room air cleaners and conducted investigatory testing and product teardowns. DOE determined that the three separate product classes, as shown in Table 3.3.2, are appropriate based on its analysis. The following paragraphs discuss DOE’s analysis and reasons for establishing the three product classes shown in Table 3.3.2.

Table 3.3.2 Air Cleaner Product Classes

Product Class	PM _{2.5} CADR Bins
PC1	$10 \leq \text{PM}_{2.5} \text{ CADR} < 100$
PC2	$100 \leq \text{PM}_{2.5} \text{ CADR} < 150$
PC3	$\text{PM}_{2.5} \text{ CADR} \geq 150$

As seen in Figure 3.3.1, the ENERGY STAR database shows that air cleaner models (shown as blue dots in the graph) at lower CADR values generally have lower efficiencies compared to models at higher CADR (as seen by the red line that is a linear regression through all data points). DOE expects that this is likely due to the smaller motor and/or filter implemented in the lower-CADR units, which are typically intended to be used in rooms with smaller areas (*e.g.*, units in Product Class 1 would be recommended for a maximum room size of 155 square feet (sq. ft.)). To achieve a certain level of cleaning performance, a smaller unit would need to include more filtration by volume in a more limited chassis space (*i.e.*, the air cleaner cabinet). This would increase the pressure drop across the filter, which would require more blower power to maintain the same air delivery performance. These factors impact the overall efficiency of the unit. At higher CADR values (*i.e.*, air cleaners designed for larger rooms), the cabinet volume is much larger, which allows the incorporation of a much larger filter (*i.e.*, the filtration can be spread across a larger filter area), thereby reducing the pressure drop across the filter and necessary blower power, and therefore improving efficiency.

**Figure 3.3.1 ENERGY STAR Qualified Air Cleaners as of December 2022**

Establishing separate product classes for units that are intended to be used in both smaller and larger rooms is necessary to maintain consumer utility. For example, Product Class 1 units have a small cabinet volume (<0.6 cubic feet (ft³)), are designed for use in a single small room

such as a bathroom or bedroom (<155 sq. ft.), and are easily portable, which can allow product configurations such as tabletop or wall plug-ins. Units with larger capacities and corresponding larger cabinet volumes provide different utility to consumers. Product Class 2 includes medium cabinet-sized units (0.6–1.2 ft³) which are designed for a larger room (155–235 sq. ft.) such as a kitchen or living space. The size and weight of these units generally allow single-person portability without necessitating the use of wheels. Finally, Product Class 3 units have a large cabinet (> 1.6 ft³), are typically less portable than lower-capacity units, in some cases being equipped with wheels to facilitate moving, and are designed to be used for an extended duration in a large room (>235 sq. ft) such as a classroom, office, or large living area. Establishing these product classes is necessary because the three ranges of capacity each provide distinct consumer utility in terms of the application based on room size and portability of the unit and are associated with inherently different efficiency due to the different filter size and configurations that can be accommodated. Further, these product class distinctions will help ensure that higher-capacity units installed in smaller-sized rooms, which achieve higher efficiencies at the same active mode power consumption than smaller-capacity units and which warrant more stringent energy conservation standards, do not lead to unnecessarily high AEC.

DOE notes that its established product classes are also consistent with the product classes proposed in the Joint Proposal by the Joint Stakeholders.^b (More information on the Joint Proposal and Joint Stakeholders is available in chapter 1 of this direct final rule TSD.) DOE additionally notes that it is establishing Product Class 1 with a PM_{2.5} CADR lower limit of 10 cfm as opposed to 30 cfm, as specified in the ENERGY STAR V. 2.0 specification, so that tabletop and desktop portable room air cleaners as well as plug-in air cleaners, which is a growing segment of the market, will be required to demonstrate compliance with the adopted standards. DOE notes that the PM_{2.5} CADR lower limit of 10 cfm for Product Class 1 is also recommended by the Joint Stakeholders in the Joint Proposal.

Additionally, although Product Class 3 as established in this direct final does not specify an upper threshold of PM_{2.5} CADR, theoretical limitations of maximum CADR measurements by the DOE test procedure (discussed in section 3.4) result in the test procedure being applicable only to air cleaners with smoke CADR and dust CADR values no greater than 600 cfm. As discussed in section 3.4, PM_{2.5} CADR is calculated as the geometric average of smoke CADR and dust CADR. Therefore, the theoretical limit of maximum PM_{2.5} CADR is 600 cfm as well. Accordingly, DOE conducted its analyses for this direct final rule and is establishing energy conservation standards only for conventional room air cleaners with a PM_{2.5} CADR of 10–600 cfm (inclusive).

3.4 PRODUCT TEST PROCEDURES

On March 6, 2023, DOE established a new test procedure for air cleaners at 10 CFR part 430, subpart B, appendix FF (appendix FF). 88 FR 14014. Appendix FF is effective starting

^b On August 23, 2022, DOE received a Joint Proposal from the Joint Stakeholders, representing manufacturers, efficiency advocates, and consumer groups, regarding energy conservation standards and related test procedures for air cleaners. The Joint Proposal specified two tiers of standard levels (*i.e.*, Tier 1 and Tier 2), with proposed compliance dates of December 31, 2023, and December 31, 2025, respectively.

April 5, 2023. Specifically, appendix FF established an IEF metric, expressed in terms of PM_{2.5} CADR per watt (PM_{2.5} CADR/W), which measures the reduction rate of PM_{2.5} particulates in a given room volume per unit power. The numerator of the IEF metric is PM_{2.5} CADR, which is the geometric average of smoke CADR and dust CADR, where each of these CADR metrics refers to the reduction rate of smoke and dust particles, respectively, in a given room volume with the air cleaner operating. The denominator of the IEF metric is the annual energy consumption (AEC) in active mode and standby mode divided by the annual operating hours in active mode, thereby representing the annual average power of the air cleaner.

Appendix FF references two Association of Home Appliance Manufacturers (AHAM) industry standards: ANSI/AHAM AC-1-2020, *Portable Electric Room Air Cleaners*, and AHAM AC-7-2022, *Energy Test Method for Consumer Room Air Cleaners*.^c AHAM AC-1-2020 provides test methods to measure the relative reduction of particulate matter, including smoke and dust, suspended in the air in a specified test chamber when an air cleaner is in operation, which is used to calculate smoke CADR and dust CADR. AHAM AC-7-2022 provides test methods to measure an air cleaner's active mode and standby mode power consumption, which is used to determine its AEC. AHAM AC-7-2022 also specifies the methods to determine PM_{2.5} CADR and IEF.

The tests to determine smoke CADR and dust CADR in AHAM AC-1-2020 specify particle size for smoke particles of 0.1 to 1 micrometers (µm) and 0.5 to 3 µm for dust particles. However, the calculation of PM_{2.5} CADR in AHAM AC-7-2022 specifies a narrower range of allowable particle sizes for smoke (0.1 to 0.5 µm) and dust (0.5 to 2.5 µm) compared to AHAM AC-1-2020.

DOE additionally noted that although there is a difference in the allowable particle size for smoke and dust between AHAM AC-1-2020 and the PM_{2.5} CADR calculation in AHAM AC-7-2022, the subset smoke CADR and dust CADR used to calculate PM_{2.5} CADR are nearly identical to the smoke CADR and dust CADR calculated according to AHAM AC-1-2020, as shown in the figures included in the Joint Proposal.^d Accordingly, section 5.1.2 of appendix FF specifies that PM_{2.5} CADR may alternatively be calculated using the full range of particles used to calculate smoke CADR and dust CADR according to AHAM AC-1-2020. In the air cleaners test procedure final rule, DOE had stated that it may revisit allowing the use of both approaches to calculate PM_{2.5} CADR in a future standards rulemaking. 88 FR 14014, 14029-14031.

In this direct final rule, DOE is explicitly stating that the full range of particles used to calculate smoke CADR and dust CADR according to AHAM AC-1-2020 may be used to determine compliance only with the Tier 1 standards established in this direct final rule. Compliance with Tier 2 standards must be determined using the smoke and dust particle size specified in the calculation of PM_{2.5} CADR in section 5.1.1 of appendix FF. This aligns with the test parameters of the Joint Proposal and allows manufacturers more time to adjust to the tighter particle size requirements specified in AHAM AC-7-2022.

^c AHAM standards are available for purchase from AHAM's website at: www.aham.org/AHAM/AuxStore.

^d See Joint Stakeholders, No. 16 at p. 6.

3.5 MANUFACTURER TRADE GROUPS

DOE recognizes the importance of trade groups in disseminating information and promoting the interests of the industry that they support. To gain insight into the air cleaner industry, DOE researched various associations available to manufacturers, suppliers, and users of such equipment.

DOE identified AHAM^c as a trade group that supports, or has an interest in, the air cleaner industry. AHAM, formed in 1967, aims to enhance the value of the home appliance industry through leadership, public education, and advocacy. AHAM provides services to its members including government relations; certification programs for room air conditioners, dehumidifiers, refrigerators and freezers, residential clothes washers, consumer clothes dryers, and room air cleaners; an active communications program; and technical services and research. In addition, AHAM conducts other market and consumer research studies. AHAM also develops and maintains technical standards for various appliances to provide uniform, repeatable procedures for measuring specific product characteristics and performance features.

3.6 MANUFACTURER INFORMATION

The following sections detail information regarding manufacturers of air cleaners, including characterization of identified manufacturers (section 3.6.1), potential small business impacts (section 3.6.2), and product distribution channels (section 3.6.3).

3.6.1 Manufacturers

DOE identified manufacturers from AHAM's Database of Certified Room Air Cleaners,³ ENERGY STAR's Product Finder dataset,⁴ and California Air Resources Board's (CARB) CARB-Certified Air Cleaning Devices,⁵ and other publicly-available information. DOE identified 43 original equipment manufacturers (OEMs) that manufacture air cleaners covered by this rulemaking. Based on these databases, OEMs commonly offer products under multiple brand names, and air cleaner manufacturing is spread out amongst manufacturers. In terms of model counts, the three OEM's with the most models accounted for only 22 percent of all unique models identified in the combined database. Additionally, almost all air cleaner manufacturing occurs outside the United States, usually in Asia. Of the 43 OEMs, four are OEMs with U.S. manufacturing of conventional air cleaners. These companies have distinct designs and manufacturing processes from companies that import air cleaners. Most of the OEMs are foreign-owned companies with manufacturing facilities outside of the United States and rely largely on injection molding.

^c More information about AHAM is available at www.aham.org.

3.6.2 Small Business Impacts

DOE evaluates the impacts of promulgated energy conservation standards on small business manufacturers of air cleaners. The Small Business Administration (SBA) considers an entity to be a small business if, together with its affiliates, it employs less than a threshold number of employees as specified in 13 CFR part 121, which relies on size standards and codes established by the North American Industry Classification System (NAICS).

Manufacturing of air cleaners is classified under NAICS code 335210, “Small Electrical Appliance Manufacturing.” In 13 CFR 121.201, the SBA sets a threshold of 1,500 employees or fewer for an entity to be considered as a small business for this category. Based on the SBA threshold applicable to air cleaners, DOE identified four small, domestic OEMs of air cleaners. See chapter 12 of this direct final rule TSD for more details.

3.6.3 Distribution Channels

Understanding the distribution channels of products covered by this rulemaking is an important facet of the market assessment. DOE gathered information regarding the distribution channels for air cleaners from publicly available sources and market reports.

The majority of consumers purchase their appliances directly from retailers. These retailers include: (1) home improvement, appliance, and department stores; (2) Internet retailers; (3) membership warehouse clubs; and (4) kitchen remodelers.

Home appliance retailers generally obtain products directly from manufacturers. Based on data from manufacturer interviews, the primary distribution channel for air cleaners for both the commercial and residential markets is through online vendors (e-commerce), accounting for around 60 percent of all air cleaner sales across the total market. Other distribution channels include manufacturer to retailer, including both in-store purchases and through retailer websites, and a direct sales channel for commercial applications, which includes air cleaner sales from manufacturer to wholesaler/dealer to the consumer. See chapter 6 of this direct final rule TSD for more information on distribution channels.

3.7 REGULATORY PROGRAMS

Air cleaners are not currently subject to U.S. Federal energy conservation standards. Additionally, at this time, DOE is not aware of any foreign regulatory programs for air cleaners. The following sections detail U.S. States’ regulatory initiatives for air cleaners.

3.7.1 State Energy Conservation Standards

The District of Columbia, Maryland, Nevada, New Jersey, and Washington have adopted standards for air cleaners that are either based on the ENERGY STAR Version 1.2 Room Air Cleaners Specification, Rev. July 2011, (ENERGY STAR V. 1.2), or ENERGY STAR V. 2.0.

Specifically, the District of Columbia adopted standards in 2020, Maryland adopted standards in 2022, and Nevada and New Jersey adopted standards in 2021, as shown in Table 3.7.1. The District of Columbia and New Jersey State standards went into effect in 2022 while the Nevada State standard is expected to go into effect in 2023 and the Maryland State standard is expected to go into effect in 2024.

Table 3.7.1 Air Cleaner Standards Adopted by the District of Columbia and the States of Maryland, Nevada, and New Jersey

Smoke CADR Bins	Minimum Smoke CADR/W
$30 \leq \text{PM}_{2.5} \text{ CADR} < 100$	1.7
$100 \leq \text{PM}_{2.5} \text{ CADR} < 150$	1.9
$\text{PM}_{2.5} \text{ CADR} \geq 150$	2.0

Note: These standards are based on smoke CADR divided by the active mode power consumption, which is different from the IEF metric specified in appendix FF.

Washington State adopted the standards shown in Table 3.7.2 in 2022 with an effective date in 2024.

Table 3.7.2 Air Cleaner Standards Adopted by Washington State

Smoke CADR Bins	Minimum Smoke CADR/W
$30 \leq \text{PM}_{2.5} \text{ CADR} < 100$	1.9
$100 \leq \text{PM}_{2.5} \text{ CADR} < 150$	2.4
$\text{PM}_{2.5} \text{ CADR} \geq 150$	2.9

Note: These standards are based on smoke CADR divided by the active mode power consumption, which is different from the IEF metric specified in appendix FF.

As noted by the Joint Stakeholders in the Joint Proposal, the Tier 1 standards proposed in the Joint Proposal are equivalent to the standards adopted by the District of Columbia, Maryland, Nevada, and New Jersey while the Tier 2 standards proposed in the Joint Proposal are equivalent to the ENERGY STAR V. 2.0 specification and standards adopted by Washington State.

3.8 VOLUNTARY PROGRAMS

ENERGY STAR, a voluntary labeling program backed by EPA and DOE, identifies energy efficient products through a qualification process. To qualify, a product must exceed Federal minimum standards by a specified amount, or if no Federal standard exists, exhibit selected energy-saving features. The ENERGY STAR program works to recognize the top quartile of products on the market, meaning that approximately 25 percent of products on the market should meet or exceed the ENERGY STAR levels. ENERGY STAR specifications exist for several products, including air cleaners.

On October 16, 2019, EPA finalized ENERGY STAR V. 2.0, which took effect on October 17, 2020, and set minimum smoke CADR/W requirements for three different CADR bins. The ENERGY STAR V. 2.0 criteria for air cleaners are listed in Table 3.8.1.

Table 3.8.1 ENERGY STAR V. 2.0 Requirements

Smoke CADR Bins	Minimum Smoke CADR/W
$30 \leq \text{CADR} < 100$	1.9
$100 \leq \text{CADR} < 150$	2.4
$\text{CADR} \geq 150$	2.9

Based on a review of the ENERGY STAR air cleaners database, DOE identified 24 air cleaner OEMs participating in the ENERGY STAR program. These 24 OEMs represent 78 different air cleaner brands that cover approximately 544 total unique basic models in the ENERGY STAR database. In lieu of DOE certification data for air cleaners, DOE relied on publicly available information on the manufacturing industry structure to match air cleaner brands with OEMs and to estimate the number of OEMs with ENERGY STAR-qualified products. Additionally, DOE estimated that there is only one OEM per brand. ENERGY STAR does not provide market penetration data for air cleaners.

Beyond its labeling program, ENERGY STAR has established Most Efficient product criteria for many household appliances, including air cleaners. ENERGY STAR began its Most Efficient criteria for air cleaners in 2023. Products must be ENERGY STAR-certified consistent with requirements in the ENERGY STAR V. 2.0 test method to be eligible for qualification or the 2023 Most Efficient distinction. As of January 2023, 48 unique basic models qualify for the ENERGY STAR Most Efficient criteria. The performance requirements are listed in Table 3.8.2.

Table 3.8.2 2023 ENERGY STAR Most Efficient Requirements for Air Cleaners

Smoke CADR Bins	Minimum Smoke CADR/W
$30 \leq \text{CADR} < 100$	5.4
$100 \leq \text{CADR} < 150$	6.6
$\text{CADR} \geq 150$	7.6

3.9 HISTORICAL SHIPMENTS

Awareness of annual product shipment trends is an important aspect of the market assessment and in the development of the standards rulemaking. DOE reviewed data collected by the U.S. Census Bureau and AHAM to evaluate residential appliance product shipment trends and the value of these shipments, which were used during the shipments analysis (chapter 9 of this direct final rule TSD.)

3.9.1 Unit Shipments

DOE used data on historical air cleaner shipments (domestic shipments plus imports) in this analysis. DOE's source for historical shipments data was EuroMonitor.⁶ Shipment information was available beginning in 2007. Figure 3.9.1 shows the historical data on air cleaner shipments in the United States. As seen in Figure 3.9.1, air cleaner shipments have steadily increased since 2013. These trends are expected to continue with the increase in incidence of natural disasters such as wildfires and the ongoing COVID-19 pandemic.

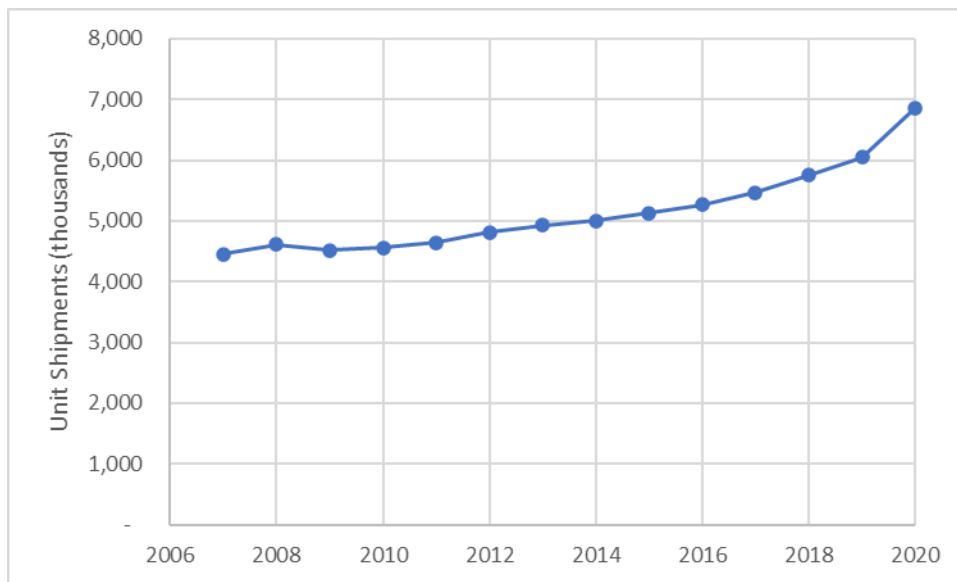


Figure 3.9.1 Industry Shipments of Air Cleaners in the United States

3.9.2 Imports

There is a large market for the import of home appliances. In this analysis, DOE assumed that imported units constitute the overwhelming majority of air cleaners sold in the United States.

3.10 PRODUCT RETAIL PRICES

DOE determined average retail prices from manufacturer production costs determined as part of the engineering analysis. Table 3.10.1 presents weighted-average retail prices (in 2021\$) by product class for the residential and commercial sectors. Retail prices were weighted by distribution channel to account for the differences in costs for online purchases compared to in-store purchases.

Table 3.10.1 Weighted Average Air Cleaner Retail Prices

Product Class	Residential Sector (2021\$)	Commercial Sector (2021\$)
$10 \leq \text{PM}_{2.5} \text{ CADR} < 100$	\$80	\$70
$100 \leq \text{PM}_{2.5} \text{ CADR} < 150$	\$100	\$100
$\text{PM}_{2.5} \text{ CADR} \geq 150$	\$160	\$150

3.11 TECHNOLOGY ASSESSMENT

This section provides a technology assessment for air cleaners. Contained in this technology assessment are details about product characteristics and operation (section 3.11.1), an examination of possible technological improvements (section 3.11.2), and a characterization of the product efficiency levels currently commercially available (section 3.11.3).

3.11.1 Product Operations and Components

In preparation for the screening and engineering analyses, DOE prepared a brief description of the characteristics and operation of air cleaners. This description provides a basis for understanding the technologies used to improve product efficiency.

An air cleaner is an appliance which combines a fan and filter, at minimum, to clean indoor air by removing, destroying, or deactivating particulates and pollutants from the air as it passes through the device. The air cleaner fan pulls in room air containing unwanted particulates and the filter captures the particulates as the air passes over them. Clean air is then circulated back into the indoor space.

Air cleaners utilize a variety of different fan and filter technologies to remove particulates from the air. The most common filter technology type are mechanical filters which are porous enough to allow air to pass, but dense enough to trap and remove particulates. Other filter types capture particulates through different methodologies, such as adsorption, electrostatic attraction, and others. In all instances, air cleaner filters are intended to capture and remove unwanted particulates from passing air. Additionally, a single air cleaner unit can utilize multiple filters to improve cleaning performance of the device. Filters are usually in a flat rectangular or cylindrical configuration.

Some air cleaners also incorporate additional particle removal technologies to the fan and filter configuration to further improve efficiency. For example, many air cleaners are equipped with an ionization device that emit negative ions that bond with the positively charged particulates in the air to make them larger, so they either are weighted down or more easily captured by the filter. There are number of specialized filtration systems that are discussed in more depth in section 3.11.2.

Air cleaner manufacturers employ different fan and blower motors that may impact the operating efficiency of the air cleaner. Commonly observed motor types are shaded-pole motors (SPM), permanent split capacitor (PSC), or brushless DC (BLDC) motors. Air cleaners usually have multiple fan speed settings for the user to select. The fan speed can be adjusted using either a mechanical or electronic control system. More advanced electronic controls typically have other features, such as air quality sensors, a timer, filter indicators, or an adjustable interface brightness setting.

Air cleaners are manufactured in a range of sizes based on CADR to serve different applications. Conventional room air cleaners are plug-in and typically available in a portable configuration. Small conventional room air cleaners at lower CADR values are often recommended to be used on a desk or tabletop, while larger CADR conventional room air cleaners are often placed on the floor and are designed for extended use in large rooms. As mentioned in section 3.3, air cleaner models at lower CADR values generally have lower efficiencies compared to models at higher CADR due to the increased blower power necessary to overcome the associated pressure drop from the smaller motor and filter system.

3.11.2 Technology Options

In order to gain a deeper understanding of the technological improvements used to increase the efficiency of air cleaners, DOE identified several possible technologies and examined the most common improvements used in today's market. Table 3.11.1 shows the list of technology options that DOE considered for this analysis. The following sections discuss the technology options associated with reducing energy consumption and improving CADR.

Table 3.11.1 Technology Options for Air Cleaners

High efficiency particulate air (HEPA)-type filter (99 percent of 0.2 µm particles)
True HEPA filter (99.97 percent of 0.3 µm particles)
Activated carbon filter
High density polyethylene (HDPE) pre-filter
Photoelectrochemical oxidation (PECO) filter
Photocatalytic oxidation (PCO) filter
Electrostatic/Polarizing media
Filter shape
Improved motor technologies
Low standby-power electronic controls
Direct double-ended blower assembly
Ionization brush
Ionization plates
Air quality sensor
Ozone generators
Thermodynamic sterilization system (TSS)
Bioreactor

3.11.2.1 HEPA-type Filter

A HEPA-type filter is a pleated mechanical air filter. HEPA-type filters use a porous filtration medium typically composed of randomly arranged polypropylene or fiberglass fibers. As air passes through the porous media, particulates in the air become trapped on the filter surface, allowing clean air to be discharged by the air cleaner. HEPA-type filters have a 99-percent efficiency rate or less at capturing particles as small as 2 microns.

3.11.2.2 True HEPA Filter

A true HEPA filter is also a type of pleated mechanical air filter, but it must remove at least 99.97 percent of all airborne particles with a size of 0.3 µm, as defined by DOE.⁷ According to the EPA, the diameter specification of 0.3 µm corresponds to the most penetrating particle size; that is, particles of 0.3 µm are the most difficult size particles to capture. Particles of smaller and larger diameters are generally captured more easily.⁸

True HEPA filters have a similar design and construction as HEPA-type filters but have a higher density; as such, they can capture smaller particles with a greater efficiency rate. Any HEPA filter that does not attain the DOE standard for HEPA filtration is not a true HEPA filter.

3.11.2.3 Activated Carbon Filter

Activated carbon filters are air filters that trap VOCs, odors, and other gaseous pollutants from the air. While HEPA-type and true HEPA filters remove pollutants by physically trapping them in the filter media, activated carbon filters rely on adsorption to cause pollutant molecules

to bind and stick to the surface of the carbon molecules. Activated carbon filters are activated by an injection of hot air, carbon, or steam. This treatment increases the surface area of the lattice structure of the carbon atoms, which greatly improves the effectiveness of the adsorption process. Activated carbon filters are typically single-use devices, and need to be replaced as they become saturated and lose their effectiveness.

3.11.2.4 High Density Polyethylene Pre-filter

Pre-filters are the first stage of filtration for air cleaners. Pre-filters work in the same manner as true HEPA or HEPA-type filters, but are coarser and are primarily used for capturing large particles. This helps to protect the main true HEPA or HEPA-type filter from getting clogged with larger debris, so those filters can more effectively capture smaller particles. The pre-filter can therefore prolong the life and effectiveness of the main filter. High density polyethylene pre-filters are typically re-usable.

3.11.2.5 Photoelectrochemical Oxidation Filter

Photoelectrochemical oxidation (PECO) is a proprietary technology that is a type of photoreactor-based air purification, similar to photocatalytic oxidation (PCO) technology (discussed in section 3.11.2.6), with some important variations. PECO processes pollutants in a photoreactor that utilizes photons to initiate a reaction that oxidizes and destroys organic pollutants in the air. The reaction converts pollutants into non-toxic substances. Specifically, PECO works by shining UV-A light on the catalytic surface of the PECO filter. Once the catalyst is activated by the UV-A light, it forms hydroxyl radicals that combine and react with airborne microbiological contaminants, which destroys them.

3.11.2.6 Photocatalytic Oxidation Filter

The PCO process is similar to PECO in that it utilizes UV radiation combined with a catalyst to break down pollutants. The major difference between the PCO and the PECO is the filter material, UV light, and subsequent byproducts. While the PECO filter is a proprietary technology, PCO uses a catalyst such as titanium dioxide. Additionally, according to the manufacturer, PECO does not emit any harmful byproducts such as ozone and formaldehyde as compared to the catalysts on PCO filters. Finally, the PECO system utilizes a UV-A light instead of a UV-C light found in PCO systems.

When the titanium dioxide used with PCO is activated by UV-C radiation, it forms oxidizing hydroxyl radicals which react with pollutants. When a pollutant comes into contact with UV-activated titanium dioxide, the reaction destroys the pollutant and releases non-toxic compounds, such as carbon dioxide and water, as byproducts, as well as certain harmful byproducts such as ozone and formaldehyde.

3.11.2.7 Electrostatic/Polarizing Media

Manufacturers that use electrostatic filter media will utilize small cotton and paper fibers to polarize the filter. The charged, electrostatic filters create static that attracts dust and other airborne particles. Over time, these filters lose effectiveness as particulates attach to the polarized surface. At that point, polarized filters act as a traditional mechanical filter.

3.11.2.8 Filter Shape

DOE has identified two primary shapes for air cleaner filters: cylindrical and rectangular. Cylindrical filters surround the inner circumference of the air cleaner, while rectangular filters provide filtration on only one side of the air cleaner. Cylindrical filters cover more surface area per unit of volume compared to rectangular filters. The larger surface area allows for more greater particle entrapment while simultaneously reducing the resistance and pressure drop from the fan. For more information about how the filter impacts the efficiency of the air cleaner, refer to chapter 5 of this direct final rule TSD.

3.11.2.9 Improved Motor Technologies

The use of more efficient motor technologies will improve the efficiency of the air cleaner. DOE identified three motor types: SPM, PSC, and BLDC. SPM motors are the least efficient, but least expensive motor type. They typically feature a three-speed system. PSC motors are single-phase AC induction motors in which the capacitor is permanently connected to the circuit in both starting and running conditions. These motors typically have 3 or 4 speeds and allow for slightly more efficient airflow compared to SPM motors.

In a PSC motor, the start-up winding is electrically connected in parallel with the main winding and in series with a capacitor. At start-up, the interactions between the magnetic field generated by the start-up winding and that generated by the main winding induce rotation. As the capacitor charges, the current flowing through the start-up winding decreases and the start-up winding becomes an auxiliary winding after the motor reaches running speed. Consequently, the current to the start-up winding is cut off once the capacitor is fully charged and the motor reaches steady-state speed. Because of this, PSC motors are substantially more efficient than their shaded-pole counterparts, with motor efficiencies ranging from 60 to 65 percent.

The most efficient motor type is the BLDC motor, which utilizes direct current to electromagnetic coils arranged radially to the rotor. The current produces magnetic fields that cause the rotor to rotate in place. These motors are highly efficient and allow variable speed control to optimize airflow through the air cleaner. These motors are less noisy and substantially more efficient than either SPM or PSC motors. However, BLDC motors require additional controls and wiring compared to PSC motors, potentially necessitating a redesign of the air cleaner chassis. In addition, BLDC motors are complex, are not currently produced in large volumes, and cost more than an equivalent PSC motor.

3.11.2.10 Low Standby-power Electronic Controls

Electronic controls may consume power even when the air cleaner is not in active mode. Depending on the implementation of the controller, standby power is required to enable the electronic controls to detect user input without the user first having to turn on a mechanical power switch or to enable displays, illuminate switches, *etc.* Reducing the standby power consumption of electronic controls will reduce the annual energy consumption of the air cleaner but will not impact its energy consumption during active mode operation.

3.11.2.11 Direct Double-ended Blower Assembly

Some high efficiency air cleaners contain a second blower assembly. Air cleaners with an additional blower assembly allow for more air through the filter, which improves performance.

3.11.2.12 Ionization Brush

Air cleaners equipped with an ionization brush or emitter work by releasing negative ions into the air. Negative ions are very unstable and seek to attach themselves to positive particles, such as dust, smoke, pollen, bacteria, or viruses. The negative ions bind to these positively charged particles and the resulting mass becomes big enough for either gravity to cause them to drop down out of the air stream or for the air cleaner filter media to capture.

3.11.2.13 Ionization Plates

Air cleaners with ionization plates utilize electrostatic force to collect and trap airborne particles. A high voltage is applied to electrodes inside the air cleaner, which ionizes the air. Airborne particles are negatively charged by the ionized air as they come into the purifier. The newly charged particles are then attracted to oppositely charged metal plates where they are trapped and collected. The metal plates must be cleaned often in order to remove the particles.

3.11.2.14 Air Quality Sensor

Built-in air quality sensors can detect air quality by measuring the number of particles in the air. These sensors usually operate together with an infrared or laser beam. Particles that move through the beam and sensor absorb a small amount of light – this change is then registered by the sensor, which provides an air quality evaluation. Some air cleaners with built-in air quality sensors feature an automatic mode that adjusts the fan speed depending on the detected air quality, thereby conserving energy when the detected air quality is clean but the air cleaner is still in operation.

3.11.2.15 Ozone Generators

Ozone is a strong oxidizer and cleaning agent. Ozone generators work by creating an electrical discharge to split oxygen molecules in ambient air into single oxygen atoms, which

then bind with existing oxygen molecules in the air to form ozone. Ozone is highly unstable and reactive, so after it is produced by the generator, it is released in the air and is claimed to chemically react with air pollutants such as chemicals, mold, viruses, bacteria, and odors.

3.11.2.16 Thermodynamic Sterilization System

DOE has identified air cleaners on the market that use a thermodynamic sterilization system (TSS) in a ceramic core to destroy microorganisms and particle pollutants. These air cleaners do not rely on filter media to trap or remove particles, but rather utilize air convection to force air through the device's internal ceramic core, which heats up to about 200 degrees Celsius (°C) (392 degrees Fahrenheit (°F)) and incinerates pollutants. The manufacturer of these air cleaners claims that TSS can kill mold, bacteria, germs, and viruses and destroy pollutants such as dust, pollen, pet dander, hair, and other airborne particulates. After the air is heated and cleaned, it is immediately cooled using heat transfer plates and released back out of the device.

3.11.2.17 Bioreactor

DOE has identified two air cleaner models on the market that utilize a bioreactor system to produce clean air. Air cleaners that use this technology option rely on convection and fans to draw large particulate matter of over 0.5 µm such as dust and dander into the bioreactor chamber. Smaller ultra-fine air pollutants and VOCs are drawn into the chamber of the air purifier by a process of molecular attraction through an electrostatic grounded air zone.

Once the various types of air contaminants are drawn into the bioreactor, an activated solution of water, oxygen, enzymes, and the trapped contaminants leads to an accelerated process of natural oxidation that digests the air contaminants and breaks them down into water, carbon dioxide, and base elements. This results in cleaner air that is released from the unit.

3.11.3 Energy Efficiency

In preparation for the screening and engineering analyses, DOE gathered data on the energy efficiency of air cleaners currently available in the marketplace, as shown in Figure 3.11.1. These data are taken from the ENERGY STAR V. 2.0 database and include only products that are ENERGY STAR certified. That is, products available at efficiencies below the ENERGY STAR V. 2.0 level are not represented in Figure 3.11.1. Additionally, Figure 3.11.2 shows the market share by product class as provided by the Joint Stakeholders in the Joint Proposal. The market share breakdown shows the distribution of capacities, as represented by PM_{2.5} CADR, for air cleaners on the market in the United States that are within the scope of this rulemaking. While this section is not intended to provide a complete characterization of the energy efficiency of all air cleaners currently available and in use, it does provide a general overview of the energy efficiency of these products as measured by industry test methods that are the basis of the DOE test procedure at appendix FF.

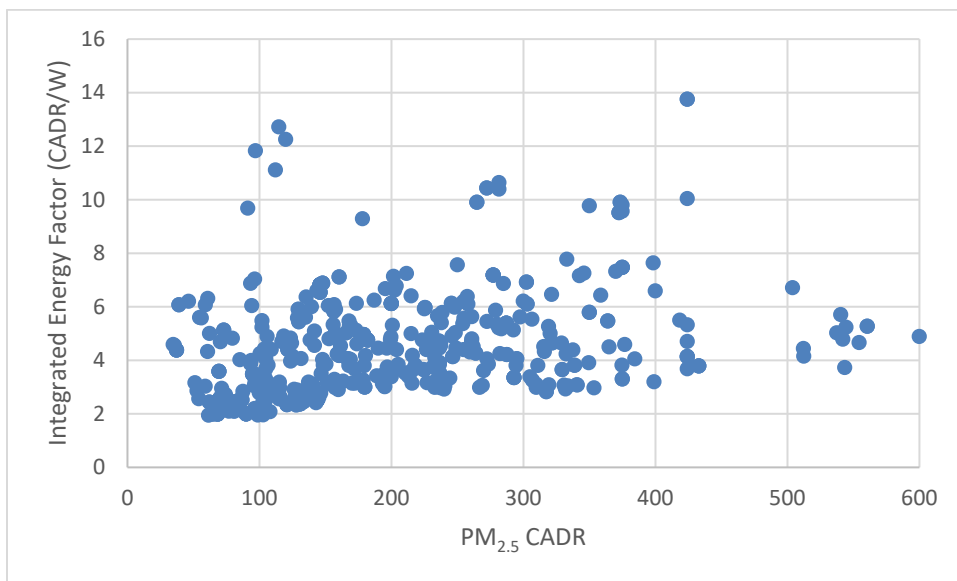


Figure 3.11.1 ENERGY STAR V. 2.0 Certified Air Cleaners as of December 2022

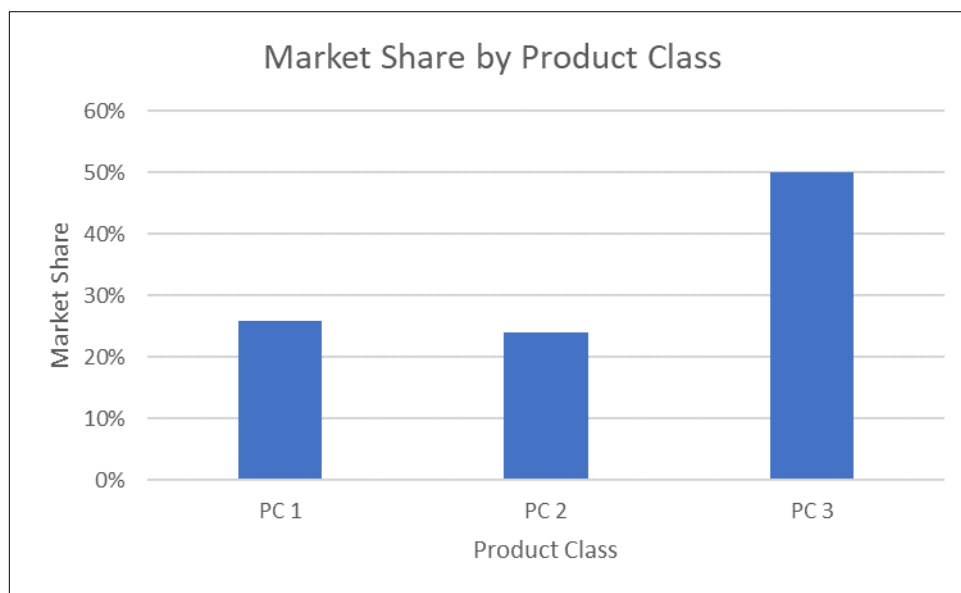


Figure 3.11.2 Air Cleaner Market Share by Product Class^f

^f Joint Statement Petition for Air Cleaners Energy Savings Calculations (Supporting data for comment EERE-2021-BT-STD-0035-0016). Available at: <https://www.regulations.gov/comment/EERE-2021-BT-STD-0035-0018>.

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CHAPTER 4. SCREENING ANALYSIS

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CHAPTER 4. SCREENING ANALYSIS

4.1 INTRODUCTION

This chapter details the screening analysis conducted by the U.S. Department of Energy (DOE) of the technology options identified in the market and technology assessment for air cleaners (chapter 3 of this direct final rule technical support document (TSD)). In the market and technology assessment, DOE presented an initial list of technologies that can be used to increase the integrated energy factor of air cleaners. The goal of the screening analysis is to identify any technology options that will be eliminated from further consideration in the rulemaking analyses.

DOE must follow specific statutory criteria for prescribing new or amended standards for covered products. The Energy Policy and Conservation Act (EPCA) requires that any new or amended energy conservation standard prescribed by the Secretary of Energy (Secretary) be designed to achieve the maximum improvement in energy or water efficiency that is technologically feasible and economically justified. (42 U.S.C. 6295(o)(2)(A)) The Secretary may not prescribe an amended or new standard that will not result in significant conservation of energy, or is not technologically feasible or economically justified. (42 U.S.C. 6295(o)(3)) As stated, DOE determines whether to eliminate certain technology options from further consideration based on the following criteria:

(1) Technological feasibility. Technologies that are not incorporated in commercial products or in commercially-viable, existing prototypes will not be considered further.

(2) Practicability to manufacture, install, and service. If it is determined that mass production of a technology in commercial products and reliable installation and servicing of the technology could not be achieved on the scale necessary to serve the relevant market at the time of the compliance date of the standard, then that technology will not be considered further.

(3) Impacts on product utility. If a technology is determined to have significant adverse impact on the utility of the product to subgroups of consumers, or results in the unavailability of any covered product type with performance characteristics (including reliability), features, size, capacities, and volumes that are substantially the same as products generally available in the United States at the time, it will not be considered further.

(4) Safety of technologies. If it is determined that a technology will have significant adverse impacts on health or safety, it will not be considered further.

(5) Unique-Pathway Proprietary Technologies. If a technology has proprietary protection and represents a unique pathway to achieving a given efficiency level, it will not be considered further, due to the potential for monopolistic concerns.

10 CFR part 430, subpart C, appendix A, 6(b)(3) and 7(b).

The candidate technology options are assessed based on DOE analysis as well as inputs from interested parties, including manufacturers, trade organizations, and energy efficiency

advocates. Technology options that are judged to be viable approaches for improving energy efficiency are retained as inputs to the subsequent engineering analysis, and are designated as “design options.”

4.2 DISCUSSION OF TECHNOLOGY OPTIONS

For air cleaners, the screening criteria specified in section 4.1 were applied to the technology options to either retain or eliminate each technology from the engineering analysis. The rationale for either screening out or retaining each technology option considered in this analysis is detailed in the following sections.

4.2.1 Screened-Out Technologies

Based on DOE’s research and consideration of comments received from interested parties, DOE screened out the technology options shown in Table 4.2.1. The following sections describe the justification for screening out each technology.

Table 4.2.1 Screened Out Technology Options

Technology Option	EPCA Criteria (X = Basis for Screening Out)				
	Technological Feasibility	Practicability to Install, Manufacture, and Service	Impacts on Product Utility	Safety of Technologies	Unique-Pathway Proprietary Technologies
Photoelectrochemical Oxidation (PECO)					X
Photocatalytic Oxidation (PCO)				X	
Ozone Generation				X	
Thermodynamic Sterilization System (TSS)					X
Bioreactor		X			

4.2.1.1 Photoelectrochemical Oxidation

PECO technology processes pollutants in a photoreactor that utilizes photons to initiate a reaction that oxidizes and destroys organic pollutants in the air. The reaction converts the pollutants into non-toxic substances. DOE identified one company that incorporates this technology into its air cleaner models. Since the company holds a patent on this technology, DOE has screened out PECO as a unique pathway proprietary technology.

4.2.1.2 Photocatalytic Oxidation

The PCO process is similar to PECO in that it utilizes ultraviolet (UV) radiation combined with a catalyst to break down pollutants. However, the PCO technology option utilizes a non-proprietary catalyst, a different UV light, and generates different subsequent byproducts from the reaction. When a pollutant contacts the UV-activated catalyst, the reaction destroys the pollutant and releases non-toxic compounds, such as carbon dioxide and water, as byproducts, as well as certain harmful byproducts such as ozone and formaldehyde.

DOE is screening out the PCO technology option due to health and safety concerns stemming from the byproducts generated by the reaction of the PCO filter. The U.S. Environmental Protection Agency (EPA) has determined that high levels of formaldehyde and ozone can have an adverse impact on human health.^{1, 2}

4.2.1.3 Ozone Generation

Ozone generator air cleaners work by creating an electrical discharge to split oxygen molecules in ambient air into single oxygen atoms, which then bind with existing oxygen molecules in the air to form ozone. The ozone that is released into the air is claimed to chemically react with air pollutants such as chemicals, mold, viruses, bacteria, and odors.

DOE has identified concerns with air cleaners that rely on ozone generation in terms of both efficacy and safety. EPA has found that ozone can chemically react with organic material inside the respiratory system, leading to harmful health effects.³ For air cleaners operating within the EPA's National Ambient Air Quality Standard for ozone, which is a maximum 8-hour average outdoor concentration of 0.08 parts per million,⁴ there is evidence to suggest that ozone is not effective at removing organic pollutants.⁵ Additionally, these air cleaners are not effective at removing particulate matter such as dust and pollen.⁶

Due to these health and safety concerns associated with ozone and lack of efficacy towards particulate removal, DOE has screened out this technology option.

4.2.1.4 Thermodynamic Sterilization System

DOE has identified air cleaners on the market that use TSS in a ceramic core to destroy microorganisms and particle pollutants. The internal ceramic core uses convection to force ambient room air into the ceramic core chamber, which heats up to about 200 degrees Celsius (°C) (392 degrees Fahrenheit (°F)). Organic pollutants and particulates in the air are incinerated inside the core and the air is then immediately cooled and then circulated back into the room.

DOE has determined that TSS is a proprietary technology implemented by a single company. Therefore, DOE has screened out this technology option as a unique pathway proprietary technology.

4.2.1.5 Bioreactor

Air cleaners that use a bioreactor system draw in air contaminants into the chamber, where an activated solution of water, oxygen, enzymes, and the trapped contaminants leads to an accelerated process of natural oxidation that digests the air contaminants and breaks them down into water, carbon dioxide, and base elements. This results in cleaner air that is released from the unit. The bioreactor is typically coupled with a convection and fan system to draw large particulate matter and electrostatic plates to attract smaller ultra-fine air pollutants and volatile organic compounds.

DOE identified two air cleaner models on the market that feature a bioreactor technology option. Given the scarcity of models on the market with this technology, DOE has screened out this technology option, as it is not proven to be practicable to manufacture, install, and service this technology on a scale necessary to serve the relevant market at the time of the compliance date of new standards.

4.2.2 Remaining Design Options

Table 4.2.2 lists the technology options for air cleaners that were retained by DOE and subsequently designated as design options. Each of these technologies were evaluated further in the subsequent engineering analysis.

Table 4.2.2 Retained Design Options for Air Cleaners

1. HEPA-type filter (99 percent of 0.2 micrometer (μm) particles)
2. True HEPA filter (99.97 percent of 0.3 μm particles)
3. Activated carbon filter
4. High density polyethylene pre-filter
5. Electrostatic/Polarizing media
6. Filter shape
7. Improved motor technologies
8. Low standby-power electronic controls
9. Direct double-ended blower assembly
10. Ionization brush
11. Ionization plates
12. Air quality sensor

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6. *Ibid.*

CHAPTER 5. ENGINEERING ANALYSIS

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CHAPTER 5. ENGINEERING ANALYSIS

This chapter provides an overview of the engineering analysis (section 5.1), discusses proposed product classes (section 5.2), establishes baseline and incremental efficiency levels (section 5.3), explains the methodology used during data gathering (section 5.4), and discusses the analysis and results (section 5.5).

5.1 INTRODUCTION

The purpose of the engineering analysis is to determine the incremental manufacturing cost associated with producing air cleaners at higher efficiency levels. The primary considerations in the engineering analysis are the selection of efficiency levels to analyze (*i.e.*, the “efficiency analysis”) and the determination of product cost at each efficiency level (*i.e.*, the “cost analysis”).

DOE conducts the efficiency analysis using either an efficiency-level approach, a design-option approach, or a combination of both. Under the efficiency-level approach, the efficiency levels to be considered in the analysis are determined based on the market distribution of existing products (in other words, observing the range of efficiency and efficiency level “clusters” that already exist on the market). This approach typically starts with compiling a comprehensive list of products available on the market, such as from DOE’s product certification database. Next, the list of models is ranked by efficiency level from lowest to highest, and DOE typically creates a scatter plot to visualize the distribution of efficiency levels. From these rankings and visual plots, efficiency levels can be identified by examining clusters of models around common efficiency levels. The maximum efficiency level currently available on the market can also be identified.

Under the design option approach, the efficiency levels to be considered in the analysis are determined through detailed engineering calculations and/or computer simulations of the efficiency improvements from implementing specific design options that have been identified in the technology assessment. In an iterative fashion, design options can also be identified during product teardowns, described below. The design option approach is typically used when a comprehensive database of certified models is unavailable (for example, if a product is not yet regulated) and therefore the efficiency-level approach cannot be used.

In certain rulemakings, the efficiency-level approach (based on actual products on the market) will be extended using the design option approach to interpolate to define “gap fill” levels (to bridge large gaps between other identified efficiency levels) and/or to extrapolate to the “max tech” level (the level that DOE determines is the maximum achievable efficiency level), particularly in cases where the “max tech” level exceeds the maximum efficiency level currently available on the market.

In this rulemaking, DOE primarily used the efficiency-level approach because, even though air cleaners are not yet regulated products, the U.S. Environmental Protection Agency (EPA) has established an ENERGY STAR® voluntary labeling program and maintains a database¹ of air cleaners which meet the current ENERGY STAR qualification criteria²

(ENERGY STAR V. 2.0). In addition, the Association of Home Appliance Manufacturers (AHAM) publishes a database of air cleaners that are certified to its Verifide program, and includes smoke, dust, and pollen CADR values only.³ Under the efficiency-level approach, DOE reviewed the ENERGY STAR V. 2.0 database to identify the market distribution of existing products, based on their capacity and efficiency. DOE also used the design-option approach, testing and physically disassembling commercially available products to fill gaps where data was not available from the efficiency-level approach (*e.g.*, to identify efficiency levels below the ENERGY STAR level, including the minimum available efficiency, which was determined to be the baseline efficiency level for this analysis). DOE then analyzed the steps manufacturers took to achieve incrementally higher product efficiencies. In this direct final rule analysis, DOE determined that manufacturers would likely rely on certain design options to reach higher efficiencies.

The cost analysis portion of the engineering analysis is conducted using one or a combination of cost approaches. The selection of the cost approach depends on a variety of factors such as the availability and reliability of information on product features and pricing, the physical characteristics of the regulated product, and the practicability of purchasing the product on the market. DOE generally uses the following cost approaches:

- Physical teardown: Under this approach, DOE physically dismantles a commercially available product, component-by-component, to develop a detailed bill of materials (BOM) for the product.
- Catalog teardown: In lieu of physically deconstructing a product, DOE identifies each component using parts diagrams (available from manufacturer websites or appliance repair websites, for example) to develop the BOM for the product.
- Price surveys: If neither a physical nor catalog teardown is feasible (for example, for tightly integrated products that are infeasible to disassemble and for which parts diagrams are unavailable), DOE conducts retail price surveys by scanning retailer websites and other marketing materials. This approach must be coupled with assumptions regarding distributor markups and retailer markups in order to estimate the actual manufacturing cost of the product.

In this direct final rule, DOE used the physical teardown approach for its cost analysis. For each product class, DOE tore down a representative sample of models spanning the entire range of efficiency levels, as well as multiple manufacturers within each product class. DOE aggregated the results so that the cost-efficiency relationship developed for each product class reflects DOE's assessment of a market-representative "path" to achieve each higher efficiency level. The resulting bill of materials provides the basis for the manufacturer production cost (MPC) estimates. In addition to determining MPCs for each efficiency level, DOE disaggregated the overall MPCs to identify the filter costs, which are used to estimate the cost of replacement filters through the life of the product in the life-cycle cost (LCC) and payback period (PBP) analyses.

The primary output of the engineering analysis is a set of tables identifying the incremental MPC, in relation to the MPC of the minimum-efficiency baseline product, required to produce products at each of the higher efficiency levels considered in the analysis (visualized as “cost-efficiency curves”). In the subsequent markups analysis (chapter 6 of this direct final rule technical support document (TSD)), DOE determines customer (*i.e.*, product purchaser) prices by applying manufacturer markups, distribution markups, and sales tax. After applying these markups, the cost-efficiency curves serve as the input to the building energy-use and end-use load characterization (chapter 7 of this direct final rule TSD), and the LCC and PBP analyses (chapter 8 of this direct final rule TSD).

5.2 PRODUCT CLASSES ANALYZED

Because air cleaners were not previously covered by the Energy Policy and Conservation Act (ECPA) of 1975, Pub. L. 94-163 (42 United States Code (U.S.C.) 6291–6309), there are no existing product classes for these products.

When evaluating energy conservation standards, DOE may establish separate standards for a group of covered products (*i.e.*, establish a separate product class) if DOE determines that separate standards are justified based on the type of energy used, or if DOE determines that a product’s capacity or other performance-related feature justifies a different standard. (42 U.S.C. 6295(q)(1)(A) and (B)) In making a determination whether a performance-related feature justifies a different standard, DOE must consider factors such as the utility to the consumer of the feature and other factors DOE determines are appropriate. (42 U.S.C. 6295(q)(1))

As discussed in chapter 3 of this direct final rule TSD, the scope of this rulemaking comprises conventional room air cleaners, as defined in 10 CFR 430.2, with capacities, expressed in terms of PM_{2.5} clean air delivery rate (CADR) in cubic feet per minute (cfm), of 10 to 600 cfm (inclusive).^a DOE determined the three product classes for air cleaners within the scope of this rulemaking, as shown in Table 5.2.1, to be appropriate based on investigatory testing, product teardowns, and a review of ENERGY STAR-qualified products.

Table 5.2.1 Air Cleaner Product Classes

Product Class*	PM_{2.5} CADR Bins (<i>cfm</i>)
PC1	10 ≤ PM _{2.5} CADR < 100
PC2	100 ≤ PM _{2.5} CADR < 150
PC3	PM _{2.5} CADR ≥ 150**

*The analysis included only conventional room air cleaners.

**The analysis included units in Product Class 3 only with PM_{2.5} CADR ≤ 600 cfm.

^a Note that the scope of this rulemaking is conventional room air cleaners with smoke CADR and dust CADR between 10 to 600 cfm (inclusive). As discussed in chapter 3 of this direct final rule TSD, PM_{2.5} CADR is calculated as the geometric mean of smoke CADR and dust CADR. Accordingly, the scope in terms of PM_{2.5} CADR is 10 to 600 cfm as well.

For this rulemaking, DOE analyzed all three air cleaner product classes to determine efficiency levels and associated incremental MPCs.

5.3 EFFICIENCY LEVELS

DOE developed representative efficiency levels using a combination of air cleaner performance data, the ENERGY STAR database, and Tier 1 and Tier 2 levels recommended in the Joint Proposal^b. (More information on the Joint Proposal and Joint Stakeholders is available in chapter 1 of this direct final rule TSD.) Each efficiency level is represented in terms of the integrated energy factor (IEF), expressed in PM_{2.5} CADR per watt (PM_{2.5} CADR/W), which is the efficiency of an air cleaner as a function of PM_{2.5} CADR relative to energy use.

As a first step to determine baseline and incremental efficiency levels, DOE selected units for testing and teardowns using the AHAM Verifide and ENERGY STAR databases and identified the PM_{2.5} CADR values at which most models were clustered. The ENERGY STAR database includes smoke CADR, dust CADR, and pollen CADR values in addition to providing CADR/W values for each pollutant, but the AHAM Verifide database includes only smoke CADR, dust CADR, and pollen CADR values. From the available data, DOE was able to calculate PM_{2.5} CADR (as the geometric mean of the smoke CADR and the dust CADR), and IEF (using the calculated PM_{2.5} CADR and annual energy consumption values provided in the ENERGY STAR database) for units in the ENERGY STAR database. Using these databases, DOE selected a representative sample of 36 products for testing and teardowns that spanned the IEF range and available product characteristics (see section 5.5.2 for additional details of the model selection). Because product design varies depending on capacity, DOE identified a representative nominal capacity for each product class based on the most commonly occurring approximate PM_{2.5} CADR value in its test sample for a given product class, which are 50 CADR, 125 CADR, and 200 CADR for Product Class 1, Product Class 2, and Product Class 3, respectively. Note that DOE prioritized selecting representative units that spanned the range of available IEF and product characteristics in evaluating candidate models in each product class for inclusion in the test sample. A representative nominal capacity was necessary when selecting units for teardown because units within the same product class that have similar IEFs can have widely different MPCs if they have significantly different capacities.

5.3.1 Baseline Efficiency Levels

For each product class, DOE selects a baseline model as a reference point for that class. To determine energy savings and changes in cost that would result from a potential new or amended energy conservation standard, DOE compares energy use and price of more energy-efficient units to the energy consumption and cost of the baseline unit. Generally, a baseline

^b On August 23, 2022, DOE received a Joint Proposal from the Joint Stakeholders, representing manufacturers, efficiency advocates, and consumer groups, regarding energy conservation standards and related test procedures for air cleaners. The Joint Proposal specified two tiers of standard levels (*i.e.*, Tier 1 and Tier 2), with proposed compliance dates of December 31, 2023, and December 31, 2025, respectively. Tier 1 standards are equivalent to the standards established by the States of Maryland, Nevada, and New Jersey, and the District of Columbia (*i.e.*, the State standards). Tier 2 standards are equivalent to the standards specified in the ENERGY STAR Version 2.0 Room Air Cleaners Specification and those adopted by the State of Washington.

model is one that exactly meets current energy conservation standards; or, if no standards are in place, the baseline is typically the most common or least efficient unit on the market.

For each product class, DOE then selected the baseline efficiency level at the nominal capacity based on a commercially available unit below the levels established by certain States and the ENERGY STAR V. 2.0 level. Given there is no database that contains energy use data for air cleaners other than the ENERGY STAR database, which provides a list of products that meet or exceed ENERGY STAR V. 2.0 levels, DOE identified the baseline efficiency levels by testing a representative sample of commercially available units that were not included in the ENERGY STAR database. Through this approach, DOE was able to identify the baseline efficiency level using the IEF of the least efficient unit tested in each product class for Product Classes 1 and 3. For Product Class 2, DOE did not identify any unit in its limited test sample with an IEF below the State (*i.e.*, proposed Tier 1 standard in the Joint Proposal) or ENERGY STAR V. 2.0 (*i.e.*, proposed Tier 2 standard in the Joint Proposal) levels. Accordingly, DOE used the baseline unit from Product Class 1, scaled to the representative PM_{2.5} CADR for Product Class 2, to determine a representative baseline IEF for Product Class 2.

Table 5.3.1 summarizes the baseline efficiency levels defined for each product class.

Table 5.3.1 Baseline Efficiency Levels

Product Class	Representative Capacity (PM_{2.5} CADR, in cfm)	IEF (PM_{2.5} CADR/W)
PC1: $10 \leq \text{PM}_{2.5} \text{ CADR} < 100$	50	1.5
PC2: $100 \leq \text{PM}_{2.5} \text{ CADR} < 150$	125	1.5
PC3: $\text{PM}_{2.5} \text{ CADR} \geq 150$	200	1.2

5.3.2 Higher Efficiency Levels

To establish higher efficiency levels for the analysis, DOE reviewed data in the ENERGY STAR database to evaluate the range of efficiencies, beyond the baseline, for air cleaners currently available on the market. To do this, DOE converted the smoke CADR and dust CADR to PM_{2.5} CADR by taking the geometric average and used the smoke CADR/W to calculate active mode power consumption which was used along with the low power mode power consumption data to calculate IEF. For all product classes, DOE considered efficiency levels higher than baseline levels based on specifications prescribed by multiple State standards and ENERGY STAR, which align with the Tier 1 and Tier 2 levels proposed in the Joint Proposal, respectively, as well as gap-fill levels.

As part of its analysis, DOE must determine the maximum improvement in efficiency that is technologically feasible (max-tech) for each product class. (42 U.S.C. 6295(p)(1)) DOE typically determines max-tech levels based on technologies that are either commercially available or have been demonstrated as working prototypes. DOE also considers consumer utility and availability of features, which may be met by a niche product. If the max-tech design meets DOE's screening criteria, DOE considers the design in further analysis.

For this direct final rule, DOE considered the maximum efficiency for each product class to correspond to the highest efficiency units currently available on the market that provide complete consumer utility and have a PM_{2.5} CADR close to that of the representative capacity for the product class. The max-tech unit for each product class is shown in Table 5.3.2.

Table 5.3.2 Max Tech Efficiency Levels

Product Class	Representative Capacity (PM_{2.5} CADR, in cfm)	IEF (PM_{2.5} CADR/W)
PC1: $10 \leq \text{PM}_{2.5} \text{ CADR} < 100$	50	5.4
PC2: $100 \leq \text{PM}_{2.5} \text{ CADR} < 150$	125	12.8
PC3: $\text{PM}_{2.5} \text{ CADR} \geq 150$	200	7.4

In addition to selecting units with a PM_{2.5} CADR close to that of the representative capacity for each product class, to determine its max-tech units DOE also selected units that utilized a true HEPA^c filter, which is a filter that is rated to remove at least 99.97 percent of particles that have a size of 0.3 micro-meters (µm). DOE selected this criterion because, according to EPA, the diameter specification of 0.3 µm corresponds to the most penetrating particle size; that is, particles between 0.2 and 0.3 µm are the most difficult size particles to capture and particles larger or smaller than 0.3 µm are generally captured more easily.⁴ Therefore, DOE selected its max-tech unit to include a true HEPA filter to ensure that there would not be any loss in product utility at the selected max-tech efficiency level. The most efficient units listed in the ENERGY STAR database for Product Class 1 and Product Class 3 do not include a true HEPA filter and instead utilize ionic plates^d or a filter that is rated to capture 98 percent of 5 µm particles, neither of which meet the rating requirement of a true HEPA filter for capturing at least 99.97 percent of particles that have a size of 0.3 µm, which DOE determined is required to maintain full consumer functionality. DOE notes that the pressure drop across a true HEPA filter would be greater than filters with less effective particle capture due to the more powerful motor required to move the same quantity of air across the true HEPA filter as compared to a less effective filter.

While the max-tech units selected by DOE for Product Class 2 and Product Class 3 are the most-efficient units at the representative PM_{2.5} CADR value, for Product Class 1, DOE observed another unit that had a higher IEF compared to its selected unit. However, DOE ultimately selected the unit shown in Table 5.3.2 because the other unit did not include a true HEPA filter; instead, it included a filter that is rated to remove only up to 97 percent of particles that have a size of 0.3 µm, which DOE determined did not maintain full consumer functionality.

To establish other incremental higher efficiency levels between the baseline and max-tech, DOE considered State standards and reviewed data in the ENERGY STAR database to evaluate the range of efficiencies above the baseline for air cleaners currently available on the market. For all three product classes, DOE considered Efficiency Level 1 (EL 1) to correspond to

^c True HEPA filters have a similar design and construction as other HEPA-type filters but have a higher density as they can capture smaller particles with a greater efficiency rate. Any HEPA filter that does not attain the DOE standard for HEPA filtration is not a true HEPA filter.

^d Ionic plates utilize electrostatic force to collect and trap airborne particles.

the level established by certain States, as discussed. EL 1 also corresponds to the Tier 1 level provided in the Joint Proposal. DOE selected EL 2 for all product classes to correspond to the ENERGY STAR V. 2.0 level, which is also the Tier 2 level provided in the Joint Proposal. Finally, DOE identified EL 3 as a “gap-fill” level between EL 2 and max-tech (*i.e.*, EL 4) based on the number of available models grouped (or “clustered”) between EL 2 and max-tech for each product class. Table 5.3.3 through Table 5.3.5 summarizes the efficiency levels analyzed for each product class.

Table 5.3.3 Product Class 1 Efficiency Levels

EL	Efficiency Level Description	IEF ($PM_{2.5}CADR/W$)
Baseline	Minimum available and tested units	1.5
1	State Standard Levels	1.7
2	ENERGY STAR v. 2.0	1.9
3	Gap Fill	3.4
4	Maximum available	5.4

Table 5.3.4 Product Class 2 Efficiency Levels

EL	Efficiency Level Description	IEF ($PM_{2.5}CADR/W$)
Baseline	Minimum available and tested units	1.5
1	State Standard Levels	1.9
2	ENERGY STAR v. 2.0	2.4
3	Gap Fill	5.4
4	Maximum available	12.8

Table 5.3.5 Product Class 3 Efficiency Levels

EL	Efficiency Level Description	IEF ($PM_{2.5}CADR/W$)
Baseline	Minimum available and tested units	1.2
1	State Standard Levels	2.0
2	ENERGY STAR v. 2.0	2.9
3	Gap Fill	6.6
4	Maximum available	7.4

5.4 METHODOLOGY OVERVIEW

DOE relied on multiple sources of information for this engineering analysis, including manufacturer interviews, internal product testing, and product teardowns. The following sections describe DOE’s methodology for product selection, product testing, and product teardowns.

5.4.1 Manufacturer Interviews

DOE understands that design strategies and cost structures may vary greatly among air cleaner manufacturers. To better understand these variances, DOE conducted confidential manufacturer interviews. These interviews provided DOE with a deeper understanding of the various combinations of technologies used to increase air cleaner efficiency and their associated manufacturing costs. DOE conducted interviews in advance of this direct final rule.

5.4.2 Product Selection

DOE generally adopts the following criteria for selecting units for testing and teardown analysis:

- The selected products should span the full range of efficiency levels for each product class under consideration;
- Within each product class, the selected products should, if possible, come from the same manufacturer and belong to the same product platform;
- The selected products should, if possible, come from manufacturers with large market shares in that product class, although the highest efficiency products are chosen irrespective of manufacturer; and
- The selected products should have non-efficiency-related features that are the same as, or similar to, features of other products in the same class and at the same efficiency level.

5.4.3 Product Testing

Much of the analysis in this chapter relies on data from publicly available sources such as ENERGY STAR databases. However, DOE also conducted its own testing according to the DOE test procedure at 10 CFR part 430, subpart B, appendix FF (appendix FF) for the following purposes:

- Verify performance trends that are apparent in the publicly available data
- Develop a better understanding of the design options and product features currently available on the market
- Develop a better understanding of the operational characteristics of air cleaners
- Establish baseline efficiency levels based on commercially available air cleaners

5.4.4 Product Teardowns

As noted, DOE used the physical teardown approach for the large majority of the cost analysis for this direct final rule.

The teardown method provides key information that is difficult to obtain from the other cost analysis approaches and that can be critical to the quality of the engineering analysis. In conducting a teardown, DOE physically disassembles a commercially available product,

component-by-component, to develop a structured BOM that estimates the costs of raw materials, machining processes, purchased parts, capital depreciation, and factory overhead costs at an assumed production volume. The teardown approach allows for the unbundling of non-efficiency related features that are often bundled with higher efficiency technologies and subsequently contribute to the higher retail price of higher-efficiency models. The teardown approach also reveals different design paths used by different manufacturers, and enables DOE to estimate manufacturer investments required to achieve each higher efficiency level. The teardown approach is time-intensive, and unit selection is critical to ensure an industry-representative cost curve. Periodically, DOE typically invites manufacturers to review the results in detail and may verify the manufacturing parameters/processes via site visits to production facilities.

A supplementary method to a physical teardown, called a catalog teardown or virtual teardown, uses published manufacturer catalogs and supplementary component data to estimate the major physical differences between a product that has been physically disassembled and another similar product. In a catalog teardown, DOE references public information to estimate manufacturing costs. Usually, these estimates leverage prior product teardowns, allowing the analysis to capture a broader range of capacities and other features within a product family. Using product teardown data as a foundation, DOE can model the costs associated with higher-efficiency models through parametric scaling, part substitutions, *etc.*, to capture the most relevant product parameters. In some instances, the incremental cost of well-defined equipment may also be characterized with a catalog teardown alone. This method is best suited for product categories where features are well-documented, with “mature” or commoditized designs, and where efficiency-related design are physically scalable. This approach allows a wide variety of capacities and features to be covered. As discussed, DOE largely relied on physical teardowns for its analysis of air cleaners for this direct final rule.

The teardown methodology is further explained in the following sections.

5.4.4.1 Generation of Bill of Materials

The end result of each teardown is a structured BOM, which describes each product part and its relationship to the other parts, in the estimated order of assembly. The BOMs describe each fabrication and assembly operation in detail, including the type of value-added equipment needed (*e.g.*, stamping presses, injection molding machines, spot-welders, *etc.*) and the estimated cycle times associated with each conversion step. The result is a thorough and explicit model of the production process.

Materials in the BOM are divided between raw materials—which require conversion steps to be made ready for assembly—and purchased parts, which are typically delivered ready for installation. The classification into raw materials or purchased parts is based on DOE’s previous industry experience, recent information in trade publications, and past discussions with original equipment manufacturers (OEMs). For purchased parts, the purchase price is based on volume-variable price quotations and detailed discussions with suppliers.

For parts fabricated in-house, the prices of the underlying “raw” metals (*e.g.*, sheet metal) are estimated on the basis of 5-year averages to smooth out spikes in demand. Other raw materials such as plastic resins, insulation materials, etc. are estimated on a current-market basis. The costs of raw materials are based on past discussions with manufacturers, quotes from suppliers, and secondary research. DOE regularly updates historical cost data to present-day prices using indices from resources such as MEPS Intl.,⁵ PolymerUpdate,⁶ the U.S. geologic survey (USGS),⁷ and the Bureau of Labor Statistics (BLS).⁸

5.4.4.2 Cost Structure of the Spreadsheet Models

The manufacturing cost assessment methodology is based on a detailed, component-focused technique for rigorously calculating the manufacturing cost of a product, including materials, labor and overhead costs. Figure 5.4.1 shows the three major steps in generating the manufacturing cost.

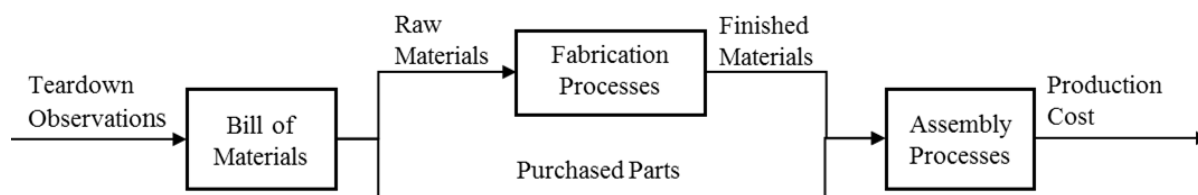


Figure 5.4.1 Manufacturing Cost Assessment Stages

The first step in the manufacturing cost assessment is the creation of a complete and structured BOM from the disassembly of the units selected for teardown. The units are dismantled, and each part is characterized according to weight, manufacturing processes used, dimensions, material, and quantity. The BOM incorporates all materials, components, and fasteners, with estimates of raw material costs and purchased part costs. Assumptions on the sourcing of parts and in-house fabrication are based on industry experience and past discussions with manufacturers.

Following the development of a detailed BOM, the major manufacturing processes are identified and developed for the spreadsheet model. Some of these processes are listed in Table 5.4.1.

Table 5.4.1 Major Manufacturing Processes for Air Cleaners

Fabrication	Finishing	Assembly/Joining	Quality Control
Injection Molding Stamping/Pressing Fixturing	De-burring Powder Coating	Adhesive Bonding Ultrasonic Welding Clinching Packaging	Inspecting & Testing

Fabrication process cycle times for each part made in-house are estimated and entered into the BOM. Based on estimated assembly and fabrication time requirements, the labor content

of each teardown unit are estimated based on typical annual wages and benefits of industry employees.

Cycle requirements for fabrication steps are similarly aggregated by fabrication machine type, while accounting for dedicated versus non-dedicated machinery and/or change-over times (*e.g.*, die swaps in a press). Once the cost estimate for each teardown unit is finalized, a detailed summary is prepared for relevant components, subassemblies, and processes. The BOM thus details all aspects of unit costs: material, labor, and overhead.

Design options used in units subject to teardown are noted in the summary sheet of each cost model and are cost-estimated individually. Thus, various implementations of design options can be accommodated, ranging from assemblies that are entirely purchased to units that are made entirely from raw materials. Hybrid assemblies, consisting of purchased parts and parts made on site are thus also accommodated.

5.4.4.3 Cost Model and Definitions

The cost model is based on production activities and divides factory costs into the following categories:

- **Materials:** Purchased parts (*i.e.*, wiring, screws, electronics, *etc.*), raw materials, (*i.e.*, plastics), and indirect materials that are used for processing and fabrication.
- **Labor:** Fabrication, assembly, supervisor, and indirect labor. Fabrication and assembly labor cost are burdened with benefits and supervisory costs.
- **Overhead:** Equipment, tooling, and building depreciation, as well as utilities, equipment and tooling maintenance, insurance, and property taxes. The equipment, tooling, and building depreciation costs are modeled as a “green-field” site; *i.e.*, a new manufacturing plant with all new equipment.

DOE defines the above terms as follows:

- **Direct material:** Purchased parts (outsourced) plus manufactured parts (made in-house from raw materials).
- **Indirect material:** Material used during manufacturing (*e.g.*, welding rods, adhesives).
- **Fabrication labor:** Labor associated with in-house piece manufacturing.
- **Assembly labor:** Labor associated with final assembly.
- **Supervisory labor:** Labor associated with fabrication and assembly basis. Assigned on a span basis ($\times$ number of employees per supervisor) that depends on the industry.
- **Indirect labor:** Labor costs that scale with fabrication and assembly labor. These included the cost of technicians, manufacturing engineering support, stocking, *etc.* that are proportional to all other labor.
- **Equipment depreciation:** Money allocated to pay for initial equipment installation and replacement as the production equipment is amortized. All depreciation is assigned in a linear fashion and affected equipment life depends on the type of equipment.

- Tooling depreciation: Cost for initial tooling (including non-recurring engineering and debugging of the tools) and tooling replacement as it wears out or is rendered obsolete.
- Building depreciation: Money allocated to pay for the building space and the conveyors that feed and/or make up the assembly line.
- Utilities: Electricity, gas, telephones, *etc.*
- Maintenance: Annual money spent on maintaining tooling and equipment.
- Insurance: Appropriated as a function of unit cost.
- Property taxes: Appropriated as a function of unit cost.

5.4.4.4 Cost Model Assumptions

As discussed in the previous section, assumptions about manufacturer practices and cost structure played an important role in estimating the final product cost. In converting physical information about the product into cost information, DOE reconstructs manufacturing processes for each component using internal expertise and knowledge of the methods used by the industry. DOE regularly confirms its cost model assumptions through various sources such as previous manufacturer interviews and reviews of current BLS data.

5.5 ANALYSIS AND RESULTS

5.5.1 Manufacturer Interviews

As discussed, following the publication of a request for information in January 2022, DOE conducted limited interviews with air cleaner manufacturers to develop a better understanding of current product features and the technologies used to improve energy efficiency. During these interviews, DOE asked manufacturers questions about the following topics related to the engineering analysis: product classes, design features of current baseline products, incremental efficiency levels and design options, manufacturer costs, impacts on consumer utility, and installation and repair costs. The interviewed manufacturers represent a wide range of U.S. market share and included both domestic and international companies that sell air cleaners in the United States.

The discussion below represents a consolidation of the manufacturer responses.

5.5.1.1 Brushless Direct Current Motors

Manufacturers generally noted that baseline and lower ELs will utilize brushed, 3 or 4-speed motor technology such as permanent split capacitor (PSC) motors. Several manufacturers noted that while these motors are less efficient, they are cheaper than brushless direct current motors (BLDC) and more suitable for lower-capacity units (*i.e.*, those with less than 150 PM_{2.5} CADR). For EL 3 and EL 4, manufacturers noted that upgrading the motor technology to variable-speed BLDC motors would cost more, but provide much more energy efficient airflow.

5.5.1.2 True HEPA and HEPA-type Filters

Manufacturers utilize true HEPA or HEPA-type filters in conjunction with a motor to move air through the filter and collect particles. Manufacturers noted that the difference between true HEPA and HEPA-type filters is significant, as true HEPA filters ensure that between 99.97 and 99.99 percent of 0.3 μm particles are collected, but are more restrictive to airflow due to these requirements. Another filter option is HEPA-type filters, which do not have to meet specific requirements and thus vary in the amount and size of particles that are collected but are less restrictive to airflow. Manufacturers noted that the filter selection alone does not improve efficiency significantly; rather, optimizing the filter/motor relationship to improve air flow rate and particle collection is key to improving efficiency.

5.5.1.3 Filter Size

Some manufacturers noted that filter surface area size is an important parameter that can be optimized. One manufacturer indicated that they could improve efficiency by using a smaller motor if a very large surface area filter were implemented that would minimize the pressure drop through the system. Manufacturers recommended a filter replacement frequency of 6 to 12 months to maintain filter performance. This is discussed further in section 8.3.3 of chapter 8 of this direct final rule TSD.

5.5.1.4 Ionization Technologies

Some manufacturers utilize ionizers, sometimes called ionization brushes, which precipitate charged ions onto or near the filter which attract small dust and smoke particles to improve particle collection. One manufacturer noted that utilizing an ionizer can allow for the use of a less restrictive filter, while maintaining a high particle collection rate. With a less restrictive filter, a smaller more efficient motor can be used to significantly improve efficiency.

5.5.1.5 Cabinet Size

Several manufacturers noted that cabinet case size and optimizing the inlets/outlets of airflow is an important factor to the overall efficiency of the unit. Manufacturers noted that cabinet size is an important factor for consumers; however, a larger cabinet allows for more airflow, which can be a factor to improve performance and thereby, IEF.

5.5.1.6 Carbon/Volatile Organic Compound Filters

Based on feedback from manufacturers, market research, and product teardowns, most air cleaners incorporate some form of carbon or activated carbon filter. These are typically used to remove gases, odors, and/or volatile organic compounds (VOC). However, some manufacturers noted that the test procedure does not account for the removal of organic compounds and therefore the carbon filters are reducing energy efficiency as they restrict airflow through the unit.

5.5.1.7 Air Quality Sensor

Some manufacturers noted that air quality sensor technology is a crucial factor for automatic mode efficiency. However, automatic mode is not currently tested and, therefore, does not impact the efficiency levels analyzed for this direct final rule.

5.5.1.8 Lower Standby Power Components

Most manufacturers stated that air cleaners use components that contribute to standby power such as Wi-Fi transceiver, air quality sensor, and LED displays. Efficiency improvements to these components improves the IEF of the test unit.

5.5.2 Selected Models for Analysis

For the direct final rule, DOE analyzed a total of 36 air cleaners representing what DOE research indicates is at least 18 manufacturers, and which incorporate a wide range of product characteristics. Table 5.5.1 shows the number of units from each potential product class that DOE analyzed for this direct final rule.

Table 5.5.1 Number of Air Cleaner Models Selected for the Test Sample

Product Class	Number of Models
PC1: $10 \leq \text{PM}_{2.5} \text{ CADR} < 100$	11
PC2: $100 \leq \text{PM}_{2.5} \text{ CADR} < 150$	8
PC3: $\text{PM}_{2.5} \text{ CADR} \geq 150$	18
TOTAL	36

DOE tested all 36 units according to Appendix FF^e and subsequently tore down the 12 units that were identified, based on testing results and observed product functions and characteristics, as representative units for each of the efficiency levels in the three product classes. Table 5.5.2 through Table 5.5.4 summarize the key attributes of each analyzed unit, including the following: IEF and PM_{2.5} CADR values (calculated using tested power consumption and the geometric average of smoke CADR and dust CADR provided by the manufacturers and/or ENERGYSTAR); whether the model has an ionizer; type of HEPA filter (applicable only to units with a filter), and whether the unit was torn down.

^e Appendix FF references the industry standard, AHAM AC-7-2022, “Energy Test Method for Consumer Room Air Cleaners” by reference. AHAM AC-7-2022 may be purchased at: <https://www.aham.org/ItemDetail?iProductCode=30014&Category=PADSTD&WebsiteKey=69a0a5fb-295a-4894-acd0-5785f146b899>

Table 5.5.2 Product Class 1 Units Selected for Testing and Teardown

Unit #	IEF (<i>PM_{2.5}</i> <i>CADR/W</i>)	PM_{2.5} CADR	True HEPA Filter?	Ionizer?	Testing	Teardown
1	1.3	66	Y	Y	X	
2	1.4	46	Y	N	X	
3	1.4	50	Y	N	X	X
4	1.6	65	N	N	X	
5	1.8	34	N	Y	X	X
6	1.9	61	Y	N	X	X
7	2.1	83	Y	Y	X	
8	2.5	74	N	N	X	
9	3.4	59	Y	N	X	X
10	5.4	61	Y	N	X	X
11	6.1	39	N	N	X	

Table 5.5.3 Product Class 2 Units Selected for Testing and Teardowns

Unit #	IEF (<i>PM_{2.5}</i> <i>CADR/W</i>)	PM_{2.5} CADR	True HEPA Filter?	Ionizer?	Testing	Teardown
1	2.0	127	Y	Y	X	
2	2.1	108	Y	N	X	
3	2.3	120	Y	N	X	
4	2.4	116	N	N	X	X
5	2.7	135	Y	N	X	
6	3.7	135	Y	N	X	
7	5.4	130	Y	N	X	X
8	12.8	115	N	Y	X	X

Table 5.5.4 Product Class 3 Units Selected for Testing and Teardowns

Unit #	IEF (<i>PM_{2.5}</i> <i>CADR/W</i>)	PM_{2.5} CADR	True HEPA Filter?	Ionizer?	Testing	Teardown
1	1.2	202	Y	N	X	X
2	1.4	245	Y	N	X	
3	2.4	190	N	N	X	
4	2.4	190	N	N	X	
5	2.6	225	Y	Y	X	
6	2.6	193	Y	Y	X	
7	3.0	353	Y	Y	X	
8	3.0	193	Y	Y	X	
9	3.0	195	Y	N	X	X
10	3.3	569	N	N	X	
11	3.4	180	Y	Y	X	
12	3.8	339	Y	N	X	
13	4.1	256	Y	Y	X	
14	5.6	255	Y	Y	X	
15	5.8	350	N	Y	X	
16	6.6	195	Y	N	X	X
17	7.2	211	Y	N	X	X
18	13.7	424	N/A	Y	X	

5.5.3 Cost Efficiency Curves

DOE conducted teardowns on the units indicated in Table 5.5.2 through Table 5.5.4, which covered the entire range of efficiency levels within each product class. As discussed, the physical teardown process was used to identify the specific technology options that manufacturers use to achieve higher levels of efficiency and the associated component MPCs. The teardown process also allowed DOE to exclude the non-efficiency related components from the MPC calculation. Also as discussed, DOE defined the incremental efficiency levels in terms of the current metric (IEF) and conducted the cost analysis using this metric.

The units selected for teardown had PM_{2.5} CADR values close to, but not necessarily identical to, the representative PM_{2.5} CADR value for its relevant product class. To improve

comparability between these representative units, relevant component costs were normalized to the representative PM_{2.5} CADR within each product class, identified in section 5.3.1. DOE used this approach because the size or type of some components, such as the motor, filter, cabinet size, *etc.* would change with capacity while certain components such as controls or wiring may not change with capacity. This allows the MPC for a 40 PM_{2.5} CADR unit and a 60 PM_{2.5} CADR unit, for example, to be compared at the same PM_{2.5} CADR value to ensure that the costs are representative for each efficiency level. Additionally, for filter costs, DOE normalized costs only using true HEPA filters. While DOE's representative units included some units that had a HEPA-type filter, MPCs for these units are based on estimates derived from true-HEPA filters.

The following sections summarize the teardown results for each product class. Chapter 3 of this direct final rule TSD provides additional technical details regarding each of the design options described below.

5.5.3.1 Product Class 1

Many of the Product Class 1 units currently on the market are small (<0.6 cubic feet (ft³)) units designed for a small single room such as a bathroom or bedroom (<155 square feet (sq. ft.)). All of the units analyzed for this product class utilized a fan and filter system to remove particulates.

Baseline: IEF of 1.5 PM_{2.5} CADR/W

The baseline Product Class 1 air cleaner is equipped with a mechanical dial that allows the user to choose between different preset fan speeds. The control system includes a single main control board. The baseline unit uses a rectangular HEPA filter and a three-speed shaded pole motor (SPM) which drives a blower assembly. The baseline case is injection molded plastic and contains a metal mesh screen in the front for outgoing airflow and a plastic filter cabinet with a grille that is detachable for filter replacement.

Efficiency Level 1: IEF of 1.7 PM_{2.5} CADR/W

EL 1 is typically achieved in Product Class 1 air cleaners through the following changes to the baseline model:

- Optimized motor-filter relationship: a typical EL 1 unit still uses an SPM, but alters the filter and motor to improve efficiency. For example, the analyzed unit reduced the size and energy consumption of the motor by half while also reducing the filter thickness to reduce airflow restriction. This reduced the total PM_{2.5} CADR, while slightly improving the overall IEF. Although the PM_{2.5} CADR for this representative unit was lower than that of the baseline representative unit, as mentioned previously, DOE normalized costs for the representative capacity. Additionally, while the altered filter was not a true-HEPA filter, DOE normalized filter costs for the representative capacity to the cost of a true-HEPA filter.

Efficiency Level 2: IEF of 1.9 PM_{2.5} CADR/W

EL 2 is typically achieved in Product Class 1 air cleaners through the following changes to the EL 1 model:

- Permanent Split Capacitor Motor: the typical EL 2 air cleaner has a PSC motor that improves the overall efficiency and capabilities of the motor. The PSC motor typically has 3 or 4 speeds and allows for more efficient airflow.

Efficiency Level 3: IEF of 3.4 PM_{2.5} CADR/W

EL 3 is typically achieved in Product Class 1 air cleaners through the following changes to the EL 2 model:

- Brushless DC Motor: the typical EL 3 air cleaner has a BLDC motor that improves the overall efficiency and capabilities of the motor. BLDC motors allow variable-speed control which further optimizes airflow.
- Cylindrical HEPA Filter: at EL 3, current models on the market use a cylindrical HEPA filter design, which increases filter surface area significantly compared to rectangular filters. A cylindrical filter design further reduces the restriction in air flow across the filter without compromising on performance because a cylindrical shape allows for a much larger surface area for the same volume of filter material. The larger surface area reduces the resistance across the filter material, which reduces the pressure drop and improves efficiency overall.

Efficiency Level 4: IEF of 5.4 PM_{2.5} CADR/W

EL 4 (max-tech) is typically achieved in Product Class 1 air cleaners through the following changes to the EL 3 model:

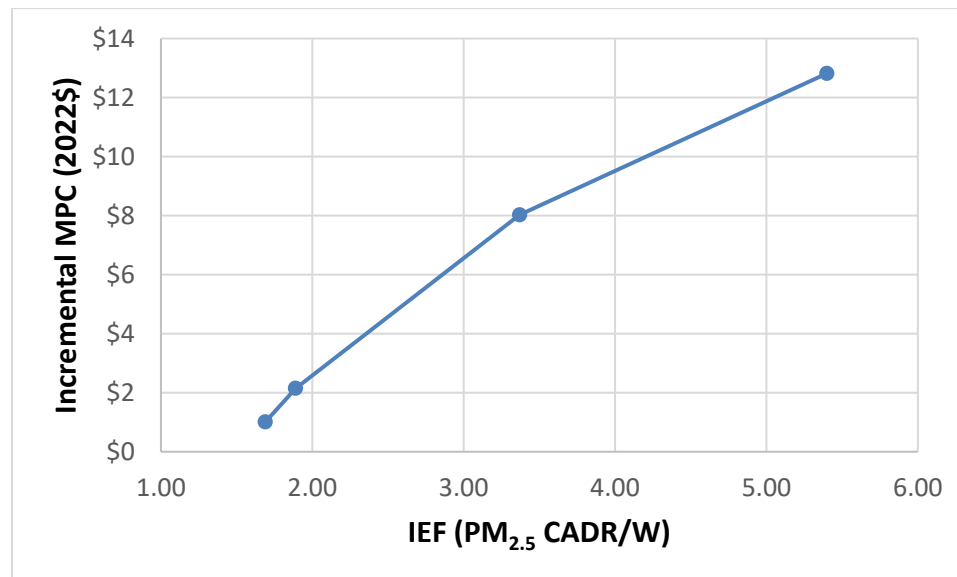
- Optimized motor-filter relationship: the typical EL 4 air cleaner changes the fan and filter size to improve airflow and efficiency. For example, the analyzed EL 4 unit utilized a smaller motor which reduced PM_{2.5} CADR and energy consumption, making the unit overall more efficient.

Summary

Table 5.5.5 and Figure 5.5.1 present the incremental MPCs at each efficiency level for Product Class 1.

Table 5.5.5 Incremental Manufacturer Production Costs for Product Class 1

EL	IEF ($PM_{2.5}$ CADR/W)	MPC (2022\$)	Incremental MPC (2022\$)
Baseline	1.5	\$31.24	-
1	1.7	\$32.25	\$1.01
2	1.9	\$33.39	\$2.15
3	3.4	\$39.27	\$8.03
4	5.4	\$44.06	\$12.82

**Figure 5.5.1 Cost-Efficiency Relationship for Product Class 1**

5.5.3.2 Product Class 2

Unit selection for Product Class 2 was based off the ENERGY STAR database for EL 2 and above. Because DOE's analysis did not directly identify units on the market at baseline and EL 1 levels for Product Class 2 that could be tested and torn down, estimates for the design options at these efficiency levels were made based on units with similar IEF from the other two product classes.

Product Class 2 air cleaners include medium cabinet-sized units (0.6–1.2 ft³) and are designed to clean a single room, such as a kitchen or shared living space (155–235 sq. ft.).

Baseline: IEF of 1.5 $PM_{2.5}$ CADR/W

As discussed, no baseline unit was identified for Product Class 2 using the available databases. Therefore, the design options for a typical baseline unit were derived for this product class from the Product Class 1 baseline unit.

DOE estimated that the typical baseline unit is a slightly larger, scaled-up version of the Product Class 1 baseline unit. Therefore, the typical baseline Product Class 2 unit is equipped with a mechanical dial with different fan speed settings, contains a single main control board, and uses an SPM motor.

The typical baseline unit has a rectangular HEPA filter and is made from an injection molded plastic case with a filter cabinet containing an additional carbon mesh filter.

Efficiency Level 1: IEF of 1.9 PM_{2.5} CADR/W

As noted, DOE was unable to source an EL 1 unit in Product Class 2 and estimated the design options for a typical EL 1 unit based on the EL 1 unit from Product Class 1.

DOE estimates that EL 1 is typically achieved in Product Class 2 air cleaners through the following changes to the baseline model:

- Permanent Split Capacitor Motor: the use of a more efficient motor technology significantly increases the airflow through the air cleaner while maintaining power consumption.

Efficiency Level 2: IEF of 2.4 PM_{2.5} CADR/W

EL 2 is typically achieved in Product Class 2 units through the following changes to the EL 1 model:

- Direct Double-Ended Blower Assembly: with an additional blower assembly, the unit is able to move more air through the filter, improving the rate at which it can deliver clean air.
- Optimized motor-filter relationship: The analyzed EL 2 unit used a thinner filter with a larger surface area, which helps reduce the pressure drop across the filter. This less restrictive filter allowed more airflow and improved efficiency. While this thinner filter in the teardown unit was HEPA-type, DOE normalized the costs for the representative capacity to the cost of a true-HEPA filter.

Efficiency Level 3: IEF of 5.4 PM_{2.5} CADR/W

EL 3 is typically achieved in Product Class 2 air cleaners through the following changes to the EL 2 model:

- Brushless DC Motor: the typical EL 3 air cleaner has a BLDC motor that improves the overall efficiency and capabilities of the motor. BLDC motors allow for variable speed control which helps further optimize airflow.
- Cylindrical HEPA Filter: at EL 3, current models on the market use a cylindrical HEPA filter design, which increases filter surface area significantly compared to rectangular filters. A cylindrical filter design further reduces the restriction in air flow across the filter without compromising on performance because a cylindrical shape allows for a much larger surface area for the same volume of filter material. The

larger surface area reduces the resistance across the filter material, which in turn reduces the pressure drop and improves efficiency overall.

Efficiency Level 4: IEF of 12.8 PM_{2.5} CADR/W

EL 4 is typically achieved in Product Class 2 air cleaners through the following changes to the EL 3 model:

- Optimized motor-filter relationship: the typical EL 4 air cleaner alters the motor and/or filter size to improve airflow and efficiency. For example, the analyzed EL 4 unit utilized a smaller motor and thinner filter which reduced PM_{2.5} CADR and energy consumption, making the unit overall more efficient.
- Lower Standby Power Components: the typical EL 4 unit has lower standby power compared to EL 3 units which can be attributed to upgrades to the standby power components such as the air quality sensor.
- Ionization Brush: Use of an ionization brush allows for a less restrictive filter to be used as the ionizer can help capture some of the smaller particles.

Summary

Table 5.5.6 and Figure 5.5.2 present the incremental MPCs at each efficiency level for Product Class 2.

Table 5.5.6 Incremental Manufacturer Production Costs for Product Class 2

EL	IEF (PM_{2.5} CADR/W)	MPC (2022\$)	Incremental MPC (2022\$)
Baseline	1.5	\$42.97	-
1	1.9	\$44.26	\$1.29
2	2.4	\$45.62	\$2.65
3	5.4	\$50.45	\$7.48
4	12.8	\$55.55	\$12.58

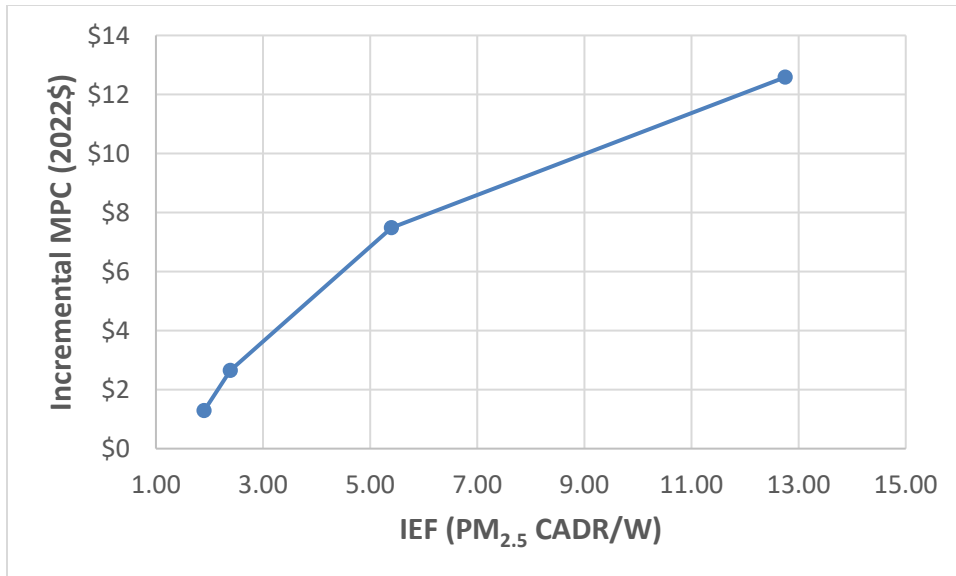


Figure 5.5.2 Cost-Efficiency Relationship for Product Class 2

5.5.3.3 Product Class 3

Product Class 3 air cleaners have a large cabinet ($> 1.6 \text{ ft}^3$), are typically less portable than lower-capacity units, in some cases being equipped with wheels to facilitate moving, and are designed to be used for an extended duration in a large room ($> 235 \text{ sq. ft}$) such as a classroom, office, or large living area. Product Class 3 units are also the most prevalent air cleaners on the market.

Baseline: IEF 1.2 PM_{2.5} CADR/W

The baseline Product Class 3 air cleaner is equipped with electronic controls that allow the user to select fan speed, set a timer, check if filter replacement is necessary, and change the interface brightness. The baseline includes a main control board, and uses a PSC motor. The typical baseline unit has a rectangular HEPA filter, encased in plastic with air inlets on the sides with high density polyethylene (HDPE) mesh screens to collect large particles.

Efficiency Level 1: IEF of 2.0 PM_{2.5} CADR/W

DOE's sample of teardown units did not include any Product Class 3 air cleaners at EL 1. In lieu of teardown data, DOE assumed that the pathway from baseline to EL 1 and higher efficiency levels for Product Class 3 would be similar to the pathway for the other two product classes. Based on this, DOE estimates the following design options to improve the efficiency from baseline to EL 1:

- Optimized motor-filter relationship: DOE estimates that an EL 1 unit would optimize the motor and filter relationship to improve efficiency, such as increasing the surface area of the filter, reducing filter thickness, and/or increasing air inlet/outlet size.

- Lower Standby Power Components: Optimization of the standby components, such as Wi-Fi transceiver and air quality sensor would further improve the efficiency of an EL 1 unit.

Efficiency Level 2: IEF of 2.9 PM_{2.5} CADR/W

EL 2 is typically achieved in Product Class 3 air cleaners through the following changes to the EL 1 model:

- Increased Filter Surface Area: The typical EL 2 unit has a rectangular filter with a larger surface area, which reduces the pressure drop across the filter, thereby improving efficiency.
- Larger Air Inlets/Outlets: A typical EL2 unit has larger openings for airflow compared to the baseline to improve air flow, which improve the efficiency of the unit.

Efficiency Level 3: IEF of 6.6 PM_{2.5} CADR/W

EL 3 is typically achieved in Product Class 3 air cleaners through the following changes to the EL 2 model:

- Brushless DC Motor: the typical EL 3 air cleaner has a BLDC motor that improves the overall efficiency and capabilities of the motor. BLDC motors allow for variable speed control which helps further optimize airflow.
- Cylindrical HEPA Filter: at EL 3, current models on the market use a cylindrical HEPA filter design, which increases filter surface area significantly compared to rectangular filters. A cylindrical filter design further reduces the restriction in air flow across the filter without compromising on performance because a cylindrical shape allows for a much larger surface area for the same volume of filter material. The larger surface area reduces the resistance across the filter material, which in turn reduces the pressure drop and improves efficiency overall.

Efficiency Level 4: IEF of 7.4 PM_{2.5} CADR/W

EL 4 (max-tech) is typically achieved in Product Class 3 air cleaners through the following changes to the EL 3 model:

- Optimized motor-filter relationship: the typical EL 4 air cleaner alters the motor and/or filter size to improve airflow and efficiency. For example, the analyzed EL 4 unit utilized a smaller motor and larger filter while reducing the filter thickness. These optimizations reduced PM_{2.5} CADR and energy consumption, making the unit overall more efficient.
- Lower Standby Power Components: the typical EL 4 unit has lower standby power compared to EL 3 units, this can be attributed to upgrades to the standby power components such as the air quality sensor.
- Ionization Brush: Use of an ionization brush allows for a less restrictive filter to be used as the ionizer can help capture some of the smaller particles.

Summary

Table 5.5.7 summarizes the MPCs at each efficiency level for Product Class 3. Figure 5.5.3 shows this information graphically.

Table 5.5.7 Incremental Manufacturer Production Cost for Product Class 3 (2022\$)

EL	IEF ($PM_{2.5}$ CADR/W)	MPC (2022\$)	Incremental MPC (2022\$)
Baseline	1.2	\$70.50	-
1	2.0	\$71.66	\$1.17
2	2.9	\$72.50	\$2.00
3	6.6	\$74.33	\$3.84
4	7.4	\$74.61	\$4.11

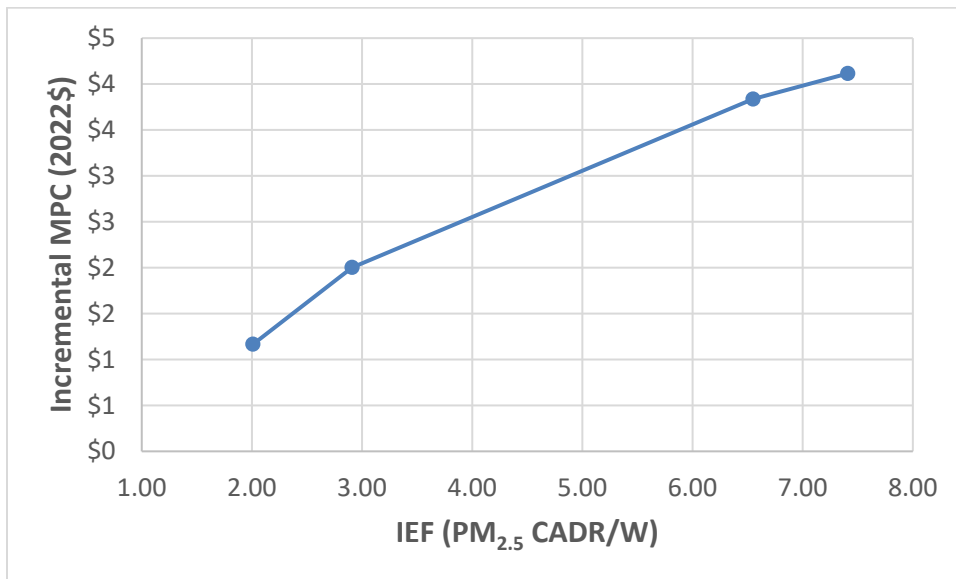


Figure 5.5.3 Cost-Efficiency Relationship for Product Class 3

5.5.3.4 Filter Costs

In addition to determining the MPCs for each representative unit at each efficiency level, DOE also disaggregated the overall MPC at each efficiency level to determine filter costs, which are used to determine the maintenance and repair costs for the LCC and PBP. These costs are shown in Table 5.5.8.

Table 5.5.8 Disaggregated Filter Costs (2022\$)

Efficiency Level	Product Class 1	Product Class 2	Product Class 3
Baseline	\$2.62	\$5.83	\$9.06
EL 1	\$1.92	\$5.00	\$8.68
EL 2	\$1.79	\$4.16	\$8.29
EL 3	\$6.71	\$10.25	\$12.10
EL 4	\$7.05	\$7.78	\$12.69

DOE observed that the filter MPC typically decreased from the baseline efficiency level to EL 2 and then increased for EL 3 and EL 4. This is because the baseline unit typically has a larger rectangular filter compared to EL 1 and EL 2 filters, leading to higher filter costs for the baseline unit. EL 3 and EL 4 units have cylindrical filters with plastic casing, compared to the paper/cardboard casing seen at baseline through EL 2, both of which lead to substantially higher filter costs at these levels.

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CHAPTER 6. MARKUPS ANALYSIS

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CHAPTER 6. MARKUPS ANALYSIS

6.1 INTRODUCTION

To carry out its analyses, the U.S. Department of Energy (DOE) needed to determine the cost to the consumer of baseline efficiency products and the cost of more efficient units the consumer would purchase under new or amended energy conservation standards. DOE calculated such costs based on engineering estimates of manufacturing production costs, a manufacturer markup to calculate the manufacturer selling price (*i.e.*, the price to the manufacturer's first customer), and appropriate additional markups for the various distribution channels to move the product to consumers.

The total markups applied to all product classes differ by their corresponding distribution channel, as discussed below. DOE adopts a total firm approach to markups, consistent with economic theory, rather than a product level cost accounting approach. In the economics-based approach, the firm is assumed to produce a single product, so marginal costs include total costs for the firm rather than cost of goods inventoried and produced. The markup in this treatment depends on marginal cost, not average cost as is the case in cost accounting. These distinctions should be kept in mind when considering the results presented. At each point in a distribution channel, companies mark up the price of a product to cover costs. In financial statements, gross margin in dollar value (GM) is the difference between the company revenue and the company cost of sales, or CGS. The GM takes account of the expenses of companies, including various operating costs; research and development (R&D); interest expenses; depreciation; and taxes—and company profits. To cover costs and to contribute positively to company cash flow, the price of products must include a markup. Products command lower or higher markups depending on company expenses associated with the product and the degree of market competition.

DOE estimates a baseline markup and an incremental markup for each market participant besides manufacturers. DOE defines a baseline markup as a multiplier that converts the manufacturer selling price (MSP) of equipment with baseline efficiency to the consumer purchase price. An incremental markup is defined as the multiplier to convert the incremental increase in MSP of higher efficiency equipment to the consumer purchase price. Because companies mark up the price at each point in the distribution channel, both overall baseline and incremental markups are dependent on the distribution channel, as described in Section 6.1.1.

6.1.1 Distribution Channel

The appropriate markups for determining consumer product prices depend on the type of distribution channel through which products move from manufacturer to consumer. At each point in the distribution channel, companies mark up the price of the product to cover costs.

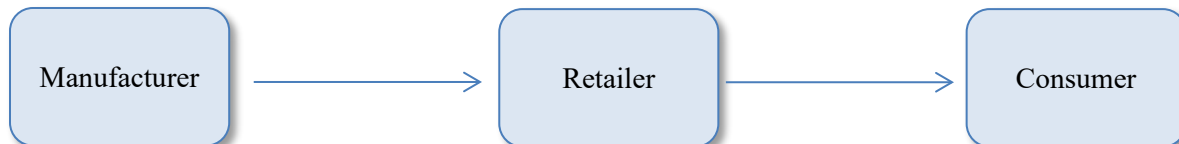
Based on data from the manufacturer interview feedback, the primary distribution channel for air cleaners is through online vendors (e-commerce), accounting for around 60 percent of all air cleaner sales across the total market. The rest of the sales are roughly equally divided between the direct sales channel and the retail channel (including both the in-store

purchase and through retailers' online sales), each representing 20 percent of the total market (Figure 6.1.1).

e-Commerce Channel:



Retail Channel:



Direct Sales Channel (in Commercial Application Only):

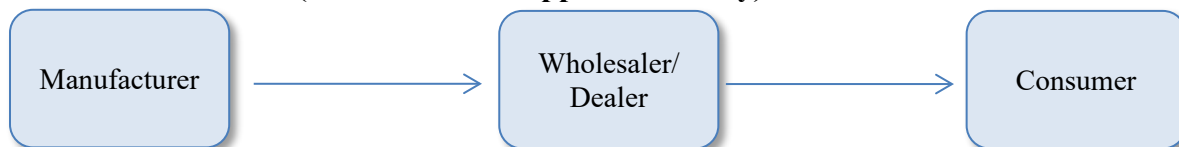


Figure 6.1.1 Distribution Channels for Air Cleaners

In this direct final rule, DOE considered both the residential and commercial application for air cleaners and assumed that the three aforementioned distribution channels are available in the commercial application but only the e-commerce channel and the retail channel are available in the residential application. Combining with data provided by TechSci Research's air purifier report,¹ DOE estimated the market shares by application and distribution channel, as shown in Table 6.1.1.

Table 6.1.1 Market Shares by Application and Distribution Channel

End Use Application	Distribution Channel	Shares in 2021 (%)
Commercial	Direct Sales	20.00
	Retail	7.86
	e-Commerce	23.59
Residential	Retail	12.14
	e-Commerce	36.41

6.2 MANUFACTURER MARKUPS

DOE uses the manufacturer markups to convert manufacturer production costs to manufacturer selling prices. The manufacturer markup covers all manufacturer non-production costs (e.g., SG&A, R&D, and interest) and profit. The derivation of manufacturer markups is described in more detail in chapter 12 of this direct final rule technical support document (TSD).

6.3 RETAILER AND WHOLESALER MARKUPS

A change in energy efficiency standards usually increases the manufacturer selling price that wholesalers or retailers pay. In the past, DOE used the same markup multiplier to estimate the prices of both the baseline and more efficient products. However, given the nature of the appliance retail and wholesale market, applying the same markup on higher MSPs of more efficient products would imply an increase in the dollar margin earned by retailers, and an increase in per-unit profit.

Based on microeconomic theory, the degree to which firms can pass along a cost increase to consumers depends on the level of market competition and market structure on both the supply and demand side (*e.g.*, supply and demand elasticity). DOE examined industry data from IBISWorld and the results suggest the industry groups involved in appliance retail and wholesaling exhibit a strong degree of competition (Table 6.3.1).² In addition, consumer demand for household appliances is relatively inelastic (*i.e.*, demand is not expected to decrease substantially with an increase in the price of equipment). In such relatively competitive markets with inelastic demands, it may be tenable for retailers to maintain a fixed markup for a short period of time after a product cost increase, but the market competition would eventually force them to readjust their markups to reach a medium-term equilibrium in which per-unit profit is relatively unchanged before and after standards are implemented.

Table 6.3.1 Competitive Environment of Appliance Retailers and Wholesalers

Sector	Industry Concentration	Competition	Barriers to Entry
TV & appliance wholesaling	Low	High and steady	Medium and steady
TV & appliance retailers	Low	High and steady	Medium and steady
Consumer electronics stores	Medium	High and increasing	Medium and steady
Department stores	High	High and increasing	Medium and steady
Home improvement stores	High	Medium and steady	Medium and steady

* Note that there is competition between the four types of appliance retailers listed in this table, as well as within each individual retailing type.

Thus, DOE concluded that applying fixed markups for both baseline products and higher-priced products meeting a new standard is not viable in the medium to long term considering the nature of the appliance retail market. DOE developed the incremental markup approach based on the widely accepted economic view that firms are not able to sustain a persistently higher dollar profit in a competitive market in the medium to long term. If the wholesale price of the product increases under standards, the only way to maintain the same dollar profit as before is for the markup (and percent gross margin) to decline.

To estimate the markup under standards, DOE derived an incremental markup that is applied to the incremental equipment costs of higher efficiency products. The overall

markup on the products meeting standards is an average of the markup on the component of the cost that is equal to the baseline product (baseline markup) and the markup on the incremental cost (incremental markup), weighted by the share of each in the total cost of the standards-compliant product.

DOE’s incremental markup approach allows the part of the cost that is thought to be affected by the standard to scale with the change in manufacturer price. The income statements DOE used to develop retailer and wholesaler markups itemize firm costs into a number of expense categories, including direct costs to purchase or install the equipment, operating labor and occupancy costs, and other operating costs and profit. Although retailers and wholesalers tend to handle multiple commodity lines, DOE contends that these aggregated data provide the most accurate available indication of the cost structure of distribution channel participants.

DOE uses these income statements to divide firm costs between those that are not likely to scale with the manufacturer price of equipment (labor and occupancy expenses, or “invariant” costs) and those that are (operating expenses and profit, or “variant” costs). For example, when the MSP of equipment increases, only a fraction of a retailer’s expenses increases (operating expenses and profit), while the remainder can be expected to stay relatively constant (labor and occupancy expenses). For example, if the unit price of a air cleaner increases by 20 percent under standards, it is unlikely that the cost of secretarial support in an administrative office or office rental expenses will increase proportionally.

6.3.1 Approach for Retailer Markups

DOE based the retailer markups for air cleaners on financial data for “electronics and appliance stores” from the 2017 U.S. Census *Annual Retail Trade Survey (ARTS)*, which is the most recent survey that includes industry-wide detailed operating expenses for that economic sector.³

The baseline markup converts the MSP of baseline products to the retailer sales price. DOE considers baseline models to be products that just meet current Federal energy conservation standards. DOE used the following equation to calculate an average baseline markup (MU_{BASE}) for retailers.

$$MU_{BASE} = \frac{SALES_{RTL}}{CGS_{RTL}} = \frac{CGS_{RTL} + GM_{RTL}}{CGS_{RTL}} = 1 + \frac{GM_{RTL}}{CGS_{RTL}} \quad \text{Eq. 6.1}$$

Where:

MU_{BASE} =	retailer’s baseline markup multiplier,
$SALES_{RTL}$ =	retailer’s sales revenue,
CGS_{RTL} =	retailer’s cost of goods sold (CGS), and
GM_{RTL} =	retailer’s gross margin (GM).

To estimate incremental retailer markups, DOE divides retailers' operating expenses into invariant and variant cost categories, as described in previous section. DOE used the following equation to calculate the incremental markup (MU_{INCR}) for retailers.

$$MU_{INCR} = \frac{CGS_{RTL} + VC_{RTL}}{CGS_{RTL}} = 1 + \frac{VA_{RTL}}{CGS_{RTL}}$$

Eq. 6.2

Where:

MU_{INCR} = retailer's incremental markup multiplier,
 CGS_{RTL} = retailer's cost of goods sold, and
 VC_{RTL} = retailer's variant costs.

6.3.2 Derivation of Retailer Markups

The 2017 *ARTS* data for electronics and appliance stores provide total sales data and detailed operating expenses. To construct a complete data set for estimating markups, DOE needed to estimate CGS and GM. The 2017 *ARTS* publishes a separate document containing historical sales and gross margin for electronics and appliance stores. DOE took the GM as a percent of sales reported for 2017 and combined that percent with detailed operating expenses data from 2017 *ARTS* to construct a complete income statement for electronics and appliance stores to estimate both baseline and incremental markups. Table 6.3.2 shows the calculation of the baseline retailer markup.

Table 6.3.2 Data for Calculating Baseline Markup: Electronics and Appliance Stores

Business Item	Amount (\$1,000,000)
Sales	99,401
Cost of goods sold (CGS)	66,897
Gross margin (GM)	32,504
Baseline Markup = (CGS+GM)/CGS	1.49

Source: U.S. Census, 2017 *Annual Retail Trade Survey*.

Table 6.3.3 shows the breakdown of operating expenses for electronics and appliance stores based on the 2017 *ARTS* data. The incremental markup is calculated as 1.24.

Table 6.3.3 Data for Calculating Incremental Markup: Electronics and Appliance Stores

Business Item	Amount (\$1,000,000)
Sales	99,401
<i>Cost of goods sold (CGS)</i>	66,897
<i>Gross margin (GM)</i>	32,504
Labor & Occupancy Expenses (invariant)	
Annual payroll	10,226
Employer costs for fringe benefit	1,574
Contract labor costs, including temporary help	157
Purchased utilities, total	459
Purchased Repairs and Maintenance to Buildings, Structures, and Offices	266
Lease and Rental Payments for Land, Buildings, Structures, Store Space, and Offices	2,686
Purchased communication services	290
Cost of purchased professional and technical services	743
Subtotal:	16,401
Other Operating Expenses & Profit (variant)	
Expensed equipment	87
Cost of purchased packaging and containers	51
Other materials and supplies not for resale	387
Cost of purchased transportation, shipping, and warehousing services	471
Cost of purchased advertising and promotional services	1,392
Cost of purchased software	93
Data Processing and Other Purchased Computer Services	66
Purchased Repairs and Maintenance to Machinery and Equipment	118
Lease and Rental Payments for Machinery, Equipment, and Other Tangible Items	89
Commission expenses	235
Depreciation and amortization charges	1,019
Taxes and license fees	382
Other operating expenses	2,312
<i>Net profit before tax (operating profit)</i>	9,401
Subtotal:	16,103
Incremental markup = (CGS + Total Other Operating Expenses and Profit)/CGS	1.24

Source: U.S. Census. 2017 Annual Retail Trade Survey.

6.3.2.1 Derivation of Online Vendor Markups

To differentiate the markups applied in the e-Commerce channel, DOE conducted a price comparison of top five best-selling models listed in TechSci Research's air purifier report between their manufacturer suggested retail prices (MSRPs) and retail prices found on the e-Commerce website. DOE estimated that the online retail prices are on average 1.1 percent lower than the MSRPs. Thus, DOE adjusted the retailer markups by applying the 1.1 percent difference to derive the markups used in the e-Commerce channel.

6.3.3 Approach for Wholesaler Markups

DOE developed baseline and incremental wholesaler markups using the firm income

statement for household appliances and electrical and electronic goods merchant wholesale sector from the 2017 U.S. Census *Annual Wholesale Trade Report* (“AWTR”)⁴. Baseline markups cover all the wholesaler’s costs (both fixed and variable). DOE calculated the baseline markup for wholesalers using the following equation.

$$MU_{BASE} = \frac{CGS_{WHOLE} + GM_{WHOLE}}{CGS_{WHOLE}}$$

Eq. 6.3

where:

MU_{BASE} = wholesaler’s baseline markup,
 CGS_{WHOLE} = wholesaler’s cost of goods sold, and
 GM_{WHOLE} = wholesaler’s gross margin,

DOE used the following equation to calculate the incremental markup (MU_{INCR}) for wholesalers.

$$MU_{INCR} = \frac{CGS_{WHOLE} + VC_{WHOLE}}{CGS_{WHOLE}}$$

Eq. 6.4

where:

MU_{INCR} = wholesaler’s incremental markup,
 CGS_{WHOLE} = wholesaler’s cost of goods sold, and
 VC_{WHOLE} = wholesaler’s variable costs.

6.3.4 Derivation of Wholesaler Markups

The 2017 AWTR data for household appliances and electrical and electronic goods merchant wholesalers provide total sales data and detailed operating expenses representing refrigerator, refrigerator-freezer, and freezer wholesalers, similar to the data used in developing retailers markups. Hence, DOE took the same approach as described in section 6.3.2 to construct a complete data set for that particular sector and estimated their baseline and incremental markups. Table 6.3.4 presents the calculation of the baseline retailer markup.

Table 6.3.4 Data for Baseline Markup Calculation: Household Appliances and Electrical and Electronic Goods Merchant Wholesale

Kind of Business Item	Amount (\$1,000,000)
Sales	583,634
Cost of Goods Sold (CGS)	433,056
Gross Margin (GM)	150,578
Baseline Markup = (CGS+GM)/CGS	1.35

Source: U.S. Census, 2017 Annual Wholesale Trade Report

Table 6.3.5 shows the breakdown of operating expenses using the 2017 AWTR data. The incremental markup is calculated as 1.20.

Table 6.3.5 Data for Incremental Markup Calculation: Household Appliances and Electrical and Electronic Goods Merchant Wholesale

	Amount (\$1,000,000)
Sales	583,634
<i>Cost of Goods Sold (CGS)</i>	<i>433,056</i>
<i>Gross Margin (GM)</i>	<i>150,578</i>
Labor & Occupancy Expenses ("Fixed")	
Total payroll, other employee wages	44,715
Total fringe benefits	10,082
Temporary staff and leased employee expenses	1,797
Rental costs of machinery and equipment	-
Rental costs of buildings	3,440
Cost of repair to building	566
Cost of repair to machinery and equipment	592
Purchased communication services	973
Purchased utilities, total	522
Subtotal:	62,687
Other Operating Expenses & Profit ("Variable")	
Purchased professional and technical services	5,087
Data processing and other purchased computer services	649
Expensed computer hardware and other equipment	1,147
Expensed purchases of software	889
Advertising and promotion services	5,627
All other expenses	-
Purchased transportation, shipping and warehousing services	-
Taxes and license fees	843
Total depreciation	4,956
Commission expenses	3,074
Purchases of packaging material and containers	-
Purchases of other materials, parts, and supplies (not for resale)	943
<i>Net profit before tax (Operating profit)</i>	<i>51,636</i>
Subtotal:	87,891
Incremental Markup = (CGS+Total Other Operating Expenses and Profit)/CGS	1.20

Source: U.S. Census, 2017 Annual Wholesale Trade Report

6.4 SALES TAXES

The sales tax represents state and local sales taxes that are applied to the consumer product price. The sales tax is a multiplicative factor that increases the consumer product price.

DOE used state and local taxes data provided by the Sales Tax Clearinghouse.⁵ DOE derived population-weighted average tax values for each census division and each state, as shown in Table 6.4.1 and Table 6.4.2. DOE then assigned division-level average tax values for each household used in the life-cycle cost analysis

Table 6.4.1 Average Sales Tax Rates by Census Division

Region	Census Division	2022 Tax Rate (%)
1	New England	5.63
2	Middle Atlantic	7.39
3	East North Central	7.07
4	West North Central	7.19
5	South Atlantic	6.61
6	East South Central	8.09
7	West South Central	8.31
8	Mountain	6.83
9	Pacific	7.98
Weighted Average		7.30

Table 6.4.2 Average Sales Tax Rates by State

State	State Abbreviation	2022 Tax Rate (%)
Alabama	AL	8.70
Alaska	AK	1.30
Arizona	AZ	7.30
Arkansas	AR	9.15
California	CA	8.80
Colorado	CO	6.40
Connecticut	CT	6.35
Delaware	DE	0.00
District of Columbia	DC	6.00
Florida	FL	7.00
Georgia	GA	7.40
Hawaii	HI	4.45
Idaho	ID	6.05
Illinois	IL	8.60
Indiana	IN	7.00
Iowa	IA	6.95

State	State Abbreviation	2022 Tax Rate (%)
Kansas	KS	8.40
Kentucky	KY	6.00
Louisiana	LA	9.40
Maine	ME	5.50
Maryland	MD	6.00
Massachusetts	MA	6.25
Michigan	MI	6.00
Minnesota	MN	7.45
Mississippi	MS	7.05
Missouri	MO	7.05
Montana	MT	0.00
Nebraska	NE	6.10
Nevada	NV	8.25
New Hampshire	NH	0.00
New Jersey	NJ	6.60
New Mexico	NM	7.05
New York	NY	8.45
North Carolina	NC	7.00
North Dakota	ND	6.25
Ohio	OH	7.20
Oklahoma	OK	8.60
Oregon	OR	0.00
Pennsylvania	PA	6.35
Rhode Island	RI	7.00
South Carolina	SC	7.45
South Dakota	SD	6.00
Tennessee	TN	9.50
Texas	TX	8.00
Utah	UT	7.15
Vermont	VT	6.10
Virginia	VA	5.75
Washington	WA	9.30
West Virginia	WV	6.15
Wisconsin	WI	5.45
Wyoming	WY	5.40
Weighted Average		7.30

6.5 SUMMARY OF MARKUPS

Table 6.5.1 summarizes the markups at each stage in the distribution channel and provides the average sales tax to arrive at overall markups for the product classes considered in this analysis.

Table 6.5.1 Summary of Markups

	Direct Sales		Retail		e-Commerce	
	Baseline	Incremental	Baseline	Incremental	Baseline	Incremental
Manufacturer	1.45		1.45		1.45	
Wholesaler/Dealer	1.35	1.20				
Retailer/Online Vendor			1.49	1.24	1.47	1.22
Sales Tax	1.073		1.073		1.073	
Overall	2.10	1.87	2.31	1.93	2.29	1.91

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CHAPTER 7. ENERGY USE ANALYSIS

7.1 INTRODUCTION

The purpose of the energy use analysis is to determine the annual energy consumption of air cleaners in representative U.S. homes and commercial buildings, and to assess the potential savings in energy use and costs that consumers would experience from more-efficient products. The U.S. Department of Energy (DOE) uses annual energy use, along with energy prices, to establish energy costs at various energy efficiency levels and to carry out the life-cycle cost (LCC) and payback period (PBP) analysis described in chapter 8. This chapter describes how DOE determined the annual energy use of air cleaners at various efficiency levels for each product class considered in this Direct Final Rule (DFR).

The engineering analysis, described in chapter 5 of this technical support document (TSD), reports energy use based on the DOE test procedure, which uses typical operating conditions in a laboratory setting. This test serves as the basis for comparing the performance of appliances under uniform conditions. Actual energy usage in the field varies depending on the conditions under which an appliance is operated. The energy use analysis seeks to estimate the range of energy consumption of the products in the field across diverse climate types, household and commercial building characteristics.

To establish a reasonable range of real-world energy consumption for air cleaners in residential applications, DOE relied on data from the Energy Information Administration's (EIA's) *2020 Residential Energy Consumption Survey (RECS 2020)*.¹ The *RECS* survey is a national sample of housing units that collects statistical information on residential energy consumption and expenditures across the nation, along with data on housing and household characteristics. *RECS 2020* has energy-related data for 18,496 housing units and represents the household population of the United States.

Based on a market research report conducted by Evergreen Economics², in general, 60 percent of air cleaners are used in residences, and 40 percent are used in commercial settings. To calculate the energy use patterns of air cleaners in commercial applications, DOE relied largely on data from the EIA's *2018 Commercial Building Energy Consumption Survey (CBECS 2018)*.³ *CBECS 2018* has energy-related data for 6,436 buildings and represents the commercial building stock in the United States.

The details of each survey sample are described in the following sections.

7.2 SURVEY SAMPLES

Although *RECS 2020* and *CBECS 2018* are the most recent national residential household and commercial building surveys available, neither of them contains information about air cleaner ownership or usage. In the absence of such data, DOE included all household samples from *RECS 2020*, and all non-vacant and operating building samples from *CBECS 2018* for energy use calculations. The weightings indicate how frequently each household or commercial

building configuration occurs in the general population. Therefore, DOE adjusted the EIA's assigned sample weights to reflect geographic and principal building activity distributions of ownership of air cleaners for residential and commercial applications respectively based on the Evergreen Economics market research report.² This step allows DOE to account for the probability of a household or a building sample being equipped with air cleaners, given its location or principal building activity. Sections 7.2.1 and 7.2.2 describe the methodology of this calibration.

7.2.1 Sample for Residential Applications

Given the lack of the air cleaner ownership information in *RECS 2020*, to adjust the EIA's assigned sample weight, to construct a national representative consumer sample pool for air cleaners, DOE first estimated the number of air cleaners per household sample based on data from the Evergreen Economics market research report,² and adjusted the sample weights accordingly. Table 7.2.1 shows the distribution of number of air cleaners owned for each household size bin provided by the Evergreen Economics report.

Table 7.2.1 Number of Air Cleaners Owned by Household Size Bin Based on Evergreen Economics Report 2021

Number of Household Members*	Number of Air Cleaners Owned				Weighted Average of Number of Air Cleaners Owned
	1	2	3	4 or more	
	Percentage of Respondents				
1-2 (n=228)	68%	24%	6%	2%	1.42
3-4 (n=155)	65%	23%	10%	2%	1.49
5-6 (n=108)	42%	40%	14%	4%	1.80

*n = survey subsample size from Evergreen Economics Report

DOE further adjusted the *RECS 2020* sample weights based on the geographic distribution of air cleaner stocks report by TechSci Research.⁴ Table 7.2.2 shows the US air cleaner market share by census region from the report.

Table 7.2.2 US Air Cleaner Market Share by Census Region

US Census Region	% of Market Share
North-East	11.28
Mid-West	21.55
South	42.30
West	24.87
Total	100.00

7.2.2 Sample for Commercial Applications

For the commercial sector air cleaner sample, due to the lack of air cleaner ownership information in *CBECS 2018*, DOE selected the subset of *CBECS 2018* records that met all of the following criteria:

- The building was not vacant.
- The building's total hours open per week are greater than 0.

Of the 6,436 commercial building samples, DOE used 6,278 samples from *CBECS 2018*.

DOE first adjusted the weights of the selected samples using their principal building activity (PBA) as reported in *CBECS 2018* and relied on data provided by 2022 TechSci Research's market report.⁴ This step allows DOE to take into account the probability of each building sample being equipped with air cleaner given the reported principal building activity.

Table 7.2.3 shows the US air cleaner market share by PBA as reported by TechSci Research.

Table 7.2.3 US Air Cleaner Market Share by Principal Building Activity

Principal Building Activity	% Market Share
Offices	23.36
Retail	14.50
Education	11.42
Food Service	7.35
Healthcare	5.14
Others	38.23
Total	100.00

CBECS 2018 provides the total climate-controlled square footage and the percentage of occupancy in each building, but it does not provide air cleaner ownership information, such as the number of air cleaners in a given building. DOE therefore used the total conditioned square

footage data in conjunction with guidance from ENERGY STAR for air cleaner sizing by room square footage⁵ to estimate the number of air cleaners per product class that each building sample would require if the building is equipped with air cleaners. Table 7.2.4 shows the ENERGY STAR recommended air cleaner CADR based on room size.

Table 7.2.4 Air Cleaner Minimum CADR for Typical Room Sizes

Room Area (ft ²)	100	200	300	400	500	600
Minimum CADR	65	130	195	260	325	390

7.3 ESTIMATES OF ENERGY USE

DOE calculated the annual energy consumption of an air cleaner using the following equation to account for operation in standby mode and four active modes – low speed mode, medium speed mode, max speed mode, and auto mode:

$$AirCleaner_{ENERGY} = (W_{stby} \times OH_{stby}) + [(W_{low} \times OH_{low}) + (W_{med} \times OH_{med}) + (W_{max} \times OH_{max}) + (W_{auto} \times OH_{auto})]$$

Eq. 7.1

Where:

$AirCleaner_{ENERGY}$	=	annual energy consumption of air cleaners (kWh/year),
W_{stby}	=	power consumption in standby mode (kW),
OH_{stby}	=	annual number of operating hours in standby mode,
W_{low}	=	power consumption in low speed mode (kW),
OH_{low}	=	annual number of operating hours in low speed mode,
W_{med}	=	power consumption in medium speed mode (kW),
OH_{med}	=	annual number of operating hours in medium speed mode,
W_{max}	=	power consumption in max speed mode (kW),
OH_{max}	=	annual number of operating hours in max speed mode,
W_{auto}	=	power consumption in auto mode (kW), and
OH_{auto}	=	annual number of operating hours in auto mode,

The power for each active mode of operation is scaled based on the max speed mode power:

$$W_i = PF_i \times W_{max}$$

Where:

i	=	low, medium, or auto mode,
W_i	=	power for mode ‘i’,
PF_i	=	power factor for mode ‘i’ relative to max speed mode power, and
W_{max}	=	power consumption in max speed mode (kW).

Based on the testing results, DOE determined the power factors shown in Table 7.3.1 to estimate the power consumption in each mode.

Table 7.3.1 Power Factor by Mode of Operation Relative to Max Speed Mode

Mode of Operation	Power Factor Relative to Max Speed Mode
Medium Speed Mode	0.610
Low Speed Mode	0.327
Auto Mode	0.491

As described in the following section, for the residential sector, DOE determined that the air cleaners, when active, would operate on auto, max speed, medium speed, and low speed modes. For the commercial sector, DOE estimated that the air cleaners would operate on auto mode and max speed mode when active.

7.3.1 Operating Hours

DOE estimated that the operating hours for residential settings would depend on consumer's age, occupancy and air quality parameters, while the operating hours in the commercial settings would depend on building schedules.

7.3.1.1 Operating Hours for Residential Applications

DOE considered that the senior-only households would exhibit usage behaviors that are different from the other households. *RECS 2020* provides information on the number of household members and of household members above the age of 65 (seniors) for each household sample, which allows DOE to identify the senior-only households in *RECS 2020*.^a DOE determined that 22.65 percent of air cleaner household samples were senior-only based on the adjusted sample weights.

RECS 2020 also provides the number of weekdays someone is at home all or most of the day.

For the senior-only households, DOE estimated the annual air cleaner operating days by assuming that the air cleaners were on:

- during the evening and the night of those weekdays that the household members were reported to be not at home, and one day of the weekend (50 percent of those days),
- the whole day during the weekdays that the household members were reported to be at home, and one day of the weekend.

^a To be qualified as a senior-only household, all household members need to be at least 65 years old.

For households that were not senior-only, DOE relied on both the occupancy data and the air quality data by county published by U.S. Environmental Protection Agency (EPA)⁶ to determine the number of operating days. DOE adopted a more conservative estimation by taking the minimum of

- number of weekdays that the *RECS 2020* household sample was occupied,
- numbers of days the 2020 EPA data indicated to be moderate, unhealthy, or hazardous based on the Air Quality Index (AQI), plus half of the days during the remainder of the year, based on the geographic location of the *RECS 2020* household samples.

Table 7.3.2 shows the state-wide population-weighted average of the number of days air cleaners were assumed to be on in 2020 based on the EPA's air quality data and the assumption mentioned above (sum of the number of days with moderate, unhealthy and hazardous AQI, plus half of the days during the remainder of the year).

Table 7.3.2 Estimated Annual Air Cleaner Operating Days by State Based on 2020 EPA's Air Quality Data

State	Days	State	Days	State	Days
Alabama	224	Kentucky	213	North Dakota	192
Alaska	201	Louisiana	206	Ohio	214
Arizona	321	Maine	190	Oklahoma	226
Arkansas	216	Maryland	201	Oregon	212
California	296	Massachusetts	200	Pennsylvania	216
Colorado	246	Michigan	208	Rhode Island	209
Connecticut	221	Minnesota	211	South Carolina	199
Delaware	209	Mississippi	213	South Dakota	204
District of Columbia	209	Missouri	214	Tennessee	198
Florida	210	Montana	211	Texas	232
Georgia	206	Nebraska	198	Utah	252
Hawaii	184	Nevada	227	Vermont	193
Idaho	215	New Hampshire	189	Virginia	193
Illinois	222	New Jersey	207	Washington	203
Indiana	212	New Mexico	255	West Virginia	194
Iowa	209	New York	203	Wisconsin	216
Kansas	221	North Carolina	205	Wyoming	221

Sum of the days with moderate, unhealthy and hazardous AQI, plus half of the days during the remainder of the year.

For the days that the air cleaner was assumed to be on, DOE relied on the Evergreen Economics study² to estimate the daily active hours by drawing from the distributions shown in Table 7.3.3 based on household size. For days DOE determined that the air cleaners were not operating, DOE assumed that the air cleaners were turned off with no power consumption.

Table 7.3.3 Usage Frequency Distribution by Household Size Based on the 2021 Evergreen Economics Report

Number of Household Members	Reported Usage Frequency						
	24 hours	17-23 hours	13-16 hours	7-12 hours	4-6 hours	1-3 hours	<1 hour
	Percentage of Respondents (n=448)						
1-2	9%	37%	2%	14%	14%	17%	7%
3-4	9%	14%	1%	14%	21%	28%	13%
5-6	13%	4%	13%	23%	31%	11%	5%

DOE used the mid-point of each usage frequency interval to assign daily active operating hours, and the remainder of the 24 hours was considered to be in standby mode. DOE further split the daily active hours into auto, low speed, medium speed, and max speed modes operating hours. DOE estimated that if an air cleaner was active for 24 hours a day, it was likely to operate on auto mode. Table 7.3.4 shows DOE's assumption of fraction of operating hours in auto, max speed, medium speed, and low speed modes based on daily active hours for a given sample household.

Table 7.3.4 Assumption of Fraction of Active Hours by Modes Based on Average Daily Operating Hours

Modes of Operation	Bins of Daily Active Hours			
	Less than or equal to 2 hours	Greater than 2 hours, less than 10 hours	Greater than or equal to 10 hours, less than 24 hours	24 hours
	Percentage of Daily Active Hours Assigned to Each Mode			
Low speed	20%	40%	80%	0%
Medium speed	50%	40%	20%	0%
Max speed	30%	20%	0%	0%
Auto	0%	0%	0%	100%

DOE estimated that the air cleaners operated on average 10.6 hours per day, and 248 days per year in the residential sector.

7.3.1.2 Operating Hours for Commercial Applications

For the commercial sector, DOE estimated that the air cleaners would operate on auto mode for 80 percent of the time and on max speed mode for 20 percent of the time when active.

CBECS 2018 provides the number of hours the sample buildings is open weekly as well as the number of months the sample building is in use. DOE used this data to estimate the annual operating hours for all commercial building samples selected from *CBECS 2018* for air cleaners. DOE assumed that the remaining hours of the year were equally split between standby mode and off mode.

DOE estimated an average of 4,198 annual operating hours, which is equivalent to 12.9 operating hours per day and 325 operating days per year.

7.3.2 Test Procedure Annual Energy Consumption by Efficiency Level

Table 7.3.5 through Table 7.3.7 provide the annual energy use estimated in the test procedure that corresponds to each efficiency level for all three product classes of air cleaners.

Table 7.3.5 Air Cleaners Product Class 1, $10 \leq \text{PM}_{2.5} \text{ CADR} < 100$: Annual Energy Consumption Based on DOE Test Procedure Energy Use Calculations

Efficiency Level	Efficiency Level ($\text{PM}_{2.5} \text{ CADR/W}$)	On Mode Power (W)*	Standby Power (W)	Annual Energy Consumption (kWh/yr)**
Baseline	1.5	32.60	0.16	191
1	1.7	29.40	0.37	173
2	1.9	26.13	0.65	155
3	3.4	14.61	0.46	87
4	5.4	9.12	0.27	54

* On Mode Power is the power at maximum speed mode.

** Based on DOE test procedure annual operating hours: 5,840 hours on maximum speed mode and 2,920 hours on standby mode, which is equivalent to 16 max speed mode operating hours per day and 365 operating days per year.

Table 7.3.6 Air Cleaners Product Class 2, $100 \leq \text{PM}_{2.5} \text{ CADR} < 150$: Annual Energy Consumption Based on DOE Test Procedure Energy Use Calculations

Efficiency Level	Efficiency Level ($\text{PM}_{2.5} \text{ CADR/W}$)	On Mode Power (W)*	Standby Power (W)	Annual Energy Consumption (kWh/yr)**
Baseline	1.5	81.65	0.10	477
1	1.9	65.64	0.29	384
2	2.4	52.09	0.42	305
3	5.4	22.80	0.70	135
4	12.8	9.69	0.22	57

* On Mode Power is the power at maximum speed mode.

** Based on DOE test procedure annual operating hours: 5,840 hours on maximum speed mode and 2,920 hours on standby mode, which is equivalent to 16 max speed mode operating hours per day and 365 operating days per year.

Table 7.3.7 Air Cleaners Product Class 3, $\text{PM}_{2.5} \text{ CADR} \geq 150$: Annual Energy Consumption Based on DOE Test Procedure Energy Use Calculations

Efficiency Level	Efficiency Level ($\text{PM}_{2.5} \text{ CADR/W}$)	On Mode Power (W)*	Standby Power (W)	Annual Energy Consumption (kWh/yr)**
Baseline	1.2	166.39	0.56	973
1	2.0	99.19	0.63	581
2	2.9	68.43	0.60	401
3	6.6	30.11	0.85	178
4	7.4	26.61	0.77	158

* On Mode Power is the power at maximum speed mode.

** Based on DOE test procedure annual operating hours: 5,840 hours on maximum speed mode and 2,920 hours on standby mode, which is equivalent to 16 max speed mode operating hours per day and 365 operating days per year.

The test procedure provides standardized results that help compare the performance of various units operating under the same conditions. An air cleaner's energy usage in the field, however, differs from the test procedure results depending on the conditions under which the appliance is operated.

7.3.3 Average Annual Energy Consumption by Efficiency Level

Table 7.3.8 through Table 7.3.10 show the average annual energy consumption for each product class at each analyzed efficiency level for the residential and commercial sector. Also shown is the overall average annual energy consumption calculated for combining energy use in both sectors, assuming that 60 percent of installations are in residential settings, and 40 percent are in commercial settings based on data from TechSci Research.⁴

Table 7.3.8 Air Cleaners Product Class 1, $10 \leq \text{PM}_{2.5} \text{ CADR} < 100$: Average Annual Energy Use

Efficiency Level	Efficiency Level (<i>PM 2.5 CADR/W</i>)	Residential Sector Energy Use (<i>kWh</i>)	Commercial Sector Energy Use (<i>kWh</i>)	Weighted Average Energy Use (<i>kWh</i>)
EL0 - Baseline	1.5	40.0	83.2	57.3
EL1 - Tier 1	1.7	36.8	75.5	52.3
EL2 - Tier 2	1.9	33.8	67.9	47.4
EL3	3.4	19.2	38.2	26.8
EL4 - Max Tech	5.4	11.9	23.8	16.9

Table 7.3.9 Air Cleaners Product Class 2, $100 \leq \text{PM}_{2.5} \text{ CADR} < 150$: Average Annual Energy Use

Efficiency Level	Efficiency Level (<i>PM 2.5 CADR/W</i>)	Residential Sector Energy Use (<i>kWh</i>)	Commercial Sector Energy Use (<i>kWh</i>)	Weighted Average Energy Use (<i>kWh</i>)
EL0 - Baseline	1.5	99.3	207.7	142.7
EL1 - Tier 1	1.9	80.5	167.5	115.3
EL2 - Tier 2	2.4	64.5	133.3	92.0
EL3	5.4	29.9	59.5	41.8
EL4 - Max Tech	12.8	12.5	25.1	17.5

Table 7.3.10 Air Cleaners Product Class 3, PM_{2.5} CADR $\geq$ 150: Average Annual Energy Use

Efficiency Level	Efficiency Level (PM 2.5 CADR/W)	Residential Sector Energy Use (kWh)	Commercial Sector Energy Use (kWh)	Weighted Average Energy Use (kWh)
EL0 - Baseline	1.2	203.5	424.1	291.7
EL1 - Tier 1	2.0	122.3	253.5	174.8
EL2 - Tier 2	2.9	84.9	175.2	121.0
EL3	6.6	39.3	78.4	54.9
EL4 - Max Tech	7.4	34.8	69.3	48.6

7.3.4 Variability of Annual Energy Consumption

Figure 7.3.1 and Figure 7.3.2 are whisker-box plots and show the range of annual energy consumption by efficiency level within the residential and commercial samples for product class 1, respectively. The whiskers show the 5 percent and 95 percent value of the range. A rectangular box shows the range of energy consumption between 25 percent and 75 percent. A horizontal line within the rectangular box shows the median and the open square within the rectangle shows the average. DOE can generate a whisker-box plot like the ones shown in Figure 7.3.1 and Figure 7.3.2 for each product class in both user sectors.

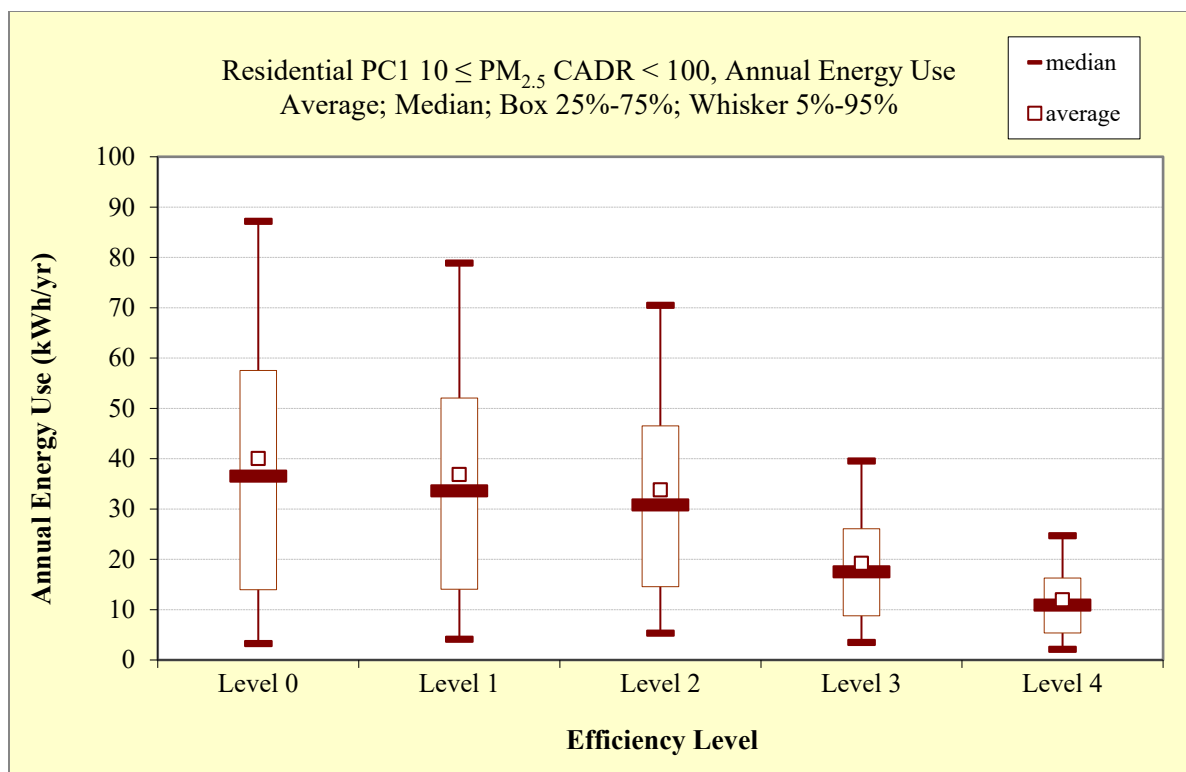


Figure 7.3.1 Air Cleaner PC 1 Residential: Range of Average Annual Energy Use

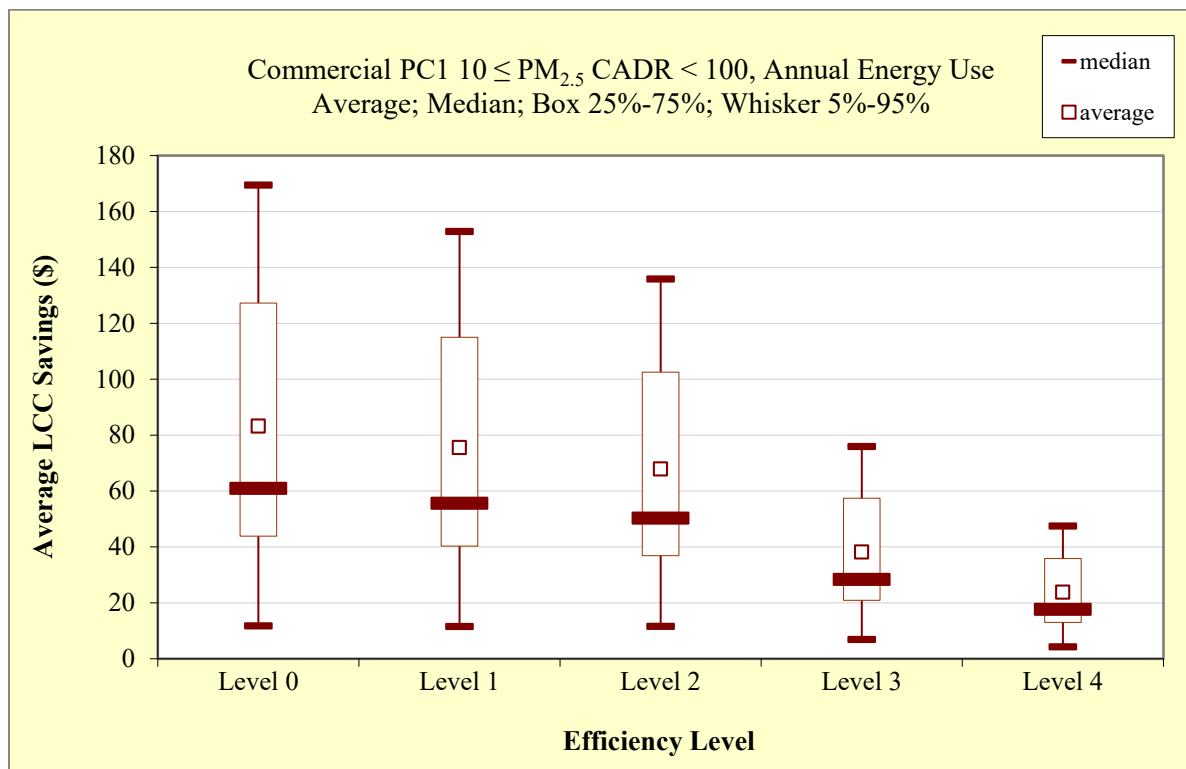


Figure 7.3.2 Air Cleaner PC 1 Commercial: Range of Average Annual Energy Use

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CHAPTER 8. LIFE-CYCLE COST AND PAYBACK PERIOD ANALYSIS

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CHAPTER 8. LIFE-CYCLE COST AND PAYBACK PERIOD ANALYSIS

8.1 INTRODUCTION

This chapter describes the U.S. Department of Energy's (DOE's) method for analyzing the economic impacts on individual consumers from potential energy efficiency standards for air cleaners. The effects of standards on individual consumers include a change in operating expense (usually a decrease) and a change in purchase price (usually an increase). This chapter describes three metrics DOE used to determine the impact of standards on individual consumers:

- **Life-cycle cost (LCC)** is the total consumer expense during the lifetime of an appliance, including purchase expense and operating costs (including electricity expenditures). DOE discounts future operating costs to the time of purchase and sums them over the lifetime of the product.
- **Payback period (PBP)** measures the amount of time it takes a consumer to recover the assumed higher purchase price of a more energy efficient product through lower operating costs. DOE calculates a simple payback period which does not discount operating costs and average values for installed cost and first-year operating costs.
- **Rebuttable payback period** is a special case of the PBP. Whereas LCC is estimated for a range of inputs that reflect real-world conditions, rebuttable payback period is based on laboratory conditions, specifically the DOE test procedure.

Inputs to the LCC and PBP calculations are described in section 8.2 and section 8.3, of this chapter. Results of the analysis are presented in section 8.5. Key variables and calculations are presented for each metric. The calculations discussed here are illustrated with a Microsoft Excel spreadsheet available on the DOE's rulemaking website at http://www.eere.energy.gov/buildings/appliance_standards/. Details and instructions for using the spreadsheets are provided in appendix 8A of this final rule technical support document (TSD).

8.1.1 General Approach to Analysis

Life-cycle cost is calculated using the following equation:

$$LCC = TIC + \sum_{t=0}^{N-1} \frac{OC_t}{(1+r)^t}$$

Eq. 8.1

Where:

LCC =	life-cycle cost (in dollars),
TIC =	total installed cost in dollars,
$\sum$ =	sum over the appliance lifetime, from year 1 to year N ,
N =	lifetime of the appliance in years,
OC =	operating cost in dollars,
r =	discount rate, and
t =	year to which operating cost is discounted.

The payback period is the ratio of the increase in total installed cost (i.e., from a less energy efficient design to a more efficient design) to the decrease in the first-year annual operating expenditures. This type of calculation results in what is termed a simple payback period because it does not take into account changes in energy expenses over time or the time value of money. The equation for PBP is:

$$PBP = \frac{\Delta TIC}{\Delta OC}$$

Eq. 8.2

Where:

ΔTIC =	difference in weighted-average total installed cost between a more energy efficient design and the baseline design for the entire sample, and
ΔOC =	difference in weighted-average first-year operating costs for the entire sample.

Payback periods are expressed in years. Payback periods greater than the life of the product indicate that the increased total installed cost is not recovered through reduced operating costs.

Recognizing that inputs to the determination of consumer LCC may be either variable or uncertain, DOE conducts the LCC analysis by modeling both the uncertainty and variability of the inputs using a Monte Carlo simulation and probability distributions for inputs. Appendix 8B provides a detailed explanation of Monte Carlo simulation and the use of probability distributions and discusses the tool used to incorporate these methods.

DOE calculates LCC impacts relative to a case without amended or new energy conservation standards (referred to as the “no-new-standards case”). In the no-new-standards case, some consumers may purchase products with energy efficiency higher than a baseline

model. For any given standard level under consideration, consumers expected to purchase a product with efficiency equal to or greater than the considered level in the no-new-standards case would be unaffected by that standard.

DOE calculates the LCC and PBP as if all consumers purchase air cleaners in the expected initial year of compliance with a new or amended standard. At this time, the expected compliance date of potential energy conservation standards for air cleaners manufactured in, or imported into, the United States is in 2028. Therefore, DOE conducted the LCC and PBP analysis assuming purchases take place in 2028. Based on the Joint Proposal's two-tier standard recommendation, DOE also added 2024 and 2026 as additional first years of compliance in one of the scenarios analyzed.

8.1.2 Overview of Inputs to Analysis

DOE categorizes inputs to the LCC and PBP analysis as: (1) inputs for establishing the purchase expense, otherwise known as *total installed cost*, and (2) inputs for calculating operating costs.

Following are the primary inputs for establishing the total installed cost:

- *Baseline manufacturer cost*: The costs incurred by the manufacturer to produce products that meet current minimum efficiency standards, if any.
- *Efficiency-level manufacturer cost increase*: The change in manufacturer cost associated with producing products to meet a particular efficiency level.
- *Markups and sales tax*: The markups and sales tax associated with converting the manufacturer cost to a consumer product cost.
- *Installation cost*: The cost to the consumer of installing the product. The installation cost represents all costs required to install the product other than the marked-up consumer product cost. The installation cost includes labor, overhead, and any miscellaneous materials and parts. Thus, the total installed cost equals the consumer product cost plus the installation cost.

Following are the primary inputs for calculating the operating cost.

- *Product energy consumption*: The product energy consumption is the site energy use associated with operating the product.
- *Product efficiency*: The product efficiency dictates the energy consumption associated with products that have efficiencies greater than the baseline efficiency.

- *Electricity prices:* The prices purchasers paid for electricity based on average and marginal price data from the Edison Electric Institute (EEI).
- *Electricity price trends:* The EIA's *Annual Energy Outlook 2022 (AEO 2022)*¹ is used to forecast electricity prices into the future. For the results presented in this chapter, DOE used the *AEO 2022* reference case to forecast future energy prices.
- *Repair and maintenance costs:* Repair costs are associated with repairing or replacing components that fail. Maintenance costs are associated with maintaining the operation of the product.
- *Lifetime:* The lifetime is the age at which the product is retired from service.
- *Discount rate:* The discount rate is the rate at which DOE discounts future expenditures to establish their present value.

The data inputs to the PBP are the average total installed cost (product cost and installation cost) to the consumer plus the average annual (first-year) operating costs for each efficiency level. The inputs to operating costs are the annual energy cost, the annual repair cost, and the annual maintenance cost. Because the PBP represents a simple payback, the energy price is required only for the year in which a potential energy conservation standard is estimated to take effect. The energy price DOE uses in the PBP calculation is the price projected for that year.

Figure 8.1.1 depicts the relationships among the inputs to installed cost and operating cost for calculating a product's LCC and PBP. In the figure, the tan boxes indicate inputs, the green boxes indicate intermediate outputs, and the blue boxes indicate final outputs. Table 8.1.1 provides a summary of inputs used in the analysis.

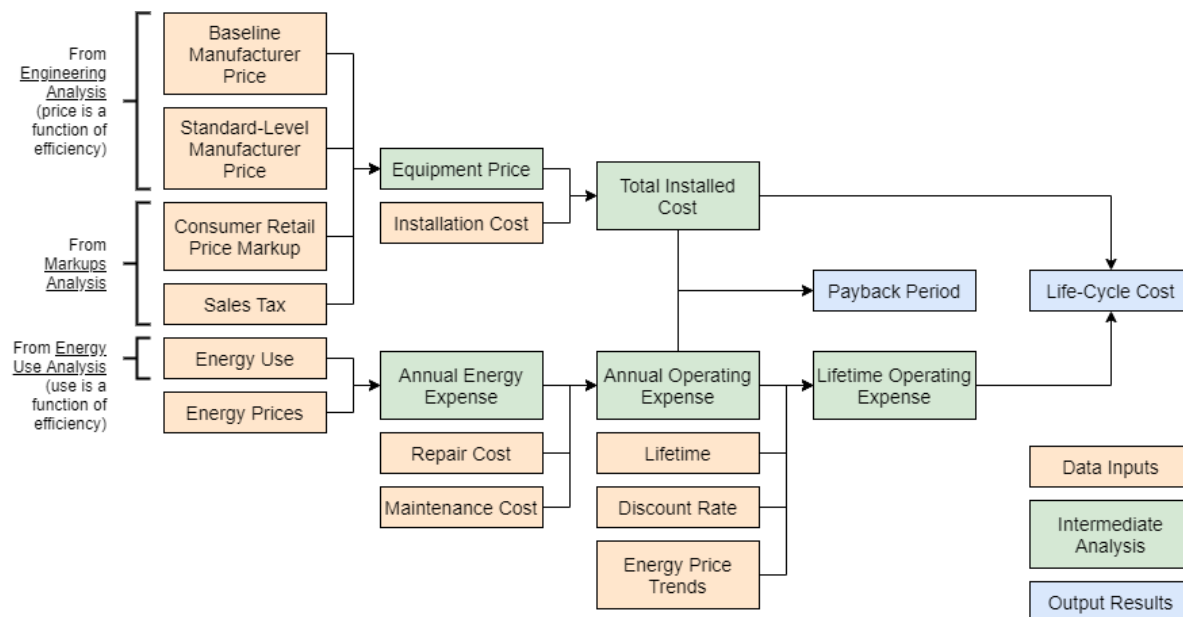


Figure 8.1.1 Flow Diagram of Inputs and Outputs for LCC and PBP

Table 8.1.1 Summary of Inputs to Life-Cycle Cost and Payback Period

Inputs	Average or Typical Value	Characterization
Total Installed Cost Inputs		
Product Price	Varies by distribution channel, efficiency level, and product class	Single-point value
Sales Tax	National average: 7.3%	Varies by state
Operating Cost Inputs		
Energy Use	Varies by efficiency level and product class	Distribution (see chapter 7 of this TSD for details)
Electricity Prices	Residential sector: 0.154 \$/kWh Commercial sector: 0.110 \$/kWh	Vary by census division for each sector
Electricity Price Trends	AEO 2022 reference case	Vary by census division for each sector
Product Lifetime	Mean: 9.0 years	Weibull distribution
Discount Rate	Residential sector: 4.30% Commercial sector: 6.67%	Residential: Varies by consumer income Commercial: Varies by building type
Assumed Date Standards Become Effective	Two-tier scenario: 2024 and 2026 Other scenarios: 2028	Single-point value

8.1.3 Sample of Air Cleaner Users

The LCC and PBP calculations detailed here are for a representative sample of individual air cleaner users. By developing consumer samples, DOE accounts for the variability in energy consumption and energy price associated with a range of consumers.

As described in chapter 7 of this final rule TSD, DOE used the DOE Energy Information Administration (EIA)’s 2020 *Residential Energy Consumption Survey (RECS 2020)* to develop weight-adjusted household samples.² *RECS 2020* consists of 18,496 housing units and is representative of the household population of the United States. DOE was able to assign a unique annual energy use and energy price to each household in the sample. The large sample of households considered in the analysis provides wide ranges of annual energy use and energy prices. (The ranges in energy consumption are presented in chapter 7 of this final rule TSD.) The variability in annual energy use and energy pricing across households contributes to the range of LCCs calculated for any particular efficiency level. DOE also used the EIA’s 2018 *Commercial Buildings Energy Consumption Survey (CBECS 2018)* to develop a sample of commercial buildings that use air cleaners.³ *CBECS 2018* consists of 6,436 commercial buildings and is representative of commercial buildings throughout the United States. As it did with the *RECS 2020*, DOE used *CBECS 2018* to establish the variability of annual energy use and energy prices.

DOE displays the LCC savings results as distributions of impacts relative to the absence of a new energy conservation standard (hereafter referred to as the “no-new-standards case”). Results, presented in section 8.5, were derived from 10,000 samples for each Monte Carlo

simulation run for the reference scenario. To illustrate the implications of the analysis, DOE generated frequency charts that depict the variation in LCC for each efficiency level considered for potential standards for air cleaners.

8.2 TOTAL INSTALLED COST INPUTS

DOE defines the total installed cost using the following equation:

$$TIC = CPC + INST \quad \text{Eq. 8.3}$$

Where:

CPC = consumer product cost (i.e., consumer cost for the product only), and
 $INST$ = installation cost (the consumer cost to install the product).

The product cost depends on how the consumer purchases the product. As discussed in chapter 6 of this final rule TSD, DOE defines markups and sales taxes to convert manufacturing costs into consumer product costs. Table 8.2.1 summarizes the inputs for determining total installed cost.

Table 8.2.1 Inputs to Total Installed Cost

Baseline manufacturer cost
Efficiency-level manufacturer cost increase
Markups
Sales tax (part of markups)
Installation cost

The *baseline manufacturer cost* is the cost incurred by the manufacturer to produce products that meet current minimum efficiency standards, if any. *Efficiency-level manufacturer cost increase* is the change in manufacturer cost associated with producing products at a higher efficiency level. *Markups and sales tax* convert the manufacturer cost to a consumer product cost. The *installation cost* represents all costs to the consumer for installing the product, other than the marked-up consumer product cost. The installation cost includes labor, overhead, and any miscellaneous materials and parts. DOE calculated the total installed cost for baseline products using the following equation:

$$\begin{aligned} IC_{BASE} &= CPC_{BASE} + INST_{BASE} \\ &= COST_{MFG} \times MU_{OVERALL_BASE} + INST_{BASE} \end{aligned} \quad \text{Eq. 8.4}$$

Where:

IC_{BASE} = baseline total installed cost,

CPC_{BASE} = consumer product cost for baseline model,
 $INST_{BASE}$ = baseline installation cost,
 $COST_{MFG}$ = manufacturer cost for baseline model, and
 $MU_{OVERALL_BASE}$ = overall baseline markup (product of manufacturer markup, baseline retailer and/or distributor markup, and sales tax).

DOE calculated the total installed cost for products at higher efficiency levels using the following equation:

$$\begin{aligned}
 IC_{EL} &= CPC_{EL} + INST_{EL} \\
 &= (CPC_{BASE} + \Delta CPC_{EL}) + (INST_{BASE} + \Delta INST_{EL}) \\
 &= (CPC_{BASE} + INST_{BASE}) + (\Delta CPC_{EL} + \Delta INST_{EL}) \\
 &= IC_{BASE} + (\Delta COST_{MFG} \times MU_{OVERALL_INCR} + \Delta INST_{EL})
 \end{aligned}$$

Eq. 8.5

Where:

IC_{EL} = total installed cost for higher-efficiency product,
 CPC_{EL} = consumer product cost for higher-efficiency product,
 $INST_{EL}$ = installation cost for higher-efficiency product,
 CPC_{BASE} = consumer product cost for baseline product,
 ΔCPC_{EL} = change in product cost for higher-efficiency product,
 $INST_{BASE}$ = baseline installation cost,
 $\Delta INST_{EL}$ = change in installation cost for higher-efficiency product,
 IC_{BASE} = baseline total installed cost,
 $\Delta COST_{MFG}$ = change in manufacturer cost for higher-efficiency product, and
 $MU_{OVERALL_INCR}$ = incremental overall markup (product of manufacturer markup, incremental retailer or distributor markup, and sales tax for higher-efficiency product).

The rest of this section provides information about each of the above input variables that DOE used to calculate the total installed cost of air cleaners.

8.2.1 Baseline Manufacturer Cost

DOE developed the baseline manufacturer costs for air cleaners as described in chapter 5 of this final rule TSD. Table 8.2.2 shows the manufacturer production cost for baseline models as well as the manufacturer cost for filters of baseline models for all the analyzed product classes.

Table 8.2.2 Air Cleaners: Baseline Manufacturer Cost

PC	Product Class	Baseline PM _{2.5} CADR/W	Manufacturer Production Cost – Air Cleaners (2021\$)	Manufacturer Production Cost – Filters (2021\$)
1	$10 \leq \text{PM}_{2.5} \text{ CADR} < 100$	1.5	\$29.59	\$2.49
2	$100 \leq \text{PM}_{2.5} \text{ CADR} < 150$	1.5	\$40.69	\$5.52
3	$\text{PM}_{2.5} \text{ CADR} \geq 150$	1.2	\$66.76	\$8.58

8.2.2 Efficiency-Level Manufacturer Cost Increases

As described in chapter 5 of this final rule TSD, DOE used a reverse-engineering analysis to develop manufacturer cost increases associated with increases in energy efficiency levels for air cleaners.

8.2.3 Overall Markup

The overall markup on a product is determined by multiplying the manufacturer and retailer markups and the sales tax together to arrive at a single markup value. Table 8.2.3 shows the overall baseline and incremental markups for air cleaners. Incremental markups are those that apply to products at higher efficiency levels manufactured under a potential energy efficiency standard. Refer to chapter 6 of this final rule TSD for details.

Table 8.2.3 Air Cleaners: Overall Markups

	Direct Sales		Retail		e-Commerce	
	Baseline	Incremental	Baseline	Incremental	Baseline	Incremental
Manufacturer	1.45		1.45		1.45	
Wholesaler/Dealer	1.35	1.20				
Retailer/Online Vendor			1.49	1.24	1.47	1.22
Sales Tax	1.073		1.073		1.073	
Overall	2.10	1.87	2.31	1.93	2.29	1.91

8.2.4 Forecasting Product Prices

Examination of historical price data for certain appliances and equipment that have had energy conservation standards indicates that the assumption of constant real prices and costs may, in many cases, overestimate long-term trends in appliance and equipment prices. Economic literature and historical data suggest that the real costs of these products may, in fact, trend downward over time according to “learning” or “experience” curves. Desroches *et al.* (2013) summarizes the data and literature currently available that is relevant to price projections for selected appliances and equipment.⁴ The extensive literature on the “learning” or “experience” curve phenomenon is typically based on observations in the manufacturing sector.^a In the experience curve method, the real cost of production is related to the cumulative production or “experience” with a manufactured product. This experience is usually measured in terms of cumulative production. A common functional relationship used to model the evolution of production costs in this case is:

$$Y = a X^{(-b)}$$

Eq. 8.6

Where:

- a = an initial price (or cost),
- b = a positive constant known as the learning rate parameter,
- X = cumulative production, and
- Y = the price as a function of cumulative production.

Thus, as experience (production) accumulates, the cost of producing the next unit decreases. The percentage reduction in cost that occurs with each doubling of cumulative production is known as the learning rate (LR), given by:

$$LR = 1 - 2^{(-b)}$$

Eq. 8.7

In typical learning curve formulations, the learning rate parameter is derived using two historical data series: cumulative production and price (or cost).

In this final rule analysis, DOE developed an experience curve to project the future air cleaner prices. An overall air cleaner price trend was applied to the non-filter portion of the unit cost. To derive the learning rate parameter for air cleaners, DOE combined historical Producer Price Index (PPI) data for “small electric household appliances, except fans” between 1982 and 2015 and for the more aggregated small electrical appliance manufacturing between 2015 and 2021 from the Bureau of Labor Statistics’ (BLS).^b The former PPI is the most disaggregated price index that represents air cleaners; however the price index is discontinued beyond 2015 due to insufficient sample size. To obtain price index beyond 2015, DOE used the aggregated small

^a In addition to Desroches (2013), see Weiss, M., Junginger, H.M., Patel, M.K., Blok, K., (2010a). A Review of Experience Curve Analyses for Energy Demand Technologies. *Technological Forecasting & Social Change*. 77:411-428.

^b Product series ID: PCU33521033521014 and PCU335210335210, see more information at: <http://www.bls.gov/ppi/>

electrical appliance manufacturing as a proxy for air cleaners. Inflation-adjusted price indices were calculated by dividing the PPI series by the implicit Gross Domestic Product price deflator for the same years. These inflation-adjusted price indices (Figure 8.2.1) were used in subsequent analysis steps.

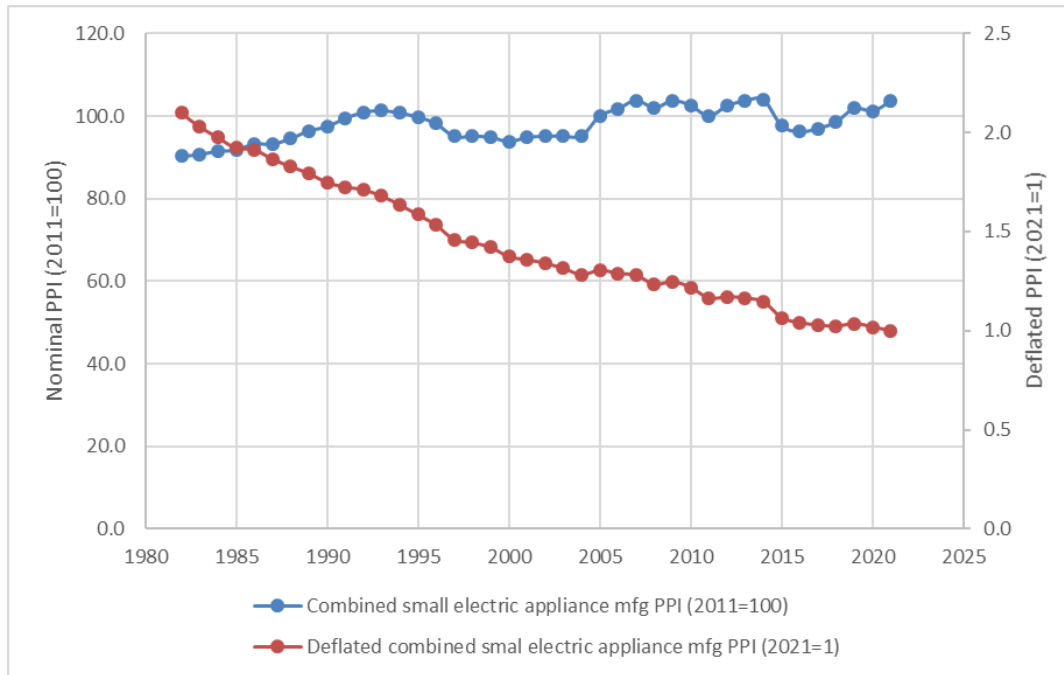


Figure 8.2.1 Nominal and Deflated Small Electric Appliance Manufacturing PPI from 1982 to 2021

DOE assembled a time-series of annual shipments for air cleaners from an EuroMonitor product sales report⁵ and a Tech Sci market research report.⁶ The annual shipments data were used to estimate cumulative shipments (production). Projected shipments after 2020 were obtained from the no-new-standards case projections made for the NIA (see chapter 9 of this final rule TSD).

To estimate learning rate parameter, a least-squares power-law fit was performed on the deflated price index versus cumulative shipments (see Figure 8.2.2.)

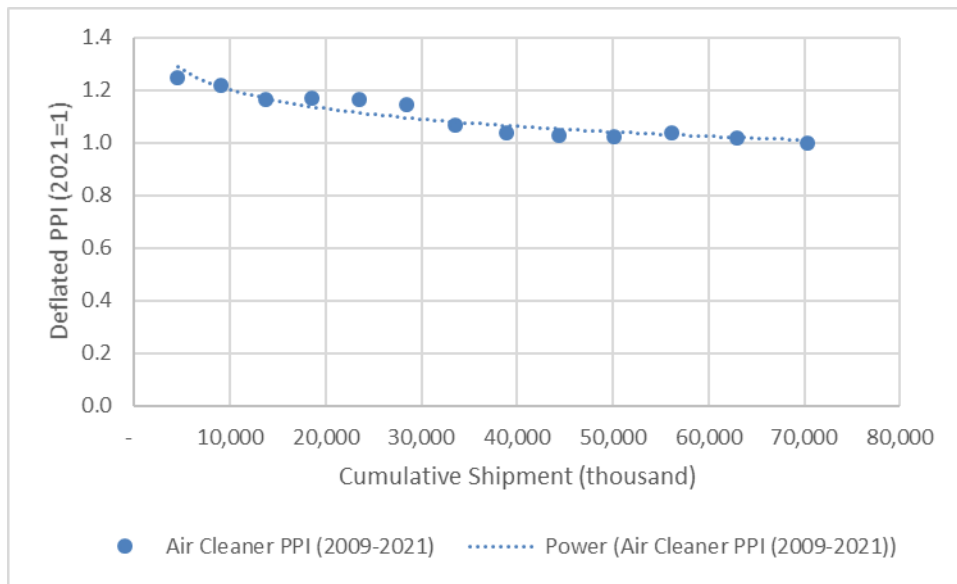


Figure 8.2.2 Relative Price versus Cumulative Shipments of Air Cleaners from 2009 to 2021, with Power Law Fit

The form of the power-law equation is:

$$P_t = P_o X_t^{(-b)}$$

Eq. 8.8

where the two parameters, b (the learning rate parameter) and P_o (the price or cost of the first unit of production), are obtained by fitting the model to the data. The variable X_t is the cumulative shipments from period zero to period t and P_t is the average deflated price of shipments in period t . DOE notes that the cumulative shipments on the right hand side of the equation can depend on price, so there is an issue with simultaneity where the independent variable X_t is not truly independent. DOE's use of a simple least squares fit is equivalent to an assumption of no significant effect of the price elasticity of supply.

For air cleaners, the parameter values obtained are:

$$P_o = 2.73^{+0.647}_{-0.523} (95\% \text{ confidence}), \text{ and}$$

$$b = 0.089 \pm 0.021 (95\% \text{ confidence}).$$

The estimated learning rate (defined as the fractional reduction in price expected from each doubling of cumulative production) is $6.0 \pm 1.3\%$ (95% confidence).

For air cleaners, the index value in a given year is a function of the experience parameter and the cumulative production forecast through that year, with 2021 equal to 1. DOE applied the same index values to project prices for each product class at each efficiency level through the analysis period. The estimated price forecast index is shown in Figure 8.2.3.

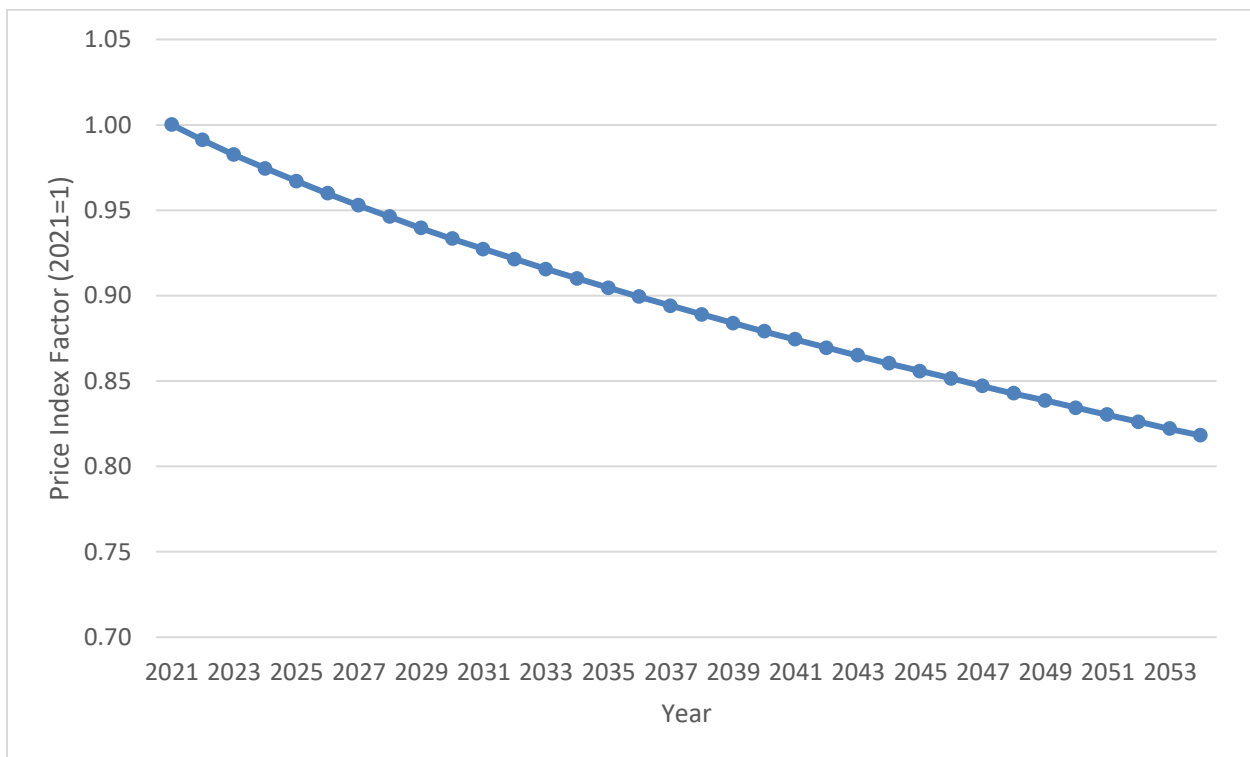


Figure 8.2.3 Price Forecast Index for Air Cleaners

8.2.5 Installation Cost

DOE determined that any installation costs would not be affected by increased efficiency levels.

8.2.6 Total Installed Cost

The total installed cost is the sum of the consumer product cost and installation cost. Table 8.2.4 through Table 8.2.6 present the total installed costs in 2028 for the air cleaners in the residential and commercial sectors, accounting for the effects of the learning curve. Costs are presented at the baseline level and each efficiency level identified in chapter 5 of this final rule TSD.

Table 8.2.4 Air Cleaners PC 1: $10 \leq \text{PM}_{2.5} \text{ CADR} < 100$: Total Installed Costs

Efficiency Level	Efficiency Level (<i>PM 2.5</i> CADR/W)	Residential Total Installed Cost (2021\$)	Commercial Total Installed Cost (2021\$)
Baseline	1.5	\$65	\$62
1	1.7	\$66	\$64
2	1.9	\$68	\$66
3	3.4	\$79	\$76
4	5.4	\$87	\$84

Table 8.2.5 Air Cleaners PC 2: $100 \leq \text{PM}_{2.5} \text{ CADR} < 150$: Total Installed Costs

Efficiency Level	Efficiency Level (<i>PM 2.5</i> CADR/W)	Residential Total Installed Cost (2021\$)	Commercial Total Installed Cost (2021\$)
Baseline	1.5	\$89	\$86
1	1.9	\$91	\$88
2	2.4	\$93	\$90
3	5.4	\$102	\$99
4	12.8	\$111	\$107

Table 8.2.6 Air Cleaners PC 3: $\text{PM}_{2.5} \text{ CADR} \geq 150$: Total Installed Costs

Efficiency Level	Efficiency Level (<i>PM 2.5</i> CADR/W)	Residential Total Installed Cost (2021\$)	Commercial Total Installed Cost (2021\$)
Baseline	1.2	\$146	\$141
1	2.0	\$148	\$143
2	2.9	\$149	\$144
3	6.6	\$153	\$148
4	7.4	\$153	\$148

8.3 OPERATING COST INPUTS

DOE defines operating cost (OC) as the addition of energy costs, repair costs, and maintenance costs for both the baseline efficiency levels and higher efficiency levels:

$$\begin{aligned}
 OC &= EC + RC + MC \\
 OC &= (AEC \times PRICE_{ENERGY}) + RC + MC
 \end{aligned}$$

Eq. 8.9

where:

OC = operating cost,
 EC = energy cost associated with operating the product,
 RC = repair cost associated with component failure, and
 MC = maintenance cost.
 AEC = annual energy consumption for baseline product,
 $PRICE_{ENERGY}$ = energy price,

Table 8.3.1 shows the inputs for determining the annual operating costs and their discounted value throughout a product's lifetime.

Table 8.3.1 Inputs for Operating Cost

Annual energy consumption
Energy prices and price trends
Repair and maintenance Costs
Product Lifetime
Discount Rate
Effective Date of Standard

The *annual energy consumption* is the site energy use associated with operating the product. The annual energy consumption changes with product efficiency. *Energy prices* are the prices paid by consumers for energy (e.g., electricity or gas). Multiplying the annual energy consumption by the energy price yields the annual energy cost. *Repair costs* are associated with repairing or replacing components that fail. *Maintenance costs* are associated with maintaining the operation of the product. DOE used energy price trends to forecast energy prices into the future and to establish the present value of lifetime energy costs.

The remainder of this section provides information about each of the above input variables that DOE used to calculate the operating costs for all product classes for air cleaners.

8.3.1 Annual Energy Consumption

As described in chapter 7 of this final rule TSD, and in section 8.1.1 of this chapter, DOE used data from *RECS 2020* and *CBECS 2018* to develop samples of residential households and commercial entities that have air cleaners. DOE performed the LCC calculation for each household and commercial buildings in both samples to account for the variability in both energy use and energy price. Table 8.3.2 through Table 8.3.4 provide the average annual energy consumption by efficiency level for air cleaners in residential and commercial applications, as well as the weighted average reflecting 60 percent of installations in the residential sector and 40 percent of installations in the commercial based on estimates of installed stock from the TechSci market research report.⁶

Table 8.3.2 Air Cleaners PC 1, $10 \leq \text{PM}_{2.5} \text{ CADR} < 100$: Average Annual Energy Use

Efficiency Level	Efficiency Level (PM 2.5 CADR/W)	Residential Sector Energy Use (kWh)	Commercial Sector Energy Use (kWh)	Weighted Average Energy Use (kWh)
EL0 - Baseline	1.5	40	83	57
EL1 - Tier 1	1.7	37	76	52
EL2 - Tier 2	1.9	34	68	47
EL3	3.4	19	38	27
EL4 - Max Tech	5.4	12	24	17

Table 8.3.3 Air Cleaners PC 2, $100 \leq \text{PM}_{2.5} \text{ CADR} < 150$: Average Annual Energy Use

Efficiency Level	Efficiency Level (PM 2.5 CADR/W)	Residential Sector Energy Use (kWh)	Commercial Sector Energy Use (kWh)	Weighted Average Energy Use (kWh)
EL0 - Baseline	1.5	99	208	143
EL1 - Tier 1	1.9	81	167	115
EL2 - Tier 2	2.4	65	133	92
EL3	5.4	30	60	42
EL4 - Max Tech	12.8	12	25	18

Table 8.3.4 Air Cleaners PC 3, PM_{2.5} CADR $\geq$ 150: Average Annual Energy Use

Efficiency Level	Efficiency Level (PM 2.5 CADR/W)	Residential Sector Energy Use (kWh)	Commercial Sector Energy Use (kWh)	Weighted Average Energy Use (kWh)
EL0 - Baseline	1.2	203	424	292
EL1 - Tier 1	2.0	122	253	175
EL2 - Tier 2	2.9	85	175	121
EL3	6.6	39	78	55
EL4 - Max Tech	7.4	35	69	49

8.3.2 Electricity Prices

Using data from EEI Typical Bills and Average Rates reports, DOE derived annual electricity prices in 2021 for all the census divisions in *RECS*.^{7,8} For each purchase sampled, DOE assigned the average and marginal electricity price for the census division in which the purchase was located. Because marginal electricity price captures more accurately the incremental costs or savings associated with a change in energy use relative to the consumer's bill in the reference case, it may provide a better representation of incremental change in consumer costs than average electricity prices. Therefore, DOE used average electricity prices for the energy use of product purchased in the no-new-standards case, and marginal electricity prices for the incremental change in energy costs associated with the other efficiency levels considered.

For the residential sector, the EEI reports provide the total bill assuming household consumption levels of 500, 750 and 1,000 kWh for the billing period for most of the major investor-owned utilities (IOUs) in the country. For the commercial sector, the reports provide typical bills for several combinations of monthly electricity peak demand and total consumption.

In the case of the residential sector, DOE used the EEI data to define a marginal price as the ratio of the change in the bill to the change in energy consumption. For the commercial sector, marginal prices cannot be estimated directly from the EEI data. Commercial building marginal prices depend on both the change in electricity consumption and the change in monthly peak-coincident demand. For a given change in electricity consumption, the corresponding change in demand is defined through the marginal load factor (MLF). The MLF is equal to the ratio of the average hourly change in electricity use during the billing period to the change in electricity use in the hour of the building peak load. For on-off loads such as lighting, the marginal load factor is equal to the ratio of the number of hours that the equipment is on divided by the total number of hours in the billing period. DOE used a previous analysis of commercial

sector electricity prices to estimate the ratio of the marginal to the average price assuming an MLF of 0.35.

DOE calculated regional weighted-average values for each type of price for the nine census divisions (for the residential and commercial sectors). Each EEI utility in a division was assigned a weight based on the number of consumers it serves; hence, the utility weight for the residential and commercial sectors may be slightly different. Consumer counts were taken from EIA Form 861 data (2021). DOE adjusted these regional weighted-average prices to account for systematic differences between IOUs and publicly-owned utilities (POUs), as the latter are not included in the EEI data set. For each region and sector, DOE estimated a correction factor based on the ratio of the average electricity price for IOUs to the average price charged by POUs (calculated using EIA form 861 data), and the percentage of consumers served by POUs.

Using the electricity prices derived by the method outlined above, DOE assigned seasonal average and marginal prices to each purchaser in the LCC sample based on its location for an average summer or winter month.

Table 8.3.5 through Table 8.3.8 shows the average and marginal prices for each geographic area by sector, respectively.

Table 8.3.5 Average Residential Electricity Prices in 2021

	Geographic Area	2021\$/kWh
1	New England Census Division	\$0.23
2	Middle Atlantic Census Division	\$0.17
3	East North Central Census Division	\$0.14
4	West North Central Census Division	\$0.13
5	South Atlantic Census Division	\$0.12
6	East South Central Census Division	\$0.13
7	West South Central Census Division	\$0.11
8	Mountain Division	\$0.12
9	Pacific Census Division	\$0.23

Source: EIA 2021.

Table 8.3.6 Marginal Residential Electricity Prices in 2021

	Geographic Area	2021\$/kWh
1	New England Census Division	\$0.21
2	Middle Atlantic Census Division	\$0.15
3	East North Central Census Division	\$0.13
4	West North Central Census Division	\$0.11
5	South Atlantic Census Division	\$0.10
6	East South Central Census Division	\$0.11
7	West South Central Census Division	\$0.09
8	Mountain Division	\$0.12
9	Pacific Census Division	\$0.26

Source: EIA 2021.

Table 8.3.7 Average Commercial Electricity Prices in 2021

	Geographic Area	2021\$/kWh
1	New England Census Division	\$0.19
2	Middle Atlantic Census Division	\$0.13
3	East North Central Census Division	\$0.11
4	West North Central Census Division	\$0.11
5	South Atlantic Census Division	\$0.11
6	East South Central Census Division	\$0.14
7	West South Central Census Division	\$0.10
8	Mountain Division	\$0.11
9	Pacific Census Division	\$0.18

Source: EIA 2021.

Table 8.3.8 Marginal Commercial Electricity Prices in 2021

	Geographic Area	2021\$/kWh
1	New England Census Division	\$0.22
2	Middle Atlantic Census Division	\$0.16
3	East North Central Census Division	\$0.12
4	West North Central Census Division	\$0.13
5	South Atlantic Census Division	\$0.14
6	East South Central Census Division	\$0.18
7	West South Central Census Division	\$0.12
8	Mountain Division	\$0.15
9	Pacific Census Division	\$0.22

Source: EIA 2021.

8.3.2.1 Electricity Price Trends

To arrive at prices in future years, DOE multiplied the 2021 electricity prices by the forecast of annual average price changes for each census division from *AEO 2022*.³ For each purchase sampled, DOE applied the projection for the census division in which the purchase was located. Figure 8.3.1 shows the projected national electricity price trends for the residential and commercial sectors as a percentage of the 2021 electricity price.

DOE used the *AEO 2022* reference case for the nine census divisions. The reference case is a business-as-usual estimate, given known market, demographic, and technological trends. To estimate price trends after 2050, DOE used a constant value based on the simple average between 2046 through 2050.

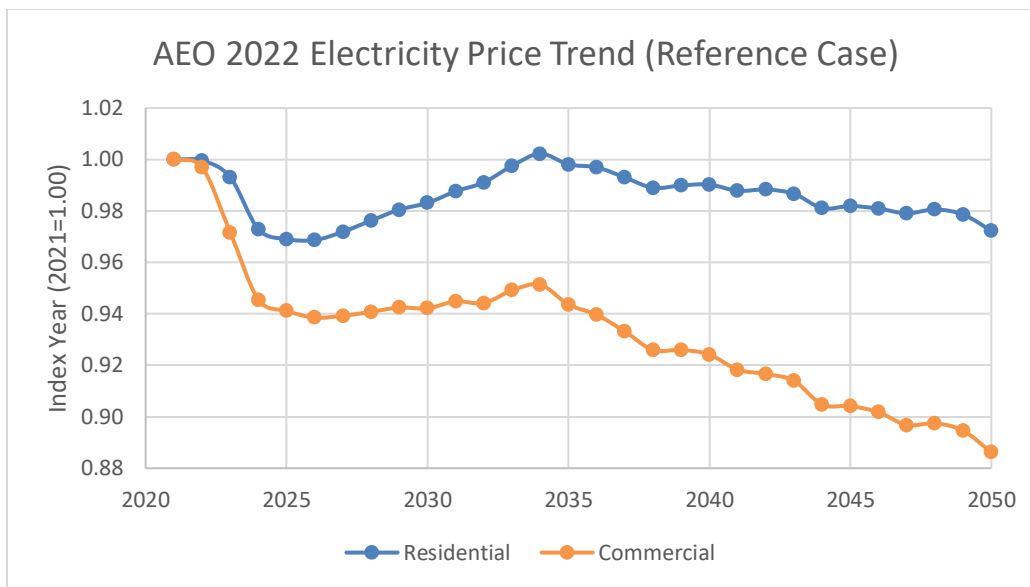


Figure 8.3.1 AEO 2022 Reference Case National Electricity Price Trends

8.3.3 Repair and Maintenance Cost

The repair cost is the cost to repair the product when it fails. The maintenance cost is the cost of regular product maintenance or the cost to replace a part of the appliance such as a filter on a routine basis.

Typically, small incremental changes in product efficiency produce no, or only slight, changes in repair and maintenance costs over baseline products. However, products having significantly higher efficiencies, compared to baseline products, are more likely to incur higher repair and maintenance costs, because their increased complexity and higher part count typically increases the cumulative probability of failure.

For air cleaners, differences in filter size, shape, and material lead to variations in filter costs at each efficiency level within each product class. Given that new air cleaner units usually

come with a new filter, the number of replacement filters required during the lifetime is the product of annual number of filter change and lifetime subtracted by 1. The annualized repair cost for filters is estimated as:

$$ARC = RC \times RF \times \frac{LT}{LT - 1}$$

Where:

ARC = annual filter change cost, in 2021\$,
RC = cost per filter change, in 2021\$,
RF = average number of filter changes per year, and
LT = average lifetime, in years.

Based on the information received from the manufacturer interviews, for commercial buildings, DOE estimated a flat filter change frequency of twice per year. For the residential sector, DOE associated the filter change frequency with the air cleaner usage (see Table 8.3.9). DOE correlated higher filter change frequency with higher operating hours with the highest frequency of once every six months and the lowest frequency of once per year. In residential settings, on an average, 1.7 filters were used per year. This filter change rate aligns with the six to twelve months range suggested by manufacturer interviews. DOE also takes into account that a small percentage of consumers may never change the air cleaner filters.

Table 8.3.9 Residential Air Cleaner Filter Change Frequency Distribution by Usage Frequency Bin

Filter Change Frequency	Number of Replacement Filters per Year	Average Daily Operating Hours		
		Less than 10 hours	More than 10 hours and less than 24 hours	24 hours
		Probability Distribution (%)		
Never	0	1	0	0
12 months	1	9	5	2
9 months	1.33	30	15	10
6 months	2	60	80	88

8.3.4 Product Lifetime

The product lifetime is the age at which a product is retired from service. Rather than use a single average value for the lifetime of air cleaners, DOE developed a lifetime distribution to characterize the probability an air cleaner will be retired from service at a given age. This section describes the survival function DOE developed for its analysis.

For air cleaners, DOE developed a distribution of lifetimes from which specific values are assigned to the appliances in the samples. DOE ensured that the average lifetime estimate of 9 years aligned with those lifetime estimates suggested by ENERGY STAR⁹ and by CA IOUs¹⁰ (who cited EPA and various State Technical Reference Manuals). NEEA¹¹ also cited an estimated lifetime of 9 years.

8.3.4.1 Estimation of Survival Function

DOE estimated the percentage of appliances of a given age still in operation using a survival function, which DOE assumed has the form of a cumulative Weibull distribution, provides an average and a median appliance lifetime.

The Weibull distribution is a probability distribution commonly used to measure failure rates.^c Its form is similar to an exponential distribution, which models a fixed failure rate, except that a Weibull distribution allows for a failure rate that changes over time in a particular fashion. The cumulative Weibull distribution takes the form:

$$P(x) = e^{-\left(\frac{x-\theta}{\alpha}\right)^\beta} \text{ for } x > \theta \text{ and}$$

$$P(x) = 1 \text{ for } x \leq \theta$$

Where:

- $P(x)$ = probability that the appliance is still in use at age x ,
- x = appliance age,
- α = scale parameter, which would be the decay length in an exponential distribution,
- β = shape parameter, which determines the way in which the failure rate changes through time, and
- θ = delay parameter, which allows for a delay before any failures occur.

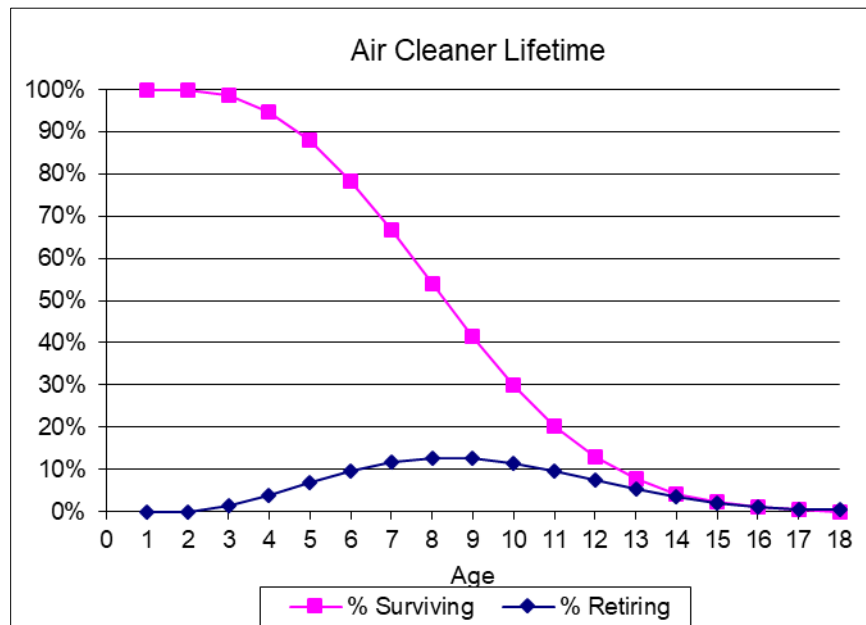
When $\beta = 1$, the failure rate is constant over time, giving the distribution the form of a cumulative exponential distribution. In the case of appliances, β commonly is greater than 1, reflecting an increasing failure rate as appliances age.

DOE estimated a maximum lifetime of 18 years and an average lifetime of 9.0 years for air cleaners. DOE then used estimated average air cleaner lifetime to fit the scale and shape parameters of Weibull distribution. Figure 8.3.2 displays the Weibull probability distribution. See appendix 8C of this final rule TSD for more details.

^c For reference on the Weibull distribution, see sections 1.3.6.6.8 and 8.4.1.3 of the *NIST/SEMATECH e-Handbook of Statistical Methods*, www.itl.nist.gov/div898/handbook/.

Table 8.3.10 Lifetime Parameters for Air Cleaners

Value			Weibull Parameters	
Minimum (years)	Average (years)	Maximum (years)	Alpha (scale)	Beta (shape)
2	9.0	18	7.89	2.50

**Figure 8.3.2 Weibull Probability Distribution for Air Cleaners**

8.3.5 Discount Rates

8.3.5.1 Discount Rates for Residential Applications

The consumer discount rate is the rate at which future operating costs of residential products are discounted to establish their present value in the LCC analysis. The discount rate value is applied in the LCC to future year energy costs and non-energy operations and maintenance costs in order to calculate the estimated net life-cycle cost of products of various efficiency levels and the life-cycle cost savings of higher-efficiency models as compared to the baseline for a representative sample of consumers.

DOE calculates the consumer discount rate using publicly available data (the Federal Reserve Board's *Survey of Consumer Finances* (SCF)) to estimate a consumer's required rate of return or opportunity cost of funds related to appliances.¹² In the economics literature, opportunity cost reflects potential foregone benefit from choosing one option over another. Opportunity cost of capital refers to the rate of return that one could earn by investing in an alternate project with similar risk; similarly, opportunity cost may be defined as the cost

associated with opportunities that are foregone when resources are not put to their highest-value use.¹³

DOE's method views the purchase of a higher efficiency appliance as an investment that yields a stream of energy cost savings. The stream of savings is discounted at a rate reflecting (1) the rates of return associated with other investments available to the consumer, and (2) the observed costs of credit options available to the consumer to reflect the value of avoided debt. DOE notes that the LCC does not analyze the appliance purchase decision, so the implicit discount rate is not relevant in this model. The LCC estimates net present value over the lifetime of the product, so the appropriate discount rate will reflect the general opportunity cost of household funds, taking this time scale into account.

Given the long time horizon modeled in the LCC, the application of a marginal interest rate associated with an initial source of funds is inaccurate. Regardless of the method of purchase, consumers are expected to continue to rebalance their debt and asset holdings over the LCC analysis period, based on the restrictions consumers face in their debt payment requirements and the relative size of the interest rates available on debts and assets. DOE estimates the aggregate impact of this rebalancing using the historical distribution of debts and assets. The discount rate is the rate at which future savings and expenditures are discounted to establish their present value.

DOE estimates separate discount rate distributions for six income groups based on income percentiles as reported in the SCF. These income groups are listed in Table 8.3.11. This disaggregation reflects the fact that low and high income consumers tend to have substantially different shares of debt and asset types, as well as facing different rates on debts and assets. Summaries of shares and rates presented in this chapter are averages across the entire population.

Table 8.3.11 Definitions of Income Groups

Income Group	Percentile of Income
1	0 – 19.9
2	20 – 39.9
3	40 – 59.9
4	60 – 79.9
5	80 – 89.9
6	90 - 100

Sources: Federal Reserve Board. *Survey of Consumer Finances (SCF)* for 1995, 1998, 2001, 2004, 2007, 2010, 2013, 2016, and 2019.

Shares of Debt and Asset Classes

DOE's approach involved identifying all household debt or equity classes in order to approximate a consumer's opportunity cost of funds over the product's lifetime. This approach assumes that in the long term, consumers are likely to draw from or add to their collection of debt and asset holdings approximately in proportion to their current holdings when future

expenditures are required or future savings accumulate. DOE now includes several previously excluded debt types (*i.e.*, vehicle and education loans, mortgages, all forms of home equity loan) in order to better account for all of the options available to consumers.

The average share of total debt plus equity and the associated rate of each asset and debt type are used to calculate a weighted average discount rate for each SCF household (Table 8.3.12). The household-level discount rates are then aggregated to form discount rate distributions for each of the six income groups.^d

DOE estimated the average percentage shares of the various types of debt and equity using data from the SCF for 1995, 1998, 2001, 2004, 2007, 2010, 2013, 2016, and 2019.^e DOE derived the household-weighted mean percentages of each source of across the twenty-one years covered by the eight survey versions. DOE posits that these long-term averages are most appropriate to use in its analysis.

^d Note that previously DOE performed aggregation of asset and debt types over households by summing the dollar value across all households and then calculating shares. Weighting by dollar value gave disproportionate influence to the asset and debt shares and rates of higher income consumers. DOE has shifted to a household-level weighting to more accurately reflect the average consumer in each income group.

^e Note that two older versions of the SCF are also available (1989 and 1992); these surveys are not used in this analysis because they do not provide all of the necessary types of data (*e.g.*, credit card interest rates, etc.). DOE feels that the time span covered by the eight surveys included is sufficiently representative of recent debt and equity shares and interest rates.

Table 8.3.12 Average Shares of Household Debt and Asset Types by Income Group (%)

Type of Debt or Equity	Income Group						
	1	2	3	4	5	6	All
Debt:							
Mortgage	14.3	22.2	33.1	43.3	47.5	37.0	31.0
Home equity loan	1.5	1.8	2.4	3.5	4.6	7.7	3.1
Credit card	15.8	12.2	9.4	6.1	4.0	1.9	9.3
Other installment loan	31.9	28.0	23.9	16.9	11.5	5.9	21.9
Other line of credit	1.4	1.8	1.5	2.0	2.5	2.3	1.8
Other residential loan	0.7	0.4	0.5	0.4	0.3	0.2	0.5
Equity:							
Savings account	19.1	15.0	11.6	9.0	8.2	7.5	12.5
Money market account	3.5	4.3	3.8	3.6	4.4	6.7	4.1
Certificate of deposit	6.0	6.4	4.6	3.8	3.1	3.3	4.8
Savings bond	1.5	1.6	1.4	1.6	1.4	1.2	1.5
State & Local bonds	0.0	0.1	0.2	0.2	0.4	1.3	0.3
Corporate bonds	0.1	0.1	0.1	0.2	0.1	0.4	0.1
Stocks	2.3	3.2	3.8	4.8	6.0	12.2	4.6
Mutual funds	1.8	3.0	3.7	4.8	6.1	12.5	4.5
Total	100.0	100.0	100.0	100.0	100.0	100.0	100.0

Sources: Federal Reserve Board. *Survey of Consumer Finances (SCF)* for 1995, 1998, 2001, 2004, 2007, 2010, 2013, 2016, and 2019.

Rates for Types of Debt

DOE estimated interest rates associated with each type of debt. The source for interest rates for mortgages, loans, credit cards, and lines of credit was the SCF for 1995, 1998, 2001, 2004, 2007, 2010, 2013, 2016, and 2019, which associates an interest rate with each type of debt for each household in the survey.

DOE adjusted the nominal rates to real rates for each type of debt by using the annual inflation rate for each year (using the Fisher formula).^f In calculating effective interest rates for home equity loans and mortgages, DOE also accounted for the fact that interest on both such loans is tax deductible. This rate corresponds to the interest rate after deduction of mortgage interest for income tax purposes and after adjusting for inflation. The specific inflation rates vary by SCF year, while the marginal tax rates vary by SCF year and income bin as shown in Table

^f Fisher formula is given by: Real Interest Rate = [(1 + Nominal Interest Rate) / (1 + Inflation Rate)] – 1. Note that for this analysis DOE used a minimum real effective debt interest rate of 0 percent.

8.3.13. For example, a 6 percent nominal mortgage rate has an effective nominal rate of 5.5 percent for a household at the 25 percent marginal tax rate. When adjusted for an inflation rate of 2 percent, the effective real rate becomes 2.45 percent.

Table 8.3.13 Data Used to Calculate Real Effective Household Debt Rates

Year	Inflation Rate (%)	Applicable Marginal Tax Rate by Income Group (%)					
		1	2	3	4	5	6
1995	2.81	15.0	15.0	15.0	28.0	28.0	39.6
1998	1.55	15.0	15.0	15.0	28.0	28.0	39.6
2001	2.83	10.0	15.0	15.0	27.5	27.5	39.1
2004	2.68	10.0	15.0	15.0	25.0	25.0	35.0
2007	2.85	10.0	15.0	15.0	25.0	25.0	35.0
2010	1.64	10.0	15.0	15.0	25.0	25.0	35.0
2013	1.46	10.0	15.0	15.0	25.0	25.0	37.3
2016	1.26	10.0	15.0	15.0	25.0	25.0	37.3
2019	1.81	10.0	12.0	12.0	22.0	22.0	36.0

Table 8.3.14 shows the household-weighted average effective real rates in each year and the mean rate across years. Because the interest rates for each type of household debt reflect economic conditions throughout numerous years and various phases of economic growth and recession, they are expected to be representative of rates in effect for all applicable compliance years.

Table 8.3.14 Average Real Effective Interest Rates for Household Debt (%)

Type of Debt	Income Group						
	1	2	3	4	5	6	All
Mortgage	4.09	3.74	3.60	2.92	2.79	2.19	3.18
Home equity loan	4.29	4.34	3.86	3.24	3.11	2.45	3.35
Credit card	9.80	11.02	11.15	11.26	10.90	10.11	10.64
Other installment loan	6.14	7.09	5.98	5.33	4.54	4.42	6.10
Other line of credit	3.73	3.67	6.23	5.47	4.89	5.33	4.97
Other residential loan	6.53	6.41	5.22	4.96	4.33	3.99	5.32

Sources: Federal Reserve Board. *Survey of Consumer Finances (SCF)* for 1995, 1998, 2001, 2004, 2007, 2010, 2013, 2016, and 2019.

Rates for Types of Assets

No similar rate data are available from the SCF for classes of assets, so DOE derived asset interest rates from various sources of national historical data (1992–2021). The rates for stocks are the annual returns on the Standard and Poor’s 500 for 1992–2021.¹⁴ The interest rates associated with AAA corporate bonds were collected from Moody’s time-series data for 1992–2021.¹⁵ Rates on Certificates of Deposit (CDs) accounts came from Cost of Savings Index (COSI) data covering 1992–2021.^{16,17,18,19,20,21,g} The interest rates associated with state and local bonds (20-bond municipal bonds) were collected from Federal Reserve Board economic data time-series for 1992–2016 and Bartel Associates for 2017–2021.^{22,h} The interest rates associated with treasury bills (30-Year treasury constant maturity rate) were collected from Federal Reserve Board economic data time-series for 1992–2021.^{23,24} Rates for money market accounts are based on three-month money market account rates reported by Organization for Economic Cooperation and Development (OECD) from 1992–2021.²⁵ Rates for savings accounts are assumed to be half the average real money market rate. Rates for mutual funds are a weighted average of the stock rates and the bond rates.ⁱ DOE adjusted the nominal rates to real rates using the annual inflation rate in each year (see appendix 8D). In addition, DOE adjusted the nominal rates to real effective rates by accounting for the fact that interest on such equity types is taxable. The capital gains marginal tax rate varies for each household based on income as shown in Table 8.3.15.

^g The Wells COSI is based on the interest rates that the depository subsidiaries of Wells Fargo & Company pay to individuals on certificates of deposit (CDs), also known as personal time deposits. Wells Fargo COSI started in November 2009. From July 2007 to October 2009 the index was known as Wachovia COSI and from January 1984 to July 2007 the index was known as GDW (or World Savings) COSI.

^h This index was discontinued in 2016. To calculate the 2017 and after values, DOE used data collected by Bartel Associates.

ⁱ SCF reports what type of mutual funds the household has (e.g. stock mutual fund, savings bond mutual fund, etc.). For mutual funds with a mixture of stocks and bonds, the mutual fund interest rate is a weighted average of the stock rates (two-thirds weight) and the savings bond rates (one-third weight).

Table 8.3.15 Average Capital Gains Marginal Tax Rate by Income Group (%)

Year	Income Group					
	1	2	3	4	5	6
1995	12.5	12.5	12.5	28.0	28.0	33.8
1998	12.5	12.5	12.5	24.0	24.0	29.8
2001	7.5	10.0	15.0	21.3	21.3	27.1
2004	7.5	10.0	15.0	21.3	21.3	27.1
2007	7.5	10.0	15.0	20.0	20.0	25.0
2010	5.0	7.5	15.0	20.0	20.0	25.0
2013	5.0	7.5	15.0	20.0	20.0	27.4
2016	5.0	7.5	15.0	20.0	20.0	27.4
2019	5.0	6.0	6.0	18.5	18.5	26.8

Average real effective interest rates for the classes of household assets are listed in Table 8.3.16. Because the interest and return rates for each type of asset reflect economic conditions throughout numerous years, they are expected to be representative of rates that may be in effect in the compliance year. The average nominal interest rates and the distribution of real interest rates by year are shown in appendix 8D.

Table 8.3.16 Average Real Interest Rates for Household Assets (%)

Equity Type	Income Group						
	1	2	3	4	5	6	All
Savings accounts	0.24	0.23	0.22	0.20	0.20	0.19	0.22
Money market accounts	0.15	0.14	0.14	0.13	0.13	0.12	0.13
Certificate of deposit	0.76	0.74	0.71	0.64	0.64	0.59	0.71
Treasury Bills (T-bills)	2.06	2.02	1.94	1.76	1.76	1.62	1.91
State/Local bonds	1.81	1.99	1.90	1.73	1.73	1.59	1.71
AAA Corporate Bonds	2.16	2.18	2.54	2.43	2.34	2.23	2.33
Stocks (S&P 500)	8.77	8.60	8.21	7.45	7.45	6.85	7.70
Mutual funds	7.22	7.29	7.06	6.31	6.39	5.60	6.45

Discount Rate Calculation and Summary

Using the asset and debt data discussed above, DOE calculated discount rate distributions for each income group as follows. First, DOE calculated the discount rate for each consumer in each of the versions of the SCF, using the following formula:

$$DR_i = \sum_j Share_{i,j} \times Rate_{i,j}$$

Where:

DR_i = discount rate for consumer i ,

$Share_{i,j}$ = share of asset or debt type j for consumer i , and

$Rate_{i,j}$ = real interest rate or rate of return of asset or debt type j for consumer i .

The rate for each debt type is drawn from the SCF data for each household. The rate for each asset type is drawn from the distributions described above.

Once the real discount rate was estimated for each consumer, DOE compiled the distribution of discount rates in each survey by income group by calculating the proportion of consumers with discount rates in bins of 1 percent increments, ranging from 0-1 percent at the low end to 30 percent and greater at the high end. Giving equal weight to each survey, DOE compiled the overall distribution of discount rates.

Table 8.3.17 presents the average real effective discount rate and its standard deviation for each of the six income groups. To account for variation among households, DOE sampled a rate for each RECS household from the distributions for the appropriate income group. (RECS provides household income data.) Appendix 8D presents the full probability distributions for each income group that DOE used in the LCC and PBP analysis.

Table 8.3.17 Average Real Effective Discount Rates

Income Group	Discount Rate (%)
1	4.71%
2	4.95%
3	4.51%
4	3.80%
5	3.44%
6	3.19%
Overall Average	4.30%

Source: Board of Governors of the Federal Reserve System, Survey of Consumer Finances (1995 – 2019).

8.3.5.2 Commercial Purchases

DOE's method views the purchase of a higher efficiency appliance as an investment that yields a stream of energy cost savings. DOE derived the discount rates for the LCC analysis by estimating the cost of capital for companies that purchase air cleaners. For private firms, the weighted average cost of capital (WACC) is commonly used to estimate the present value of cash flows to be derived from a typical company project or investment. Most companies use debt and equity capital to fund investments, so their cost of capital is the weighted average of the cost to the firm of equity and debt financing, as estimated from financial data for publicly traded firms in the sectors that purchase air cleaners.²⁶ As discount rates can differ across industries, DOE estimates separate discount rate distributions for a number of aggregate sectors with which elements of the LCC building sample can be associated.

Damodaran Online, the primary source of data for this analysis, is a widely used source of information about debt and equity financing for most types of firms.²⁷ The nearly 200 detailed industries included in the Damodaran Online data (shown in a table in Appendix 8E) were assigned to the aggregate sectors shown in Table 8.3.18, which also shows the mapping between the aggregate sectors and CBECS Principal Building Activities (PBAs).^j

Damodaran Online data for manufacturing and other similar industries were assigned to the aggregate Industrial sector, while data for farming and agriculture were assigned to the Agriculture sector. Public entities are included in the sectors Federal Government and State/Local Government, but Damodaran data are not used for these sectors.

^j Previously, Damodaran Online provided firm-level data, but now only industry-level data is available, as compiled from individual firm data, for the period of 1998-2020. The data sets note the number of firms included in the industry average for each year.

Table 8.3.18 Mapping of Sectors to CBECS Categories

Sector Name in DR Analysis	Applied to CBECS PBAs: (Sector Name and PBA number)
Education ^k	Education (14)
Food Sales	Food sales (6)
Food Service	Food service (15)
Health Care	Outpatient health care (8); Inpatient health care (16); Nursing (17); Nursing (17); Laboratory (4)
Lodging	Lodging (18)
Mercantile	Enclosed mall (24); Strip shopping mall (23); Retail other than mall (25)
Office	Office (2)
Public Assembly	Public Assembly (13)
Service	Service (26)
All Commercial	All CBECS PBAs, including those specified above
Industrial	Not in CBECS
Agriculture	Not in CBECS
Federal Government	Not in CBECS
State/Local Government	Not in CBECS

Note: CBECS only includes buildings used by firms in “commercial” sectors, so Industrial, Agriculture, Federal Government, and State/Local Government have no associated PBA identifier. However, discount rate distributions are required for these sectors because they are significant consumers of some types of appliances and energy-consuming equipment.

For private firms, DOE estimated the cost of equity using the capital asset pricing model (CAPM).²⁸ CAPM assumes that the cost of equity (k_e) for a particular company is proportional to the systematic risk faced by that company, where high risk is associated with a high cost of equity and low risk is associated with a low cost of equity. In CAPM, the systematic risk facing a firm is determined by several variables: the risk coefficient of the firm (β), the expected return on risk-free assets (R_f), and the equity risk premium (ERP). The cost of equity can be estimated at the industry level by averaging across constituent firms. The risk coefficient of the firm indicates the risk associated with that firm relative to the price variability in the stock market. The expected return on risk-free assets is defined by the yield on long-term government bonds. The ERP represents the difference between the expected stock market return and the risk-free rate. The cost of equity financing is estimated using the following equation, where the variables are defined as above:

$$k_{ei} = R_f + (\beta_i \times ERP)$$

Eq. 8.10

^k This sector applies to private education, while public education is covered under the later discussion of buildings operated by state and local government entities.

Where:

k_{ei} = cost of equity for industry i ,
 R_f = expected return on risk-free assets,
 β_i = risk coefficient of industry i , and
 ERP = equity risk premium.

Several parameters of the cost of capital equations can vary substantially over time, and therefore the estimates can vary with the time period over which data is selected and the technical details of the data averaging method. For guidance on the time period for selecting and averaging data for key parameters and the averaging method, DOE used Federal Reserve methodologies for calculating these parameters. In its use of the CAPM, the Federal Reserve uses a 40-year period for calculating discount rate averages, utilizes the gross domestic product price deflator for estimating inflation, and considers the best method for determining the risk-free rate as one where “the time horizon of the investor is matched with the term of the risk-free security.”²⁹

By taking a forty-year geometric average of Federal Reserve data on annual nominal returns for 10-year Treasury bonds, as provided by Damodaran Online, DOE estimated the following risk-free rates (Table 8.3.19).^{30,31} DOE also estimated the ERP by calculating the difference between risk free rate and stock market return for the same time period, as estimated using Damodaran Online data on the historical return to stocks.

Table 8.3.19 Risk Free Rate and Equity Risk Premium

Year	Risk-Free Rate (%)	ERP (%)	Year	Risk-Free Rate (%)	ERP (%)
1998	7.15	4.76	2010	7.47	2.51
1999	6.62	5.83	2011	7.80	1.75
2000	6.98	4.52	2012	7.78	2.62
2001	6.98	4.42	2013	7.46	4.59
2002	7.32	2.80	2014	7.65	3.86
2003	7.23	3.16	2015	7.27	3.67
2004	7.33	3.02	2016	7.26	4.21
2005	7.33	3.45	2017	7.36	4.49
2006	7.43	3.16	2018	7.34	3.89
2007	7.61	2.84	2019	7.67	3.55
2008	8.25	1.15	2020	7.75	4.08
2009	7.50	2.46	2021	6.85	5.17

The cost of debt financing (k_d) is the interest rate paid on money borrowed by a company. The cost of debt is estimated by adding a risk adjustment factor (R_a) to the risk-free rate. This risk adjustment factor depends on the variability of stock returns represented by standard

deviations in stock prices. This same calculation can alternatively be performed with industry-level data. For industry i , the cost of debt financing is:

$$k_{di} = (R_f + R_{ai}) \times (1 - tx_i)$$

Eq. 8.11

Where:

k_{di} = (after-tax) cost of debt financing for industry, i ,
 R_f = expected return on risk-free assets,
 R_{ai} = risk adjustment factor to risk-free rate for industry, i , and
 tx_i = tax rate of industry, i .

DOE estimates the weighted average cost of capital using the following equation:

$$WACC = k_{ei} \times w_{ei} + k_{di} \times w_{di}$$

Eq. 8.12

Where:

$WACC_i$ = weighted average cost of capital for industry i ,
 k_{ei} = cost of equity for industry i ,
 k_{di} = cost of debt financing for industry, i ,
 w_e = proportion of equity financing for industry i , and
 w_d = proportion of debt financing for industry i .

DOE accounts for inflation using the all items Gross Domestic Product deflator, as published by the Bureau of Economic Analysis.³² Table 8.3.20 shows the real average WACC values for the major sectors that purchase air cleaners. Tables providing full discount rate distributions by sector are included in appendix 8E. While WACC values for any sector may trend higher or lower over substantial periods of time, these values represent a cost of capital that is averaged over major business cycles.

For each entity in the consumer sample for air cleaners, a discount rate is drawn from the distribution calculated for the appropriate sector.

Table 8.3.20 Weighted Average Cost of Capital for Commercial/Industrial Sectors

Sector	Observations	Total Firms	Mean WACC (%) of Regular-Sized Businesses
Education	24	836	7.21%
Food Sales	44	896	5.66%
Food Service	24	1,910	6.51%
Health Care	57	5,720	6.90%
Lodging	24	1,685	6.55%
Mercantile	104	5,718	6.99%
Office	471	47,625	6.89%
Public Assembly	48	3,866	7.27%
Service	161	16,068	6.25%
All Commercial	971	84,482	6.77%
Industrial	1,352	81,480	7.25%
Agriculture	9	306	7.15%
Utilities	107	2,162	4.18%
R.E.I.T/Property	57	4,631	6.62%

Note: “Observations” reflect the number of Damodaran Online detailed industries included in DOE’s aggregate sector calculation, while “Total Firms” presents a sum of the number of individual companies represented by those detailed industries. These are two measures of the comprehensiveness of the data used in the WACC calculation.

For publicly owned and operated buildings, the cost of capital can be derived using state and local bond rates and U.S. Treasury bond rates.^{33,34,35} State and local bond rates are used for buildings identified as owned and/or occupied by state or local government entities, such as public schools or local government administrative buildings. Treasury bond rates are used for buildings identified as occupied by federal government entities. Table 8.3.21 presents the average values of discount rates used for public sectors. As for private firms, a discount rate is drawn from the distribution calculated for the appropriate sector.

Table 8.3.21 Discount Rates for Public Sectors that Purchase Air Cleaners

Sector	Observations	Mean Discount Rate (%)
State/Local Government	133 quarters	2.67
Federal Government	399 months	2.06

Small Business Commercial Discount Rate

Even after accounting for systematic risk through the β coefficient, CAPM underestimates the cost of equity for small firms; this phenomenon is known as the size effect (see, e.g., Fama & French, 1992; Ibbotson Associates, 2009). To account for the size effect, a size premium can be incorporated into the CAPM equation to provide an alternative estimate of the small company cost of equity, and thus, the weighted average cost of capital specific to small

businesses. The size effect is most pronounced for the smallest firms, in terms of market capitalization. In order to provide a conservative estimate of the value of discounted future energy cost savings, we focus on size effect of “microcap” companies (i.e., companies within the smallest two deciles of the overall market as measured by market capitalization).

The capital asset pricing model (CAPM) underestimates the cost of capital for small companies. In CAPM, the risk premium β is used to account for the higher returns associated with greater risk. However, for small companies, particularly very small companies, historic returns have been significantly higher than the CAPM equation predicts. This additional return can be accounted for by adding a size premium to the cost of equity for small firms:

$$k_e = R_f + (\beta \times ERP) + S$$

Eq. 8.13

Where:

k_e = Cost of equity,
 R_f = Expected return on risk-free assets,
 β = Risk coefficient of the firm,
 ERP = Equity risk premium, and
 S = Size Premium.

DOE obtained size premium data from the Stocks, Bonds, Bills, and Inflation Yearbook for 1998-2017; for 2018-2020, size premium data were extracted from the Duff & Phelps online “Cost of Capital Navigator” system (Table 8.3.22).^{46,47,48,49} For each year, the size premium is the historical average difference in performance between small companies and the market average. The size effect is most pronounced for the smallest firms, in terms of market capitalization. In order to provide a conservative estimate of the value of discounted future energy cost savings, we focus on size effect of “microcap” companies (i.e., companies within the smallest two deciles of the overall market as measured by market capitalization). For example, for the period of 1926-2007, the average size premium for the smallest companies in all industries is 3.74%, implying that on average, historic performance of small companies was 3.74% higher than the CAPM estimate of the small company cost of equity over this period.

Table 8.3.22 Size Premium by Year

Year	Market Cap. of Largest Firm (Decile 10, \$million)	Market Cap. of Largest Firm (Decile 9, \$million)	Size Premium (Deciles 9,10 Microcap , %)
1998	--	252.0	2.60
1999	97.9	214.6	2.21
2000	84.5	192.6	2.62
2001	141.5	314.0	3.53
2002	166.4	330.6	4.01
2003	262.7	505.4	4.02
2004	264.9	586.4	3.95
2005	314.4	626.9	3.88
2006	363.5	723.3	3.65
2007	218.5	456.3	3.74
2008	214.1	431.3	3.99
2009	235.6	477.5	4.07
2010	206.8	422.8	3.89
2011	253.8	514.2	3.81
2012	253.7	514.2	3.81
2013	338.8	632.8	3.84
2014	300.7	548.8	3.74
2015	--	--	3.58
2016	--	--	3.67
2017	--	--	5.40
2018			3.39
2019	--	--	3.16
2020			3.16

Note: The size premium value for 2019 is also applied to because the 2020 value was not released at the time of analysis.

DOE calculated the real weighted average cost of capital by sector (as described in chapter 8) using the cost of equity including a size premium for small companies instead of the CAPM cost of equity.^{39,50} Table 8.3.23 presents DOE's estimates of the discount rates for entire sectors, small companies specifically, and the small company discount rate premium.

To estimate the impact of standards specifically on small businesses, DOE applied the small company discount rate distributions each sector in the Life-Cycle Cost and Payback Period calculation, instead of the distributions described in chapter 8.

Table 8.3.23 Discount Rate Difference between Small Company and Sector Average

Sector	All Company WACC (%)	Small Company WACC (%)	Small Company DR Premium (%)
Education	7.12%	10.32%	3.20%
Food Sales	5.60%	8.13%	2.54%
Food Service	6.34%	9.27%	2.93%
Health Care	6.78%	9.60%	2.82%
Lodging	6.35%	8.67%	2.32%
Mercantile	6.88%	9.75%	2.87%
Office	6.78%	9.50%	2.72%
Public Assembly	7.17%	9.96%	2.79%
Service	6.22%	8.43%	2.21%
All Commercial	6.67%	9.31%	2.64%
Industrial	7.16%	10.04%	2.88%
Agriculture	6.94%	9.30%	2.36%
Utilities	4.14%	6.15%	2.01%
R.E.I.T./Property	6.43%	8.78%	2.35%

8.3.6 Effective Date of Potential Standard

New standards apply to air cleaners manufactured five years after the date on which any new standard is published. (42 U.S.C. 6295(g)(10)(B)) Therefore, DOE used 2028 as the first year of compliance with any new standards for air cleaners. On August 23, 2022, DOE received a Joint Proposal from the Joint Negotiators regarding energy conservation standards for air cleaners. Based on the Joint Proposal’s two-tier standard recommendation, DOE added 2024 and 2026 as additional first years of compliance in one of the scenarios analyzed.

8.4 ENERGY EFFICIENCY DISTRIBUTIONS

To estimate the percentage of consumers who would be affected by a potential standard at any of the efficiency levels identified in chapter 5 of this final rule TSD, DOE’s LCC analysis first considered the projected distribution of efficiencies for products that consumers purchase under the no-new-standards case (the case without new energy efficiency standards). DOE refers to this distribution of product energy efficiencies as the *no-new-standards case efficiency distribution*. Using the projected distribution of efficiencies for air cleaners, DOE randomly assigned a product efficiency to each household and a commercial user drawn from the *RECS 2020* and *CBECS 2018* data sets. If a consumer is assigned a product efficiency that is greater than or equal to the efficiency under consideration, the LCC savings calculation shows that the consumer would not be affected by that efficiency level. The energy efficiency distributions that DOE used in the LCC analysis are described below.

To estimate the energy efficiency distribution of air cleaners for 2028, DOE combined market share information submitted by manufacturers and model efficiency distribution from the ENERGY STAR database. Due to the lack of historical efficiency distributions and evidence that demonstrates air cleaner efficiency improvements across the years in the absence of the standards, DOE assumed no annual efficiency improvement for the no-new-standards case. The estimated market shares for the no-new-standards case for air cleaners are shown in Table 8.4.1.

Table 8.4.1 Air Cleaners: No-New-Standards Case Market Shares in 2028

Efficiency Level	PC 1		PC 2		PC 3	
	$10 \leq \text{PM}_{2.5} \text{ CADR} < 100$		$100 \leq \text{PM}_{2.5} \text{ CADR} < 150$		$\text{PM}_{2.5} \text{ CADR} \geq 150$	
	Efficiency	Market Share %	Efficiency	Market Share %	Efficiency	Market Share %
	$\text{PM}_{2.5} \text{ CADR/W}$		$\text{PM}_{2.5} \text{ CADR/W}$		$\text{PM}_{2.5} \text{ CADR/W}$	
Baseline	1.5	28.0%	1.5	24.4%	1.2	22.2%
1	1.7	42.1%	1.9	36.6%	2.0	33.3%
2	1.9	19.1%	2.4	28.1%	2.9	37.7%
3	3.4	7.5%	5.4	10.5%	6.6	3.1%
4	5.4	3.3%	12.8	0.4%	7.4	3.8%

8.5 LIFE-CYCLE COST AND PAYBACK PERIOD RESULTS

This section presents the LCC and PBP results for air cleaners. As discussed in section 8.1.1, DOE's approach to the LCC analysis relied on developing samples of households that use each of the product classes. DOE also used probability distributions to characterize the uncertainty in many of the inputs to the analysis. DOE used a Monte Carlo simulation to perform the LCC calculations on the households in the sample. For each set of sample households that use the product in each product class, DOE calculated the average LCC and LCC savings and the median and average PBP for each of the trial standard levels (TSLs).

DOE calculated LCC savings and PBPs relative to the no-new-standards case products that it assigned to sample households. For some consumers DOE assigned a no-new-standards case product that is more efficient than some of the TSLs. For that reason, the average LCC impacts are not equal to the difference between the LCC of a specific standard level and the LCC of the baseline product. DOE calculated the average LCC savings and the median PBP values by excluding the households that are not impacted by a standard at a given efficiency level.

LCC and PBP calculations were performed 10,000 times on the sample of consumers established for each product class. Each LCC and PBP calculation was performed on a single household selected from the sample. A household was selected based on its weight (*i.e.*, how representative it was of other households in the distribution). Each LCC and PBP calculation also sampled from the probability distributions that DOE developed to characterize many of the inputs to the analysis.

Using the Monte Carlo simulations for each efficiency level, DOE calculated the percent of consumers who experience a net LCC benefit, a net LCC cost, and no effect. DOE considered a consumer to receive no effect at a given standard level if DOE assigned it a baseline product having the same or higher efficiency than the standard level. The following sections present figures that illustrate the range of LCC and PBP effects among sample consumers.

In the absence of air cleaner ownership and usage information in in both *RECS 2020* and *CBECS 2018* datasets, for the residential sector, DOE included all household samples but adjusted the residential sample weights based on the geographic distribution of air cleaner stocks and the average number of air cleaners per sample based on household size reported by TechSci Research.⁶ For the commercial sector, DOE excluded the vacant and non-used buildings from the *CBECS 2018* samples and adjusted the remaining building sample weights based on the building occupancy, the square footage of the climate-controlled space, and the stock distribution by building principal activity reported by TechSci Research.⁶ This is described in detail in chapter 7 of this TSD. In all shipments projection years, based on the TechSci Research data, DOE assumed that 40 percent of shipments were directed to the commercial sector, and 60 percent were directed to the residential sector.

8.5.1 Summary of Results

Table 8.5.1 through Table 8.5.6 show the LCC and simple PBP results by TSL for each air cleaner product class. The TSLs correspond to the energy efficiency levels included in the engineering analysis. The average operating cost is the discounted sum.

Table 8.5.1 Average LCC and PBP Results for Product Class 1: $10 \leq \text{PM}_{2.5} \text{ CADR} < 100$

TSL*	Efficiency Level	Average Costs 2021\$				Simple Payback years	Average Lifetime years
		Installed Cost	First Year's Operating Cost	Lifetime Operating Cost	LCC		
--	Baseline	\$64	\$13	\$117	\$181	--	9.0
1	1	\$65	\$11	\$98	\$163	0.9	9.0
2	2	\$67	\$10	\$91	\$158	1.4	9.0
3**	1	\$65	\$11	\$98	\$163	0.9	9.0
	2	\$67	\$10	\$91	\$158	1.4	9.0
4	3	\$78	\$15	\$178	\$255	NA	9.0
5	4	\$86	\$14	\$176	\$262	NA	9.0

Note: The results for each TSL are calculated assuming that all consumers use products at that efficiency level. The PBP is measured relative to the baseline product.

*All TSLs except TSL 3 have a compliance year of 2028.

** For TSL 3, the first results row has a 2024 compliance year. The second results row has a 2026 compliance year.

Table 8.5.2 Average LCC Savings Relative to the No-New-Standards Case for Product Class 1: $10 \leq \text{PM}_{2.5} \text{ CADR} < 100$

TSL**	Efficiency Level	Life-Cycle Cost Savings	
		Average LCC Savings* 2021\$	Percent of Consumers that Experience Net Cost
1	1	\$18	0%
2	2	\$12	6%
3***	1	\$18	0%
	2	\$12	6%
4	3	(\$87)	88%
5	4	(\$87)	94%

* The savings represent the average LCC for affected consumers.

** All TSLs except TSL 3 have a compliance year of 2028.

*** For TSL 3, the first results row has a 2024 compliance year. The second results row has a 2026 compliance year.

Table 8.5.3 Average LCC and PBP Results for Product Class 2: $100 \leq \text{PM}_{2.5} \text{ CADR} < 150$

TSL*	Efficiency Level	Average Costs 2021\$				Simple Payback years	Average Lifetime years
		Installed Cost	First Year's Operating Cost	Lifetime Operating Cost	LCC		
--	Baseline	\$88	\$31	\$273	\$361	--	9.0
1	1	\$90	\$26	\$232	\$322	0.4	9.0
2	2	\$92	\$22	\$195	\$287	0.5	9.0
3**	1	\$90	\$26	\$232	\$322	0.4	9.0
	2	\$92	\$22	\$195	\$287	0.5	9.0
4	3	\$101	\$24	\$280	\$381	NA	9.0
5	4	\$109	\$17	\$207	\$317	1.6	9.0

Note: The results for each TSL are calculated assuming that all consumers use products at that efficiency level. The PBP is measured relative to the baseline product.

* All TSLs except TSL 3 have a compliance year of 2028.

** For TSL 3, the first results row has a 2024 compliance year. The second results row has a 2026 compliance year.

Table 8.5.4 Average LCC Savings Relative to the No-New-Standards Case for Product Class 2: $100 \leq \text{PM}_{2.5} \text{ CADR} < 150$

TSL**	Efficiency Level	Life-Cycle Cost Savings	
		Average LCC Savings* 2021\$	Percent of Consumers that Experience Net Cost
1	1	\$38	0%
2	2	\$50	0%
3***	1	\$38	0%
	2	\$50	0%
4	3	(\$60)	75%
5	4	\$11	54%

* The savings represent the average LCC for affected consumers.

**All TSLs except TSL 3 have a compliance year of 2028.

*** For TSL 3, the first results row has a 2024 compliance year. The second results row has a 2026 compliance year.

Table 8.5.5 Average LCC and PBP Results for Product Class 3: $\text{PM}_{2.5} \text{ CADR} \geq 150$

TSL*	Efficiency Level	Average Costs 2021\$				Simple Payback years	Average Lifetime years
		Installed Cost	First Year's Operating Cost	Lifetime Operating Cost	LCC		
--	Baseline	\$144	\$57	\$485	\$629	--	9.0
1	1	\$146	\$41	\$377	\$523	0.1	9.0
2	2	\$147	\$34	\$323	\$470	0.1	9.0
3**	1	\$146	\$41	\$377	\$523	0.1	9.0
	2	\$147	\$34	\$323	\$470	0.1	9.0
4	3	\$151	\$31	\$347	\$497	0.3	9.0
5	4	\$151	\$31	\$354	\$505	0.3	9.0

Note: The results for each TSL are calculated assuming that all consumers use products at that efficiency level. The PBP is measured relative to the baseline product.

*All TSLs except TSL 3 have a compliance year of 2028.

** For TSL 3, the first results row has a 2024 compliance year. The second results row has a 2026 compliance year.

Table 8.5.6 Average LCC Savings Relative to the No-New-Standards Case for Product Class 3: PM_{2.5} CADR $\geq$ 150

TSL**	Efficiency Level	Life-Cycle Cost Savings	
		Average LCC Savings* 2021\$	Percent of Consumers that Experience Net Cost
1	1	\$105	0%
2	2	\$94	0%
3***	1	\$105	0%
	2	\$94	0%
4	3	\$29	50%
5	4	\$20	56%

* The calculation considers only affected consumers. It excludes purchasers whose purchasing decision would not change under a standard set at the corresponding EL, i.e., those with zero LCC savings.

8.5.2 Distribution of Impacts

The figures in this section show the distribution of LCCs in the no-new-standards case for each product class. The figures are presented as frequency charts that show the distribution of LCCs, and LCC impacts with their corresponding probability of occurrence. DOE generated the figures for the distributions from a Monte Carlo simulation run based on 10,000 samples.

8.5.2.1 No-New-Standards Case Distribution of Life-Cycle Cost

Figure 8.5.1 and Figure 8.5.2 show the no-new-standards case LCC distributions for product class 1 of air cleaners for residential and commercial consumers. DOE can generate a frequency chart like the one shown in Figure 8.5.1 for each product class and sector.

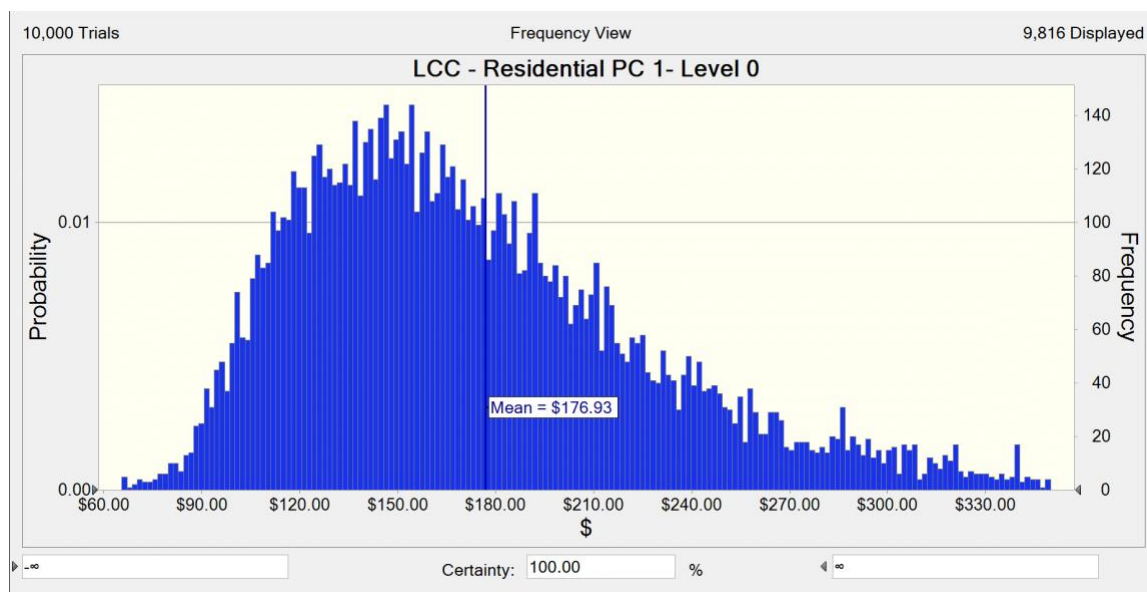


Figure 8.5.1 Air Cleaner, Residential PC 1: No-New-Standards Case LCC Distribution

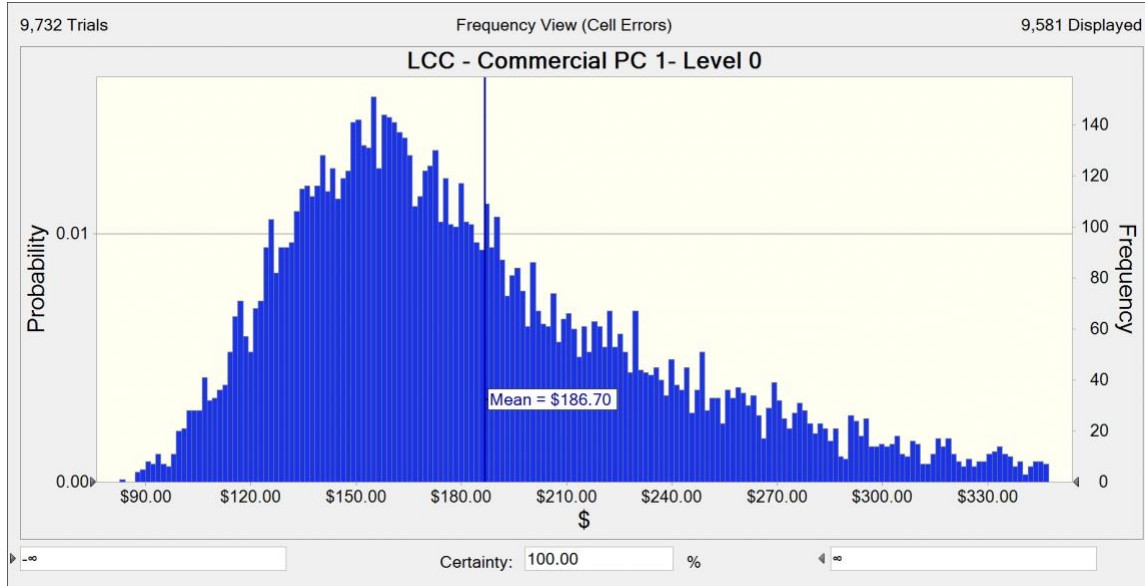


Figure 8.5.2 Air Cleaner, Commercial PC 1: No-New-Standards Case LCC Distribution

8.5.2.2 Efficiency Level Distributions of LCC Impacts

Figure 8.5.3 is an example of a frequency chart that shows the distribution of LCC differences for the case of a standard set at EL 2 for PC 1 for commercial consumers of air cleaners. In the figure, a text box next to a vertical line at a given value on the x-axis shows the mean change in LCC (a savings of \$13.81 in the example here). The note, “Certainty is 100.00% from -Infinity to +Infinity,” means that 100 percent of owners of air cleaner PC 1 commercial consumers will have LCC savings or not be affected by the efficiency level compared to the no-new-standards case. For more details, refer to section 8.4 on the distribution of product efficiencies under the no-new-standards case. DOE can generate a frequency chart like the one shown in Figure 8.5.3 for each efficiency level, product class, and user sector.

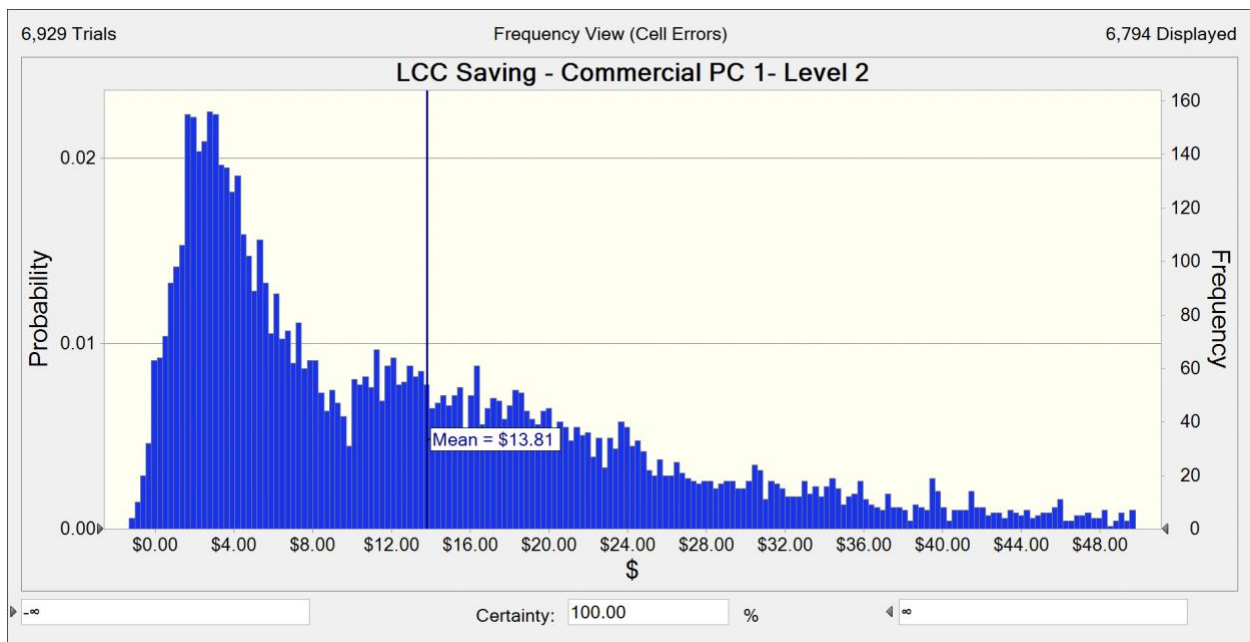


Figure 8.5.3 Air Cleaner, Commercial PC 1: Distribution of LCC Impacts for Standard at Efficiency Level 2

8.5.3 Range of Impacts

Figure 8.5.4 through Figure 8.5.9 show the range of LCC savings for all efficiency levels considered for product class 1 of air cleaners for residential and commercial users, respectively. For each efficiency level, the top and the bottom of the box indicate the 75th and 25th percentiles, respectively. The bar at the middle of the box indicates the median: 50 percent of households have LCC savings in excess of that value. The “whiskers” at the bottom and the top of the box indicate the 5th and 95th percentiles. The small box shows the average LCC savings for each standard level.

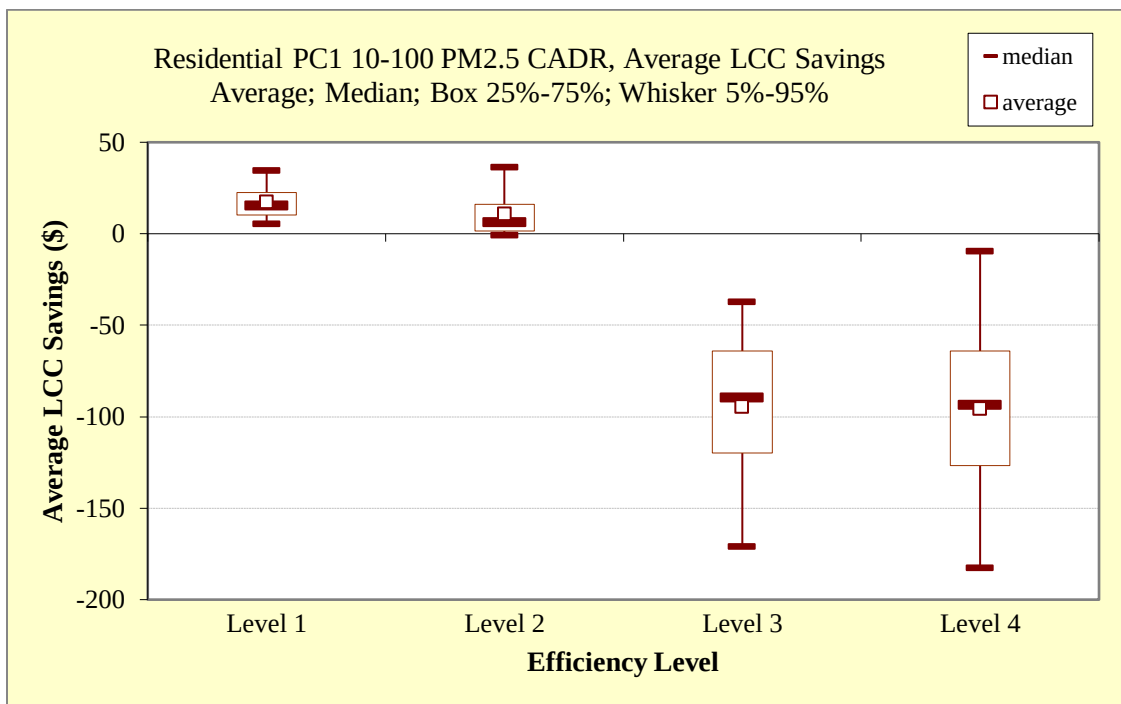


Figure 8.5.4 Air Cleaner PC 1 Residential: Range of Average LCC Savings

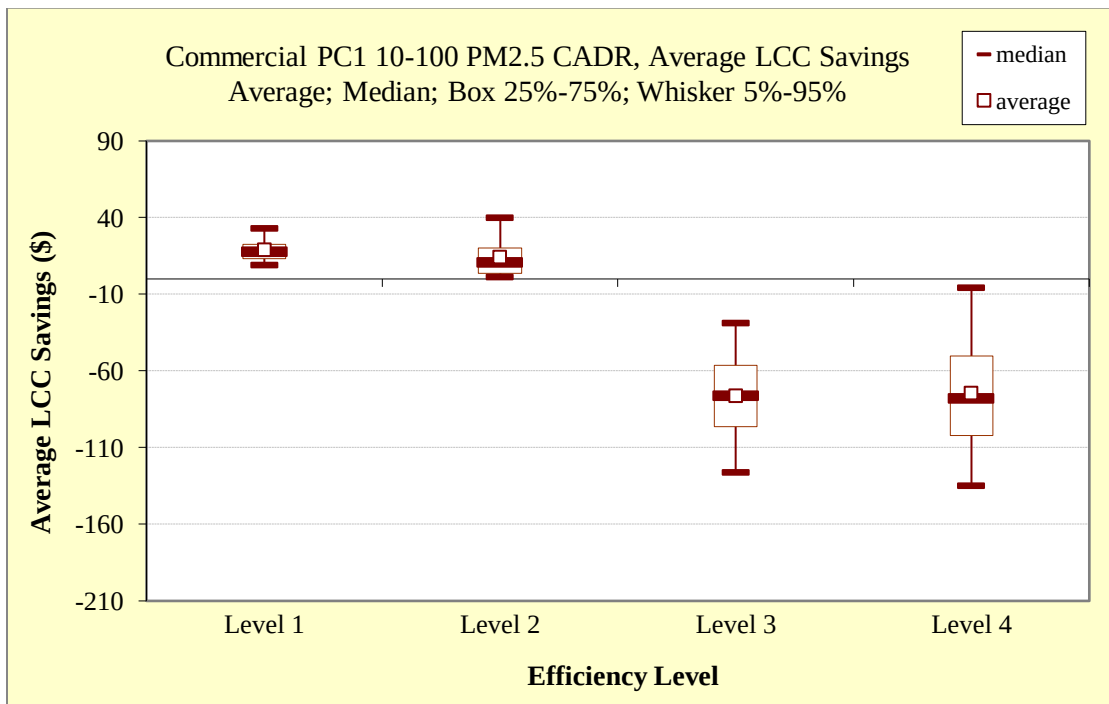


Figure 8.5.5 Air Cleaner PC 1 Commercial: Range of Average LCC Savings

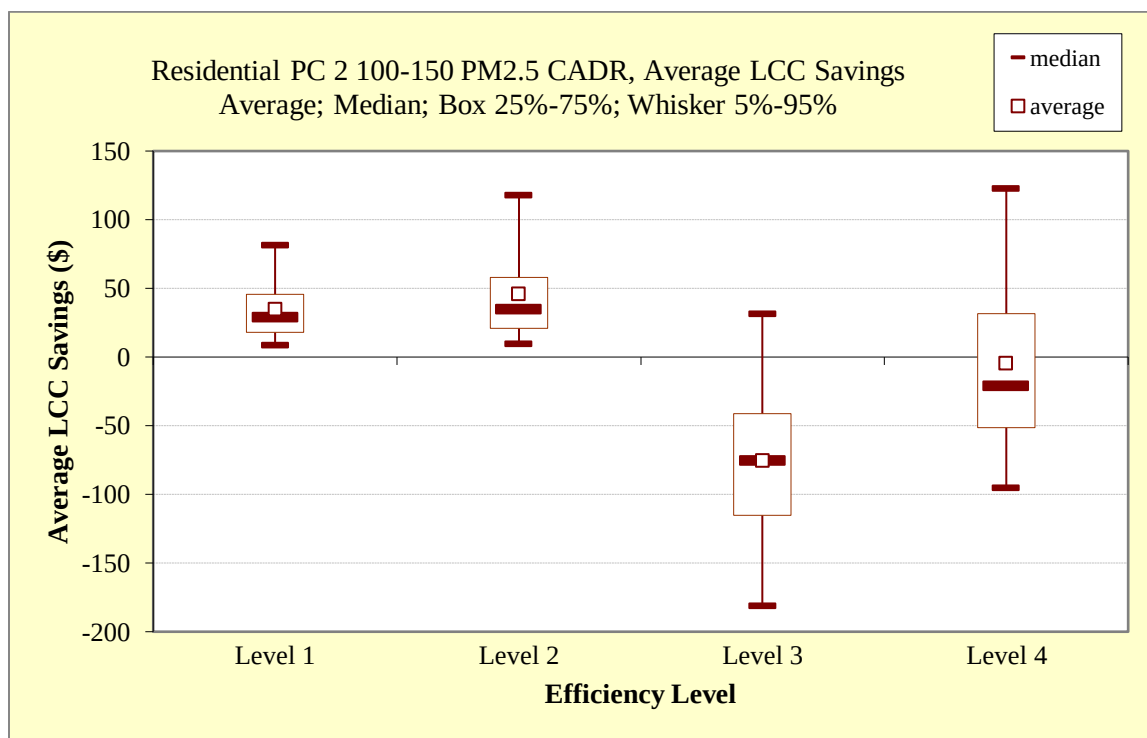


Figure 8.5.6 Air Cleaner PC 2 Residential: Range of Average LCC Savings

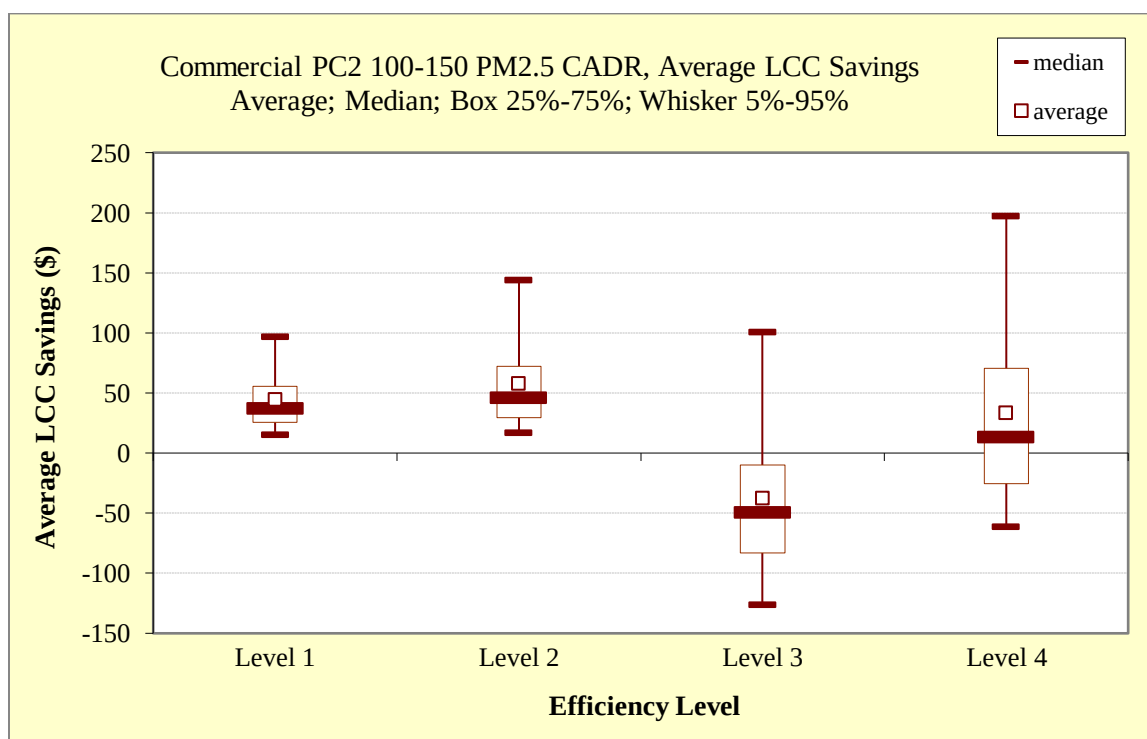


Figure 8.5.7 Air Cleaner PC 2 Commercial: Range of Average LCC Savings

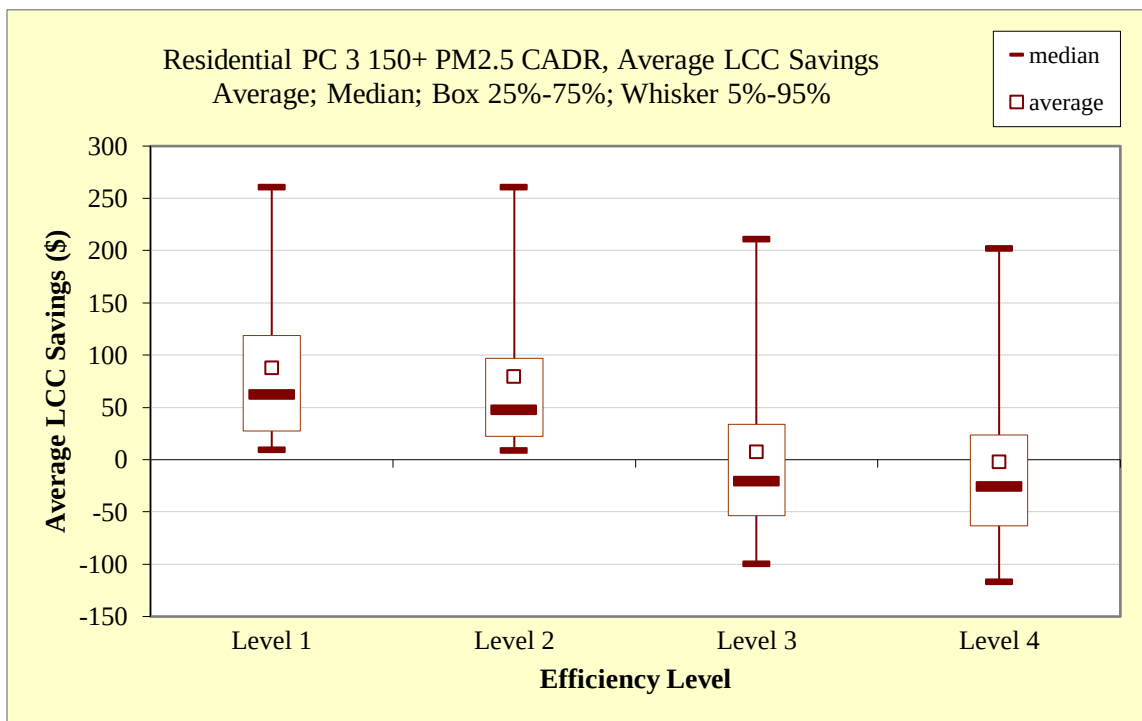


Figure 8.5.8 Air Cleaner PC 3 Residential: Range of Average LCC Savings

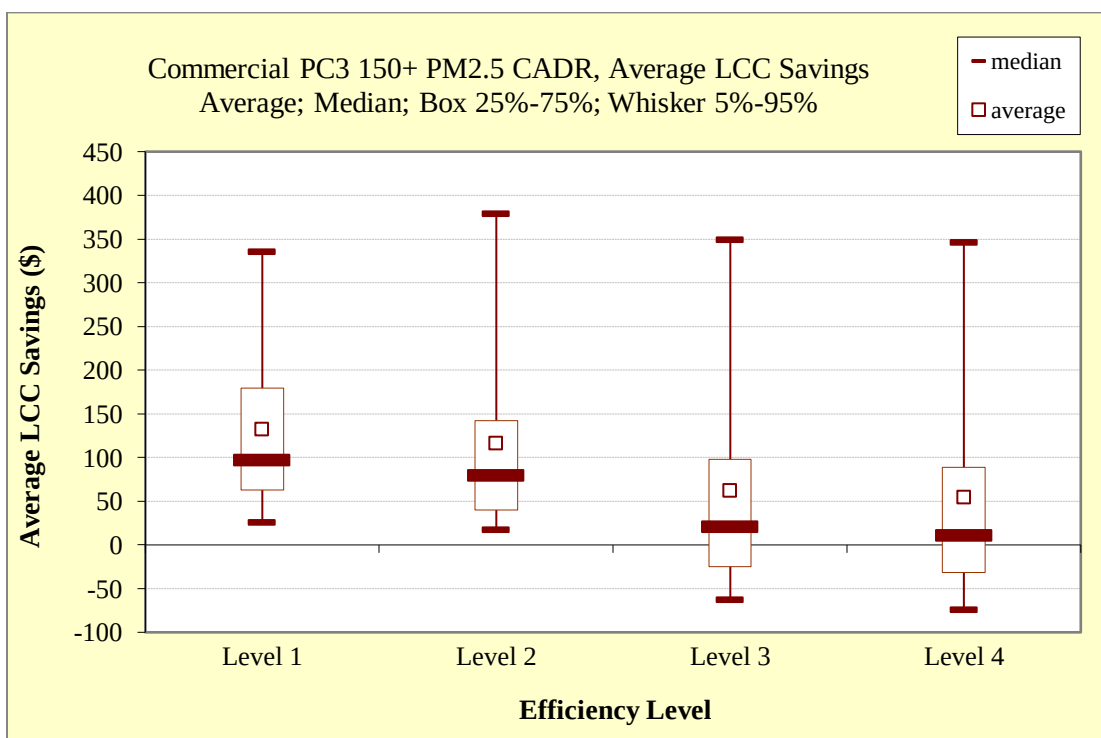


Figure 8.5.9 Air Cleaner PC 3 Commercial: Range of Average LCC Savings

8.6 REBUTTABLE PAYBACK PERIOD

DOE presents rebuttable PBPs to provide the legally established rebuttable presumption that an energy efficiency standard is economically justified, if the additional product costs attributed to the standard are less than three times the value of the first-year energy cost savings. (42 U.S.C. §6295 (o)(2)(B)(iii))

The basic equation for rebuttable PBP is the same as that shown in section 8.3, Payback Period Inputs. Unlike the analyses described in sections 8.2 and 8.3, however, the rebuttable PBP is not based on the use of household samples and probability distributions. Rather than distributions, the rebuttable PBP is based on discrete single-point values. For example, while DOE uses a probability distribution of regional energy prices in the distributional PBP analysis, it uses only the national average energy price from the probability distribution to determine the rebuttable PBP.

Other than the use of single-point values, the most notable difference between the distribution PBP and the rebuttable PBP is the latter's reliance on the DOE test procedure to determine a product's annual energy consumption. The following sections identify the differences, if any, between the annual energy consumptions determined for the distribution PBP and the rebuttable PBP.

DOE based the annual energy consumption values that it used to determine the rebuttable PBP for room air cleaners on the number of operating hours per year specified in the DOE test procedure.

8.6.1 Inputs to the Rebuttable Payback Period Analysis

The following summarizes the single-point values that DOE used in the determination of the rebuttable PBP:

- Manufacturing costs, markups, sales taxes, and installation costs were all based on the single-point values used in the distributional LCC and PBP analyses.
- Annual energy consumption is based on the DOE test procedure.
- Energy prices are based on national average values for the year that new standards are assumed to take effect.
- An average discount rate or lifetime is not required in the rebuttable PBP calculation.
- The effective date of the standard is assumed to be 2028.

8.6.2 Results of the Rebuttable Payback Period Analysis

DOE calculated rebuttable PBPs for each standard level relative to the distribution of product energy efficiencies estimated for the no-new-standards case. In other words, DOE did not determine the rebuttable PBP relative to the baseline energy efficiency level, but relative to the distribution of product energy efficiencies DOE determined for the no-new-standards case (*i.e.*, the case without new energy conservation standards).

Table 8.6.1 presents the rebuttable payback periods for air cleaners.

Table 8.6.1 Air Cleaners: Rebuttable PBPs

Efficiency Level	PC 1		PC 2		PC 3	
	$10 \leq \text{PM}_{2.5} \text{ CADR} < 100$		$100 \leq \text{PM}_{2.5} \text{ CADR} < 150$		$\text{PM}_{2.5} \text{ CADR} \geq 150$	
	Efficiency Level ($\text{PM}_{2.5}$ CADR/W)	Payback Periods (yr)	Efficiency Level ($\text{PM}_{2.5}$ CADR/W)	Payback Periods (yr)	Efficiency Level ($\text{PM}_{2.5}$ CADR/W)	Payback Periods (yr)
1	1.7	0.6	1.9	0.2	2.0	0.0
2	1.9	0.7	2.4	0.2	2.9	0.0
3	3.4	0.9	5.4	0.3	6.6	0.1
4	5.4	1.1	12.8	0.4	7.4	0.1

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CHAPTER 9. SHIPMENTS ANALYSIS

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CHAPTER 9. SHIPMENTS ANALYSIS

9.1 INTRODUCTION

Estimates of future product shipments are a necessary input to calculations of the national energy savings (NES) and net present value (NPV) related to potential energy efficiency standards. Estimated shipments also are required for the manufacturer impact analysis. This chapter describes the data and methods the U.S. Department of Energy (DOE) used to project annual shipments air cleaners under no-new-standards case and standards-case efficiency levels.

DOE estimated shipments for air cleaners based on historical shipments data. First, the shipments model estimates shipments for both of the residential and commercial sectors, and then disaggregates results to estimate the shipments for each product class within each sector. To estimate the effect of potential standard levels on product shipments, the shipments model accounts for the effects of changes in both purchase price and annual operating cost on the consumer purchase decision. The shipments model was developed as a part of the spreadsheet for the national impacts analysis (NIA). Appendix 10A of this final rule technical support document (TSD) describes how to access the NIA workbook and provides basic instructions for its use.

The rest of this chapter explains the shipments model in more detail. Section 9.2 presents the approach to building the model; section 9.3 describes the data inputs; section 9.4 discusses impacts on shipments from changes in product purchase price and operating cost; section 9.5 discusses the affected stock of air cleaners; and section 9.6 presents the model results for various trial standard levels (TSLs) for air cleaners.

9.2 APPROACH TO SHIPMENTS MODEL

DOE developed a national stock model to estimate annual shipments of products under potential energy efficiency standards. The model considers market segments as distinct inputs to projected shipments. The primary market segments are modeled as new purchases and replacements. The following is the general equation for calculating product shipments to these market segments.

$$Ship_p(j) = Rpl_p(j) + N_p(j)$$

Where:

- $Ship_p(j)$ = total shipments of product p in year j ,
- $Rpl_p(j)$ = units of product p retired and replaced in year j , and
- $N_p(j)$ = number of air cleaners purchased that are not replacing retiring units of product p in year j .

DOE calculates the total in-service stock of products by integrating historical shipments data starting from a specified year. The start year depends on the historical data available for each product. As units are added to the in-service stock, some older units retire and exit the stock according to the prescribed lifetime distribution. To estimate future shipments, DOE developed a series of equations that define the dynamics and accounting of in-service stocks. For stock units, the equation is:

$$Stock(j, age = 1) = Ship(j - 1)$$

Where:

$Stock(j, age)$ = the population of in-service units of a particular age,
 j = year for which the in-service stock is being estimated, and
 $Ship(j)$ = number of units shipped in year j .

The above equation states that the number of one-year-old units is equal to the number of new units shipped the previous year. The following equation describes the accounting of the existing in-service stock of units.

$$Stock(j + 1, age + 1) = Stock(j, age) \times (1 - prob_{Rtr}(age))$$

In the above equation, as the year is incremented from j to $j+1$, the age of units also is incremented from age to $age+1$. With time, a fraction of the in-service stock is removed, a fraction that is determined by a retirement probability function, $prob_{Rtr}(age)$, which is described in section 9.3. Over time, some units will be retired and removed from the stock, triggering the shipment of new units.

9.3 DATA INPUTS AND MARKET SEGMENTS

The following sections describe the data inputs and market segments considered for air cleaners. Those data inputs are used in the general equations presented in section 9.2 for estimation of shipments for air cleaners.

9.3.1 Historical Shipments

DOE used historical air cleaner shipments to instantiate its shipments model with a vintage distribution of present stock. To establish historical shipments, DOE used a EuroMonitor product sales report for the years 2007- 2020.¹

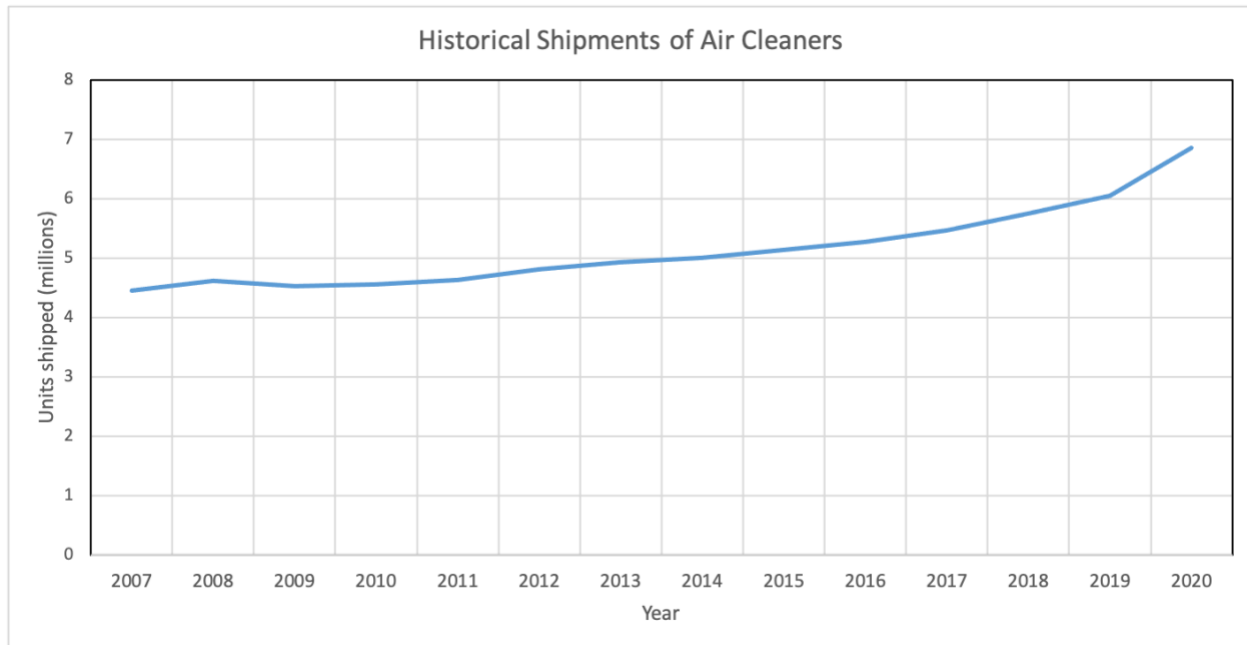


Figure 9.3.1 Historical Shipments of Air Cleaners (2007 – 2020)

9.3.2 Markets and Model Calibration

The market for air cleaners primarily consists of replacement units for equipment that have been retired. DOE's shipments model also assumes that some new units are purchased in each year in addition to those replacing retiring ones. Total air cleaner shipments are represented by the following equation:

$$Ship_p(j) = Rpl_p(j) + N_p(j)$$

Where:

$Ship_p(j)$ = Total shipments of air cleaners of class p in year j ,
 $Rpl_p(j)$ = Replacement shipments of class p in year j , and
 $N_p(j)$ = Shipments of new air cleaners of class p not replacing old ones in year j .

The sections below discuss these markets in further detail.

9.3.2.1 Replacements

DOE determined shipments to the replacement market using an accounting method that tracks the total stock of units by vintage. Over time, some units will be retired and removed from stock, thereby triggering the shipment of a replacement unit. Depending on the vintage, a certain percentage of each type of unit will fail and need to be replaced. To determine when a unit fails, DOE used a survival function based on a product lifetime distribution with an average lifetime of 9.0 years. For a more complete discussion of air cleaner lifetimes, refer to chapter 8 of this direct

final rule TSD. Figure 9.3.2 shows the retirement function that DOE used to estimate replacement shipments.

DOE determines the number of units to be replaced according to the equation:

$$Rpl_p(j) = \sum_{y=j-L}^j Ship_p(y) * Pr_{ret}(j-y)$$

Where:

$Rpl_p(j)$ = Replacement shipments of class p in year j , and
 $Ship_p(j)$ = Total shipments of air cleaners of class p in year j ,
 $Pr_{ret}(a)$ = The probability that an air cleaner requires replacement at age a .
 L = The maximum lifetime of air cleaners

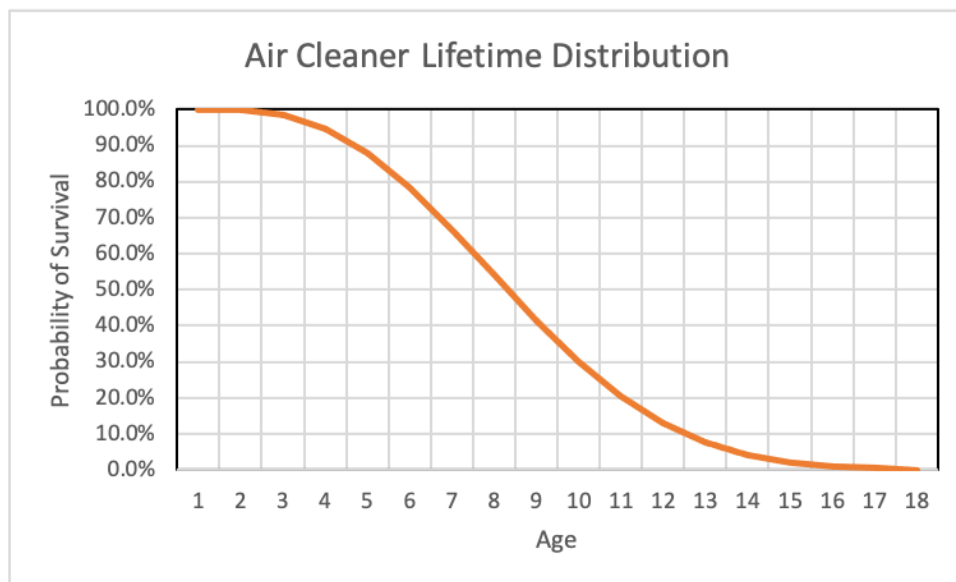


Figure 9.3.2 Air Cleaners Retirement Function

9.3.2.2 New Purchases

DOE assumed that in each year, some units are shipped that represent air cleaners not being purchased as replacements. DOE projected this new demand based on an initial estimate and an average annual growth rate for both residential and commercial sectors.

In order to derive an initial estimate of new demand, DOE calculated total new demand in 2020 as the difference between the EuroMonitor estimate of all units shipped that year, and the retirement demand estimated using the methodology in 9.3.2.1. DOE estimated this initial total new demand as 2.44 million units. DOE assumed that these initial shipments to new demand

were mostly in the residential sector (60 percent, 1.47 million units), and the remainder were to the commercial sector (40 percent, 0.98 million).

Separately, DOE estimated an average annual shipments growth rate of 4.87 percent from the 2021-2028 shipments projection provided by EuroMonitor.¹ Compared to the 7 percent annual shipments growth rate in the TechSci Research report,² this represents a more conservative estimate. New demand was projected using this annual growth rate. In the reference case growth scenario, this annual rate of increase is applied as new demand growth in both the residential and commercial sectors. Based on this approach, DOE estimated a total stock of 54 million air cleaners in 2022.

See appendix 10D for a discussion of the impacts of economic growth scenarios on air cleaner shipments forecasts.

9.3.3 No-New-Standards Case Shipments

Figure 9.3.3 presents projected shipments of air cleaners in the no-new-standards case.

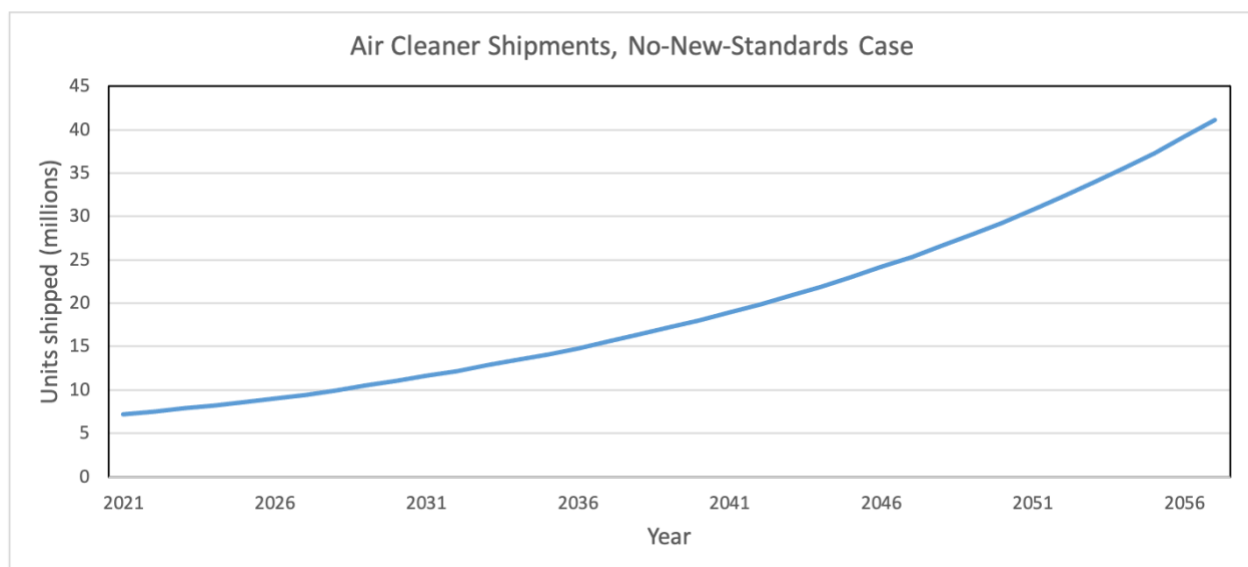


Figure 9.3.3 Air Cleaners: Projected Shipments for No-New-Standards Case

9.4 IMPACT OF INCREASED PURCHASE PRICE ON SHIPMENTS

DOE conducted a literature review and an analysis of appliance price and efficiency data to estimate the effects on product shipments from increases in product purchase price and product energy efficiency.³

Existing studies of appliance markets suggest that the demand for durable goods, such as appliances, is price-inelastic. Other information in the literature suggests that appliances are a normal good, so that rising incomes increase the demand for appliances, and that consumer

behavior reflects relatively high implicit discount rates^a when comparing appliance prices and appliance operating costs.

DOE used the available data for the period 1989 – 2009 on household appliance purchases to evaluate broad market trends and conduct simple regression analyses. These data indicate that there has been a rise in appliance shipments and a decline in appliance purchase prices and operating costs over the time period. Other relevant variables include household income, which has also risen during this time, new residential construction, and stock failures of existing appliances. Using these data, DOE performed a regression analysis to estimate two parameters, the price elasticity of appliance demand and the shipments response to appliance efficiency, defined as follows:

$$\varepsilon_d = \frac{\frac{\Delta q}{q}}{\frac{\Delta p}{p}}$$

Where:

ε_d = price elasticity of demand;
 q = quantity of shipments;
 p = price

$$\varepsilon_e = \frac{\frac{\Delta q}{q}}{\frac{\Delta e}{e}}$$

Where:

ε_e = “efficiency elasticity;”
 q = quantity of shipments;
 e = product efficiency.

DOE’s regression analysis suggests that the price elasticity of demand, based on aggregated data for five residential appliances, is -0.45. Thus, for example, a price increase of 10 percent would result in a shipments decrease of 4.5 percent, *all other factors held constant*. The efficiency elasticity is estimated to be +0.2 (*e.g.*, a 10 percent efficiency improvement would result in a shipments increase of 2%, *all other factors being equal*).^b

The price elasticity estimate of -0.45 is consistent with estimates of appliance and durables price effects in the literature. Nevertheless, DOE stresses that the measure is based on a

^a An implicit discount rate refers to a rate that can be inferred from observed consumer behavior with regard to future operating cost savings realized from more-efficient appliances. An implicit discount rate is not a true discount rate because the observed consumer behavior is affected by lack of information, high transaction costs, and other market barriers. However, implicit discount rates can predict consumer purchase behavior with respect to energy-efficient appliances. A high implicit discount rate with regard to operating costs means that consumer behavior reflects a high discounting of future operating cost savings realized from more-efficient appliances. In other words, consumers are much more concerned with higher purchase prices.

^b Note that DOE previously combined these impacts in a variable termed “relative price elasticity.” Price and efficiency impacts are now separated for greater consistency with price elasticity measures reported in the literature.

small data set, using simple statistical analysis. More importantly, the measure is based on the assumption that economic variables, including purchase price, operating costs, and household income, explain most of the trend in appliances per household in the United States between 1989 and 2009. Changes in appliance quality and consumer preferences may have occurred during this period, but DOE did not account for them in this analysis. Despite the uncertainties, DOE believes that its estimates provide a reasonable assessment of the effect that purchase price and efficiency have on product shipments.

Because DOE's projections of shipments and national impacts from potential standards consider a 30-year period, DOE needed to consider how price elasticity evolves in the years after a new standard takes effect. DOE considered the price elasticity developed above to be a short-term value, but was unable to identify sources specific to appliances that would be sufficient to model differences in short- and long-term price elasticities. Therefore, to estimate how the price elasticity changes through time, DOE relied on a study pertaining to automobiles.⁴ This study shows that the price elasticity of demand for automobiles changes in the years following a change in purchase price, a trend also observed in appliances and other durables.^{5,c} As time passes from the change in purchase price, the price elasticity becomes more inelastic until it reaches a terminal value around the tenth year after the price change. Table 9.4.1 shows the relative change over time in the price elasticity of demand for automobiles. As shown in the table, DOE developed a time series of price elasticity for residential appliances based on the relative change over time in the price elasticity of demand for automobiles. For years not shown in the table, DOE performed a linear interpolation to obtain the price elasticity.

Table 9.4.1 Change in Relative Price Elasticity following a Change in Purchase Price

	Years Following Price Change					
	1	2	3	5	10	20
Change in elasticity relative to first year	1.00	0.78	0.63	0.46	0.35	0.33
Price elasticity	-0.45	-0.35	-0.28	-0.21	-0.16	-0.15

Using the following equation, DOE estimated standards-case shipments by incorporating the effect of the relative price into the no-new-standards case shipments projection. Note that in the equation below, the *relative price* and the *relative price* elasticity are functions of the year because both change with time.

$$Ship_{STD_p}(j) = Ship_{BASE_p}(j) \times (1 - MI(j)), \quad MI(j) > 0$$

^c DOE relies on Hymens *et al.* (1970) for efficiency scaling factors because it provides the greatest detail out of all the available studies on price elasticity over time.

Where:

$Ship_{STD_p}(j)$ = total shipments of product p in year j under the standards case,
 $Ship_{BASE_p}(j)$ = total shipments of product p in year j under the no-new-standards case,
and
 $MI(j)$ = market impact in year j .

The market impact is defined below.

$$MI(j) = \varepsilon_d(j) \times \Delta P(j)$$

Where:

$\varepsilon_d(j)$ = price elasticity of demand in year j (equals -0.45 for year 1), and
 $\Delta P(j)$ = fractional change in average price due to a standard level in year j .

9.5 AFFECTED STOCK

For any energy efficiency rulemaking, the affected stock is the in-service stock of the product that is affected by a standard level. In addition to the projection of product shipments under the no-new-standards case and each standards case, the affected stock (which represents the difference in the product stock for the no-new-standards case and the standards case) is a key output of DOE's shipments models. The affected stock quantifies the effect that shipments of new products under a standard level have on the product stock. Therefore, the affected stock consists of those in-service units that are purchased in or after the year a standard takes effect, as described by the following equation.

$$Aff\ Stock_p(j) = Ship_p(j) + \sum_{age=1}^{j-Std_yr} Stock_p(age)$$

Where:

$Aff\ Stock_p(j)$ = affected stock of units of product p of all vintages that are operational in year j ,
 $Ship_p(j)$ = shipments of product p in year j ,
 $Stock_p(j)$ = stock of units of product p of all vintages that are operational in year j ,
 age = age of units (years), and
 Std_yr = effective date of standard.

As required for the above equation, to calculate affected stock DOE must define the effective date of a standard. For the NES and NPV results presented in chapter 10 of this direct final rule TSD, DOE assumed that any new energy efficiency standards for air cleaners would first become effective in 2024, before the Tier 2 standard becomes effective in 2026. For all other standards considered in this analysis, DOE assumed a compliance year of 2028.

9.6 RESULTS

This section compares the shipments projected under the no-new-standards case with those projected for all the considered TSLs for air cleaners.

Figure 9.6.1 shows total projected shipments of air cleaners in the no-new-standards case and under each TSL. At higher standard levels, there is a slightly larger drop in initial shipments, which reflects a larger difference in average first-cost. As established above in section 9.4, the elasticity parameter decreases over time, so the impact of the standards on shipments diminishes.

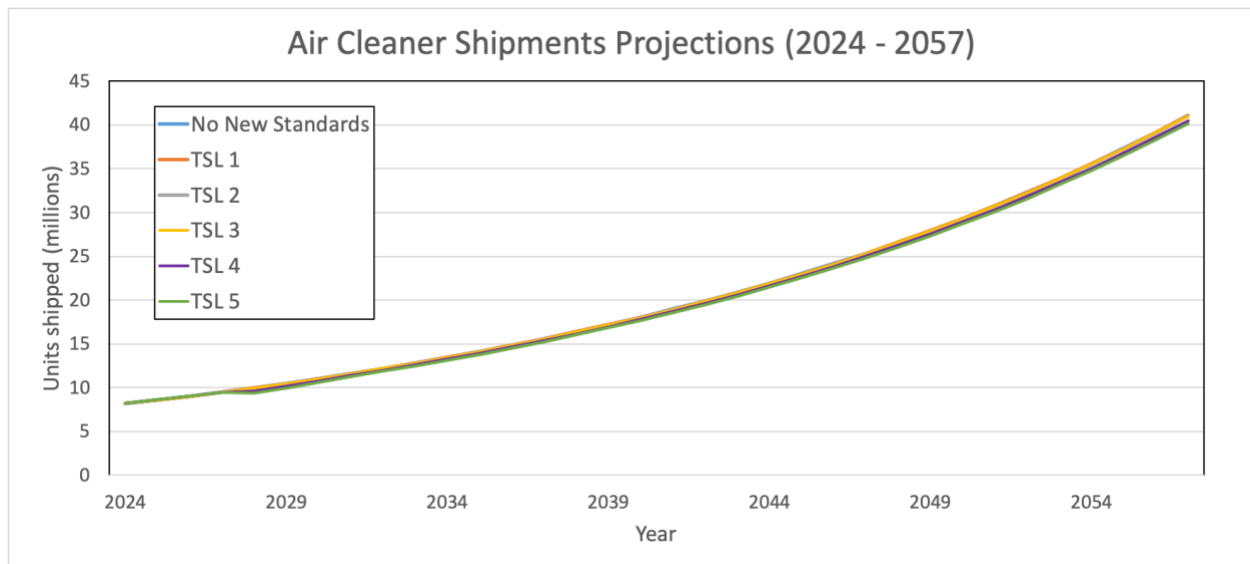


Figure 9.6.1 Air Cleaner Shipments Projected for No-New-Standards Case and Each TSL

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CHAPTER 10. NATIONAL IMPACT ANALYSIS

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CHAPTER 10. NATIONAL IMPACT ANALYSIS

10.1 INTRODUCTION

This chapter describes the methods the U.S. Department of Energy (DOE) used to conduct a national impacts analysis (NIA) of potential energy efficiency standard levels for air cleaners. DOE evaluated the following impacts: (1) national energy savings (NES) attributable to each standard, (2) monetary value of the energy savings for consumers of air cleaners, (3) increased total installed costs of air cleaners because of standards, and (4) the net present value (NPV), which is the difference between the value of energy savings and of increased total installed costs for each standard considered.

DOE determined the NES and NPV for all the trial standard levels (TSLs) considered for air cleaners. DOE performed all calculations using a Microsoft Excel spreadsheet model, which is accessible on the Internet at www.eere.energy.gov/buildings/appliance_standards/. The spreadsheet combines the calculations for determining the NES and NPV for each considered EL with input from the appropriate shipments model. Details and instructions for using the NIA model are provided in appendix 10A of this direct final rule technical support document (TSD).

The NIA calculation starts with the shipments model. Chapter 9 of this direct final rule TSD provides a detailed description of the shipments model that DOE used to project future purchases of air cleaners. Chapter 9 also includes descriptions of consumers' sensitivities to total installed cost (purchase price plus installation costs) and operating costs, and how DOE captured those sensitivities within the model using elasticity. DOE used the annual shipments projection to produce an accounting of annual NES, annual national energy cost savings, and annual national incremental non-energy costs resulting from purchasing, installing and operating the units projected to be shipped in each year of the analysis period during their estimated lifetime.

The two models used in the NIA—the NES model and the NPV model—are described more fully in subsequent sections. The descriptions include overviews of how DOE performed each model's calculations and summaries of the major inputs. After the technical model descriptions, this chapter presents the results of the NIA calculations.

Table 10.1.1 summarizes inputs to the NIA model.

Table 10.1.1 Inputs to Calculating National Energy Savings and Net Present Value

Input	Data Description
Shipments	Annual shipments from shipments model (chapter 9).
Effective date of standard	2028 (TSLs 1, 2, 4, 5), 2024 and 2026 (TSL 3).
Energy efficiency in no-new-standards	Fixed efficiency distribution, no projected efficiency increase.
Energy efficiency in standards cases	“Roll-up” scenario assumed in compliance year for each standard level.
Annual energy consumption per unit	Annual weighted-average values as a function of shipments-weighted UEC.
Total installed cost per unit	Annual weighted-average values as a function of efficiency distribution.
Energy cost per unit	Annual weighted-average values as a function of the annual UEC and energy prices (see chapter 8 for energy prices).
Repair and maintenance costs per unit	Annual values as a function of efficiency level (see chapter 8).
Trend in energy prices	Based on EIA’s <i>AEO 2022</i> reference case (to 2050). Fixed to constant value thereafter based on the 2050 value.
Energy site-to-power plant conversion	A time-series conversion factor that includes losses due to electricity generation, transmission, and distribution. Conversion, which changes yearly, generated by DOE/EIA’s NEMS* program.
Full-fuel-cycle multiplier	Developed to include the energy consumed in extracting, processing, and transporting or distributing primary fuels.
Discount rate	3 percent and 7 percent real.
Present year	Future expenses are discounted to 2022.

10.1.1 Trial Standard Levels

In the analysis conducted for this direct final rule (DFR), DOE analyzed the benefits and burdens of five TSLs for air cleaners. DOE developed TSLs that combine efficiency levels for each of the analyzed air cleaners product classes.

TSL 1 corresponds to a minimum qualifying EL 1 for all product classes beginning in 2028, and represents the least practical increment in efficiency possible. TSL 2 corresponds to EL 2 for all product classes, with a compliance year in 2028. On August 23, 2022, DOE received a Joint Proposal from the Joint Stakeholders regarding energy conservation standards for consumer room air cleaners. Based on the Joint Proposal’s two-tier standard recommendation, DOE analyzed TSL 3 which corresponds to a minimum qualifying ELs 1 in 2024 as Tier 1 and minimum qualifying ELs 2 in 2026 as Tier 2 for all product classes. TSL 4 corresponds to EL 3 for all product classes beginning in 2028, and TSL 5 represents the max-tech energy efficiency

for all product classes. Table 10.1.2 presents the TSLs and the corresponding efficiency levels that DOE has identified for potential amended energy conservation standards for air cleaners.

Table 10.1.2 TSL to EL mapping

TSL		TSL 1	TSL 2	TSL 3	TSL 4	TSL 5
PC	Compliance Year(s)	2028	2028	2024, 2026	2028	2028
10 ≤ PM_{2.5} CADR < 100	Selected EL	EL 1	EL 2	EL 1, EL 2	EL 3	EL 4
	PM _{2.5} CADR/W	1.7	1.9	1.7, 1.9	3.4	5.4
100 ≤ PM_{2.5} CADR < 150	Selected EL	EL 1	EL 2	EL 1, EL 2	EL 3	EL 4
	PM _{2.5} CADR/W	1.9	2.4	1.9, 2.4	5.4	12.8
PM_{2.5} CADR ≥ 150	Selected EL	EL 1	EL 2	EL 1, EL 2	EL 3	EL 4
	PM _{2.5} CADR/W	2.0	2.9	2.0, 2.9	6.6	7.4

10.2 PROJECTED ENERGY EFFICIENCIES

A key component of DOE's estimates of NES and NPV for air cleaners is the energy efficiency level projected for the no-new-standards case (without new energy conservation standards) and each standards case (with new energy conservation standards). The projected energy efficiency levels represent the annual shipments-weighted energy efficiency of air cleaners during the shipments analysis period (that is, from the assumed effective date of a new standard to 30 years after that date). Because standard levels were modeled with different compliance years, DOE assumed a shipments analysis period ending in 2057 for all TSLs to ensure the resulting savings are comparable and the difference is not due to the change in shipments.

For calculating the NES, per-unit average annual energy consumption is a direct function of product energy efficiency. For the NPV, the per-unit total installed cost is a direct function of energy efficiency; the per-unit annual operating cost, because it is a function of per-unit annual energy consumption, is indirectly dependent on product energy efficiency. The NES and NPV inputs described above, as well as all other inputs for the calculations of NES and NPV, are discussed further in sections 10.4 and 10.5, respectively.

To project the no-new-standards case energy efficiency for air cleaners, DOE used methodologies described in chapter 8 of this direct final rule TSD. Without data available to inform an efficiency trend in the absence of standards, DOE assumed that the efficiency distribution of each product class would remain constant over time.

Under a standards case, market share for ELs below the standard level “roll up” to the minimum EL that meets the standard in the year of compliance.

Table 10.2.1 presents the no-new-standards case efficiency distribution and shipments-weighted CADR/W.

Table 10.2.1 No-New-Standards Case Energy Efficiency Distribution for air cleaners

Product Class	Market Share Efficiency Level (%)					Shipments-weighted CADR/W (cfm/w)
	Baseline	EL 1	EL 2	EL 3	EL 4	
$10 \leq \text{PM}_{2.5} \text{ CADR} < 100$	28.0	42.1	19.1	7.5	3.3	1.9
$100 \leq \text{PM}_{2.5} \text{ CADR} < 150$	24.4	36.6	28.1	10.5	0.4	2.4
$\text{PM}_{2.5} \text{ CADR} \geq 150$	22.2	33.3	37.7	3.1	3.8	2.5

10.3 PRICE TRENDS

As discussed in chapter 8 of this DFR TSD, DOE combined historical Producer Price Index (PPI) data for “small electric household appliances, except fans” between 1982 and 2015 and for the more aggregated small electrical appliance manufacturing between 2015 and 2021 from the Bureau of Labor Statistics’ (BLS).^a Figure 10.3.1 shows the estimated price forecast index over the analysis period.

^a Product series ID: PCU33521033521014 and PCU335210335210, see more information at: <http://www.bls.gov/ppi/>

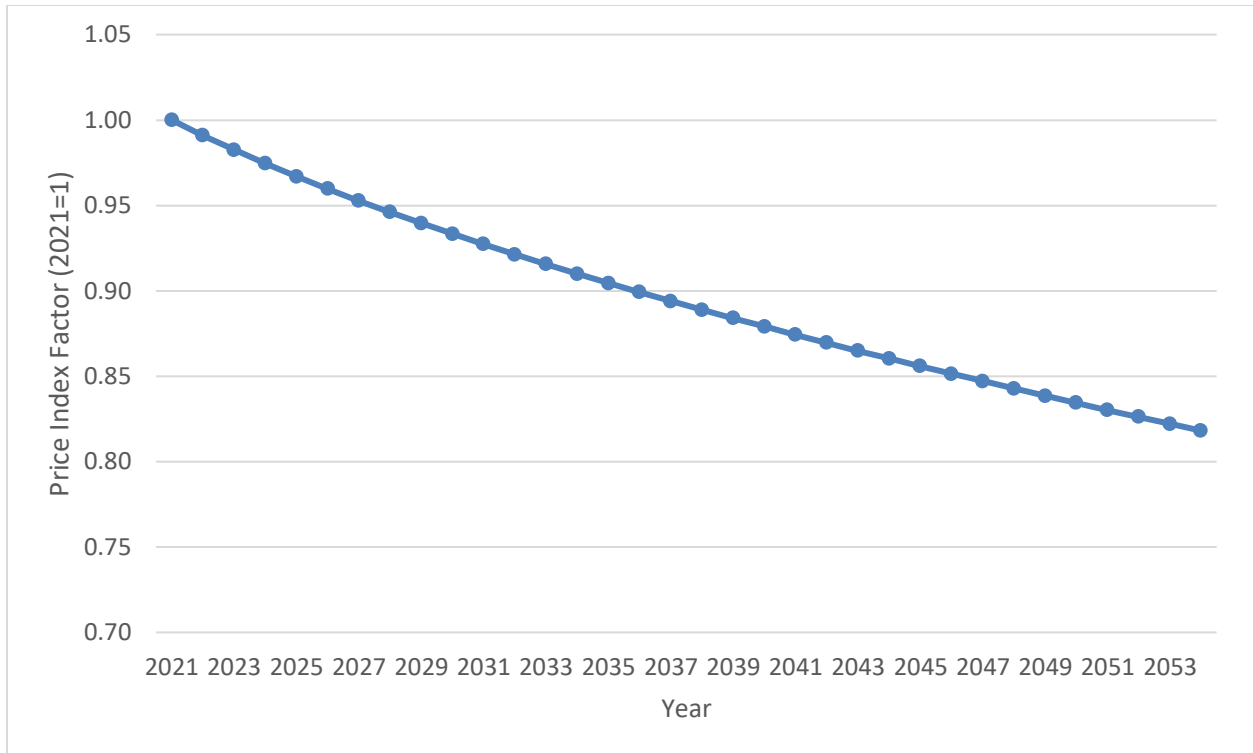


Figure 10.3.1 Price Factor Index for Air Cleaners

10.4 NATIONAL ENERGY SAVINGS

DOE calculated the NES associated with the difference between the no-new-standards case and each TSL for air cleaners. DOE calculated cumulative energy savings accumulated from shipments between the years of 2024-2057. Positive values of NES represent energy savings, meaning national energy consumption under the analyzed standards is lower than in the no-new-standards case.

DOE calculates NES and NWS expressed as:

- Site energy: Accounts for the energy used at the air cleaner point of use,
- Primary energy: Includes the energy used to generate electricity, and
- Full-fuel-cycle (FFC) energy: Includes the energy consumed in extracting, processing, and transporting or distributing primary fuels.

10.4.1 Definition of National Energy Savings

DOE calculates annual NES (NES_y) for a given year as the difference between the national annual energy consumption (AEC) in a no-new-standards case and a higher efficiency case. The following equation shows that DOE calculated national annual energy savings as the difference between two projections: a no-new-standards case and a standards case (with a higher efficiency level). Positive values of NES represent energy savings (*i.e.*, national annual energy consumption under a higher efficiency case is less than the no-new-standards case).

$$NES_y = AEC_{NNS} - AEC_{STD}$$

Cumulative energy savings are the sum of annual NES from products shipped between the years 2024 through 2057.

DOE calculated the national annual site energy consumption by multiplying the number or stock of the product (by vintage) by its unit energy consumption (UEC ; also by vintage). National annual energy consumption is calculated using the following equation.

$$AEC_y = \sum STOCK_v \times UEC_v$$

AEC	=	annual national energy consumption each year in quadrillion Btus (quads), summed over vintages of the product stock, $STOCK_v$;
$STOCK_v$	=	stock of product (millions of units) of vintage V that survive in the year for which DOE calculated annual energy consumption;
UEC_v	=	annual energy consumption of air cleaners in kilowatt-hours (kWh);
V	=	year in which the product was purchased as a new unit; and
y	=	year in the forecast.

The stock of a product depends on annual shipments and the lifetime of the product. As described in chapter 9 of this direct final rule TSD, DOE projected product shipments under the no-new-standards case and standards cases. To avoid including savings attributable to shipments displaced (units not purchased) because of standards, DOE used the projected standards-case shipments and, in turn, the standards-case stock, to calculate the AEC for the no-new-standards.

10.4.2 Shipments and Product Stock

DOE projected shipments of each product class under the no-new-standards case and each standards case. The method DOE used to calculate and generate the shipments projections for each considered product class is described in detail in chapter 9 of this DFR TSD.

The product stock in a given year is the number of products shipped from earlier years that survive in that year. The NIA model tracks the number of units shipped each year. DOE assumes that products have an increasing probability of retiring as they age. The probability of survival as a function of years since purchase is called the survival function. Chapter 8 of this DFR TSD provides additional details on the survival function that DOE used air cleaners.

DOE estimated a total stock of 54 million air cleaners in 2022 with a projection based on the 2021 EuroMonitor¹ and 2022 TechSci² market research reports. The 2021 Evergreen market research report³ indicated that those U.S. households owning at least one air cleaner would have on average 1.5 air cleaners. Considering the 60 percent market share of the residential sector overall the total installed stocks reported by the 2022 TechSci study,² DOE estimated that 21.8 million U.S. households owned at least one air cleaners in 2022.

10.4.3 Annual Energy Consumption per Unit

DOE developed per-unit annual energy consumption as a function of product energy efficiency for air cleaners (see chapter 7 and chapter 8 of this DFR TSD). Because per-unit annual energy consumption depends directly on energy efficiency, DOE used the shipments-weighted energy efficiencies for the no-new-standards case and standards cases (presented in section 10.2), along with the annual energy use data presented in chapter 8, to estimate the shipments-weighted annual average per-unit energy consumption (SWAEU) under the no-new-standards and standards cases. Table 10.4.1 shows the values of AEU and average SWAEUs applied to shipments in the residential and commercial sectors.

Table 10.4.1 Shipments-Weighted Average Per-Unit Annual Energy Consumption in the No-New-Standards Case

Product Class	Annual Energy Consumption (kWh/yr)					Shipments-weighted AEU (kWh/yr)
	Baseline	EL 1	EL 2	EL 3	EL 4	
Residential Sector						
10 ≤ PM _{2.5} CADR < 100	40.0	36.8	33.8	19.2	11.9	35.0
100 ≤ PM _{2.5} CADR < 150	99.3	80.5	64.5	29.9	12.5	75.0
PM _{2.5} CADR ≥150	203.5	122.3	84.9	39.3	34.8	120.3
Commercial Sector						
10 ≤ PM _{2.5} CADR < 100	83.2	75.5	67.9	38.2	23.8	71.7
100 ≤ PM _{2.5} CADR < 150	207.7	167.5	133.3	59.5	25.1	155.7
PM _{2.5} CADR ≥150	242.1	253.5	175.2	78.4	69.3	249.5

10.4.4 National Annual Energy Consumption

The national AEC is the product of the UEC and the number of units of each vintage (V). This method of calculation accounts for differences in unit energy consumption from year to year. DOE used the equation below (initially presented in section 10.3.1) to calculate AEC.

$$AEC = \sum STOCK_v \times UEC_v$$

In determining national AEC, DOE first calculated AEC at the site, then applied a conversion factor, described below, to calculate primary energy consumption.

10.4.5 Site-to-Power-Plant Energy Conversion Factors

In determining annual NES, DOE initially considered the AEC at a residence (for electricity, the energy, expressed in kWh, consumed by a household). DOE then calculated primary (source) energy use from site energy consumption by applying a conversion factor to account for losses associated with the generation, transmission, and distribution of electricity. The site-to-source conversion factor is a multiplicative factor used to convert site energy consumption into primary, or source, energy consumption, expressed in quadrillion Btus (quads).

DOE used annual site-to-power-plant conversion factors based on the version of the national energy modeling system (NEMS)^b that corresponds to the DOE Energy Information Administration's (EIA's) *Annual Energy Outlook 2022 (AEO 2022)*.⁴ The factors are marginal values, which represent the response of the national power system to incremental changes in consumption. For electricity, the conversion factors change over time in response to projected changes in generation sources (the types of power plants projected to provide electricity).

Figure 10.4.1 shows the site-to-power-plant conversion factors for electricity between 2024 and 2050 for air cleaners. The conversion factors were generated by NEMS based on the load shape for air cleaners.

^b For more information on NEMS, refer to the U.S. Department of Energy, Energy Information Administration documentation. A useful summary is *National Energy Modeling System: An Overview 2018*, DOE/EIA-0581(2018), April 2019. EIA approves use of the name NEMS to describe only an official version of the model with no modification to code or data.

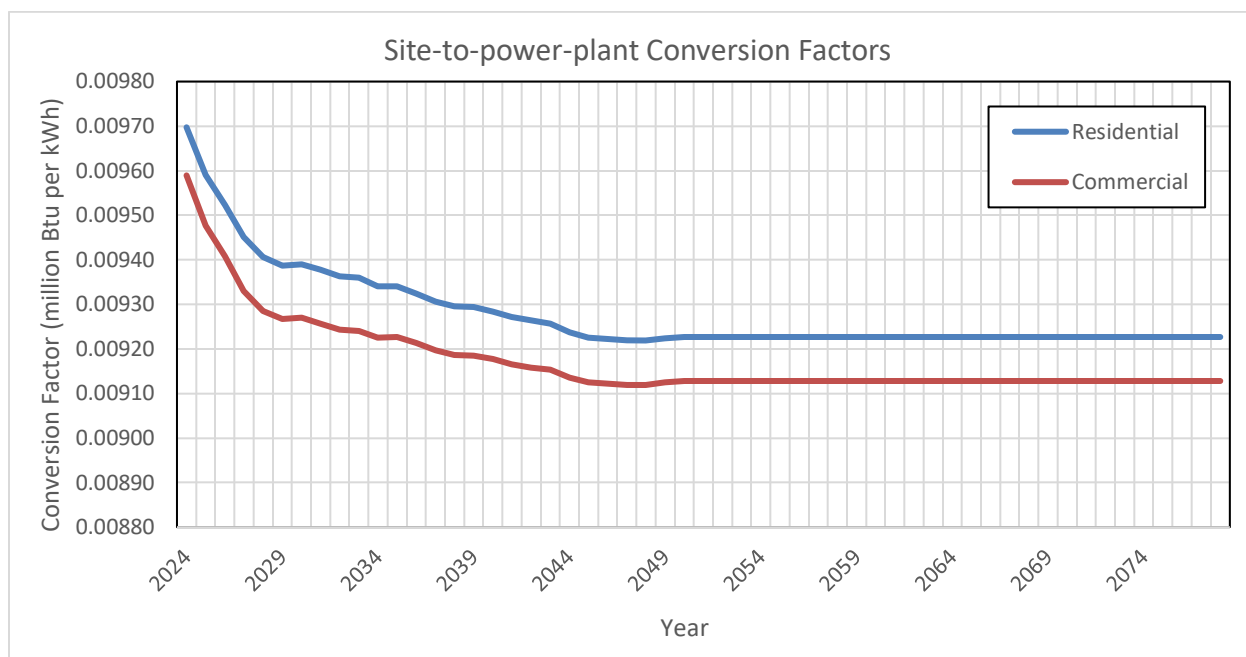


Figure 10.4.1 Site-to-Power Plant Conversion Factors for Electricity

10.4.6 Full-Fuel-Cycle Multipliers

The FFC encompasses point-of-use (site) energy; the energy losses associated with generation, transmission, and distribution of electricity; and the energy consumed in extracting, processing, and transporting or distributing primary fuels. To estimate the FFC by including the energy consumed in extracting, processing, and transporting or distributing primary fuels, referred to as upstream activities, DOE developed FFC multipliers^c using the data and projections generated for AEO 2022. The AEO 2022 provides extensive information about the energy system, including projections of future oil, natural gas, and coal supplies; energy use for oil and gas field and refinery operations; and fuel consumption and emissions related to electric power production. The information can be used to define a set of parameters that represent the energy intensity of energy production. The method used to calculate FFC energy multipliers is described in appendix 10B of this direct final rule TSD.

Table 10.4.2 shows the FFC energy multipliers used for air cleaners for selected years.

Table 10.4.2 Full-Fuel-Cycle Energy Multipliers (Based on AEO 2022)

Electricity	2026	2030	2035	2040	2045	2050	2055
Residential	1.044	1.041	1.039	1.039	1.039	1.039	1.039
Commercial	1.044	1.041	1.039	1.039	1.039	1.039	1.039

^c FFC multipliers discussed in this chapter relate to the upstream part of the FFC process.

10.5 NET PRESENT VALUE

DOE calculated the NPV of the increased product cost and reduced operating cost associated with the difference between the no-new-standards and each EL for air cleaners.

10.5.1 Definition of Net Present Value

The NPV is the value in the present of a time-series of costs and savings. The NPV is described by the equation:

$$NPV = PVS - PVC$$

Where:

PVS = present value of operating cost savings, and
 PVC = present value of increased total installed costs (including purchase price and installation costs).

DOE determined the PVS and PVC according to the following expressions.

$$PVS = \sum OCS_y \times DF_y$$

$$PVC = \sum TIC_y \times DF_y$$

Where:

OCS = total annual-savings in operating costs each year summed over vintages of the product stock, $STOCK_v$;
 DF = discount factor in each year;
 TIC = total annual increases in installed cost each year summed over vintages of the product stock, $STOCK_v$; and
 y = year in the forecast.

DOE calculated the total annual consumer savings in operating costs by multiplying the number or stock of the product (by vintage) by its per-unit operating cost savings (also by vintage). DOE calculated the total annual increases in consumer product price by multiplying the number or shipments of the product (by vintage) by its per-unit increase in consumer cost (also by vintage). Total annual operating cost savings and total annual product price increases are calculated by the following equations.

$$OCS_y = \sum STOCK_v \times UOCS_v$$

$$TIC_y = \sum SHIP_y \times UTIC_y$$

Where:

OCS_y	=	operating cost savings per unit in year y ,
$STOCK_V$	=	stock of products of vintage V that survive in the year for which DOE calculated annual energy consumption,
$UOCS_V$	=	annual operating cost savings per unit of vintage V , including the incremental cost of filter changes,
V	=	year in which the product was purchased as a new unit;
TIC_y	=	total increase in installed product cost in year y .
$SHIP_y$	=	shipments of the product in year y ; and
$UTIC_y$	=	annual per-unit increase in installed product cost in year y .

DOE determined the total increased product cost for each year from 2024 to 2057. DOE determined the present value of operating cost savings for each year from 2024 to the year when all units purchased in 2057 are estimated to retire (2086). DOE calculated costs and savings as the difference between a standards case and a no-new-standards without new standards.

DOE developed a discount factor from the national discount rate and the number of years between the “present” (year to which the sum is being discounted) and the year in which the costs and savings occur. The NPV is the sum over time of the discounted net savings.

10.5.2 Inputs to Calculation of Net Present Value

The inputs to the calculation of NPV are:

- total installed cost per unit,
- total annual increases in installed cost,
- annual savings in operating cost per unit,
- total annual operating costs,
- discount factor,
- present value of costs, and
- present value of savings.

The increase in total annual installed cost is equal to the annual change in the per-unit total installed cost (difference between no-new-standards and standards cases) multiplied by the shipments forecasted for each case.

The total annual savings in operating costs are equal to the change in annual operating costs (difference between no-new-standards and standards case) per unit multiplied by the shipments forecasted for each case. The annual operating cost includes energy, repair, and maintenance costs, as described in chapter 8 of this direct final rule TSD.

10.5.3 Total Installed Cost per Unit

The per-unit total installed cost of air cleaners, described in chapter 8 of this direct final rule TSD, is a function of product energy efficiency. Because the per-unit total annual installed cost depends directly on energy efficiency, DOE used the shipments-weighted energy efficiencies distribution of the no-new-standards case and standards cases described in section 10.2, in combination with the total installed costs developed in chapter 8, to estimate the shipments-weighted average annual per-unit total installed cost under the no-new-standards case and standards cases. Table 10.5.1 shows the shipment-weighted average consumer price for air cleaners in the residential and commercial sector in 2028 based on the efficiency distributions that correspond to the no-new-standards case. Due to the different distribution channels and consumer samples, the shipments-weighted average total installed costs vary by sector.

Table 10.5.1 Shipments-Weighted Average Total Installed Cost in 2028

Product Class	Total Installed Cost (2021\$)					Shipments-weighted Cost (2021\$)
	Baseline	EL 1	EL 2	EL 3	EL 4	
Residential Sector						
10 ≤ PM _{2.5} CADR < 100	64.50	66.17	68.10	78.64	86.88	67.69
100 ≤ PM _{2.5} CADR < 150	88.98	91.10	93.35	102.22	110.71	92.46
PM _{2.5} CADR ≥150	145.92	147.88	149.27	152.78	153.32	148.33
Commercial Sector						
10 ≤ PM _{2.5} CADR < 100	62.31	63.95	65.87	73.32	84.49	65.46
100 ≤ PM _{2.5} CADR < 150	85.95	88.05	90.28	90.08	107.49	89.40
PM _{2.5} CADR ≥150	140.95	142.89	144.27	147.75	148.28	143.33

As discussed in chapter 8 of this direct final rule TSD, DOE developed an experience curve to project the future air cleaner prices. This trend was applied to the component of air cleaner price that wasn't filter. DOE used this price trends to project the prices of air cleaners sold in each year of the forecast period (2024–2057). To investigate the effect of different product price projections on the consumer NPV for various standard levels, DOE also considered two alternative price trends. Details on how those alternative price trends were developed are documented in appendix 10C of this direct final rule TSD, which also presents the results of DOE's analysis of price sensitivities.

10.5.3.1 Increase in Total Annual Installed Cost

The increase in total annual installed cost for any given EL is the product of the total installed cost increase per unit under that standard and the number of units of each vintage. This approach accounts for differences in total installed cost from year to year. DOE used the following equation (also presented in section 10.5.1) to determine the increase in total annual installed cost for a given EL.

$$TIC(y) = \sum UTIC(y) \times S(y)$$

10.5.4 Annual Savings in Operating Costs per Unit

Per-unit annual operating costs encompass the annual costs for energy, repair, and maintenance. DOE determined the savings in per-unit annual energy cost by multiplying the savings in per-unit annual energy consumption by the appropriate energy price. Estimates of per-unit annual energy consumption for the no-new-standards and each standards case were presented in section 10.4.3. To estimate energy prices in future years, DOE multiplied the recent electricity prices by a projection of annual national average residential and commercial electricity prices consistent with the *AEO 2022* reference case electricity price trends. The energy prices and price trends are described in chapter 8 of this direct final rule TSD.

10.5.4.1 Total Savings in Annual Operating Costs

The total savings in annual operating costs for an EL is the product of the annual operating cost savings per unit under that standard and the number of units of each vintage. This approach accounts for differences in savings in annual operating costs from year to year. DOE used the following equation (also presented in section 10.5.1) to determine the total savings in annual operating cost for a given standards case.

$$OCS(y) = \sum UOCS(y) \times AffStock(y)$$

10.5.5 Discount Factors

DOE multiplies monetary values in future years by a discount factor to determine present values. The discount factor (DF) is described by the equation:

$$DF = \frac{1}{(1 + r)^{(y - y_p)}}$$

Where:

- r = discount rate,
- y = year of the monetary value, and
- y_p = year in which the present value is being determined.

DOE uses both a 3-percent and a 7-percent real discount rate when estimating national impacts. Those discount rates were applied to product prices of air cleaners in accordance with the Office of Management and Budget (OMB)'s guidance to Federal agencies on developing regulatory analyses (OMB Circular A-4, September 17, 2003, and section E., "Identifying and Measuring Benefits and Costs," therein). DOE defines the present year as 2022.

10.5.5.1 Present Value of Costs

The present value of increased installed costs is the annual increase in installed cost for each year (*i.e.*, the difference between the standards case and no-new-standards), discounted to the present and summed over the period for which DOE is considering the installed products (2024–2057). The increase in total installed cost refers to both product and installation costs associated with the higher energy efficiency of products purchased under a standards case compared to the no-new-standards. DOE calculated annual increases in installed cost as the difference in total cost of new products installed each year, multiplied by the shipments in the standards case.

10.5.5.2 Present Value of Savings

The present value of savings in operating costs is the annual savings in operating cost (*i.e.*, the difference between the no-new-standards case and standards case), discounted to the present and summed over the period that begins with the effective date of standards and ends when the last installed unit is retired from service. Savings represent decreases in operating costs (including costs for energy, repair, and maintenance) associated with the higher energy efficiency of products purchased in a standards case compared to the no-new-standards case. Total annual savings in operating costs are the savings per unit multiplied by the number of units of each vintage that survive in a particular year. Because a product consumes energy throughout its lifetime, the energy consumption for units installed in a given year includes energy consumed until the unit is retired from service.

10.6 RESULTS

The NIA model provides estimates of the NES and NPV associated with conservation standards at different efficiency levels. The inputs to the NIA model were discussed in sections 10.4.2 (NES Inputs) and 10.5.2 (NPV Inputs). DOE generated the NES and NPV results using a Microsoft Excel spreadsheet, which is accessible on the internet at www.eere.energy.gov/buildings/appliance_standards/. Details and instructions for using the spreadsheet are provided in appendix 10A of this direct final rule TSD, in the section titled, “Descriptions of National Impact Analysis Worksheets.”

10.6.1 National Energy Savings by TSL

This section provides results of the NES calculations for each TSL analyzed for air cleaners. NES results, which are cumulative to 2086 (the last year in which products purchased in 2057 would be expected to be retired), are shown as savings in both primary and FFC energy. Because DOE based the inputs to the NIA model on weighted-average values, results are discrete point values, rather than a distribution of values as produced by the life-cycle cost and payback period analysis (chapter 8 of this direct final rule TSD). Table 10.6.1 through Table 10.6.3 show the site, primary, and FFC energy savings, respectively, for the TSLs analyzed for air cleaners. National energy savings for high and low economic growth scenarios are presented in appendix

10D of this direct final rule TSD. National energy savings for high and low market share growth rate of variable-speed technologies are presented in appendix 10E of this direct final rule TSD.

Table 10.6.1 Cumulative Site National Energy Savings of Air Cleaners

TSL	Residential (Quads)	Commercial (Quads)	Total (Quads)
1	0.11	0.16	0.27
2	0.26	0.36	0.62
3	0.27	0.38	0.64
4	0.60	0.85	1.45
5	0.68	0.96	1.64

Table 10.6.2 Cumulative Primary National Energy Savings of Air Cleaners

TSL	Residential (Quads)	Commercial (Quads)	Total (Quads)
1	0.31	0.43	0.73
2	0.70	0.97	1.67
3	0.72	1.01	1.73
4	1.63	2.27	3.90
5	1.85	2.56	4.42

Table 10.6.3 National Full-Fuel-Cycle Energy Savings of Air Cleaners

TSL	Residential (Quads)	Commercial (Quads)	Total (Quads)
1	0.32	0.44	0.76
2	0.72	1.01	1.73
3	0.75	1.05	1.80
4	1.69	2.36	4.05
5	1.93	2.66	4.59

10.6.2 Annual Costs and Savings

Figure 10.6.1 illustrates the basic inputs to calculating the NPV under TSL 3 for the non-discounted annual increases in installed cost and annual savings in operating cost for air cleaners. The annual product cost is the sum of the increase in total installed cost for products purchased

each year during the analysis period. The annual savings in operating cost is the savings for products operating in each year. The figure also shows net savings, which is the difference between the savings and costs for each year. The NPV is the difference between the cumulative annual discounted savings and the cumulative annual discounted costs. DOE is able to create figures similar to Figure 10.6.1 for each TSL.

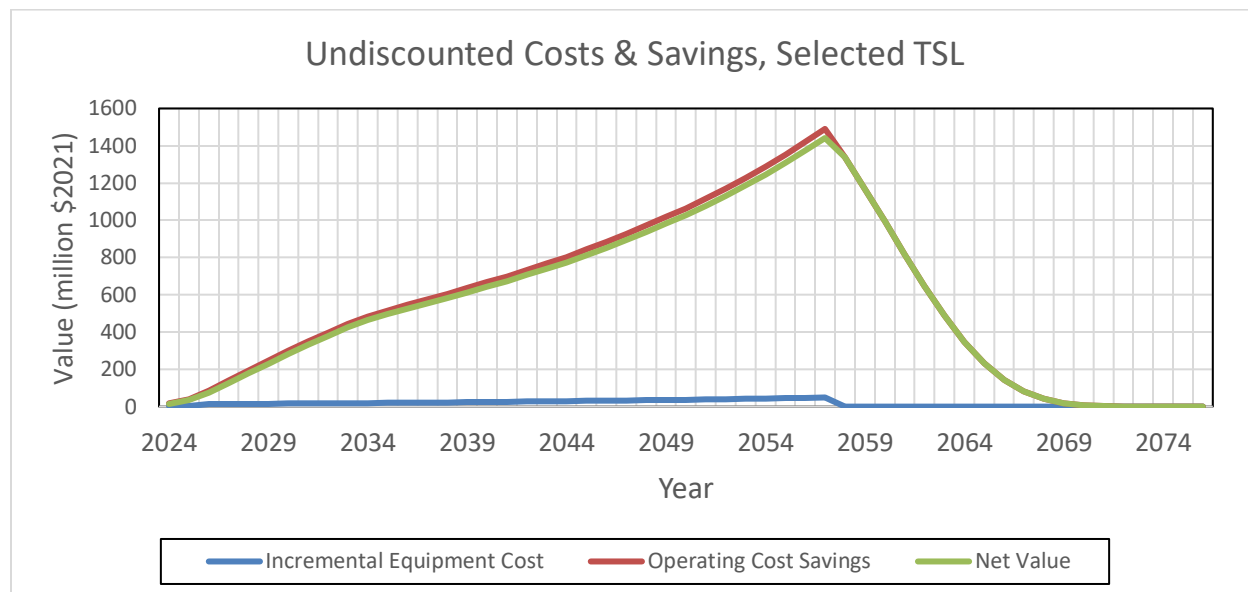


Figure 10.6.1 Non-Discounted Changes in Annual Installed Cost and Operating Costs under TSL 3

10.6.3 Consumer Net Present Value by TSL

This section provides results of calculating the NPV of consumer benefits for each TSL considered for air cleaners. Cumulative results are shown as the discounted value of the net savings in dollar terms. DOE based the inputs to the NIA model on weighted-average values, yielding results that are discrete point values, rather than a distribution of values as in the life-cycle cost and payback period analysis.

Table 10.6.4 shows the results of calculating the NPV for the TSLs analyzed for air cleaners. The table presents the NPV at both a 3-percent and a 7-percent discount rate. The NPV for high and low economic growth scenarios are presented in appendix 10D of this direct final rule TSD. The NPV for high and low price decline scenarios are presented in appendix 10E of this direct final rule TSD.

Table 10.6.4 Cumulative Consumer Net Present Value for Each TSL

TSL	NPV at 3% Discount Rate Billion, 2021\$			NPV at 7% Discount Rate Billion, 2021\$		
	Residential	Commercial	Total	Residential	Commercial	Total
1	2.85	2.60	5.44	1.13	1.04	2.18
2	6.70	6.11	12.81	2.66	2.44	5.11
3	7.16	6.53	13.69	3.00	2.76	5.76
4	(5.94)	(2.44)	(8.39)	(2.41)	(0.99)	(3.40)
5	(3.90)	(0.59)	(4.50)	(1.64)	(0.28)	(1.92)

Parentheses indicate negative values.

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CHAPTER 11. CUSTOMER SUBGROUP ANALYSIS

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CHAPTER 11. CUSTOMER SUBGROUP ANALYSIS

11.1 INTRODUCTION

Chapter 8 of this TSD describes the life-cycle cost (LCC) and payback period (PBP) analysis that examines energy savings and cost impacts of energy conservation standards on the U.S. population. In analyzing the potential impact of new or amended standards on consumers, the U.S. Department of Energy (DOE) further evaluates the impacts on identifiable groups of consumers (subgroups) that may be disproportionately affected by a national standard level. The consumer subgroup analysis evaluates effects by analyzing the LCC and PBPs for subgroups of residential and commercial consumers to determine if standards could potentially have a disproportionate impact. For air cleaners, DOE identified three consumer subgroups that warranted further study:

For the residential sector:

- Households composed of people 65 years of age or older (senior-only);
- Households at or below the poverty line (low-income).

For the commercial sector:

- Small businesses.

DOE determined the impact on consumer subgroups for air cleaners using the LCC spreadsheet model, which enables DOE to analyze the LCC for any subgroup by sampling only the data that apply to that subgroup. (Chapter 8 explains in detail the inputs to the model used in determining LCC and PBPs.) As described in section 11.3, the energy use characteristics of the two residential subgroups (senior-only and low-income) and of the commercial subgroup (small business) differ from those for the general population and for the general commercial buildings.

This chapter describes the identification of the three subgroups and gives the results of the LCC and PBP analyses for those subgroups.

11.2 IDENTIFIED SUBGROUPS

The following three sections describe how DOE defined the three consumer subgroups identified for further examination.

11.2.1 Senior-Only Households

Senior-only households comprise occupants who are all at least 65 years of age. Based on DOE's Energy Information Administration's Residential Energy Consumption Survey of 2020 (*RECS 2020*), senior-only households represent 23.5 percent of the U.S. households.¹

The weightings of the RECS samples indicate how frequently each household occurs in the general population. As described in chapter 7 of this TSD, DOE adjusted the EIA's assigned sample weights to reflect geographic distributions of ownership of air cleaners for residential applications. After the weight adjustment of the selected samples for air cleaners, senior-only households represent 22.6 percent of the U.S. residential air cleaner consumers.

11.2.2 Low-Income Households

As defined in the RECS survey, low-income household residents are living at or below the poverty line. The poverty line varies with household size, age of head of household, and family income. The RECS survey classifies 13.4 percent of the country's households as low-income.

After the weight adjustment of the selected samples for air cleaners, low-income households represent 13.8 percent of the U.S. residential air cleaner consumers.

11.2.3 Small-Businesses

DOE identified small businesses within EIA's 2018 Commercial Buildings Energy Consumption Survey (*CBECS 2018*)² database by using threshold levels for maximum number of employees within each building type. DOE estimated the fraction of commercial buildings that could house a small business based on the small business employment share by industry for 2018. For this analysis, the CBECS Principal Building Activity (PBA) categories were mapped to the Commercial Discount Rate (CDR) categories (see Table 11.2.1). The PBA categories of all the *CBECS 2018* samples were used to identify small businesses (see Table 11.2.2).³

Table 11.2.1 Mapping of CBECS 2018 Principal Building Activity and Commercial Discount Rate Categories

Principal Building Activity Category	Commercial Discount Rate Category	Principal Building Activity Category	Commercial Discount Rate Category
Office	Office	Food service	Food Sales
Laboratory	Health Care	Inpatient health care	Health Care
Nonrefrigerated warehouse	Other	Nursing	Health Care
Food sales	Food Service	Lodging	Lodging
Public order and safety	Other	Strip shopping mall	Mercantile
Outpatient health care	Health Care	Enclosed mall	Mercantile
Refrigerated warehouse	Other	Retail other than mall	Mercantile
Religious worship	Other	Service	Service
Public assembly	Public Assembly	Other	Other
Education	Education	Food service	Food Sales

Table 11.2.2 Maximum Employees for Small Business Identification by Industry, 2018

Commercial Discount Rate Category	Maximum Employees for Small Businesses
Education	267
Food Sales	43
Food Service	699
Health Care	266
Lodging	358
Mercantile	28
Office	61
Public Assembly	240
Service	132
Other	67

In general, smaller businesses tend to discount future stream of monetary flows at higher rates. Table 11.2.3 presents DOE's estimates of the discount rates for entire sectors and small companies specifically. DOE conducted a subgroup analysis for small businesses, and subsequently the results of the subgroup analysis were compared to the results from all customers. To estimate the impact of standards specifically on small businesses, the small company discount rates for each sector were used in the life-cycle cost and payback period analysis instead of the sector average discount rates.

Table 11.2.3 Discount Rate Difference between Small Company and Sector Average

Sector	All Company WACC (%)	Small Company WACC (%)	Small Company DR Premium (%)
Education	7.12%	10.32%	3.20%
Food Sales	5.60%	8.13%	2.54%
Food Service	6.34%	9.27%	2.93%
Health Care	6.78%	9.60%	2.82%
Lodging	6.35%	8.67%	2.32%
Mercantile	6.88%	9.75%	2.87%
Office	6.78%	9.50%	2.72%
Public Assembly	7.17%	9.96%	2.79%
Service	6.22%	8.43%	2.21%
All Commercial	6.67%	9.31%	2.64%
Industrial	7.16%	10.04%	2.88%
Agriculture	6.94%	9.30%	2.36%
Utilities	4.14%	6.15%	2.01%
R.E.I.T./Property	6.43%	8.78%	2.35%

11.3 INPUTS TO THE CONSUMER SUBGROUP ANALYSIS

Table 11.3.1 and Table 11.3.2 summarize the weighted-average annual energy use for the households and the commercial buildings analyzed in the consumer subgroup analysis respectively. These values are compared against the weighted-average values for the national sample.

**Table 11.3.1 Weighted-Average Annual Energy Use for Residential Air Cleaners
(kWh/yr)**

	Product Class	Efficiency Level	All Households	Senior-Only	Low-Income
1	$10 \leq \text{PM}_{2.5} \text{ CADR} < 100$	Baseline	40.0	59.2	37.9
		1	36.8	54.3	34.9
		2	33.8	49.4	32.1
		3	19.2	28.0	18.3
		4	11.9	17.4	11.3
2	$100 \leq \text{PM}_{2.5} \text{ CADR} < 150$	Baseline	99.3	147.1	94.0
		1	80.5	119.1	76.2
		2	64.5	95.2	61.1
		3	29.9	43.7	28.4
		4	12.5	18.3	11.8
3	$\text{PM}_{2.5} \text{ CADR} \geq 150$	Baseline	203.5	301.2	192.6
		1	122.3	180.7	115.8
		2	84.9	125.3	80.4
		3	39.3	57.4	37.3
		4	34.8	50.8	33.0

**Table 11.3.2 Weighted-Average Annual Energy Use for Commercial Air Cleaners
(kWh/yr)**

	Product Class	Efficiency Level	All Commercial Buildings	Small Business
1	$10 \leq \text{PM}_{2.5} \text{ CADR} < 100$	Baseline	83.2	83.5
		1	75.6	75.8
		2	67.9	68.1
		3	38.2	38.3
		4	23.8	23.9
2	$100 \leq \text{PM}_{2.5} \text{ CADR} < 150$	Baseline	207.7	208.3
		1	167.5	168.0
		2	133.3	133.7
		3	59.5	59.7
		4	25.1	25.2
3	$\text{PM}_{2.5} \text{ CADR} \geq 150$	Baseline	424.1	425.3
		1	253.5	254.2
		2	175.2	175.8
		3	78.4	78.6
		4	69.3	69.5

11.4 RESULTS

Table 11.4.1 through Table 11.4.6 summarize the LCC results for senior-only households. Table 11.4.7 through Table 11.4.12 summarize the LCC results for low-income households. Table 11.4.13 through Table 11.4.18 summarize the LCC results for small businesses (commercial sector). The LCC results provide the average installed price, first year's operating cost, average lifetime operating cost (discounted), average life-cycle cost, simple payback period, average lifetime, the percentage of consumers that are burdened with net costs, and average life-cycle cost savings. DOE did not model life expectancy of air cleaner owners in this population subgroup, and recognizes that some consumers will be outlived by their air cleaners.

11.4.1 LCC Results for Senior-Only Households

Table 11.4.1 Air Cleaner PC 1 ($10 \leq \text{PM}_{2.5} \text{ CADR} < 100$): Average LCC and PBP Results for Senior-Only Households

TSL*	Efficiency Level	Average Costs 2021\$				Simple Payback years	Average Lifetime years
		Installed Cost	First Year's Operating Cost	Lifetime Operating Cost	LCC		
--	Baseline	\$64	\$13	\$132	\$196	--	9.0
1	1	\$66	\$11	\$112	\$178	1.0	9.0
2	2	\$68	\$10	\$104	\$172	1.5	9.0
3**	1	\$66	\$11	\$112	\$178	1.0	9.0
	2	\$68	\$10	\$104	\$172	1.5	9.0
4	3	\$79	\$14	\$191	\$270	NA	9.0
5	4	\$87	\$14	\$188	\$275	NA	9.0

Note: The results for each TSL are calculated assuming that all consumers use products at that efficiency level. The PBP is measured relative to the baseline product.

*All TSLs except TSL 3 have a compliance year of 2028.

** For TSL 3, the first results row has a 2024 compliance year. The second results row has a 2026 compliance year.

Table 11.4.2 Air Cleaner PC 1 ($10 \leq \text{PM}_{2.5} \text{ CADR} < 100$): Average LCC Savings Relative to the No-New-Standards Case Efficiency Distribution for Senior-Only Households

TSL**	Efficiency Level	Life-Cycle Cost Savings	
		Average LCC Savings* 2021\$	Percent of Consumers that Experience Net Cost
1	1	\$19	0%
2	2	\$13	7%
3***	1	\$19	0%
	2	\$13	7%
4	3	(\$87)	89%
5	4	(\$85)	94%

* The savings represent the average LCC for affected consumers.

** All TSLs except TSL 3 have a compliance year of 2028.

*** For TSL 3, the first results row has a 2024 compliance year. The second results row has a 2026 compliance year.

Table 11.4.3 Air Cleaner PC 2 ($100 \leq \text{PM}_{2.5} \text{ CADR} < 150$): Average LCC and PBP Results for Senior-Only Households

TSL*	Efficiency Level	Average Costs 2021\$				Simple Payback years	Average Lifetime years
		Installed Cost	First Year's Operating Cost	Lifetime Operating Cost	LCC		
--	Baseline	\$89	\$30	\$308	\$397	--	9.0
1	1	\$91	\$26	\$262	\$353	0.4	9.0
2	2	\$93	\$21	\$221	\$314	0.5	9.0
3**	1	\$91	\$26	\$262	\$353	0.4	9.0
	2	\$93	\$21	\$221	\$314	0.5	9.0
4	3	\$102	\$23	\$303	\$405	NA	9.0
5	4	\$111	\$16	\$224	\$335	1.5	9.0

Note: The results for each TSL are calculated assuming that all consumers use products at that efficiency level. The PBP is measured relative to the baseline product.

* All TSLs except TSL 3 have a compliance year of 2028.

** For TSL 3, the first results row has a 2024 compliance year. The second results row has a 2026 compliance year.

Table 11.4.4 Air Cleaner PC 2 ($100 \leq \text{PM}_{2.5} \text{ CADR} < 150$): Average LCC Savings Relative to the No-New-Standards Case Efficiency Distribution for Senior-Only Households

TSL**	Efficiency Level	Life-Cycle Cost Savings	
		Average LCC Savings* 2021\$	Percent of Consumers that Experience Net Cost
1	1	\$43	0%
2	2	\$56	0%
3***	1	\$43	0%
	2	\$56	0%
4	3	(\$54)	74%
5	4	\$23	46%

* The savings represent the average LCC for affected consumers.

**All TSLs except TSL 3 have a compliance year of 2028.

*** For TSL 3, the first results row has a 2024 compliance year. The second results row has a 2026 compliance year.

Table 11.4.5 Air Cleaner PC 3 ($\text{PM}_{2.5} \text{ CADR} \geq 150$): Average LCC and PBP Results for Senior-Only Households

TSL*	Efficiency Level	Average Costs 2021\$				Simple Payback years	Average Lifetime years
		Installed Cost	First Year's Operating Cost	Lifetime Operating Cost	LCC		
--	Baseline	\$146	\$57	\$551	\$696	--	9.0
1	1	\$148	\$41	\$424	\$572	0.1	9.0
2	2	\$149	\$33	\$361	\$510	0.1	9.0
3**	1	\$148	\$41	\$424	\$572	0.1	9.0
	2	\$149	\$33	\$361	\$510	0.1	9.0
4	3	\$153	\$29	\$377	\$530	0.2	9.0
5	4	\$153	\$29	\$384	\$537	0.3	9.0

Note: The results for each TSL are calculated assuming that all consumers use products at that efficiency level. The PBP is measured relative to the baseline product.

*All TSLs except TSL 3 have a compliance year of 2028.

** For TSL 3, the first results row has a 2024 compliance year. The second results row has a 2026 compliance year.

Table 11.4.6 Air Cleaner PC 3 ($PM_{2.5}$ CADR ≥ 150): Average LCC Savings Relative to the No-New-Standards Case Efficiency Distribution for Senior-Only Households

TSL **	Efficiency Level	Life-Cycle Cost Savings	
		Average LCC Savings* 2021\$	Percent of Consumers that Experience Net Cost
1	1	\$127	0%
2	2	\$111	0%
3***	1	\$127	0%
	2	\$111	0%
4	3	\$47	44%
5	4	\$38	49%

* The savings represent the average LCC for affected consumers.

** All TSLs except TSL 3 have a compliance year of 2028.

*** For TSL 3, the first results row has a 2024 compliance year. The second results row has a 2026 compliance year.

11.4.2 LCC Results for Low-Income Households

Table 11.4.7 Air Cleaner PC 1 ($10 \leq PM_{2.5}$ CADR < 100): Average LCC and PBP Results for Low-Income Households

TSL *	Efficiency Level	Average Costs 2021\$				Simple Payback years	Average Lifetime years
		Installed Cost	First Year's Operating Cost	Lifetime Operating Cost	LCC		
--	Baseline	\$65	\$10	\$108	\$173	--	9.0
1	1	\$66	\$8	\$90	\$156	1.2	9.0
2	2	\$68	\$8	\$85	\$153	1.9	9.0
3**	1	\$66	\$8	\$90	\$156	1.2	9.0
	2	\$68	\$8	\$85	\$153	1.9	9.0
4	3	\$79	\$13	\$177	\$256	NA	9.0
5	4	\$87	\$12	\$178	\$265	NA	9.0

Note: The results for each TSL are calculated assuming that all consumers use products at that efficiency level. The PBP is measured relative to the baseline product.

* All TSLs except TSL 3 have a compliance year of 2028.

** For TSL 3, the first results row has a 2024 compliance year. The second results row has a 2026 compliance year.

Table 11.4.8 Air Cleaner PC 1 ($10 \leq \text{PM}_{2.5} \text{ CADR} < 100$): Average LCC Savings Relative to the No-New-Standards Case Efficiency Distribution for Low-Income Households

TSL**	Efficiency Level	Life-Cycle Cost Savings	
		Average LCC Savings* 2021\$	Percent of Consumers that Experience Net Cost
1	1	\$17	0%
2	2	\$10	10%
3***	1	\$17	0%
	2	\$10	10%
4	3	(\$95)	89%
5	4	(\$97)	96%

* The savings represent the average LCC for affected consumers.

**All TSLs except TSL 3 have a compliance year of 2028.

*** For TSL 3, the first results row has a 2024 compliance year. The second results row has a 2026 compliance year.

Table 11.4.9 Air Cleaner PC 2 ($100 \leq \text{PM}_{2.5} \text{ CADR} < 150$): Average LCC and PBP Results for Low-Income Households

TSL*	Efficiency Level	Average Costs 2021\$				Simple Payback years	Average Lifetime years
		Installed Cost	First Year's Operating Cost	Lifetime Operating Cost	LCC		
--	Baseline	\$89	\$23	\$250	\$339	--	9.0
1	1	\$91	\$19	\$215	\$306	0.6	9.0
2	2	\$93	\$16	\$182	\$275	0.7	9.0
3**	1	\$91	\$19	\$215	\$306	0.6	9.0
	2	\$93	\$16	\$182	\$275	0.7	9.0
4	3	\$102	\$20	\$281	\$383	NA	9.0
5	4	\$111	\$15	\$211	\$322	NA	9.0

Note: The results for each TSL are calculated assuming that all consumers use products at that efficiency level. The PBP is measured relative to the baseline product.

*All TSLs except TSL 3 have a compliance year of 2028.

** For TSL 3, the first results row has a 2024 compliance year. The second results row has a 2026 compliance year.

Table 11.4.10 Air Cleaner PC 2 ($100 \leq \text{PM}_{2.5} \text{ CADR} < 150$): Average LCC Savings Relative to the No-New-Standards Case Efficiency Distribution for Low-Income Households

TSL**	Efficiency Level	Life-Cycle Cost Savings	
		Average LCC Savings* 2021\$	Percent of Consumers that Experience Net Cost
1	1	\$34	0%
2	2	\$44	0%
3***	1	\$34	0%
	2	\$44	0%
4	3	(\$78)	82%
5	4	(\$9)	64%

* The savings represent the average LCC for affected consumers.

**All TSLs except TSL 3 have a compliance year of 2028.

*** For TSL 3, the first results row has a 2024 compliance year. The second results row has a 2026 compliance year.

Table 11.4.11 Air Cleaner PC 3 ($\text{PM}_{2.5} \text{ CADR} \geq 150$): Average LCC and PBP Results for Low-Income Households

TSL*	Efficiency Level	Average Costs 2021\$				Simple Payback years	Average Lifetime years
		Installed Cost	First Year's Operating Cost	Lifetime Operating Cost	LCC		
--	Baseline	\$146	\$42	\$435	\$581	--	9.0
1	1	\$148	\$31	\$351	\$499	0.2	9.0
2	2	\$149	\$26	\$308	\$457	0.2	9.0
3**	1	\$148	\$31	\$351	\$499	0.2	9.0
	2	\$149	\$26	\$308	\$457	0.2	9.0
4	3	\$153	\$25	\$347	\$500	0.4	9.0
5	4	\$154	\$26	\$356	\$510	NA	9.0

Note: The results for each TSL are calculated assuming that all consumers use products at that efficiency level. The PBP is measured relative to the baseline product.

*All TSLs except TSL 3 have a compliance year of 2028.

** For TSL 3, the first results row has a 2024 compliance year. The second results row has a 2026 compliance year.

Table 11.4.12 Air Cleaner PC 3 ($PM_{2.5}$ CADR ≥ 150): Average LCC Savings Relative to the No-New-Standards Case Efficiency Distribution for Low-Income Households

TSL **	Efficiency Level	Life-Cycle Cost Savings	
		Average LCC Savings* 2021\$	Percent of Consumers that Experience Net Cost
1	1	\$85	0%
2	2	\$76	0%
3***	1	\$85	0%
	2	\$76	0%
4	3	\$2	61%
5	4	(\$7)	67%

* The savings represent the average LCC for affected consumers.

** All TSLs except TSL 3 have a compliance year of 2028.

*** For TSL 3, the first results row has a 2024 compliance year. The second results row has a 2026 compliance year.

11.4.3 LCC Results for Small Businesses (Commercial Sector)

Table 11.4.13 Air Cleaner PC 1 ($10 \leq PM_{2.5}$ CADR < 100): Average LCC and PBP Results for Small Businesses (Commercial Sector)

TSL *	Efficiency Level	Average Costs 2021\$				Simple Payback years	Average Lifetime years
		Installed Cost	First Year's Operating Cost	Lifetime Operating Cost	LCC		
--	Baseline	\$62	\$17	\$124	\$186	--	9.0
1	1	\$64	\$15	\$104	\$168	0.7	9.0
2	2	\$66	\$14	\$96	\$162	1.0	9.0
3**	1	\$64	\$15	\$104	\$168	0.7	9.0
	2	\$66	\$14	\$96	\$162	1.0	9.0
4	3	\$76	\$19	\$173	\$249	NA	9.0
5	4	\$84	\$17	\$169	\$253	NA	9.0

Note: The results for each TSL are calculated assuming that all consumers use products at that efficiency level. The PBP is measured relative to the baseline product.

* All TSLs except TSL 3 have a compliance year of 2028.

** For TSL 3, the first results row has a 2024 compliance year. The second results row has a 2026 compliance year.

Table 11.4.14 Air Cleaner PC 1 ($10 \leq \text{PM}_{2.5} \text{ CADR} < 100$): Average LCC Savings Relative to the No-New-Standards Case Efficiency Distribution for Small Businesses (Commercial Sector)

TSL**	Efficiency Level	Life-Cycle Cost Savings	
		Average LCC Savings* 2021\$	Percent of Consumers that Experience Net Cost
1	1	\$18	0%
2	2	\$14	1%
3***	1	\$18	0%
	2	\$14	1%
4	3	(\$77)	87%
5	4	(\$75)	92%

* The savings represent the average LCC for affected consumers.

** All TSLs except TSL 3 have a compliance year of 2028.

*** For TSL 3, the first results row has a 2024 compliance year. The second results row has a 2026 compliance year.

Table 11.4.15 Air Cleaner PC 2 ($100 \leq \text{PM}_{2.5} \text{ CADR} < 150$): Average LCC and PBP Results for Small Businesses (Commercial Sector)

TSL*	Efficiency Level	Average Costs 2021\$				Simple Payback years	Average Lifetime years
		Installed Cost	First Year's Operating Cost	Lifetime Operating Cost	LCC		
--	Baseline	\$86	\$41	\$292	\$378	--	9.0
1	1	\$88	\$35	\$245	\$333	0.3	9.0
2	2	\$90	\$29	\$203	\$294	0.3	9.0
3**	1	\$88	\$35	\$245	\$333	0.3	9.0
	2	\$90	\$29	\$203	\$294	0.3	9.0
4	3	\$99	\$30	\$271	\$370	NA	9.0
5	4	\$107	\$21	\$197	\$304	1.1	9.0

Note: The results for each TSL are calculated assuming that all consumers use products at that efficiency level. The PBP is measured relative to the baseline product.

* All TSLs except TSL 3 have a compliance year of 2028.

** For TSL 3, the first results row has a 2024 compliance year. The second results row has a 2026 compliance year.

Table 11.4.16 Air Cleaner PC 2 ($100 \leq \text{PM}_{2.5} \text{ CADR} < 150$): Average LCC Savings Relative to the No-New-Standards Case Efficiency Distribution for Small Businesses (Commercial Sector)

TSL**	Efficiency Level	Life-Cycle Cost Savings	
		Average LCC Savings* 2021\$	Percent of Consumers that Experience Net Cost
1	1	\$44	0%
2	2	\$57	0%
3***	1	\$44	0%
	2	\$57	0%
4	3	(\$38)	67%
5	4	\$32	41%

* The savings represent the average LCC for affected consumers.

**All TSLs except TSL 3 have a compliance year of 2028.

*** For TSL 3, the first results row has a 2024 compliance year. The second results row has a 2026 compliance year.

Table 11.4.17 Air Cleaner PC 3 ($\text{PM}_{2.5} \text{ CADR} \geq 150$): Average LCC and PBP Results for Small Businesses (Commercial Sector)

TSL*	Efficiency Level	Average Costs 2021\$				Simple Payback years	Average Lifetime years
		Installed Cost	First Year's Operating Cost	Lifetime Operating Cost	LCC		
--	Baseline	\$141	\$78	\$528	\$669	--	9.0
1	1	\$143	\$55	\$395	\$538	0.1	9.0
2	2	\$144	\$44	\$329	\$474	0.1	9.0
3**	1	\$143	\$55	\$395	\$538	0.1	9.0
	2	\$144	\$44	\$329	\$474	0.1	9.0
4	3	\$148	\$38	\$335	\$482	0.2	9.0
5	4	\$148	\$38	\$340	\$488	0.2	9.0

Note: The results for each TSL are calculated assuming that all consumers use products at that efficiency level. The PBP is measured relative to the baseline product.

*All TSLs except TSL 3 have a compliance year of 2028.

** For TSL 3, the first results row has a 2024 compliance year. The second results row has a 2026 compliance year.

Table 11.4.18 Air Cleaner PC 3 (PM_{2.5} CADR ≥ 150): Average LCC Savings Relative to the No-New-Standards Case Efficiency Distribution for Small Businesses (Commercial Sector)

TSL**	Efficiency Level	Life-Cycle Cost Savings	
		Average LCC Savings* 2021\$	Percent of Consumers that Experience Net Cost
1	1	\$133	0%
2	2	\$117	0%
3***	1	\$133	0%
	2	\$117	0%
4	3	\$61	37%
5	4	\$54	43%

* The savings represent the average LCC for affected consumers.

** All TSLs except TSL 3 have a compliance year of 2028.

*** For TSL 3, the first results row has a 2024 compliance year. The second results row has a 2026 compliance year.

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CHAPTER 12. MANUFACTURER IMPACT ANALYSIS

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CHAPTER 12. MANUFACTURER IMPACT ANALYSIS

12.1 INTRODUCTION

In determining whether a standard is economically justified, the U.S. Department of Energy (DOE) is required to consider the economic impact of the standard on the manufacturers and consumers of product subject to such a standard. (42 U.S.C. 6295(o)(2)(B)(i) The law also calls for an assessment of the impact of any lessening of competition as determined in writing by the Attorney General. DOE conducted a manufacturer impact analysis (MIA) to estimate the financial impact of energy conservation standards on manufacturers of air cleaners and assessed the impact of such standards on direct employment and manufacturing capacity.

The MIA has both quantitative and qualitative aspects. The quantitative part of the MIA primarily relies on the Government Regulatory Impact Model (GRIM), an industry cash-flow model adapted for the product in this rulemaking. The GRIM inputs include information on industry cost structure, shipments, and pricing strategies. The GRIM's key output is the industry net present value (INPV). The model estimates the financial impact of more stringent energy conservation standards for each product class by comparing changes in INPV between a no-new-standards-case and the various trial standard levels (TSLs) in the standards case. The qualitative part of the MIA addresses product characteristics, manufacturer characteristics, and market and product trends, as well as the impact of standards on subgroups of manufacturers.

12.2 METHODOLOGY

DOE conducted the MIA in phases. Phase I, "Industry Profile," consisted of preparing an industry characterization for the air cleaner industry. DOE collects pertinent qualitative and quantitative information about the market and manufacturer financials. In phase II, "Industry Cash Flow Analysis and Interview Guide," DOE used the GRIM to assess the potential impacts of energy conservation standards on manufacturers. In phase III, "Industry and Subgroup Analysis," DOE developed additional analyses for subgroups that may be affected in various ways. Each phase of the MIA is described in greater detail in the following sections.

12.2.1 Phase I: Industry Profile

In phase I of the MIA, DOE prepared a profile of the air cleaner industry that built on the market and technology assessment prepared for this rulemaking (refer to chapter 3 of this NOPR technical support document (TSD)). Before initiating detailed impact analyses, DOE collected information on past and present market characteristics of the air cleaner industry. This information included wages, employment, cost structure, manufacturer markups, manufacturer market shares, and consolidation trends. DOE also collected information on industry financial parameters, such as net plant, property, and equipment (PPE); selling, general and administrative (SG&A) expenses; research and development (R&D) expenses, depreciation, revenue, cost of goods sold, etc. These parameters allowed DOE to derive preliminary industry financial inputs for the GRIM as discussed in section 12.4.3.

DOE used public and private information to develop its initial characterization of the industry, including reports published by industry groups, trade journals, prior DOE rulemakings, market research tools (*i.e.*, D&B Hoovers¹ and ZoomInfo²), the U.S. Census Bureau's *Annual*

Survey of Manufactures (ASM),³ and U.S. Census Bureau's *Quarterly Survey of Plant Capacity Utilization*.⁴ DOE also used information from its engineering analysis and prior rulemakings to enhance its industry profile.

12.2.2 Phase II: Industry Cash-Flow Analysis and Interview Guide

Phase II focused on the financial impacts of energy conservation standards on manufacturers of air cleaners. More stringent energy conservation standards can affect manufacturer cash flows in three distinct ways. These include: (1) creating a need for increased investment; (2) raising production costs per unit; and (3) altering revenue due to higher per-unit prices and/or possible changes in sales volumes. To quantify these impacts, DOE used the GRIM to perform a cash-flow analysis for the air cleaner industry. DOE used the financial values derived during phase I and the shipment scenarios used in the national impact analysis (NIA) to perform these analyses. The GRIM modeled impacts for all the analyzed TSLs.

12.2.2.1 Industry Cash-Flow Analysis

DOE uses the GRIM to analyze the financial impacts of potential energy conservation standards. The GRIM uses several factors to determine a series of annual cash flows for the years leading up to the expected compliance date of energy conservation standards and for several years after implementation. These factors include annual expected revenues, costs of goods sold, SG&A costs, taxes, and capital expenditures. Inputs to the GRIM include financial information, manufacturer production costs (MPCs), shipments forecasts, and price forecasts developed in other analyses. Financial information is developed based on publicly available data and any confidentially submitted manufacturer information. MPCs are estimated in the engineering analysis as presented in chapter 5 of this NOPR TSD. The GRIM uses the NIA shipment projections, presented in chapter 9 of this NOPR TSD. DOE compares the GRIM results for potential standard levels against the results for the no-new-standards case, in which energy conservation standards are not amended. The financial impact of analyzed amended energy conservation standards is the difference between the two sets of discounted annual cash flows.

12.2.2.2 Interview Guide

During Phase II of the MIA, DOE typically develops an interview guide to distribute to manufacturers of air cleaners in order to develop other key GRIM inputs, including product and capital conversion costs, and to gather additional information on the anticipated effects of energy conservation standards on revenue, direct employment, capital assets, industry competitiveness, and subgroups. The interview guide typically provides a starting point to identify relevant issues and help identify the potential impacts of energy conservation standards on individual manufacturers. The information from these interviews is protected by nondisclosure agreements (NDAs) and resides with DOE's contractors. Prior to each interview, DOE typically provides company representatives with an interview guide that included the topics for which DOE sought input. The MIA interview topics included (1) key issues to this rulemaking; (2) product classes; (3) efficiency levels; (4) technology options; (5) cost efficiency curves; (6) organization and market share; (7) conversion costs; (8) financial parameters; (9) markups and profitability; (10) cumulative regulatory burden; (11) capacity, foreign competition, and outsourcing; (12)

consolidation; (13) impacts on small manufacturers; and (14) repair and maintenance costs. The interview guide is presented in Appendix 12A.

Due to DOE pursuing the Direct Final Rule and explicit time constraints written as part of the Joint Proposal recommendation, DOE conducted preliminary interviews that were an abridged version of the process described above. The preliminary interviews emphasized content related to key issues, product classes, technology options, engineering, manufacturer markups, distribution channels, and shipments.

12.2.3 Phase III: Industry and Subgroup Analysis

In Phase III, DOE typically interviews a cross-section of air cleaner manufacturers. The interviews provide DOE with valuable information for evaluating the impacts of energy conservation standards. DOE sought to isolate key issues and concerns, to obtain feedback from industry on the approaches used in DOE analytical processes.

12.2.3.1 Manufacturer Interviews

The information gathered in Phase I and the cash-flow analysis performed in Phase II is typically supplemented with information gathered from manufacturer interviews in Phase III. The interview process provides an opportunity for interested parties to express their views on important issues privately, allowing confidential or sensitive information to be considered in the rulemaking process.

DOE contacted companies from its database of manufacturers for preliminary interviews. DOE interviewed a cross-section of air cleaner manufacturers, accounting for approximately seven percent of industry models. Participants included domestic-based and foreign-based manufacturers. Interviews were scheduled well in advance to provide every opportunity for key individuals to be available for comment. Although a written response to the questionnaire was acceptable, DOE sought interactive interviews, which help clarify responses and identify additional issues. The resulting information provides valuable inputs to the GRIM and other analyses.

12.2.3.2 Revised Industry Cash Flow Analysis

In Phase II of the MIA, DOE typically provides manufacturers with preliminary GRIM input financial figures for review and evaluation. During the interviews, DOE typically requests feedback on the preliminary financial parameters. DOE typically revises the GRIM inputs based on manufacturer responses. Section 12.4.3 provides additional details on how DOE calculated the parameters. DOE requested manufacturer feedback on industry markup during preliminary interviews.

DOE relied on industry parameters from the Portable Air Conditioners Final Rule published in January 2020, given the lack of publicly-listed OEMs of air cleaners. 85 FR 1378 (Jan. 9, 2020). In reviewing other appliance standards rulemakings where DOE had sufficient data to estimate product-specific manufacturer markups and other financial parameters, DOE found portable air conditioners to be the most recent rulemaking covering a product similar to air cleaners in terms of product and market attributes.

12.2.3.3 Manufacturer Subgroup Analysis

Using average cost assumptions to develop an industry cash flow estimate is not adequate for assessing differential impacts among subgroups of manufacturers. Small, low-volume manufacturers, or manufacturers exhibiting a cost structure that differs significantly from the industry average could be more negatively affected. Ideally, DOE would consider the impact on every manufacturer individually; however, it typically uses the results of the industry characterization to group manufacturers exhibiting similar characteristics. During the interview process, DOE discussed the potential subgroups and subgroup members that have been identified for the analysis. DOE looked to stakeholders to suggest what subgroups or characteristics are the most appropriate for the analysis.

Small Business Manufacturers

DOE used the U.S. Small Business Administration (SBA) small business size standards, effective December 19, 2022, as amended, and the North American Industry Classification System (NAICS) code, presented in Table 12.2.1 to determine whether any small entities would be affected by the rulemaking.⁵ For the product classes under review, the SBA bases its small business definition on the total number of employees for a business, including the total number of employees of its parent company and any subsidiaries. An aggregated business entity with fewer employees than the listed limit is considered a small business.

DOE began its assessment by reviewing its AHAM's Database of Certified Room Air Cleaners,⁶ and ENERGY STAR's Product Finder dataset,⁷ and California Air Resources Board's (CARB) CARB-Certified Air Cleaning Devices⁸ to create a list of original equipment manufacturers (OEMs) that manufacture the products covered by this proposal. DOE consulted publicly available data, such as manufacturer websites, manufacturer specifications and product literature, import/export logs (e.g., bills of lading from Panjiva⁹), and basic model numbers, to identify OEMs of covered air cleaners. DOE further relied on public data and subscription-based market research tools (e.g., Dun & Bradstreet reports) to determine company, location, headcount, and annual revenue. DOE typically asks industry representatives if they were aware of any small manufacturers during manufacturer interviews. DOE screened out companies that do not offer products covered by this rulemaking, do not meet the SBA's definition of a "small business," or are foreign-owned and operated.

DOE initially identified 43 OEMs that sell air cleaners in the United States. Of the 43 OEMs identified, DOE identified four OEMs that qualify as small, domestic manufacturers.

Table 12.2.1 SBA and NAICS Classifications of Small Businesses Potentially Affected by this Rulemaking

Industry Description	Revenue Limit	Employee Limit	NAICS
Small Electrical Appliance Manufacturing	N/A	1,500	335210

12.2.3.4 Manufacturing Capacity Impact

Energy conservation standards could result in the obsolescence of existing manufacturing assets, including tooling and capital investments. DOE analyzed manufacturer capacity

utilization and plant location decisions in the United States, with and without standards; the ability of manufacturers to upgrade or remodel existing facilities to accommodate the new efficiency requirements; the nature and value of any stranded assets that might result from standards; and estimates for any one-time changes to existing PPE that be necessitated by standards. DOE's estimates of the one-time capital changes and stranded assets affect the cash flow estimates in the GRIM. These estimates can be found in section 12.4.8. DOE's discussion of the manufacturing capacity impact can be found in section 12.7.2.

12.2.3.5 Direct Employment Impact

The impact of energy conservation standards on employment is an important consideration in the rulemaking process. To assess how domestic direct employment patterns might be affected, the interviews explored current employment trends in the air cleaner industry. The employment questions in the interview guide focused on domestic versus foreign manufacturing decisions with and without energy conservation standards. DOE also obtained data from the U.S. Census Bureau's *ASM* on current direct employment trends in the air cleaner industry. The direct employment impacts are reported in section 12.7.1.

12.2.3.6 Cumulative Regulatory Burden

DOE seeks to mitigate the overlapping effects on manufacturers due to energy conservation standards and other regulatory actions affecting the same products. DOE analyzed the impact on manufacturers of multiple Federal, product-specific regulatory actions. Based on its own research, DOE identified ongoing rulemakings relevant to air cleaner manufacturers, including other Federal regulations that affect other products made by the same manufacturers. A discussion of the cumulative regulatory burden of energy conservation standards and the impact on manufacturers of multiple, product-specific regulatory actions can be found in section 12.7.3.

12.3 GRIM INPUTS AND ASSUMPTIONS

The GRIM serves as the main tool for assessing the impacts on industry due to energy conservation standards. DOE relies on several sources to obtain inputs for the GRIM. Data and assumptions from these sources are then fed into an accounting model that calculates the industry cash flow both with and without energy conservation standards.

12.3.1 Overview of the Government Regulatory Impact Model

The basic structure of the GRIM, illustrated in Figure 12.4.1, is an annual cash-flow analysis that uses manufacturer prices, manufacturing costs, shipments, and industry financial information as inputs, and accepts a set of regulatory conditions such as changes in costs, investments, and associated margins. The GRIM spreadsheet uses these and other inputs to calculate a series of annual cash flows, beginning with the publication year of the NOPR, 2023, and continuing to 2057, 30 years after the compliance year of the rulemaking. The model calculates the INPV by summing the stream of annual discounted cash flows during this period and adding a discounted terminal value.¹⁰

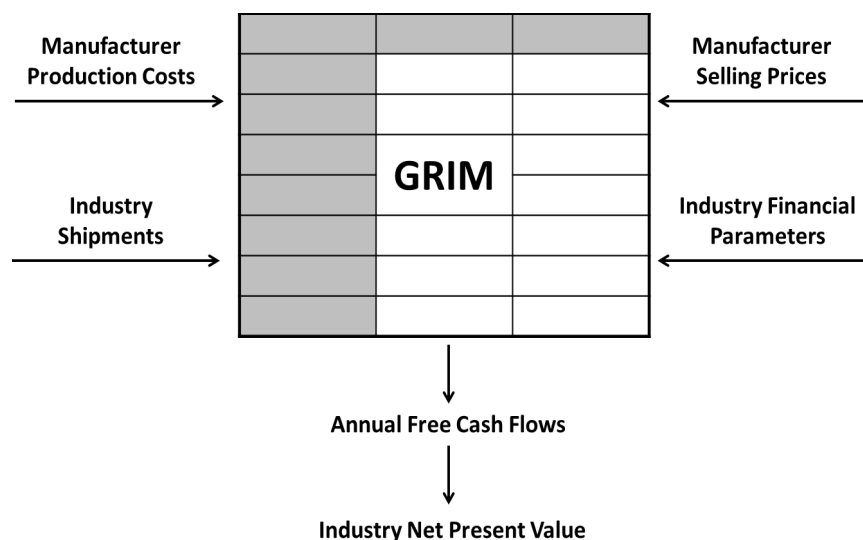


Figure 12.4.1 Using the GRIM to Calculate Cash Flow

The GRIM projects cash flows using standard accounting principles and compares INPV between the no-new-standards-case and the standard-case scenarios. The difference in INPV between the no-new-standards case and the standard case(s) represents the estimated financial impact of the energy conservation standards on manufacturers. Appendix 12A provides more technical details and user information for the GRIM.

12.3.2 Sources for GRIM Inputs

The GRIM uses several different sources for data inputs in determining industry cash flow. These sources include prior rulemakings, corporate annual reports, company profiles, U.S. Census data, the shipments model, the engineering analysis, and the manufacturer interviews.

12.3.2.1 Prior Rulemakings

Given the lack of publicly-listed OEMs of air cleaners, DOE relied on industry parameters from the Portable Air Conditioners Final Rule published in January 2020. 85 FR 1378 (Jan. 9, 2020). In reviewing other appliance standards rulemakings where DOE had sufficient data to estimate product-specific manufacturer markups and other financial parameters, DOE found portable air conditioners to be the most recent rulemaking covering a product similar to air cleaners in terms of product and market attributes. DOE used the industry parameters from the Portable Air Conditioners Final Rule to derive the following initial inputs to the GRIM:

- Working capital;
- SG&A;
- R&D;
- Depreciation;
- Capital expenditures;
- Net PPE;
- Industry WACC.

During confidential preliminary interviews, DOE received feedback on manufacturer markup.

12.3.2.2 Shipments Model

The GRIM used shipments projections derived from DOE's shipments model in the NIA. The shipments model takes an accounting approach, tracking market shares of each product class and the vintage of units in the stock. Stock accounting uses product shipments as inputs to estimate the age distribution of in-service product stocks for all years. Total shipments for air cleaners are developed by considering the demand from replacements for units in stock that fail and the demand from first-time owners, which are the households without existing air cleaners. Chapter 9 of the NOPR TSD describes the methodology and analytical model DOE used to forecast shipments.

12.3.2.3 Engineering Analysis

The engineering analysis develops the relationship between the manufacturer selling price (MSP) and energy efficiency for the products analyzed in this rulemaking. This relationship serves as the basis for the cost-benefit calculations for consumers, manufacturers, and the nation. In determining the cost-efficiency relationship, DOE estimates the increase in manufacturing costs associated with increasing the efficiency of product above the baseline up to the maximum technologically feasible (max-tech) efficiency level for each product class.

For this analysis, DOE used a combination of the efficiency-level approach and the design option approach. This analysis involved physically disassembling commercially available products, reviewing publicly available cost information, and modeling equipment cost. From this information, DOE estimated the MPCs for a range of products currently available on the market. DOE then considered the incremental steps manufacturers may take to reach higher efficiency levels. In its modeling, DOE started with the baseline MPC and added the expected design options at each higher efficiency level to estimate incremental MPCs. Chapter 5 of the NOPR TSD describes the methodology DOE used to conduct the engineering analysis.

12.3.2.4 Manufacturer Interviews

As part of the MIA, DOE typically conducts interviews with a representative cross-section of manufacturers. Through these discussions, DOE typically obtains information to determine and verify GRIM input assumptions. In its preliminary interviews, DOE asked for feedback on key issues, product classes, technology options, engineering, manufacturer markup, distribution channels, and shipments.

12.3.3 Financial Parameters

DOE used the financial parameters from the January 2020 Portable Air Conditioners Final Rule for determining the air cleaner financial parameters. 85 FR 1378 (Jan. 9, 2020). The January 2020 Portable Air Conditioners Final Rule financial parameters were vetted by multiple manufacturers in confidential interviews and went through public notice and comment. The results are the most robust product-specific estimates that are publicly available for air cleaner

products. Given the lack of publicly-listed OEMs of air cleaners, DOE reviewed other appliance standards rulemakings where DOE had sufficient data to estimate product-specific manufacturer markups and other financial parameters. DOE found portable air cleaners to be the most recent rulemaking covering a product similar to air cleaners in terms of product and market attributes.

Table 12.3.1 below shows the initial financial parameter estimates for air cleaners.

Table 12.3.1 Initial Financial Parameters Based on the January 2020 Portable Air Conditioners Final Rule

Parameter	Industry Weighted Average
Tax Rate (<i>% of Taxable Income</i>)	27.4
Working Capital (<i>% of Revenue</i>)	16.6
SG&A (<i>% of Revenue</i>)	18.0
R&D (<i>% of Revenues</i>)	1.7
Depreciation (<i>% of Revenues</i>)	2.1
Capital Expenditures (<i>% of Revenues</i>)	5.0
Net Property, Plant, and Equipment (<i>% of Revenues</i>)	16.7
Discount Rate (%)	6.6

12.3.4 Manufacturer Production Costs

Manufacturing a higher efficiency product is typically more expensive than manufacturing a baseline product due to the use of more complex components, which are typically more costly than baseline components. The changes in the MPCs of the analyzed product can affect the revenues, gross margins, and cash flow of the industry, making these product cost data key GRIM inputs for DOE's analysis. The GRIM relied on MPCs derived in the engineering analysis (chapter 5 of the NOPR TSD).

In this rulemaking, DOE applied a hybrid approach of efficiency-level and design-option approaches described above. This approach involved reviewing publicly available efficiency data and physically disassembling commercially available products. From this information, DOE estimated the MPCs for a range of products available at that time on the market. DOE then analyzed the steps manufacturers took to improve product efficiencies. In its analysis, DOE determined that manufacturers would likely rely on certain design options to reach higher efficiencies.

To account for manufacturers' non-production costs (*i.e.*, SG&A, R&D, and interest) and profit, DOE applies a manufacturer multiplier to the MPC. The resulting MSP is the price at which manufacturers distribute units into commerce. Based on publicly available financial information for manufacturers of air cleaners, prior rulemaking analyses, and comments from manufacturer interviews, DOE assumed the industry average no-new-standards-case manufacturer markup to be 1.45.

Table 12.3.2 Manufacturer Production Cost Breakdown (2021\$)

Product Class	EL	MPC	Manufacturer Markup	MSP
10 ≤ PM_{2.5} CADR < 100	Baseline	\$28.24	1.45	\$45.30
	EL 1	\$27.15	1.45	\$46.76
	EL 2	\$37.68	1.45	\$48.42
	EL 3	\$41.15	1.45	\$56.94
	EL 4	\$42.88	1.45	\$63.89
100 ≤ PM_{2.5} CADR < 150	Baseline	\$28.24	1.45	\$62.31
	EL 1	\$37.68	1.45	\$64.18
	EL 2	\$47.15	1.45	\$66.15
	EL 3	\$42.88	1.45	\$73.15
	EL 4	\$53.59	1.45	\$80.55
PM_{2.5} CADR ≥ 150	Baseline	\$69.93	1.45	\$102.23
	EL 1	\$59.15	1.45	\$103.91
	EL 2	\$73.45	1.45	\$105.13
	EL 3	\$74.72	1.45	\$107.78
	EL 4	\$74.20	1.45	\$108.18

DOE also incorporated learning over time into the analysis, which affects the MPC over time. For the MIA, DOE used the same product price trend as outlined in the NIA from the publication year of the NOPR, 2023, through the end of the analysis period. DOE applied the learning over time in both the no-standards case and standards cases. See chapter 8 of the NOPR TSD for a description of how DOE derived and incorporated product price trend into the analysis.

12.3.5 Trial Standard Levels

DOE developed efficiency levels for each analyzed product class. TSLs were then developed by selecting groupings of efficiency levels for all analyzed product classes. Table 12.3.3 presents the efficiency levels analyzed as part of this rulemaking. Table 12.3.4 presents the TSLs examined as part of this rulemaking.

Table 12.3.3 Analyzed Efficiency Levels for Air cleaners

Product Class	Efficiency Level	Estimated Annual Energy Use (CADR/Watt)
$10 \leq \text{PM}_{2.5} \text{ CADR} < 100$	Baseline	1.53
	EL 1	1.69
	EL 2	1.89
	EL 3	3.37
	EL 4 (Max-Tech)	5.40
$100 \leq \text{PM}_{2.5} \text{ CADR} < 150$	Baseline	1.53
	EL 1	1.90
	EL 2	2.39
	EL 3	5.40
	EL 4 (Max-Tech)	12.75
$\text{PM}_{2.5} \text{ CADR} \geq 150$	Baseline	1.20
	EL 1	2.01
	EL 2	2.91
	EL 2	6.55
	EL 4 (Max-Tech)	7.41

Table 12.3.4 Trial Standard Levels for Air cleaners

Product Class	TSL 1	TSL 2	TSL 3	TSL 4	TSL 5
	Efficiency Level - Maximum allowable estimated annual energy use (CADR/Watt)				
$10 \leq \text{PM}_{2.5} \text{ CADR} < 100$	EL 1 – 1.69	EL 2 – 1.89	EL 1* – 1.69 EL 2** – 1.89	EL 3 – 3.37	EL 4 – 5.40
$100 \leq \text{PM}_{2.5} \text{ CADR} < 150$	EL 1 – 1.90	EL 2 – 2.39	EL 1* – 1.90 EL 2** – 2.39	EL 3 – 5.40	EL 4 – 12.75
$\text{PM}_{2.5} \text{ CADR} \geq 150$	EL 1 – 2.01	EL 2 – 2.91	EL 1* – 2.01 EL 2** – 2.91	EL 3 – 6.55	EL 4 – 7.41

*Tier 1, with an effective date of December 31, standards are equivalent to the standards established by the States of Maryland, Nevada, and New Jersey, and the District of Columbia.

**Tier 2, with an effective date of December 31, 2025, standards are equivalent to the standards specified in the ENERGY STAR Version 2.0 Room Air Cleaners Specification and those adopted by the State of Washington.

12.3.6 NIA Shipment Forecasts

The GRIM estimates manufacturer revenues based on total-unit-shipment forecasts and the distribution of these values by product class and efficiency level. Changes in the efficiency mix at each standard level are a key driver of manufacturer finances. For this analysis, the GRIM used the air cleaner shipment data from the NIA from 2023 (the NOPR publication year) through 2057 (the end year of the analysis period). Chapter 9 of the NOPR TSD contains additional information on the shipments analysis.

Total shipments for air cleaners are developed by considering the demand from replacements for units in stock that fail and the demand from first-time owners, which are the households without existing air cleaners. DOE calculated shipments due to replacements using the retirement function developed for the LCC analysis and historical data from Association of Home Appliance Manufacturers (AHAM).

Table 12.3.5 shows the market share and total shipments estimates for each product class of air cleaners in 2023, the expected year of the NOPR publication.

Table 12.3.5 Shipments and Market Share by Product Class in 2023

Product Class	Annual Shipments (millions)	Percent of Total Shipments
$10 \leq \text{PM}_{2.5} \text{ CADR} < 100$	2.056	26.0%
$100 \leq \text{PM}_{2.5} \text{ CADR} < 150$	1.898	24.0%
$\text{PM}_{2.5} \text{ CADR} \geq 150$	3.955	50.0%
Total	7.909	100.0%

12.3.6.1 No-New-Standards-Case Shipments Forecast

As part of the shipment analysis, DOE estimated the distribution of shipments by efficiency level for each product class of air cleaners. To estimate the energy efficiency distribution of air cleaners for 2028, DOE used data from the engineering analysis, and DOE's dataset of models from AHAM's Database of Certified Room Air Cleaners,¹¹ and ENERGY STAR's Product Finder dataset,¹² and CARB's Certified Air Cleaning Devices¹³. For this analysis, DOE assumed no annual efficiency improvement for the no-new-standards case based on the current market evaluation.

See chapter 8 of the NOPR TSD for further information on the derivation of the efficiency distributions and chapter 9 of the NOPR TSD DOE's shipments analysis.

Table 12.3.6 No-New-Standards Case Efficiency Distribution in 2028

Product Class	2028 Market Share (%)					
	Baseline	EL 1	EL 2	EL 3	EL 4	Total*
$10 \leq \text{PM}_{2.5} \text{ CADR} < 100$	28	42	19	7	3	100
$100 \leq \text{PM}_{2.5} \text{ CADR} < 150$	24	37	28	11	0	100
$\text{PM}_{2.5} \text{ CADR} \geq 150$	22	33	38	3	4	100

12.3.6.2 Standards-Case Shipments Forecast

To examine the impact of amended energy conservation standards on shipments, which in turn affects the INPV, DOE used the no-new-standards-case shipments described in the previous section as a point of comparison for shipments forecast in the standards case. For each TSL described in the standards case, DOE used the shipments forecasts developed in the NIA for air cleaners.

For the standards cases, DOE used a “roll up” scenario to establish the shipment-weighted efficiency for the year that standards are assumed to become effective (2029). In this scenario, the market shares of products in the no-new-standards case that do not meet the standard under consideration would “roll up” to meet the new standard level, and the market share of products above the standard would remain unchanged. See chapter 9 of the NOPR TSD for the additional information regarding the shipments analysis.

12.3.7 Conversion Costs and Stranded Assets

Energy conservation standards could cause manufacturers to incur conversion costs to bring their production facilities and equipment designs into compliance. DOE evaluated the level of conversion-related expenditures that would be needed to comply with each considered efficiency level in each product class. For the MIA, DOE classified these conversion costs into two major groups: (1) product conversion costs; and (2) capital conversion costs. Product conversion costs are investments in research, development, testing, marketing, and other non-capitalized costs necessary to make product designs comply with energy conservation standards. Capital conversion costs are investments in property, plant, and equipment necessary to adapt or change existing production facilities such that new compliant product designs can be fabricated and assembled.

Table 12.3.7 Conversion Costs by Product Class

	Efficiency Level	Product Conversion Costs (2021\$ Millions)	Capital Conversion Costs (2021\$ Millions)	Stranded Assets (2021\$ Millions)
10 ≤ PM_{2.5} CADR < 100	Baseline	\$0.0	\$0.0	\$0.0
	EL 1	\$6.1	\$3.6	\$0.7
	EL 2	\$8.4	\$9.0	\$1.8
	EL 3	\$14.2	\$19.0	\$3.8
	EL 4	\$15.1	\$20.6	\$4.1
100 ≤ PM_{2.5} CADR < 150	Baseline	\$0.0	\$0.0	\$0.0
	EL 1	\$5.6	\$3.1	\$0.6
	EL 2	\$7.6	\$7.8	\$1.6
	EL 3	\$13.9	\$26.7	\$5.3
	EL 4	\$15.0	\$29.8	\$6.0
PM_{2.5} CADR ≥ 150	Baseline	\$0.0	\$0.0	\$0.0
	EL 1	\$5.5	\$6.9	\$1.4
	EL 2	\$7.3	\$17.2	\$3.4
	EL 3	\$14.3	\$48.5	\$9.7
	EL 4	\$14.7	\$50.1	\$10.0

12.3.7.1 Product Conversion Costs

DOE relied on information derived from the engineering analysis, product teardowns, and air cleaner test procedure final rule to evaluate the level of product conversion costs manufacturers would likely incur at the various efficiency levels. DOE evaluated the testing costs for manufacturers to certify models to DOE and the investments necessary to update product designed to comply with standards as part of product conversion costs. DOE estimated cost of \$6,000 for 3rd party lab testing per basic model, consistent with the test procedure final rule. Manufacturers with in-house testing capabilities would likely have lower testing costs per basic model. DOE multiplied the 3rd party lab testing costs by the number of basic models identified in the industry to estimate total cost of be \$17.75 million for first time DOE compliance testing. Additionally, DOE calculated a recurring “Annual testing” cost of \$1.8 million for the industry.

To estimate produce redesign costs at each efficiency level, DOE took into account annual spending on R&D; complexity of design options; and percentage of industry shipments that would require redesign. Each efficiency level reflects the portion of shipments that would not meet the proposed standard, R&D to redesign failing units, and the cost to re-test updated product designs.

12.3.7.2 Capital Conversion Costs

DOE relied on information derived from the engineering analysis and product teardowns to evaluate the level of capital conversion costs manufacturers would likely incur at the various efficiency levels. As part of the engineering analysis, DOE determined the likely equipment and tooling required for manufacturers to produce products that were physically disassembled. DOE evaluating the incremental product changes, and associated manufacturing equipment and

tooling changes, that affected energy usage between efficiency levels. The most significant drivers of capital conversion costs in the industry were the potential need for mold alterations or entirely new molds for injection molded parts. To avoid underestimating capital conversion costs, DOE did not estimate the cost to alter mold dies. Rather, DOE's modeling assumed that OEMs would fabricate or purchase new mold dies. Also, manufacturers using soft tooling or that do not rely on injection molding would have lower capital conversion costs than modeled by DOE.

DOE acknowledges that manufacturers may follow different design paths to reach the various efficiency levels analyzed. An individual manufacturer's investments depend on a range of factors, including the company's current product offerings and product platforms, existing production facilities and infrastructure, and make vs. buy decisions for products. DOE's conversion cost methodology incorporated industry shipments from the NIA analysis and the tooling and testing costs from engineering analysis. While industry average values may not represent any single manufacturer, DOE's modeling provides reasonable estimates of industry-level investments.

DOE assumes all conversion-related investments occur between the year of publication of the final rule and the year by which manufacturers must comply with the new standard.

12.3.7.3 Stranded Assets

DOE relies on information derived from manufacturer interviews and the engineering analysis to estimate potential levels of stranded assets incurred at various efficiency levels. From the results engineering analysis, DOE aligned stranded assets with capital investments as manufacturers replace injection mold dies.

12.3.8 Manufacturer Markup Scenarios

DOE modeled two standards-case scenarios to represent uncertainty regarding the potential impacts on prices and profitability for manufacturers following the implementation of energy conservation standards: (1) a preservation of gross margin percentage scenario; and (2) a tiered scenario. These scenarios lead to different markup values that, when applied to the MPCs, result in varying revenue and cash flow impacts.

12.3.8.1 Preservation of Gross Margin Scenario

Under the preservation of gross margin percentage scenario, DOE applied a single uniform "gross margin percentage" across all efficiency levels, which assumes that manufacturers would be able to maintain the same amount of profit as a percentage of revenues at all efficiency levels within a product class.

As manufacturer production costs increase with efficiency, this scenario implies that the per-unit dollar profit will increase. DOE assumed a gross margin percentage of 31 percent for all product classes.^a Manufacturers tend to believe it is optimistic to assume that they would be able to maintain the same gross margin percentage as their production costs increase, particularly for

^a The gross margin percentage of 31 percent is based on manufacturer markups of 1.45 percent.

minimally efficient products. Therefore, this scenario represents a high bound of industry profitability under an energy conservation standard.

12.3.8.2 Preservation of Operating Profit Scenario

DOE also modeled the preservation of operating profit scenario to estimate a lower bound of profitability for the industry. Under this scenario, as the cost of production goes up under a standards case, manufacturers are generally required to reduce their manufacturer markups to a level that maintains base-case operating profit. DOE implemented this scenario in the GRIM by lowering the manufacturer markups at each TSL to yield approximately the same earnings before interest and taxes in the standards case as in the no-new-standards case in the year after the expected compliance date of the amended standards. The implicit assumption behind this scenario is that the industry can only maintain its operating profit in absolute dollars after the standard takes effect. Table 12.3.8 lists the calibrated manufacturer markups used in the preservation of operating profit scenario.

Table 12.3.8 Preservation of Operating Profit Manufacturer Markups by TSL

	Efficiency Level	No-New-STDs	TSL 1	TSL 2	TSL 3*	TSL 4	TSL 5
10 ≤ PM_{2.5} CADR < 100	Baseline	1.450					
	EL 1	1.450	1.448				
	EL 2	1.450	1.450	1.443	1.442		
	EL 3	1.450	1.450	1.450	1.450	1.416	
	EL 4	1.450	1.450	1.450	1.450	1.450	1.399
100 ≤ PM_{2.5} CADR < 150	Baseline	1.450					
	EL 1	1.450	1.448				
	EL 2	1.450	1.450	1.444	1.444		
	EL 3	1.450	1.450	1.450	1.450	1.428	
	EL 4	1.450	1.450	1.450	1.450	1.450	1.411
PM_{2.5} CADR ≥ 150	Baseline	1.450					
	EL 1	1.450	1.449				
	EL 2	1.450	1.450	1.448	1.448		
	EL 3	1.450	1.450	1.450	1.450	1.443	
	EL 4	1.450	1.450	1.450	1.450	1.450	1.443

*TSL 3 reflects the manufacturer markup at Tier 2 of the Joint Proposal after the standard goes into effect.

12.4 INDUSTRY FINANCIAL IMPACTS

Using the inputs and scenarios described in the previous sections, DOE used the GRIM to estimate the financial impacts on the air cleaner industry. The MIA uses two key financial metrics: INPV and annual cash flows. The main results of the MIA are reported in this section.

12.4.1 Impacts on Industry Net Present Value

The INPV measures the industry value and is used in the MIA to compare the economic impacts of different TSLs. The INPV is the sum of all net cash flows discounted at the industry's cost of capital or discount rate. The GRIM for this rulemaking estimates cash flows from 2023 to 2057. This timeframe models both the short-term impacts on the industry from the announcement of the standard until the compliance date, and a long-term assessment over the 30-year analysis period immediately thereafter.

In the MIA, DOE compares the INPV in the no-new-standards-case to the INPVs that result at each TSL in the standards case. The difference between these estimates represents the economic impacts implementing a particular TSL would have on the industry. For the air cleaner industry, DOE examined the two scenarios described in section 12.4.9: the preservation of gross margin percentage scenario and the preservation of operating profit. DOE's estimates of INPV for the full analysis period (2023–2057) for the no-new-standards case and at each TSL in the standards case are presented in Table 12.4.1 and Table 12.4.2 below.

Table 12.4.1 MIA Results – Preservation of Gross Margin Percentage Scenario

	Units	No-New-Standards Case	Trial Standard Level				
			1	2	3*	4	5
INPV	<i>2021\$ millions</i>	1,565.9	1,535.7	1,528.0	1,525.2	1,535.8	1,574.0
Change in INPV	<i>2021\$ millions</i>	-	(30.2)	(37.9)	(40.7)	(30.2)	8.1
	%	-	(1.9)	(2.4)	(2.6)	(1.9)	0.5

*TSL 3 represents the standards case presented in the Joint Proposal which corresponds to a two-tiered approach. Conversion costs reflect the sum of Tier 1 and Tier 2 standards.

Table 12.4.2 MIA Results – Preservation of Operating Profit Scenario

	Units	No-New-Standards Case	Trial Standard Level				
			1	2	3*	4	5
INPV	<i>2021\$ millions</i>	1,565.9	1,528.3	1,503.5	1,499.2	1,422.3	1,394.4
Change in INPV	<i>2021\$ millions</i>	-	(37.7)	(62.4)	(66.7)	(143.7)	(171.5)
	%	-	(2.4)	(4.0)	(4.3)	(9.2)	(11.0)

*TSL 3 represents the standards case presented in the Joint Proposal which corresponds to a two-tiered approach. Conversion costs reflect the sum of Tier 1 and Tier 2 standards.

12.4.2 Impacts on Annual Cash Flow

While INPV is useful for evaluating the long-term effects of energy conservation standards, short-term changes in cash flow are also important indicators of the industry's financial situation. For example, a large investment over one or two years could strain the industry's capital reserves and cash flow. Consequently, the sharp drop in financial performance

could cause investors to flee, even if recovery is possible. Thus, a short-term disturbance can have long-term effects that the INPV cannot capture.

Between 2023 and the 2028 compliance date, cash flows are driven by the level of conversion costs and the portion of these investments made each year. After the standard announcement date (*i.e.*, the publication date of the final rule), industry cash flows begin to decline as companies use their financial resources to prepare for the energy conservation standard. Generally, the more stringent the energy conservation standard, the greater the impact on industry cash flows in the years leading up to the compliance date, as product conversion costs lower cash flows from operations and capital conversion costs increase outlays of cash for capital expenditures.

Free cash flow in the year the energy conservation standards take effect is driven by two competing factors. In addition to capital and product conversion costs, energy conservation standards could create stranded assets, *i.e.*, the residual un-depreciated value of tooling and equipment that would have enjoyed longer use if the energy conservation standard had not made them obsolete. In this year, manufacturers write down the remaining book value of existing tooling and equipment, the value of which is affected by the energy conservation standards. This one time write down acts as a tax shield that mitigates decreases in cash flow from operations in the year of the write-down. In this year, there is also an increase in working capital that reduces cash flow from operations. A large increase in working capital can be attributed to more costly production components and materials, higher inventory carrying to sell more expensive products, and higher accounts receivable for more expensive products. Depending on these two competing factors, cash flow can either be positively or negatively affected in the year the standard takes effect.

Table 12.4.3 presents free cash flow impacts in the year before the standard takes effect. While free cash flows vary over the course of the analysis period depending on the manufacturer markup scenario analyzed, they do not vary by manufacturer markup scenario in the years prior to the standard, as a shift in product mix and markup structure triggered by a standard has not yet taken effect.

Table 12.4.3 Industry Free Cash Flow Impacts in the Year before Compliance (2027)

	Units	No-New-STDs Case	Trial Standard Level				
			1	2	3	4	5
Free Cash Flow (2027)	2021\$ MM	53.8	42.1	30.9	40.1	(2.4)	(6.0)
Change in Free Cash Flow	2021\$ MM	-	(11.74)	(22.91)	(9.86)	(56.23)	(59.83)
Change in Free Cash Flow	%	-	(21.8)	(-42.6)	(19.7)	(104.5)	(111.2)

*Values in parentheses indicate negative numbers.

12.5 IMPACTS ON SMALL BUSINESS MANUFACTURERS

For manufacturers of air cleaners, the SBA has set a size threshold, which defines those entities classified as “small businesses” for the purposes of the statute. DOE used the SBA’s

small business size standards to determine whether any small entities would be subject to the requirements of the rule. (See 13 CFR part 121.) The size standards are listed by NAICS code and industry description and are available at www.sba.gov/document/support--table-size-standards. Manufacturing of air cleaners is classified under NAICS code 335210, “Small Electrical Appliance Manufacturing.” In 13 CFR 121.201, the SBA sets a threshold of 1,500 employees or fewer for an entity to be considered as a small business for this category.

As discussed in section 12.2.3.3, DOE reviewed AHAM’s Database of Certified Room Air Cleaners, and ENERGY STAR’s Product Finder dataset, and California Air Resources Board’s (CARB) CARB-Certified Air Cleaning Devices to create a list of OEMs that manufacture the products covered by this proposal. DOE relied on public data and subscription-based market research tools (e.g., Dun & Bradstreet) to determine company location, headcount, and annual revenue. DOE screened out companies that do not offer products covered by this rulemaking, do not meet SBA’s definition of a “small business,” or are foreign-owned and operated. DOE identified 43 OEMs that sell air cleaners in the United States. Of the 43 OEMs identified, DOE tentatively determined four are small, domestic companies.

The four companies had limited technical specifications available in their public documents. However, in some cases, DOE was able to determine likely product performance based on the available specifications, component information, and filter design.

For the first small business, DOE believes the company’s range of products are likely within the scope of the test procedure and subject to the energy conservation standard. These products would meet Tier 2 levels based on the available design information. The second small business has two models that are likely within the scope of the test procedure and subject to the energy conservation standard. Again, DOE has reviewed the publicly available information and determined that both models would likely meet Tier 2 levels.

DOE determined that the third small business has two models that are within the scope of the test procedure and subject to the energy conservation standard. DOE suspects these two models would likely meet Tier 1 but not Tier 2 standards. DOE determined the fourth small business likely has five models that are within the scope of the test procedure and subject to the energy conservation standard. Based on the product specifications, three of those models may need redesign to meet Tier 2 standards.

To meet the required efficiencies, DOE estimated conversion costs for the third small business by using model counts to scale the industry conversion costs. The third small business accounts for 0.1 percent of models on the market that DOE identified. Based on a review of publicly available information, DOE believes the third small business utilizes soft tooling and flexible manufacturing techniques for production. Therefore, DOE anticipates this small manufacturer would have limited capital expenditures. To be conservative, DOE assumes this small manufacturer accounts for 0.1 percent of industry capital conversion costs at TSL 3, totaling \$10,350. Product conversion costs may be necessary for developing, qualifying, sourcing, and testing new components. To be conservative, DOE assumed the manufacturer would incur 1 percent of industry product conversion costs. DOE estimates that the third small business may incur \$10,350 in capital conversion costs and \$18,000 in product conversion costs to meet Tier 2 standards for those two models. Based on subscription-based market research

reports, the third small business has an annual revenue of approximately \$1.31 million. The total conversion costs of \$28,350 are approximately 0.7 percent of the third small business’s revenue over the 3-year conversion period.

Based on a review of publicly available information, DOE estimated conversion costs for the fourth small business by using model counts to scale the industry conversion costs. The fourth small business accounts for 0.4 percent of models on the market that DOE identified. To be conservative, DOE assumed 1 percent of industry capital conversion costs and 1 percent of industry product conversion costs for the relevant product classes at TSL 3 would be attributable to this small business. The conversion costs total \$121,500. Based on subscription-based market research reports, the fourth small business has an annual revenue of approximately \$272.64 million. The total conversion costs of \$121,500 are approximately 0.01 percent of the fourth small business’s revenue over the 3-year conversion period.

12.6 OTHER IMPACTS

12.6.1 Direct Employment

DOE quantitatively assessed the impacts of energy conservation standards on direct employment in the air cleaner industry. DOE used the GRIM to estimate the domestic labor expenditures and number of domestic production and non-production workers in the no-new-standards case and at each TSL from 2023 to 2057. DOE used statistical data from the U.S. Census Bureau’s 2020 *ASM*,³ the U.S. Bureau of Labor Statistics’ (BLS) employee compensation data,¹⁴ the results of the engineering analysis, and interviews with manufacturers to determine the inputs necessary to calculate industry-wide labor expenditures and domestic employment levels. Labor expenditures involved with the manufacturing of the product are a function of the labor intensity of the product, the sales volume, and an assumption that wages remain fixed in real terms over time. The total labor expenditures in each year are calculated by multiplying the total MPCs by the labor percentage of MPCs.

$$\text{Total Labor Expenditures} = \text{Total MPCs} \times \text{Labor \% of MPCs}$$

The total labor expenditures in the GRIM were then converted to production employment levels by dividing production labor expenditures by the average annual wages of a production worker. To do this, DOE relied on the *ASM* inputs: Production Workers Annual Wages, Production Workers Annual Hours, Production Workers for Pay Period (*i.e.*, the number of production workers), and Number of Employees. DOE also relied on the BLS employee compensation data to determine the fringe benefits multiplier. The fringe benefits multiplier factors in paid leave, supplemental pay, insurance, retirement and savings, and legally required benefits.

$$\text{Total Production Employees} = \frac{\text{Total Labor Expenditures}}{(\text{Production Worker Avg. Annual Wage} \times \text{Fringe Benefits Multiplier})}$$

Where:

$$\text{Production Worker Average Annual Wage} = \frac{\text{Production Workers Annual Wages}}{\text{Number of Production Workers}}$$

Fringe Benefits Multiplier = 1.43

Total Production Employees is then multiplied by the U.S. labor percentage value to convert total production employment to total domestic production employment.

Total Domestic Production Employees = Total Production Employees × U.S. Labor Percentage

The U.S. labor percentage represents the industry fraction of domestic manufacturing production capacity for the covered product. This value is derived from manufacturer interviews, product database analysis, and publicly available information. DOE estimates that approximately 78 percent of standard-size air cleaners are produced domestically. DOE estimates that no compact-size air cleaners are produced domestically. Therefore, overall, DOE estimates that approximately 76 percent of all covered air cleaners sold in the United States are produced domestically.

The domestic production employees estimate covers production line workers, including line supervisors, who are directly involved in fabricating and assembling products within the OEM facility. Workers performing services that are closely associated with production operations, such as materials handling tasks using forklifts, are also included as production labor.^b DOE's estimates only account for production workers who manufacture the specific equipment covered by this rulemaking.

Non-production workers account for the remainder of the direct industry employees affected by this rulemaking. The estimate of domestic non-production employees covers workers who are not directly involved in the production process such as: sales, engineering, human resources, and management.^c Using the number of domestic production workers calculated above, non-production domestic employees are extrapolated by multiplying the ratio of non-production workers in the industry compared to production employees. DOE assumes that this employee distribution ratio remains constant between the no-new-standards case and standards cases.

$$\begin{aligned} \text{Domestic Non Production Employees} \\ &= \text{Domestic Production Employees} \\ &\times \left(\frac{(\text{Number of Employees} - \text{Number of Production Workers})}{\text{Number of Production Workers}} \right) \end{aligned}$$

The direct employment impacts shown in Table 12.6.1 represent the potential domestic employment changes that could result following the compliance date for standard-size air cleaner

^b According to the “Definitions and Instructions for the *Annual Survey of Manufactures*, MA-10000,” (2021) production workers include “Workers (direct labor including first-line supervisors) engaged in fabricating, processing, assembling, inspecting, receiving, packing, warehousing, shipping (but not delivering), maintenance, repair, janitorial, guard services, product development, auxiliary production for plant’s own use (e.g., power plant), record keeping, delivering of ready-mixed concrete, and other closely associated services” Available at: www2.census.gov/programs-surveys/asm/technical-documentation/questionnaire/2021/instructions/MA_10000_Instructions.pdf

^c Non-production workers are defined as, “Supervision above line-supervisor level, sales employees, including driver salesperson, sales delivery, include truck drivers and helpers, advertising, clerical, credit, collection, purchasing, finance, legal, executive, and technical employees, and employees installing and servicing the establishments products.” *Id.*

industry. As previously noted, DOE did not identify any U.S. manufacturing facilities producing compact-size air cleaners for the domestic market, and therefore does not present a range of direct employment impacts. The upper bound estimate corresponds to an increase in the number of domestic workers that would result from energy conservation standards if manufacturers continue to produce the same scope of covered products within the United States after standards take effect. The lower bound estimate represents the maximum decrease in production workers if manufacturing moved to lower labor-cost countries or if companies shifted to importing finished goods (versus manufacturing domestically in-house). At lower TSLs (*i.e.*, TSL 1 through TSL 4), DOE believes the likelihood of changes in production location due to standards are low due to the relatively minor production line updates required. However, at max-tech, both the complexity and cost of production facility updates increases, manufacturers are more likely to revisit their production location. At max-tech, one manufacturer representing a large portion of the market noted concerns about the level of investment and indicated the potential need to relocate production lines to remain competitive.

The domestic employment impacts shown are independent of the employment impacts from the broader U.S. economy, documented in chapter 16 of the NOPR TSD.

DOE estimates that in the absence of energy conservation standards, there would be approximately 58 domestic workers involved in manufacturing standard-size air cleaners in 2028. Table 12.6.1 presents the range of potential impacts of amended energy conservation standards on U.S. workers in the air cleaner industry.

Table 12.6.1 Domestic Direct Employment Impacts in 2028

	No-New-Standards Case	Trial Standard Level				
		1	2	3**	4	5
Domestic Production Workers in 2028	58	59	59	59	59	59
Domestic Non-Production Workers in 2028	25	26	26	26	26	26
Total Direct Employment in 2028	83	85	85	85	85	85
Potential Changes in Total Direct Employment in 2028	-	(58) to 1	(58) to 1	(58) to 1	(58) to 1	(58) to 1

*DOE presents a range of potential employment impacts. Numbers in parentheses indicate negative numbers.

**For TSL 3, Tier 2 standard goes into effect in 2026. DOE presents 2028 Direct Employment for consistent comparison in this table.

The direct employment impacts shown in Table 12.6.1 represent the potential domestic employment changes that could result following the compliance date of the air cleaner standards considered. The upper bound estimate corresponds to an increase in the number of domestic workers that would result from energy conservation standards if manufacturers continue to produce the same scope of covered equipment within the United States after compliance takes

effect. The lower bound estimate represents the maximum decrease in production workers if manufacturing moved to lower labor-cost countries. Most manufacturers currently produce their air cleaners in countries with lower labor costs.

Of the 300 air cleaner brands DOE identified, the vast majority are produced outside of the U.S. DOE identified 4 companies that have U.S. manufacturing. These companies have distinct designs and manufacturing processes from companies that import air cleaners. DOE found these companies largely do not rely on injection molding, the production process that drives capital expenditures resulting from the standard. Additionally, DOE found many of these companies focus on air cleaners for commercial applications. These companies leverage design and production processes used for their commercial air cleaner models to offer conventional air cleaners. Additionally, when product literature with technical detail were available, DOE found that most conventional air cleaners from these domestic manufacturers would likely meet standards for TSLs 1, 2, and 3. DOE concludes it is unlikely these companies would relocate production overseas solely due to the adoption of this final rule.

Additionally, the employment impacts discussed in this section are independent of the employment impacts from the broader U.S. economy, which are documented in chapter 16 of the direct final rule TSD.

12.6.2 Production Capacity

Implementing the different design options analyzed for this Final Rule would require varying levels of resources and investment. DOE did not observe any design options at the adopted level that would require changes to the fundamental construction or manufacturing of air cleaners. Generally, DOE observed incremental increases in cabinet dimension, incremental changes in filter volume and dimension, and improved motors or optimized motor/filter relationship in the more efficient products meeting the adopted level. Changes in cabinet and filter dimensions could require tooling adjustments and replacement, which DOE accounted for in its analysis of conversion costs. However, DOE's analysis does not suggest there would be design changes that could lead to insufficient availability of product to meet market demand.

12.6.3 Cumulative Regulatory Burden

While any one regulation may not impose a significant burden on manufacturers, the combined effects of existing and impending regulations may have serious consequences for some manufacturers, subgroups of manufacturers, or an entire industry. Assessing the impact of a single regulation may overlook this cumulative regulatory burden. In addition to energy conservation standards, other regulations can significantly affect manufacturers' financial operations. Multiple regulations affecting the same manufacturer can strain profits and lead companies to abandon product lines or markets with lower expected future returns than competing products. For these reasons, DOE conducts an analysis of cumulative regulatory burden as part of its rulemakings pertaining to appliance efficiency.

DOE evaluates product-specific regulations that will take effect approximately 5 years before or after the estimated 2028 compliance date of any energy conservation standards for air cleaners. This information is presented in Table 12.6.2.

Table 12.6.2 Compliance Dates and Expected Conversion Expenses of Federal Energy Conservation Standards Affecting Air Cleaner Original Equipment Manufacturers

Federal Energy Conservation Standard	Number of OEMs*	Number of OEMs Affected from Today's Rule**	Approx. Standards Year	Industry Conversion Costs (Millions \$)	Industry Conversion Costs / Product Revenue***
Residential Central Air Conditioners and Heat Pumps 82 FR 1786 (January 6, 2017)	30	1	2023	\$342.6 (2015\$)	0.50%
Portable Air Conditioners 85 FR 1378 (January 10, 2020)	11	1	2025	\$320.90 (2015\$)	6.70%
Room Air Conditioners [†]	8	1	2026	\$24.80 (2021\$)	0.44%

* This column presents the total number of manufacturers identified in the energy conservation standard rule contributing to cumulative regulatory burden.

** This column presents the number of manufacturers producing air cleaners that are also listed as manufacturers in the listed energy conservation standard contributing to cumulative regulatory burden.

*** This column presents industry conversion costs as a percentage of product revenue during the conversion period. Industry conversion costs are the upfront investments manufacturers must make to sell compliant products/equipment. The revenue used for this calculation is the revenue from just the covered product/equipment associated with each row. The conversion period is the time frame over which conversion costs are made and lasts from the publication year of the final rule to the compliance year of the energy conservation standard. The conversion period typically ranges from 3 to 5 years, depending on the rulemaking.

[†] At the time of issuance of this consumer air cleaners direct final rule, the room air conditioner final rule has been issued and is pending publication in the Federal Register. Once published, the room air conditioners final rule will be available at: <https://www.regulations.gov/docket/EERE-2014-BT-STD-0059>.

12.7 CONCLUSION

This section summarizes the likely range of financial impacts air cleaner manufacturers will experience as a result of energy conservation standards. DOE also notes that while these scenarios bound the range of most plausible impacts on manufacturers, circumstances could potentially cause manufacturers to experience impacts outside of this range.

Table 12.7.1 summarizes the overall INPV impacts and conversion costs projected to result from each of the TSLs analyzed. Table 12.8.2 summarizes the INPV impacts and conversion costs for the standard-size air cleaner industry. Table 12.8.3 summarizes the INPV impacts and conversion costs for the compact-size air cleaner industry.

Table 12.7.1 Manufacturer Impact Analysis Results

	Units	No-New-Standards Case	Trial Standard Level				
			1	2	3*	4	5
INPV	<i>2021\$ millions</i>	1,565.9	1,535.7	1,528.0	1,525.2	1,535.8	1,574.0
Change in INPV	<i>2021\$ millions</i>	-	(30.2)	(37.9)	(40.7)	(30.2)	8.1
	%	-	(1.9)	(2.4)	(2.6)	(1.9)	0.5
Free Cash Flow (2027)	<i>2021\$ millions</i>	53.8	42.1	30.9	20.8 and 40.1 **	(2.4)	(6.0)
Change in Free Cash Flow (2027)	%	-	(21.8)	(42.6)	(55.7) and (19.7) **	(104.5)	(111.2)
Product Conversion Costs	<i>2021\$ millions</i>	-	17.2	23.2	23.2	42.4	44.7
Capital Conversion Costs	<i>2021\$ millions</i>	-	13.6	34.1	34.1	94.1	100.5
Total Conversion Costs	<i>2021\$ millions</i>	-	30.8	57.3	57.3	136.6	145.2

*TSL 3 represents the standards case presented in the Joint Proposal which corresponds to a two-tiered approach. Conversion costs reflect the sum of Tier 1 and Tier 2 standards.

**The Free Cash Flow and % Change in Free Cash Flow for TSL 3 is presented to the years 2023 and 2025 due to the 2-step structure of the Joint Proposal. DOE presents FCF in the year before the standard years.

TSL 1 corresponds to EL 1 for all product classes and are equivalent to the standards established by the States of Maryland, Nevada, and New Jersey, and the District of Columbia. At TSL 1, DOE estimates that impacts on INPV will range from -\$30.2 million to -\$37.7 million, or a change in INPV of -2.4 to -1.9 percent. At TSL 1, industry free cash-flow is \$42.1 million, which is a decrease of approximately \$11.7 million compared to the no-new-standards case value of \$53.8 million in 2027, the year leading up to the proposed standards. Currently, approximately 36 percent of shipments meet the efficiencies required.

DOE noted in the engineering analysis, section IV.C.3, the efficiency improvements at EL 1 are achievable by optimizing the fan motor-filter relationship. In evaluating the design paths for optimization, DOE noted that increasing the surface area of the filter would improve test performance but could also require changes to the injection molded component of air cleaners. DOE estimated capital conversion costs based on the costs for manufacturer to purchase new injection mold dies to accommodate filters with greater surface area. Manufacturers using soft tooling or that do not rely on injection molding would have lower capital conversion costs than modeled by DOE. DOE estimated the product conversion costs for testing all models, identifying product that would not meet the standard, and redesigning that portion of market offerings. DOE estimates capital conversion costs of \$13.6 million and product conversion costs of \$17.2 million for the industry. Conversion costs total \$30.8 million.

At TSL 1, the shipment-weighted average MPC for all air cleaners is expected to increase by 1 percent relative to the no-new-standards case shipment-weighted average MPC for all air cleaners in 2028. Given this relatively small increase in production costs, DOE does not project a notable drop in shipments in the year the standard takes effect. In the preservation of gross margin percentage scenario, the slight increase in MSP is outweighed by the \$30.8 million in conversion costs, causing a negative change in INPV at TSL 1 under this scenario. Under the preservation of operating profit scenario, the reduction in the manufacturer markup and the \$30.8 million in conversion costs incurred by manufacturers cause a slightly negative change in INPV.

At TSL 2, the standard corresponds to EL 2 for all product classes and to current ENERGY STAR V.2.0 efficiency levels for air cleaners in all product classes. DOE estimates that impacts on INPV will range from -\$62.4 million to -\$37.9 million, or a change in INPV of -4.0 to -2.4 percent. At TSL 2, industry free cash-flow is \$30.9 million, which is a decrease of approximately \$22.9 million compared to the no-new-standards case value of \$53.8 million in 2027, the year leading up to the proposed standards. Currently, approximately 31 percent of shipments meet the efficiencies required.

A sizeable portion of the market, approximately 31 percent, can currently meet the TSL 2 level. Additionally, a substantial portion of existing models can be updated to meet TSL 2 through optimization and improved components rather than a full product redesign. In particular, manufacturers may be able to leverage their existing cabinet designs. However, the product interior may require updates to accommodate more efficient motors and larger filters. Some manufacturers may be able to alter existing tooling to accommodate minor changes in internal dimensions. To avoid underestimating costs to industry, DOE estimated capital conversion costs based on the cost to replace tooling – specifically injection molding dies. Also, DOE estimated the product conversion costs for testing all models, identifying product that would not meet the standard, and redesigning that portion of market offerings. Capital conversion costs may reach \$34.1 million and product conversion costs may reach \$23.2 million for the industry. Conversion costs total \$57.3 million.

At TSL 2, the shipment-weighted average MPC for all air cleaners is expected to increase by 2 percent relative to the no-new-standards case shipment-weighted average MPC for all air cleaners in 2028. Given the relatively small increase in production costs, DOE does not project a notable drop in shipments in the year the standard takes effect. In the preservation of gross margin percentage scenario, the slight increase in MSP is outweighed by the \$57.3 million in conversion costs, causing a negative change in INPV at TSL 2 under this scenario. Under the preservation of operating profit scenario, the manufacturer markup decreases in 2029, the year after the analyzed compliance year. This reduction in the manufacturer markup and the \$57.3 million in conversion costs incurred by manufacturers cause a negative change in INPV at TSL 2 under the preservation of operating profit scenario.

At TSL 3, DOE estimates that impacts on INPV will range from -\$66.7 million to -\$40.7 million, or a change in INPV of -4.3 to -2.6 percent. At TSL 3, industry free cash-flow is \$40.1 million in 2027, which is a decrease of approximately \$9.9 million compared to the no-new-standards case value of \$53.8 million in 2027, the year leading up to the proposed standards. Currently, approximately 31 percent of shipments meet the efficiencies required.

For TSL 3, DOE analyzed the standards case presented in the Joint Proposal which corresponds to a two-tier approach of the lowest efficiency level (EL 1) for Tier 1 standards (going to effect in 2024) and the current ENERGY STAR V.2.0 efficiency level (EL 2) for Tier 2 standards (going to effect in 2026) for all the product classes. The industry impacts at TSL 3 are very similar to the impacts at TSL 2 because both scenarios result in standards at the Tier 2 level. However, TSL 3 is a two-tier standard with earlier compliance dates. While conversion costs for TSL 3 and TSL 2 are identical, the timing of the costs are different. As a result, the earlier timing of conversion costs result in lower INPV values at TSL 3 than at TSL 2. However, industry may benefit from a national standard at Tier 1 in the 2024 timeframe in the form of potential reductions in stock keeping units (SKUs), marketing and sales complexity, and reduced consumer confusion associated with a patchwork of state-level energy performance standards for air cleaners. The MIA does not attempt to calculate the cost savings from industry that results from single national standard.

At TSL 3, the shipment-weighted average MPC for all air cleaners is expected to increase by 2 percent relative to the no-new-standards case shipment-weighted average MPC for all air cleaners in 2028. Given the relatively small increase in production costs, DOE does not project a notable drop in shipments in the year the standard takes effect. In the preservation of gross margin percentage scenario, the increase in MSP is outweighed by the \$57.3 million in conversion costs, causing a negative change in INPV at TSL 3 under this scenario. Under the preservation of operating profit scenario, the manufacturer markup decreases in 2029, the year after the analyzed compliance year. This reduction in the manufacturer markup and the \$57.3 million in conversion costs incurred by manufacturers cause a negative change in INPV at TSL 3 under the preservation of operating profit scenario.

TSL 4 corresponds to EL 3 for all product classes. At TSL 4, DOE estimates that impacts on INPV will range from -\$143.7 million to -\$30.2 million, or a change in INPV of -9.2 to -1.9 percent. At TSL 4, industry free cash-flow is -\$2.4 million, which is a decrease of approximately \$56.2 million compared to the no-new-standards case value of \$53.8 million in 2027, the year leading up to the proposed standards. Currently, approximately six percent of shipments meet the efficiencies required.

At TSL 4, all three product classes would likely incorporate cylindrical shaped filters and BLDC motors without an optimized motor-filter relationship. The cylindrical filter, which reduces the pressure drop across the filter because it allows for a larger surface area for the same volume of filter material, provides the improvement in efficiency at TSL 4 compared to TSL 3, which utilizes rectangular shaped filters. However, most models on the market today do not use BLDC motors and cannot accommodate cylindrical filters. Manufacturers would incur conversion costs to redesign the product to incorporate a different filter shape and more efficient components. Additionally, manufacturers that own tooling would incur conversion costs for updated cabinet designs. DOE estimates capital conversion costs of \$94.1 million and product conversion of costs of \$42.4 million. Conversion costs total \$136.6 million.

At TSL 4, the shipment-weighted average MPC for all air cleaners is expected to increase by 8 percent relative to the no-new-standards case shipment-weighted average MPC for all air cleaners in 2028. Given the projected increase in production costs, DOE expects an estimated 4 percent drop in shipments in the year the standard takes effect. In the preservation of gross

margin percentage scenario, the increase in MSP is outweighed by the \$136.6 million in conversion costs, causing a negative change in INPV at TSL 4 under this scenario. Under the preservation of operating profit scenario, the manufacturer markup decreases in 2029, the year after the analyzed compliance year. This reduction in the manufacturer markup and the \$136.6 million in conversion costs incurred by manufacturers cause a negative change in INPV at TSL 4 under the preservation of operating profit scenario.

TSL 5 corresponds to maximum technology options (EL 4) for all product classes. At TSL 5, DOE estimates that impacts on INPV will range from -\$171.5 million to \$8.1 million, or a change in INPV of -11.0 to 0.5 percent. At TSL 5, industry free cash-flow is -\$6.0 million, which is a decrease of approximately \$59.8 million compared to the no-new-standards case value of \$53.8 million in 2027, the year leading up to the proposed standards. Currently, approximately three percent of shipments meet the efficiencies required.

At TSL 5, DOE's expected design path for TSL 5 incorporates cylindrical shaped filters and BLDC motors with an optimized motor-filter relationship. As noted for TSL 4, the adoption of cylindrical filters would necessitate platform level redesign for most products on the market. Additionally, the move to cylindrical filters could necessitate significantly different cabinet designs. DOE estimates capital conversion costs of \$100.5 million and product conversion of costs of \$44.7 million. Conversion costs total \$145.2 million.

At TSL 5, the shipment-weighted average MPC for all air cleaners is expected to increase by 13 percent relative to the no-new-standards case shipment-weighted average MPC for all air cleaners in 2028. Given the projected increase in production costs, DOE expects an estimated 6 percent drop in shipments in the year the standard takes effect. In the preservation of gross margin percentage scenario, INPV remains roughly the same as in the no-new-standards scenario. Under the preservation of operating profit scenario, reduction in the manufacturer markup, reduction in shipments, and the \$145.2 million in conversion costs incurred by manufacturers cause a negative change in INPV at TSL 5.

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CHAPTER 13. EMISSIONS IMPACT ANALYSIS

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CHAPTER 13. EMISSIONS IMPACT ANALYSIS

13.1 INTRODUCTION

The emissions analysis consists of two components. The first component estimates the effect of potential energy conservation standards on power sector emissions and, if present, site combustion emissions, of carbon dioxide (CO₂), nitrogen oxides (NO_x), sulfur dioxide (SO₂) and mercury (Hg). The second component estimates the impacts of potential standards on emissions of two additional greenhouse gases, methane (CH₄) and nitrous oxide (N₂O), as well as the impacts to emissions of all species due to “upstream” activities in the fuel production chain, which are included in accordance with DOE’s FFC Statement of Policy. 76 FR 51282 (Aug. 18, 2011). These upstream activities comprise extraction, processing, and transporting fuels to the site of combustion.

The analysis of power sector emissions of CO₂, NO_x, SO₂, and Hg uses emissions intensity factors intended to represent the marginal impacts of the change in electricity consumption associated with amended or new standards. The methodology is based on results published for the *Annual Energy Outlook (AEO)* prepared by the Energy Information Administration, including a set of side cases that implement a variety of efficiency-related policies. The methodology is described in appendix 13A in this TSD, and in the report “Utility Sector Impacts of Reduced Electricity Demand” (Coughlin, 2014; Coughlin 2019).^{1,2} The analysis presented in this chapter uses projections from *AEO 2022*.³

Power sector combustion emissions of CH₄ and N₂O are derived using Emission Factors for Greenhouse Gas Inventories published by the EPA, as are site combustion emissions of CO₂, CH₄ and N₂O.^a

The FFC upstream emissions are estimated based on the methodology described in appendix 10B and in Coughlin (2013).⁴ The upstream emissions include emissions from fuel combustion during extraction, processing, and transportation of fuels, and “fugitive” emissions (direct leakage to the atmosphere) of CH₄ and CO₂.

The emissions intensity factors are expressed in terms of physical units per MWh or MMBtu of site energy savings. Total emissions reductions are estimated by multiplying the emissions intensity factor by the energy savings calculated in the national impact analysis (chapter 10). The emissions factors used in the calculations are provided in appendix 13A. For power sector emissions, the factors depend on the sector and end use. The results presented here use factors for the power plant types that supply electricity for homes and commercial buildings in general.

Each annual version of the *AEO* incorporates the projected impacts of existing air quality regulations on emissions. The *AEO* generally represents current Federal and State legislation and final implementation regulations in place as at the time of its preparation. For details, see

^a https://www.epa.gov/sites/default/files/2021-04/documents/emission-factors_apr2021.pdf.

Summary of Legislation and Regulations Included in the *AEO 2022*, Appendix, Electric power sector.^b

13.2 EMISSIONS IMPACT RESULTS

Table 13.2.1 presents the estimated cumulative emissions reductions for the lifetime of products sold in 2024-2057 for each TSL. Negative values indicate that emissions increase.

Table 13.2.1 Cumulative Emissions Reduction for Potential Standards for Air Cleaners

	TSL				
	1	2	3	4	5
Power Sector Emissions					
CO ₂ (million metric tons)	22.3	50.8	53.4	119	135
CH ₄ (thousand tons)	1.62	3.69	3.90	8.63	9.79
N ₂ O (thousand tons)	0.225	0.513	0.542	1.20	1.36
NO _x (thousand tons)	10.8	24.6	25.9	57.4	65.1
SO ₂ (thousand tons)	9.86	22.5	23.9	52.6	59.6
Hg (tons)	0.065	0.148	0.154	0.345	0.392
Upstream Emissions					
CO ₂ (million metric tons)	1.81	4.13	4.31	9.65	10.9
CH ₄ (thousand tons)	171	391	407	914	1036
N ₂ O (thousand tons)	0.009	0.021	0.022	0.048	0.054
NO _x (thousand tons)	27.4	62.6	65.2	146	166
SO ₂ (thousand tons)	0.123	0.280	0.296	0.655	0.742
Hg (tons)	0.000	0.001	0.001	0.001	0.002
Total Emissions					
CO ₂ (million metric tons)	24.1	55.0	57.7	128	146
CH ₄ (thousand tons)	173	395	411	923	1046
N ₂ O (thousand tons)	0.234	0.533	0.563	1.25	1.41
NO _x (thousand tons)	38.2	87.2	91.2	204	231
SO ₂ (thousand tons)	10.0	22.8	24.2	53.2	60.4
Hg (tons)	0.065	0.148	0.155	0.347	0.393

Figure 13.3.1 through Figure 13.3.6 show the annual reductions for total emissions for each type of emission from each TSL. The reductions reflect the lifetime impacts of products sold in 2024-2057.

^b <https://www.eia.gov/outlooks/aeo/assumptions/pdf/summary.pdf>

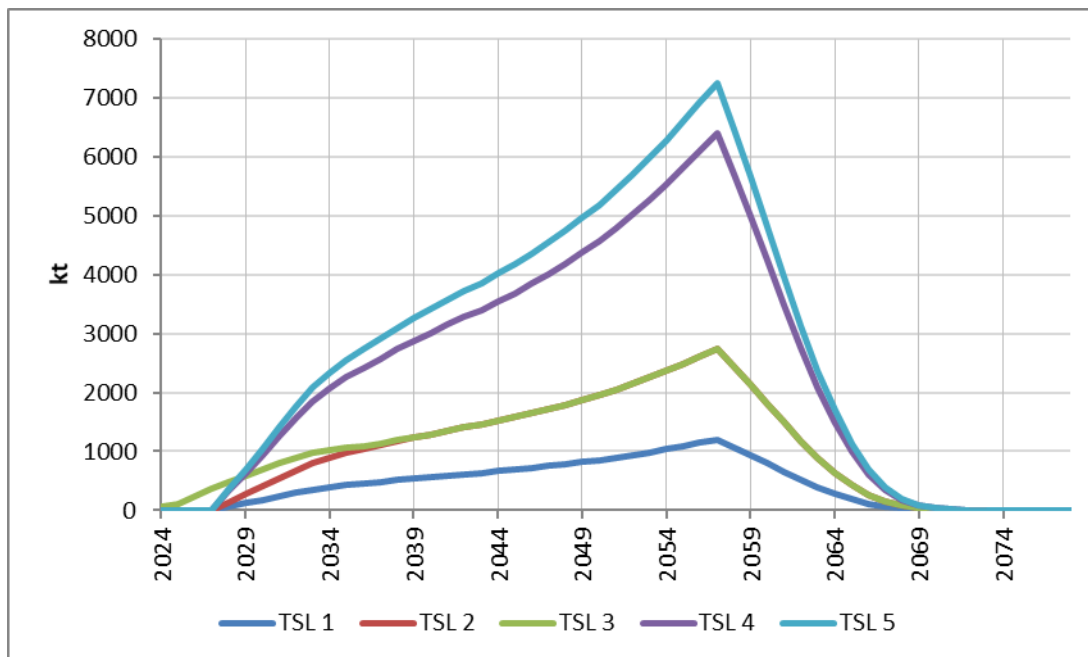


Figure 13.2.1 Air Cleaners: CO₂ Total Emissions Reduction

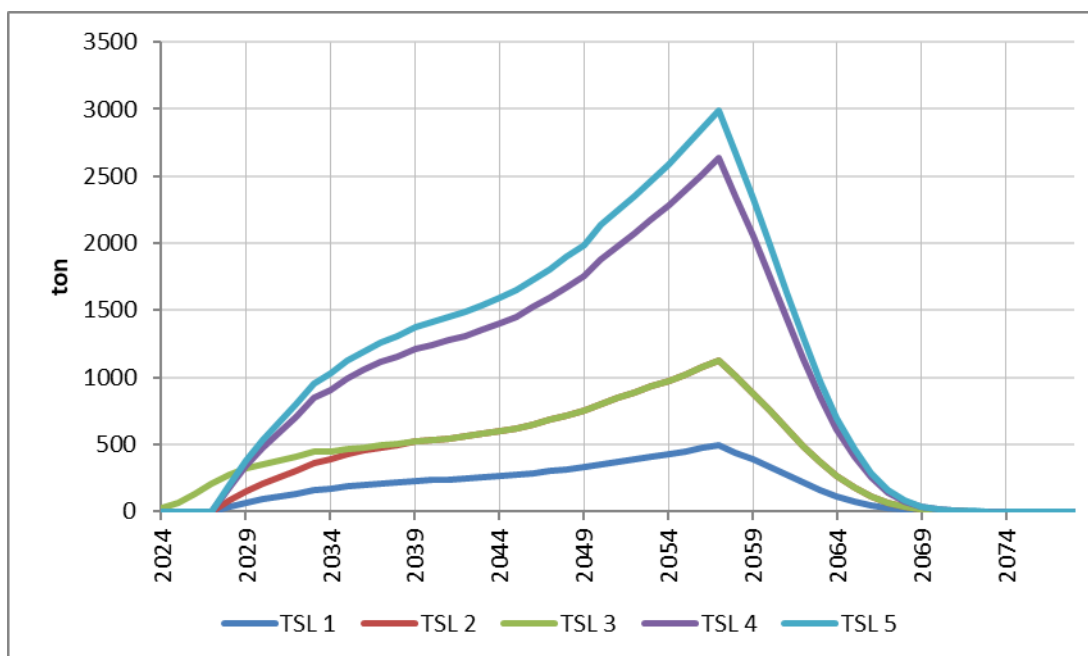


Figure 13.2.2 Air Cleaners: SO₂ Total Emissions Reduction

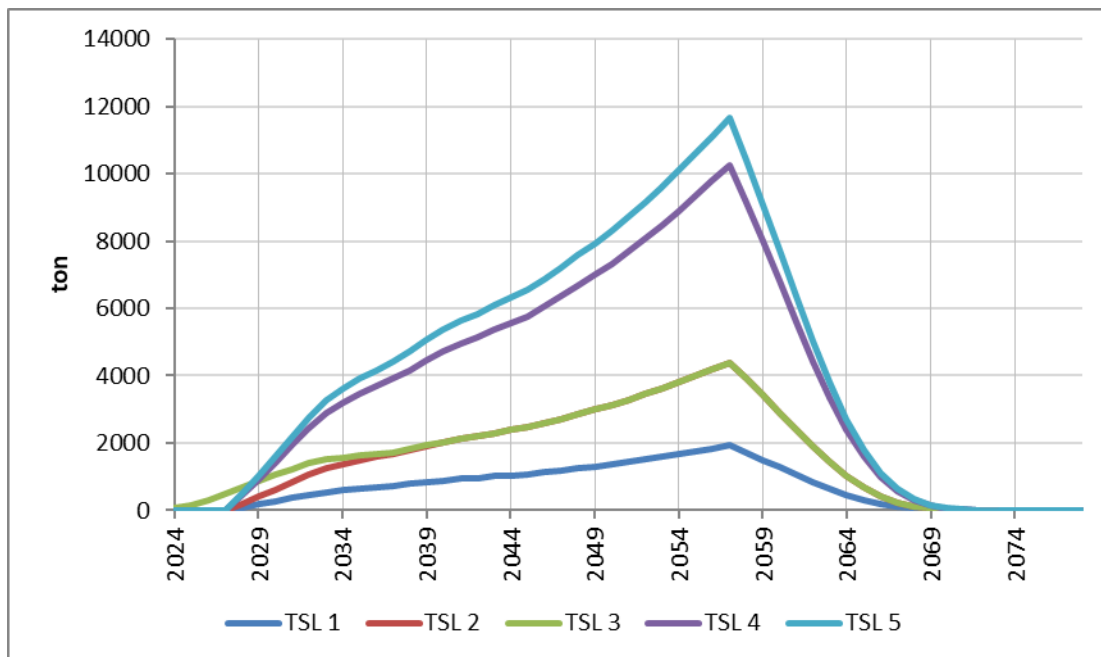


Figure 13.2.3 Air Cleaners: NO_x Total Emissions Reduction

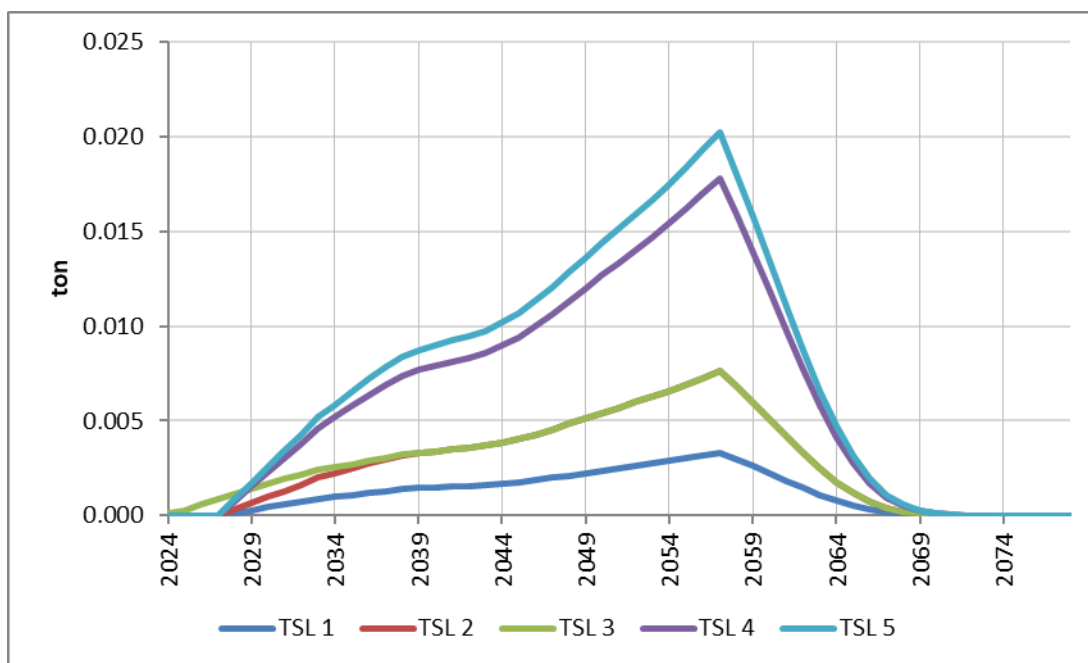


Figure 13.2.4 Air Cleaners: Hg Total Emissions Reduction

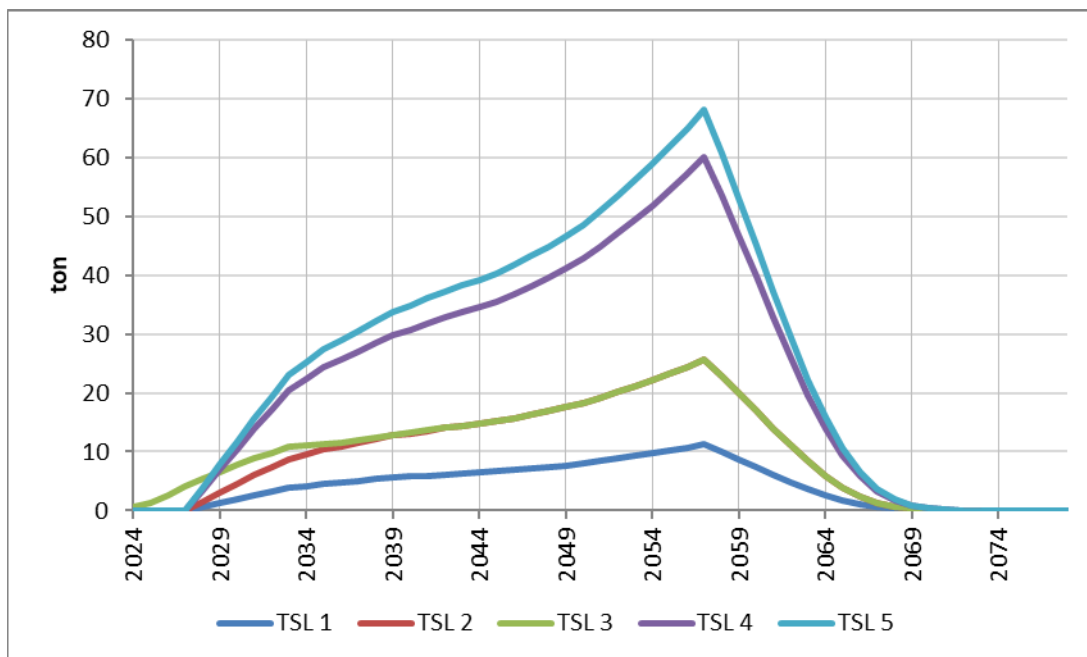


Figure 13.2.5 Air Cleaners: N₂O Total Emissions Reduction

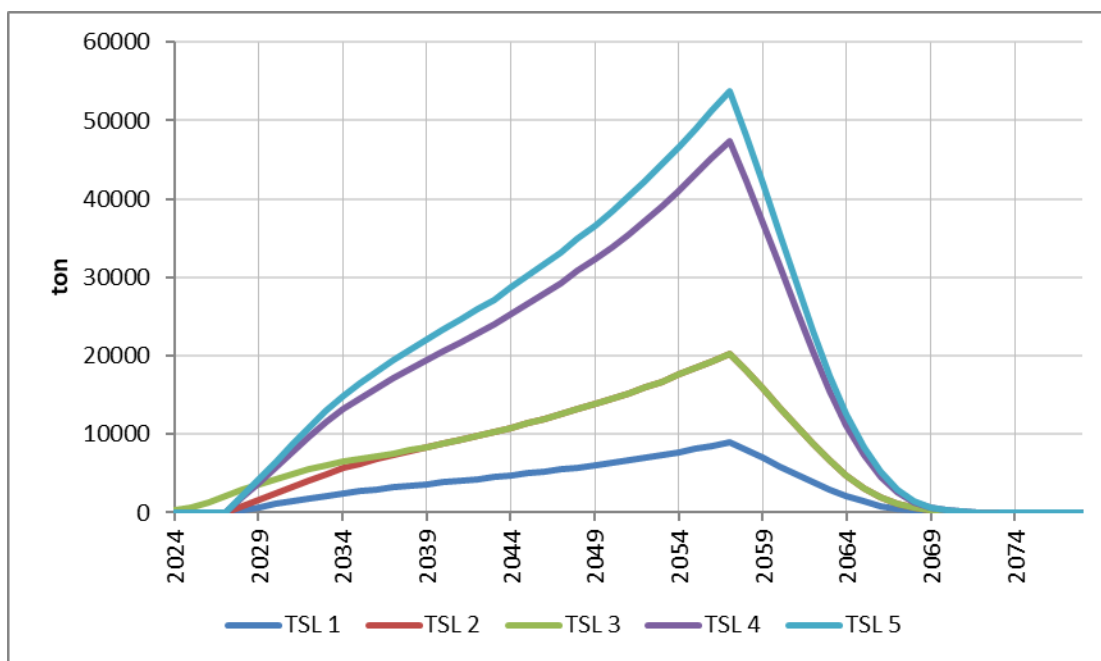


Figure 13.2.6 Air Cleaners: CH₄ Total Emissions Reduction

Table 13.2.2 displays annual emissions reductions from the selected standards (TSL 3).

Table 13.2.2 Emissions Reduction at Selected Standard Level (TSL 3)

Emissions Year	CO₂ (Million Metric tons)	CH₄ ('000 tons)	N₂O ('000 tons)	NO_x ('000 tons)	SO₂ ('000 tons)	Hg (tons)
2024	0.06	0.3	0.0006	0.1	0.03	0.00014
2025	0.11	0.6	0.0012	0.1	0.06	0.00026
2026	0.23	1.4	0.0026	0.3	0.13	0.00057
2027	0.36	2.1	0.0040	0.5	0.21	0.00087
2028	0.48	2.9	0.0054	0.7	0.28	0.00117
2029	0.59	3.6	0.0066	0.9	0.32	0.00144
2030	0.70	4.2	0.0078	1.0	0.35	0.00171
2031	0.80	4.9	0.0089	1.2	0.38	0.00194
2032	0.89	5.5	0.0098	1.4	0.41	0.00215
2033	0.98	6.1	0.0108	1.5	0.44	0.00243
2034	1.02	6.5	0.0110	1.6	0.45	0.00255
2035	1.06	6.8	0.0114	1.6	0.47	0.00271
2036	1.09	7.2	0.0116	1.7	0.48	0.00288
2037	1.14	7.6	0.0119	1.7	0.49	0.00306
2038	1.19	8.0	0.0124	1.8	0.50	0.00320
2039	1.24	8.4	0.0128	1.9	0.52	0.00333
2040	1.29	8.8	0.0132	2.0	0.53	0.00340
2041	1.35	9.3	0.0136	2.1	0.55	0.00349
2042	1.40	9.8	0.0140	2.2	0.56	0.00357
2043	1.46	10.3	0.0144	2.3	0.58	0.00368
2044	1.52	10.8	0.0148	2.4	0.60	0.00384
2045	1.58	11.4	0.0152	2.5	0.62	0.00402
2046	1.64	11.9	0.0158	2.6	0.65	0.00427
2047	1.71	12.5	0.0163	2.7	0.68	0.00453
2048	1.79	13.2	0.0169	2.9	0.71	0.00483
2049	1.87	13.8	0.0176	3.0	0.75	0.00512
2050	1.95	14.5	0.0183	3.1	0.80	0.00543
2051	2.05	15.2	0.0192	3.3	0.84	0.00570
2052	2.15	15.9	0.0202	3.4	0.89	0.00598
2053	2.26	16.7	0.0212	3.6	0.93	0.00628
2054	2.37	17.6	0.0222	3.8	0.98	0.00659
2055	2.48	18.4	0.0233	4.0	1.02	0.00691
2056	2.61	19.3	0.0245	4.2	1.07	0.00726
2057	2.74	20.3	0.0257	4.4	1.13	0.00761
2058	2.44	18.1	0.0229	3.9	1.01	0.00679
2059	2.13	15.8	0.0200	3.4	0.88	0.00593
2060	1.81	13.5	0.0170	2.9	0.75	0.00505
2061	1.49	11.1	0.0140	2.4	0.62	0.00416

Emissions Year	CO₂ (Million Metric tons)	CH₄ (‘000 tons)	N₂O (‘000 tons)	NO_x (‘000 tons)	SO₂ (‘000 tons)	Hg (tons)
2062	1.18	8.8	0.0111	1.9	0.49	0.00329
2063	0.89	6.6	0.0084	1.4	0.37	0.00248
2064	0.64	4.7	0.0060	1.0	0.26	0.00177
2065	0.42	3.1	0.0040	0.7	0.17	0.00118
2066	0.26	1.9	0.0025	0.4	0.11	0.00073
2067	0.15	1.1	0.0014	0.2	0.06	0.00041
2068	0.08	0.6	0.0007	0.1	0.03	0.00021
2069	0.03	0.3	0.0003	0.1	0.01	0.00010
2070	0.01	0.1	0.0001	0.02	0.01	0.00004
Cumulative	57.7	411.4	0.563	91.2	24.2	0.15508

Note: While emissions extend to 2076, values after 2070 are very low and, therefore, not listed here.

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CHAPTER 14. MONETIZATION OF EMISSIONS REDUCTION BENEFITS

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CHAPTER 14. MONETIZATION OF EMISSIONS REDUCTION BENEFITS

14.1 INTRODUCTION

As part of its assessment of energy conservation standards for air cleaners, the U.S. Department of Energy (DOE) considered the estimated monetary benefits likely to result from the reduced emissions of carbon dioxide (CO₂), methane (CH₄), nitrous oxide (N₂O), sulfur dioxide (SO₂) and nitrogen oxides (NO_x) that are expected to result from each of the potential standard levels considered. This chapter summarizes the basis for the benefit-per-ton values used for each of these emissions and presents the estimated total benefits for each TSL. To monetize the benefits of reducing greenhouse gas emissions this analysis uses the interim estimates presented in the *Technical Support Document: Social Cost of Carbon, Methane, and Nitrous Oxide Interim Estimates Under Executive Order 13990* published in February 2021 by the Interagency Working Group on the Social Cost of Greenhouse Gases (IWG).

14.2 MONETIZING AVOIDED GREENHOUSE GAS EMISSIONS

DOE estimates the monetized benefits of the reductions in greenhouse gas (GHG) emissions of CO₂, CH₄, and N₂O by using a measure of the social cost (SC) of each pollutant (e.g., SC-CO₂). These estimates represent the monetary value of the net harm to society associated with a marginal increase in emissions of these pollutants in a given year, or the benefit of avoiding that increase. These estimates are intended to include (but are not limited to) climate-change-related changes in net agricultural productivity, human health, property damages from increased flood risk, disruption of energy systems, risk of conflict, environmental migration, and the value of ecosystem services. DOE exercises its own judgment in presenting monetized climate benefits as recommended by applicable Executive Orders, and DOE would reach the same conclusion presented in the Notice in the absence of the social cost of greenhouse gases, including the February 2021 interim estimates presented by the Interagency Working Group on the Social Cost of Greenhouse Gases.

DOE estimated the global social benefits of CO₂, CH₄, and N₂O reductions (i.e., SC-GHGs) using the estimates presented in the Technical Support Document: Social Cost of Carbon, Methane, and Nitrous Oxide Interim Estimates under Executive Order (EO) 13990, published in February 2021 by the Interagency Working Group on the Social Cost of Greenhouse Gases (IWG).¹ The SC-GHG is the monetary value of the net harm to society associated with a marginal increase in emissions in a given year, or the benefit of avoiding that increase. In principle, SC-GHGs includes the value of all climate change impacts, including (but not limited to) changes in net agricultural productivity, human health effects, property damage from increased flood risk and natural disasters, disruption of energy systems, risk of conflict, environmental migration, and the value of ecosystem services. The SC-GHGs therefore, reflects the societal value of reducing emissions of the gas in question by one metric ton. The SC-GHGs is the theoretically appropriate value to use in conducting benefit-cost analyses of policies that affect CO₂, N₂O and CH₄ emissions. As a member of the IWG involved in the development of the February 2021 SC-GHG TSD, DOE agrees that the interim SC-GHG estimates represent the most appropriate estimate of the SC-GHG until revised estimates have been developed reflecting the latest peer-reviewed science.

The SC-GHGs estimates presented here were developed over many years, using transparent process, peer-reviewed methodologies, the best science available at the time of that process, and with input from the public. Specifically, in 2009, an IWG that included the DOE and other executive branch agencies and offices was established to ensure that agencies were using the best available science and to promote consistency in the social cost of carbon (SC-CO₂) values used across agencies. The IWG published SC-CO₂ estimates in 2010 that were developed from an ensemble of three widely cited integrated assessment models (IAMs) that estimate global climate damages using highly aggregated representations of climate processes and the global economy combined into a single modeling framework. The three IAMs were run using a common set of input assumptions in each model for future population, economic, and CO₂ emissions growth, as well as equilibrium climate sensitivity – a measure of the globally averaged temperature response to increased atmospheric CO₂ concentrations. These estimates were updated in 2013 based on new versions of each IAM. In August 2016 the IWG published estimates of the social cost of methane (SC-CH₄) and nitrous oxide (SC-N₂O) using methodologies that are consistent with the methodology underlying the SC-CO₂ estimates. The modeling approach that extends the IWG SC-CO₂ methodology to non-CO₂ GHGs has undergone multiple stages of peer review. The SC-CH₄ and SC-N₂O estimates were developed by Marten et al. (2015) and underwent a standard double-blind peer review process prior to journal publication.²

In 2015, as part of the response to public comments received to a 2013 solicitation for comments on the SC-CO₂ estimates, the IWG announced a National Academies of Sciences, Engineering, and Medicine review of the SC-CO₂ estimates to offer advice on how to approach future updates to ensure that the estimates continue to reflect the best available science and methodologies. In January 2017, the National Academies released their final report, *Valuing Climate Damages: Updating Estimation of the Social Cost of Carbon Dioxide*, and recommended specific criteria for future updates to the SC-CO₂ estimates, a modeling framework to satisfy the specified criteria, and both near-term updates and longer-term research needs pertaining to various components of the estimation process.³ Shortly thereafter, in March 2017, President Trump issued Executive Order 13783, which disbanded the IWG, withdrew the previous TSDs, and directed agencies to ensure SC-CO₂ estimates used in regulatory analyses are consistent with the guidance contained in OMB's Circular A-4, "including with respect to the consideration of domestic versus international impacts and the consideration of appropriate discount rates" (EO 13783, Section 5(c)). Benefit-cost analyses following E.O. 13783 used SC-GHG estimates that attempted to focus on the U.S.-specific share of climate change damages as estimated by the models and were calculated using two discount rates recommended by Circular A-4, 3 percent and 7 percent. All other methodological decisions and model versions used in SC-GHG calculations remained the same as those used by the IWG in 2010 and 2013, respectively.

On January 20, 2021, President Biden issued Executive Order 13990, which re-established the IWG and directed it to ensure that the U.S. Government's estimates of the social cost of carbon and other greenhouse gases reflect the best available science and the 2017 recommendations of the National Academies. The IWG was tasked with first reviewing the SC-GHG estimates currently used in Federal analyses and publishing interim estimates within 30 days of the EO that reflect the full impact of GHG emissions, including by taking global damages into account. The interim SC-GHG estimates published in February 2021 are used here to estimate the climate benefits for this direct final rule. The EO instructs the IWG to undertake a

fuller update of the SC-GHG estimates that takes into consideration the advice of the National Academies (2017) and other recent scientific literature.

The February 2021 SC-GHG TSD provides a complete discussion of the IWG's initial review conducted under EO 13990. In particular, the IWG found that the SC-GHG estimates used under EO 13783 fail to reflect the full impact of GHG emissions in multiple ways. First, the IWG found that the SC-GHG estimates used under E.O. 13783 fail to fully capture many climate impacts that affect the welfare of U.S. citizens and residents, and those impacts are better reflected by global measures of the SC-GHG. Examples of omitted effects from the EO 13783 estimates include direct effects on U.S. citizens, assets, and investments located abroad, supply chains, U.S. military assets and interests abroad, and tourism, and spillover pathways such as economic and political destabilization and global migration that can lead to adverse impacts on U.S. national security, public health, and humanitarian concern. In addition, assessing the benefits of U.S. GHG mitigation activities requires consideration of how those actions may affect mitigation activities by other countries, as those international mitigation actions will provide a benefit to U.S. citizens and residents by mitigating climate impacts that affect U.S. citizens and residents. A wide range of scientific and economic experts have emphasized the issue of reciprocity as support for considering global damages of GHG emissions. If the United States does not consider impacts on other countries, it is difficult to convince other countries to consider the impacts of their emissions on the United States. The only way to achieve an efficient allocation of resources for emissions reduction on a global basis—and so benefit the U.S. and its citizens—is for all countries to base their policies on global estimates of damages. As a member of the IWG involved in the development of the February 2021 SC-GHG TSD, DOE agrees with this assessment and, therefore, in this proposed rule DOE centers attention on a global measure of SC-GHG. This approach is the same as that taken in DOE regulatory analyses from 2012 through 2016. A robust estimate of climate damages that accrue only to U.S. citizens and residents does not currently exist in the literature. As explained in the February 2021 TSD, existing estimates are both incomplete and an underestimate of total damages that accrue to the citizens and residents of the U.S. because they do not fully capture the regional interactions and spillovers discussed above, nor do they include all of the important physical, ecological, and economic impacts of climate change recognized in the climate change literature. As noted in the February 2021 SC-GHG TSD, the IWG will continue to review developments in the literature, including more robust methodologies for estimating a U.S.-specific SC-GHG value, and explore ways to better inform the public of the full range of carbon impacts. As a member of the IWG, DOE will continue to follow developments in the literature pertaining to this issue.

Second, the IWG found that the use of the social rate of return on capital (7 percent under current OMB Circular A-4 guidance) to discount the future benefits of reducing GHG emissions inappropriately underestimates the impacts of climate change for the purposes of estimating the SC-GHG. Consistent with the findings of the National Academies (2017) and the economic literature, the IWG continued to conclude that the consumption rate of interest is the theoretically appropriate discount rate in an intergenerational context, and recommended that discount rate uncertainty and relevant aspects of intergenerational ethical considerations be accounted for in selecting future discount rates.^{4,5,6,7} Furthermore, the damage estimates developed for use in the SC-GHG are estimated in consumption-equivalent terms, and so an application of OMB Circular A-4's guidance for regulatory analysis would then use the consumption discount rate to calculate the SC-GHG. DOE agrees with this assessment and will continue to follow developments in the

literature pertaining to this issue. DOE also notes that while OMB Circular A-4, as published in 2003, recommends using 3% and 7% discount rates as "default" values, Circular A-4 also reminds agencies that "different regulations may call for different emphases in the analysis, depending on the nature and complexity of the regulatory issues and the sensitivity of the benefit and cost estimates to the key assumptions." On discounting, Circular A-4 recognizes that "special ethical considerations arise when comparing benefits and costs across generations," and Circular A-4 acknowledges that analyses may appropriately "discount future costs and consumption benefits...at a lower rate than for intragenerational analysis." In the 2015 Response to Comments on the Social Cost of Carbon for Regulatory Impact Analysis, OMB, DOE, and the other IWG members recognized that "Circular A-4 is a living document" and "the use of 7 percent is not considered appropriate for intergenerational discounting. There is wide support for this view in the academic literature, and it is recognized in Circular A-4 itself." Thus, DOE concludes that a 7% discount rate is not appropriate to apply to value the social cost of greenhouse gases in the analysis presented in this analysis. In this analysis, to calculate the present and annualized values of climate benefits, DOE uses the same discount rate as the rate used to discount the value of damages from future GHG emissions, for internal consistency. That approach to discounting follows the same approach that the February 2021 TSD recommends "to ensure internal consistency—i.e., future damages from climate change using the SC-GHG at 2.5 percent should be discounted to the base year of the analysis using the same 2.5 percent rate." DOE has also consulted the National Academies' 2017 recommendations on how SC-GHG estimates can "be combined in RIAs with other cost and benefits estimates that may use different discount rates." The National Academies reviewed "several options," including "presenting all discount rate combinations of other costs and benefits with [SC-GHG] estimates."

As a member of the IWG involved in the development of the February 2021 SC-GHG TSD, DOE agrees with this assessment and will continue to follow developments in the literature pertaining to this issue.

While the IWG works to assess how best to incorporate the latest, peer reviewed science to develop an updated set of SC-GHG estimates, it set the interim estimates to be the most recent estimates developed by the IWG prior to the group being disbanded in 2017. The estimates rely on the same models and harmonized inputs and are calculated using a range of discount rates. As explained in the February 2021 SC-GHG TSD, the IWG has recommended that agencies revert to the same set of four values drawn from the SC-GHG distributions based on three discount rates as were used in regulatory analyses between 2010 and 2016 and subject to public comment. For each discount rate, the IWG combined the distributions across models and socioeconomic emissions scenarios (applying equal weight to each) and then selected a set of four values recommended for use in benefit-cost analyses: an average value resulting from the model runs for each of three discount rates (2.5 percent, 3 percent, and 5 percent), plus a fourth value, selected as the 95th percentile of estimates based on a 3 percent discount rate. The fourth value was included to provide information on potentially higher-than-expected economic impacts from climate change. As explained in the February 2021 SC-GHG TSD, and DOE agrees, this update reflects the immediate need to have an operational SC-GHG for use in regulatory benefit-cost analyses and other applications that was developed using a transparent process, peer-reviewed methodologies, and the science available at the time of that process. Those estimates were subject to public comment in the context of dozens of proposed rulemakings as well as in a dedicated public comment period in 2013.

There are a number of limitations and uncertainties associated with the SC-GHG estimates. First, the current scientific and economic understanding of discounting approaches suggests discount rates appropriate for intergenerational analysis in the context of climate change are likely to be less than 3 percent, near 2 percent or lower.¹ Second, the IAMs used to produce these interim estimates do not include all of the important physical, ecological, and economic impacts of climate change recognized in the climate change literature and the science underlying their “damage functions” – i.e., the core parts of the IAMs that map global mean temperature changes and other physical impacts of climate change into economic (both market and nonmarket) damages – lags behind the most recent research. For example, limitations include the incomplete treatment of catastrophic and non-catastrophic impacts in the integrated assessment models, their incomplete treatment of adaptation and technological change, the incomplete way in which inter-regional and intersectoral linkages are modeled, uncertainty in the extrapolation of damages to high temperatures, and inadequate representation of the relationship between the discount rate and uncertainty in economic growth over long time horizons. Likewise, the socioeconomic and emissions scenarios used as inputs to the models do not reflect new information from the last decade of scenario generation or the full range of projections. The modeling limitations do not all work in the same direction in terms of their influence on the SC-CO₂ estimates. However, as discussed in the February 2021 TSD, the IWG has recommended that, taken together, the limitations suggest that the interim SC-GHG estimates used in this final rule likely underestimate the damages from GHG emissions. DOE concurs with this assessment.

DOE's derivations of the SC-GHGs (*i.e.*, SC-CO₂, SC-N₂O, and SC-CH₄) values are discussed in the following sections.

14.2.1 Social Cost of Carbon

The SC-CO₂ values used for DOE's analysis were generated using the values presented in the 2021 TSD from the IWG. Table 14.2.1 shows the four sets of SC-CO₂ estimates in five-year increments from 2020 to 2070.^a DOE expects additional climate benefits to accrue for any longer-life air cleaners, but a lack of available SC-CO₂ estimates for emissions years beyond 2070 prevents DOE from monetizing these additional benefits in this analysis. The case labeled “95th percentile” refers to values in the 95th percentile of simulations. Appendix 14A provides the full set of SC-CO₂ estimates. For purposes of capturing the uncertainties involved in regulatory impact analysis, the interagency group emphasizes the importance of including all four sets of SC-CO₂ values.

^a The values for the years after 2050 are based on modeling conducted by EPA for the “Revised 2023 and Later Model Year Light-Duty Vehicle GHG Emissions Standards: Regulatory Impact Analysis” published by EPA in December 2021. See Appendix 14A.

Table 14.2.1 Annual SC-CO₂ Value Based on 2021 Interagency Update and 2021 EPA Analysis, 2020–2070 (2020 dollars per metric ton)*

Year	Discount Rate and Statistic			
	5%	3%	2.5%	3%
	Average	Average	Average	95 th Percentile
2020	14	51	76	151
2025	17	56	83	169
2030	19	62	89	186
2035	22	67	96	205
2040	25	73	103	224
2045	28	79	109	242
2050	32	84	116	259
2055	35	89	122	265
2060	38	93	128	275
2065	44	100	135	300
2070	49	108	143	326

* For 2020-2050, there are slight differences from the IWG report in a few cases that are likely due to the GDP deflator used.

DOE multiplied the CO₂ emissions reduction estimated for each year by the SC-CO₂ value for that year in each of the four cases. To calculate a present value of the stream of monetary values, DOE discounted the values in each of the four cases using the specific discount rate that had been used to obtain the SC-CO₂ values in each case.

14.2.2 Social Cost of Methane and Nitrous Oxide

The SC-CH₄ and SC-N₂O values used for the present analysis were generated using the values presented in the 2021 TSD from the IWG. Table 14.2.2 shows the four sets of SC-CH₄ and SC- N₂O estimates from the latest interagency update in 5-year increments from 2020 to 2070. For purposes of capturing the uncertainties involved in regulatory impact analysis, the interagency group emphasizes the importance of including all four sets of SC-CH₄ and SC- N₂O values. The full set of annual values is reported in appendix 14A of this TSD.

DOE multiplied the CH₄ and N₂O emissions reduction estimated for each year by the SC-CH₄ and SC-N₂O estimates for that year in each of the cases. To calculate a present value of the stream of monetary values, DOE discounted the values in each of the cases using the specific discount rate that had been used to obtain the SC-CH₄ and SC-N₂O estimates in each case.

Table 14.2.2 Annual SC-CH₄ and SC-N₂O Values Based on 2021 Interagency Update and 2021 EPA Analysis, 2020–2070 (2020\$ per metric ton)*

Year	SC-CH ₄				SC-N ₂ O			
	Discount Rate and Statistic				Discount Rate and Statistic			
	5%	3%	2.5%	3%	5%	3%	2.5 %	3%
	Average	Average	Average	95 th percentile	Average	Average	Average	95 th percentile
2020	663	1,480	1,946	3,893	5,760	18,342	27,037	48,090
2025	799	1,714	2,223	4,533	6,766	20,520	29,811	54,108
2030	935	1,948	2,499	5,173	7,772	22,698	32,585	60,125
2035	1,106	2,224	2,817	5,939	9,007	25,149	35,632	66,898
2040	1,277	2,500	3,136	6,705	10,241	27,600	38,678	73,670
2045	1,464	2,778	3,450	7,426	11,687	30,238	41,888	80,766
2050	1,651	3,057	3,763	8,147	13,133	32,875	45,098	87,863
2055	1,772	3,221	3,942	8,332	14,758	35,539	48,236	94,117
2060	1,899	3,395	4,130	8,539	16,424	38,300	51,507	100,845
2065	2,508	4,163	4,960	11,177	19,687	42,625	56,397	115,590
2070	3,130	4,976	5,867	14,079	23,018	47,072	61,428	130,928

* For 2020-2050, there are slight differences from the IWG report in a few cases that are likely due to the GDP deflator used.

14.3 VALUATION OF OTHER EMISSIONS REDUCTIONS

As noted in chapter 13, new or amended energy conservation standards would reduce SO₂ emissions from electricity generation, and NO_x emissions in those States that are not affected by caps. For each of the considered TSLs, DOE estimated monetized values of NO_x and SO₂ emissions reductions from electricity generation using the latest benefit-per-ton estimates for that sector from the EPA's Benefits Mapping and Analysis Program.^b DOE used EPA's values for PM_{2.5}-related benefits associated with NO_x and SO₂ and for ozone-related benefits associated with NO_x for 2025, 2030, 2035 and 2040, calculated with discount rates of 3 percent and 7 percent. DOE used linear interpolation to define values for the years not given in the 2025 to 2040 period; for years beyond 2040 the values are held constant.

The ozone-related benefits associated with NO_x occur only in the ozone-season (May to September). EPA data indicate that ozone-season NO_x emissions from electricity generation are slightly less than half of all-year NO_x emissions. DOE accounted for this characteristic in its methodology.

DOE combined the EPA data with data from *AEO2022* to estimate benefit-per-ton values by sector. Appendix 14B provides methodological details and values that DOE used. The results presented in this chapter use benefit-per-ton values for the residential and commercial sectors. DOE multiplied the emissions reduction (in tons) in each year by the associated \$/ton values, and then discounted each series using discount rates of 3 percent and 7 percent as appropriate.

^b *Estimating the Benefit per Ton of Reducing PM_{2.5} Precursors from 21 Sectors.*

<https://www.epa.gov/benmap/estimating-benefit-ton-reducing-directly-emitted-pm25-pm25-precursors-and-ozone-precursors>

14.4 ESTIMATED BENEFITS

14.4.1 Benefits for Considered TSLs

The tables in this section show the emissions monetization results for each considered TSL.

Table 14.4.1 Present Social Value of Cumulative CO₂ Emissions Reduction from Potential Standards for Air Cleaners

TSL	SC-CO ₂ Case			
	Discount Rate and Statistic			
	5%	3%	2.5%	3%
	Average	Average	Average	95 th percentile
	<i>million 2021\$</i>			
1	209	930	1,469	2,818
2	476	2,122	3,353	6,430
3	514	2,262	3,562	6,851
4	1,112	4,959	7,835	15,026
5	1,261	5,621	8,881	17,033

Table 14.4.2 Present Social Value of Cumulative Methane Emissions Reduction from Potential Standards for Air Cleaners

TSL	SC-CH ₄ Case			
	Discount Rate and Statistic			
	5%	3%	2.5%	3%
	Average	Average	Average	95 th Percentile
	<i>million 2021\$</i>			
1	69	214	302	564
2	157	489	690	1,287
3	167	513	722	1,352
4	367	1,143	1,612	3,009
5	416	1,296	1,827	3,411

Table 14.4.3 Present Social Value of Cumulative Nitrous Oxide Emissions Reduction from Potential Standards for Air Cleaners

TSL	SC-N ₂ O Case			
	Discount Rate and Statistic			
	5%	3%	2.5%	3%
	Average	Average	Average	95 th Percentile
	<i>million 2021\$</i>			
1	0.774	3.22	5.03	8.56
2	1.77	7.34	11.5	19.5
3	1.92	7.86	12.3	20.9
4	4.13	17.2	26.8	45.6
5	4.68	19.5	30.4	51.7

Table 14.4.4 Present Social Value of Cumulative NO_x and SO₂ Emissions Reduction from Potential Standards for Air Cleaners

TSL	NO _x		SO ₂	
	7% discount rate	3% discount rate	7% discount rate	3% discount rate
	<i>million 2021\$</i>		<i>million 2021\$</i>	
1	508	1,413	184	501
2	1,159	3,223	419	1,142
3	1,304	3,440	489	1,245
4	2,706	7,532	979	2,669
5	3,065	8,536	1,109	3,026

14.4.2 Annual and Cumulative Benefits for Selected Standards (TSL 3)

The tables in this section present climate and health benefits estimated for the selected standards. The benefits of reduced CO₂, CH₄, and N₂O emissions are collectively referred to as climate benefits. The benefits of reduced SO₂ and NO_x emissions are collectively referred to as health benefits.

The annual values reflect the benefits from reduced emissions in each year. The associated benefits accrue over very many years in the case of GHG emissions, and over several years in the case of NO_x and SO₂ emissions. The time stream of benefits has been discounted to estimate the benefit-per-ton values for each year, but the total benefits associated with each emissions year are not discounted in these tables. The cumulative present value does reflect discounting at the noted discount rates.

Table 14.4.5 Climate Benefits from GHG Emissions Reduction (CO₂, CH₄, and N₂O) at Selected Standards (TSL 3) for Air Cleaners (million 2021\$)

Emissions Year*	Discount Rate and Statistic			
	5% Average	3% Average	2.5% Average	3% 95th percentile
2024	1.19	3.74	5.4	11.0
2025	2.39	7.42	10.7	21.9
2030	17.9	52.9	75.3	157
2035	31.8	89.2	125	266
2040	44.7	120	165	359
2045	62.7	160	218	480
2050	87.2	214	289	641
2055	122	287	387	836
2060	96.1	221	295	632
2065	26.9	57.1	75.0	167
2070	1.03	2.07	2.68	6.17
Cumulative PV**	683	2,783	4,296	8,224
Annualized	44	136	194	401

* Annual benefits shown are undiscounted values.

** The same discount rate used to discount the value of damages from future emissions in each SC-GHG case is used to calculate the present value of avoided emissions in that case for internal consistency.

Table 14.4.6 Climate Benefits from Changes in CO₂ Emissions from Selected Standards (TSL 3) for Air Cleaners (million 2021\$)

Emissions Year*	Discount Rate and Statistic			
	5%	3%	2.5%	3%
	Average	Average	Average	95th Percentile
2024	0.958	3.23	4.76	9.66
2025	1.91	6.39	9.39	19.1
2026	4.24	14.0	20.5	42.1
2027	6.71	22.0	32.1	66.1
2028	9.27	30.0	43.8	90.6
2029	11.6	37.4	54.4	113
2030	14.1	44.9	65.0	136
2031	16.6	52.5	75.8	159
2032	19.1	59.5	85.6	181
2033	21.5	66.3	95.1	201
2034	23.0	70.4	101	214
2035	24.6	74.5	106	227
2036	26.0	78.2	111	239
2037	27.8	82.6	117	253
2038	29.8	87.9	124	269
2039	31.8	93.2	132	286
2040	33.9	98.4	139	302
2041	36.3	104	147	321
2042	38.8	110	155	339
2043	41.2	116	162	357
2044	43.9	123	171	378
2045	46.7	130	180	398
2046	49.8	137	190	421
2047	53.0	145	201	445
2048	56.5	153	212	471
2049	60.3	162	224	499
2050	64.3	172	237	528
2051	69.4	182	252	556
2052	74.1	193	267	586
2053	79.2	205	283	617
2054	84.6	217	299	650
2055	90.3	230	317	688
2056	96.4	244	336	728
2057	103	259	355	770
2058	93.3	233	320	692
2059	82.8	206	282	609
2060	71.6	177	242	522
2061	60.8	148	202	438
2062	49.5	119	161	352
2063	38.5	91.0	123	271
2064	28.2	65.8	88.9	196
2065	19.3	44.5	60.0	133

Emissions Year*	Discount Rate and Statistic			
	5%	3%	2.5%	3%
	Average	Average	Average	95th Percentile
2066	12.3	27.9	37.5	83.7
2067	7.13	16.0	21.5	48.2
2068	3.74	8.31	11.1	25.1
2069	1.75	3.85	5.14	11.7
2070	0.724	1.57	2.09	4.77
Cumulative Present Value (all years)**	514	2,262	3,562	6,851
Annualized (all years)	33.3	110	161	334

* Annual benefits shown are undiscounted values.

** The same discount rate used to discount the value of damages from future emissions in each SC-GHG case is used to calculate the present value of avoided emissions in that case for internal consistency.

Table 14.4.7 Climate Benefits from Changes in Methane Emissions from Selected Standards (TSL 3) for Air Cleaners (million 2021\$)

Emissions Year*	Discount Rate and Statistic			
	5%	3%	2.5%	3%
	Average	Average	Average	95th Percentile
2024	0.233	0.502	0.653	1.33
2025	0.471	1.01	1.31	2.67
2026	1.06	2.26	2.93	5.99
2027	1.70	3.60	4.65	9.55
2028	2.42	5.09	6.55	13.5
2029	3.08	6.45	8.30	17.1
2030	3.76	7.83	10.0	20.8
2031	4.49	9.29	11.9	24.7
2032	5.24	10.7	13.7	28.6
2033	5.98	12.2	15.5	32.5
2034	6.60	13.4	17.0	35.7
2035	7.16	14.4	18.2	38.4
2036	7.76	15.5	19.6	41.4
2037	8.43	16.8	21.1	44.8
2038	9.15	18.1	22.8	48.4
2039	9.87	19.4	24.4	52.0
2040	10.7	20.9	26.2	56.0
2041	11.6	22.5	28.2	60.4
2042	12.5	24.2	30.2	64.8
2043	13.5	25.9	32.3	69.4
2044	14.6	27.9	34.7	74.6
2045	15.8	29.9	37.2	80.0
2046	17.0	32.1	39.7	85.6
2047	18.3	34.3	42.4	91.6
2048	19.7	36.8	45.5	98.3
2049	21.1	39.3	48.4	105

Emissions Year*	Discount Rate and Statistic			
	5%	3%	2.5%	3%
	Average	Average	Average	95th Percentile
2050	22.6	41.9	51.6	112
2051	24.2	44.6	54.8	118
2052	25.7	47.2	58.0	124
2053	27.4	50.1	61.4	131
2054	29.1	53.1	65.0	138
2055	30.9	56.3	68.8	146
2056	32.9	59.7	72.9	153
2057	35.1	63.3	77.3	162
2058	31.7	57.1	69.6	145
2059	28.1	50.4	61.3	127
2060	24.2	43.3	52.7	109
2061	21.2	37.3	45.1	95.2
2062	17.8	30.8	37.1	79.7
2063	14.2	24.2	29.0	63.5
2064	10.7	17.9	21.4	47.6
2065	7.48	12.4	14.8	33.3
2066	4.85	7.97	9.48	21.7
2067	2.88	4.68	5.55	12.9
2068	1.54	2.48	2.94	6.90
2069	0.734	1.18	1.39	3.30
2070	0.308	0.489	0.577	1.38
Cumulative Present Value (all years)**	167	513	722	1,352
Annualized (all years)	10.8	25.0	32.6	65.9

* Annual benefits shown are undiscounted values.

** The same discount rate used to discount the value of damages from future emissions in each SC-GHG case is used to calculate the present value of avoided emissions in that case for internal consistency.

Table 14.4.8 Climate Benefits from Changes in N₂O Emissions from Selected Standards (TSL 3) for Air Cleaners (million 2021\$)

Emissions Year*	Discount Rate and Statistic			
	5%	3%	2.5%	3%
	Average	Average	Average	95th Percentile
2024	0.004	0.012	0.017	0.031
2025	0.008	0.023	0.034	0.062
2026	0.017	0.052	0.075	0.137
2027	0.027	0.082	0.118	0.216
2028	0.038	0.111	0.161	0.295
2029	0.047	0.139	0.200	0.368
2030	0.057	0.167	0.240	0.444
2031	0.068	0.196	0.280	0.519
2032	0.077	0.221	0.315	0.585
2033	0.087	0.247	0.351	0.655
2034	0.091	0.257	0.366	0.684

Emissions Year*	Discount Rate and Statistic			
	5%	3%	2.5%	3%
	Average	Average	Average	95th Percentile
2035	0.097	0.271	0.385	0.722
2036	0.102	0.281	0.398	0.749
2037	0.107	0.295	0.416	0.785
2038	0.114	0.312	0.439	0.832
2039	0.122	0.330	0.464	0.880
2040	0.128	0.345	0.483	0.920
2041	0.136	0.363	0.507	0.968
2042	0.144	0.381	0.532	1.02
2043	0.152	0.399	0.555	1.06
2044	0.160	0.417	0.578	1.11
2045	0.169	0.436	0.604	1.17
2046	0.179	0.460	0.636	1.23
2047	0.190	0.484	0.667	1.29
2048	0.201	0.510	0.702	1.36
2049	0.214	0.539	0.741	1.44
2050	0.228	0.570	0.782	1.52
2051	0.245	0.609	0.833	1.61
2052	0.264	0.649	0.886	1.72
2053	0.283	0.692	0.942	1.83
2054	0.304	0.737	1.00	1.95
2055	0.326	0.785	1.07	2.08
2056	0.350	0.837	1.13	2.21
2057	0.375	0.892	1.21	2.36
2058	0.342	0.807	1.09	2.13
2059	0.305	0.716	0.964	1.89
2060	0.265	0.618	0.831	1.63
2061	0.227	0.520	0.697	1.38
2062	0.186	0.421	0.562	1.12
2063	0.146	0.324	0.432	0.870
2064	0.108	0.236	0.313	0.637
2065	0.074	0.161	0.213	0.436
2066	0.047	0.101	0.134	0.277
2067	0.028	0.059	0.077	0.161
2068	0.015	0.031	0.040	0.084
2069	0.007	0.014	0.019	0.040
2070	0.003	0.006	0.008	0.016
Cumulative Present Value (all years)**	1.92	7.86	12.3	20.9
Annualized (all years)	0.124	0.383	0.553	1.02

* Annual benefits shown are undiscounted values.

** The same discount rate used to discount the value of damages from future emissions in each SC-GHG case is used to calculate the present value of avoided emissions in that case for internal consistency.

Table 14.4.9 Health Benefits from Changes in NO_x and SO₂ Emissions from Selected Standards (TSL 3) for Air Cleaners (million 2021\$)

Emissions Year*	NO_x (as PM_{2.5} and Ozone)		SO₂	
Discount Rate	7%	3%	7%	3%
2024	3.99	4.45	1.96	2.18
2025	7.87	8.79	4.50	5.01
2026	17.9	20.0	10.3	11.5
2027	28.6	31.9	16.4	18.2
2028	40.8	45.5	21.9	24.4
2029	52.0	58.0	25.6	28.4
2030	64.2	71.7	29.0	32.3
2031	76.6	85.5	31.7	35.2
2032	89.2	100	34.7	38.7
2033	100	112	39.1	43.5
2034	105	118	39.9	44.4
2035	111	123	42.4	47.2
2036	115	129	44.2	49.2
2037	121	135	46.3	51.5
2038	130	145	48.2	53.6
2039	140	156	51.0	56.7
2040	149	166	52.6	58.5
2041	156	174	54.1	60.1
2042	162	181	55.6	61.7
2043	169	189	57.7	64.1
2044	176	196	60.0	66.7
2045	182	203	62.3	69.2
2046	191	213	65.5	72.8
2047	201	225	68.5	76.1
2048	212	236	71.8	79.8
2049	222	248	75.6	84.1
2050	232	259	80.8	89.8
2051	244	272	84.8	94.2
2052	256	286	89.0	98.9
2053	269	300	93.4	104
2054	282	315	98.0	109
2055	296	330	103	114
2056	311	346	108	120
2057	326	364	113	126
2058	291	324	101	112
2059	254	283	88.3	98.1
2060	216	241	75.1	83.5
2061	178	198	61.9	68.8
2062	141	157	49.0	54.4

Emissions Year*	NOx (as PM_{2.5} and Ozone)		SO₂	
Discount Rate	7%	3%	7%	3%
2063	106	119	36.9	41.1
2064	75.8	84.5	26.3	29.3
2065	50.6	56.4	17.6	19.5
2066	31.2	34.8	10.9	12.1
2067	17.7	19.7	6.15	6.83
2068	9.06	10.1	3.15	3.50
2069	4.14	4.62	1.44	1.60
2070	1.67	1.86	0.579	0.644
2071	0.580	0.646	0.202	0.224
2072	0.168	0.187	0.058	0.065
2073	0.036	0.040	0.012	0.014
Cumulative Present Value (all years)	1304	3440	21.9	24.4
Annualized (all years)	109	168	25.6	28.4

* Annual benefits shown are undiscounted values.

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CHAPTER 15. UTILITY IMPACT ANALYSIS

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CHAPTER 15. UTILITY IMPACT ANALYSIS

15.1 INTRODUCTION

In the utility impact analysis, the U.S. Department of Energy (DOE) analyzes several aggregate impacts on electric utilities that DOE projects would result for each trial standard level (TSL).

15.2 ELECTRIC UTILITIES

The electric utility impact analysis is based on output of the DOE/Energy Information Administration (EIA)'s National Energy Modeling System (NEMS).^a NEMS is a public domain, multi-sectored, partial equilibrium model of the U.S. energy sector. Each year, DOE/EIA uses NEMS to produce an energy forecast for the United States, the *Annual Energy Outlook (AEO)*. The EIA publishes a Reference case, which incorporates all existing energy-related policies at the time of publication, and a variety of side cases which analyze the impact of different policies, energy price and market trends. The current analysis is based on results published for the *AEO 2022*.²

DOE's *AEO*-based methodology has a number of advantages:

- The assumptions used in the *AEO* reference case and side cases are fully documented and receive detailed public scrutiny.
- NEMS is updated each year, with each edition of the *AEO*, to reflect changes in energy prices, supply trends, regulations, *etc.*
- The comprehensiveness of NEMS permits the modeling of interactions among the various energy supply and demand sectors.
- Using EIA published reference and side cases to estimate the utility impacts enhances the transparency of DOE's analysis.

The details of the methodology vary based on the number and type of side cases published with each edition of the *AEO*. The approach adopted for this analysis is described in appendix 15A. A more detailed discussion of the general approach is presented in K. Coughlin, "Utility Sector Impacts of Reduced Electricity Demand."^{3,4}

This chapter presents the results for air cleaners.

15.2.1 Methodology

DOE estimates the marginal impacts of reduction in energy demand on the energy supply sector. In principle, marginal values should provide a better estimate of the actual impact of

^a For more information on NEMS, refer to the U.S. Department of Energy, Energy Information Administration documentation. A useful summary is *National Energy Modeling System: An Overview*.¹

energy conservation standards. DOE represents these marginal impacts using time series of *impact factors*.

The impact factors are calculated based on output from NEMS for *AEO 2022*. NEMS uses predicted growth in demand for each end use to build up a projection of the total electric system load growth. The system load shapes are converted internally to load duration curves, which are then used to estimate the most cost-effective additions to capacity. When electricity demand deviates from the *AEO* reference case, in general there are three inter-related effects: the annual generation (TWh) from the stock of electric generating capacity changes, the total generation capacity itself (GW) may change, and the mix of capacity types and technologies may change. Technology changes lead to a change in the proportion of fuel consumption to electricity generated (referred to as the heat rate). Each of these effects can vary for different types of end use. The change in total generating capacity is sensitive to the degree to which the end-use is peak coincident, while the capacity mix is sensitive to the hourly load shape associated with the end use. Changes in generation by fuel type lead in turn to changes in total power sector emissions of SO₂, NO_x, Hg and CO₂.

DOE defined impact factors describing the change in emissions, installed capacity, and fuel consumption per unit reduction of site electricity demand. The impact factors vary by sector and end-use, as well as by year. DOE multiplied the impact factors by the stream of site energy savings calculated in the NIA (chapter 10) to produce estimates of the utility impacts. The utility impact factors are presented in appendix 15A. For air cleaners DOE used the impact factors for other uses in homes and commercial buildings.

15.2.2 Utility Impact Results

15.2.2.1 Installed Capacity

The figures in this section show the changes in U.S. electricity installed capacity that result for each TSL by major plant type for selected years. The changes have been calculated based on the impact factors for capacity presented in appendix 15A. Units are megawatts of capacity per gigawatt-hour of site electricity use (MW/GWh).^b Note that a negative number means an increase in capacity under a TSL.

^b These units are identical to GW/TWh.

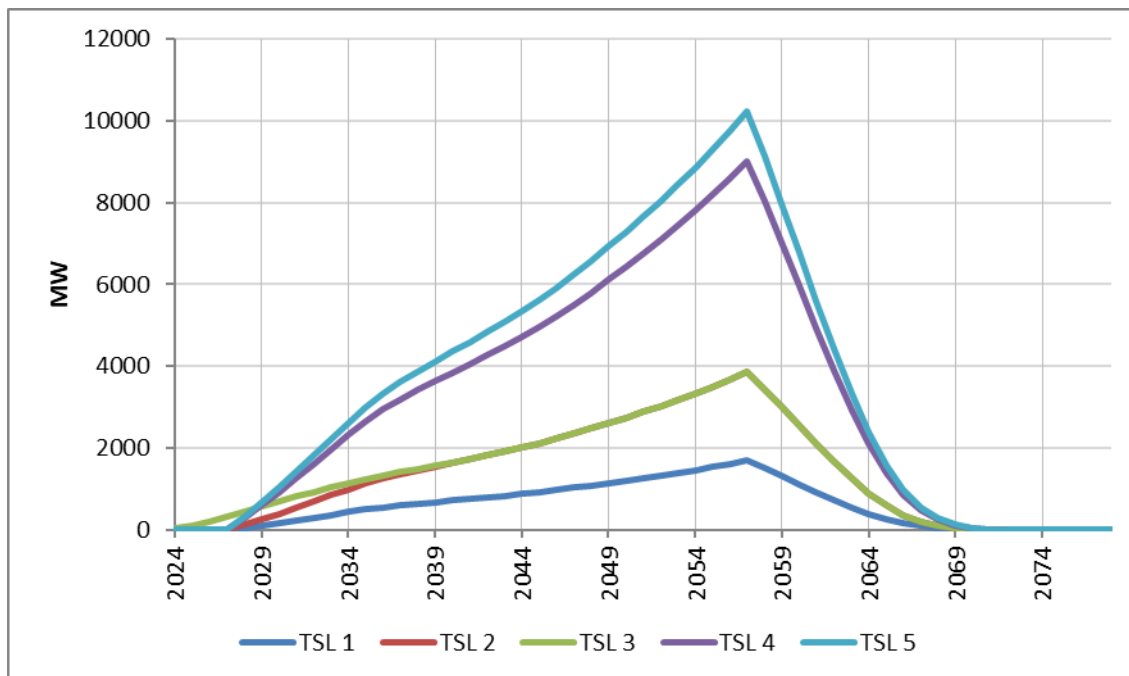


Figure 15.2.1 Air Cleaners: Total Electric Capacity Reduction

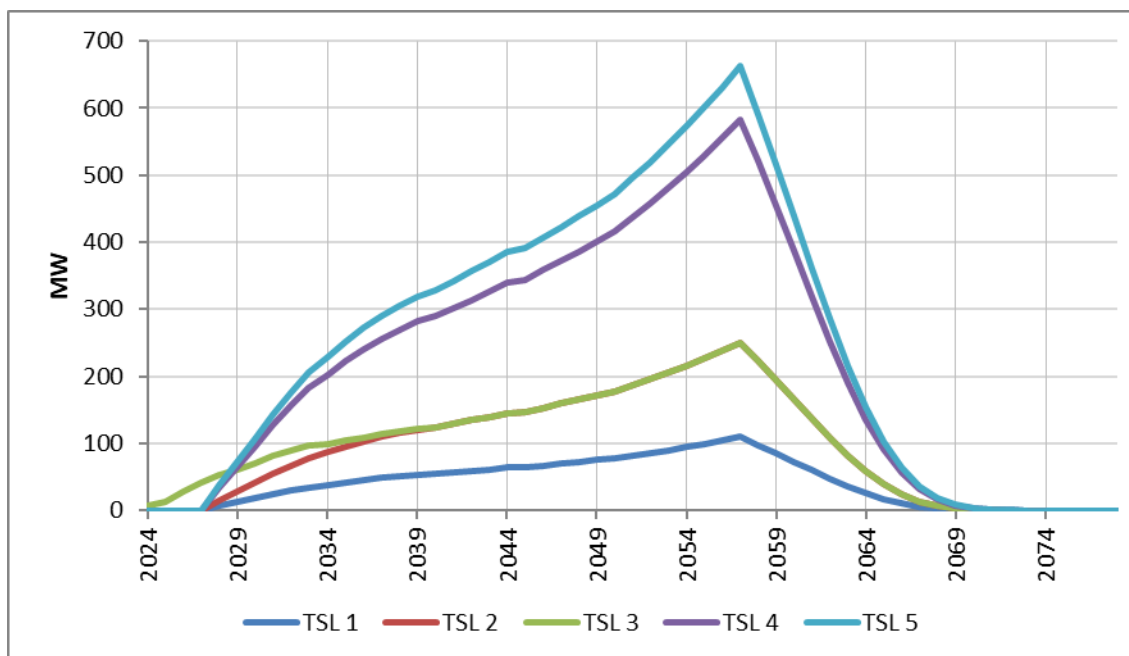


Figure 15.2.2 Air Cleaners: Coal Capacity Reduction

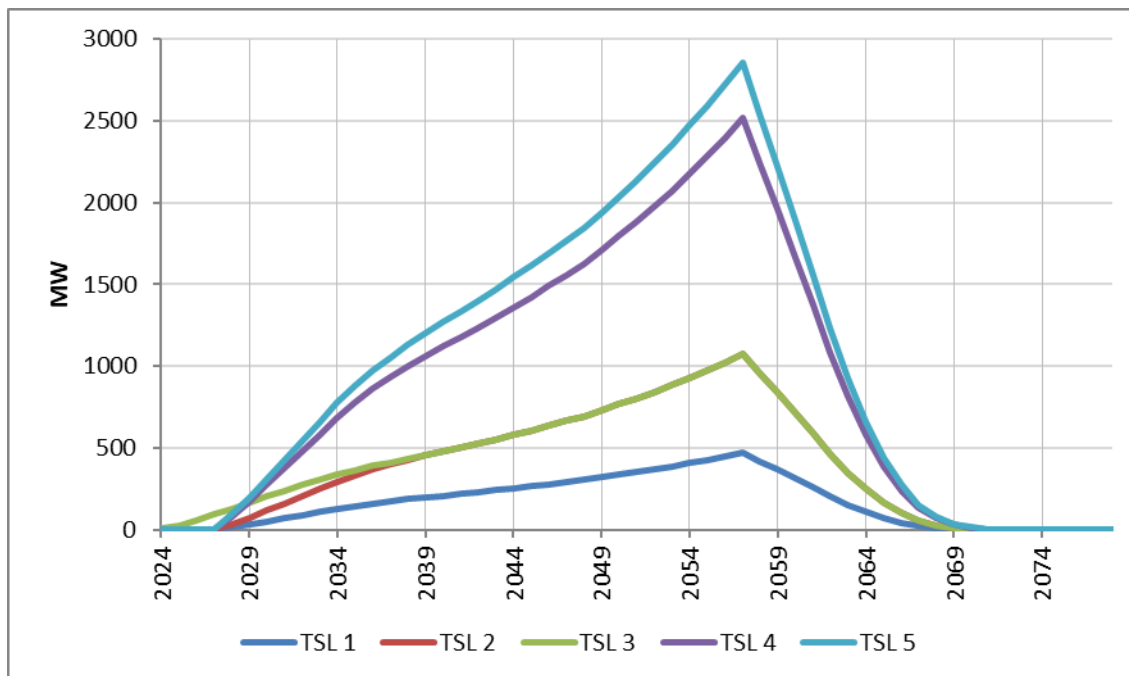


Figure 15.2.3 Air Cleaners: Gas Combined Cycle Capacity Reduction

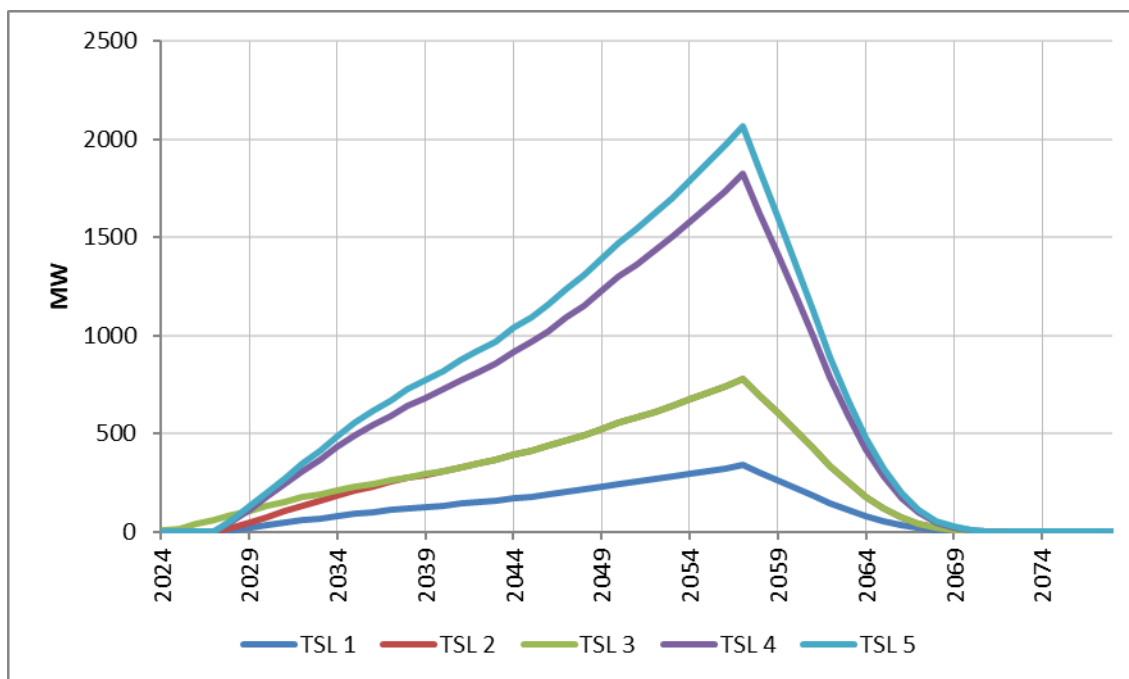


Figure 15.2.4 Air Cleaners: Peaking Capacity Reduction

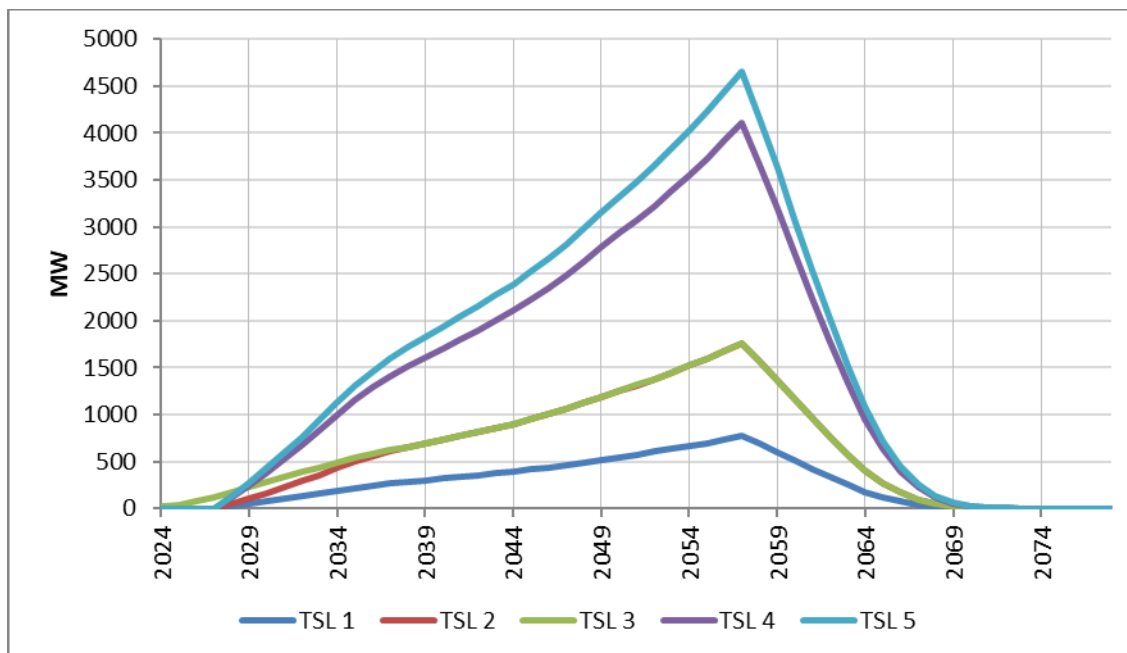


Figure 15.2.5 Air Cleaners: Renewables Capacity Reduction

15.2.2.2 Electricity Generation

The figures in this section show the annual change in electricity generation that result for each TSL by fuel type. The change by fuel type has been calculated based on factors calculated as described in appendix 15A.

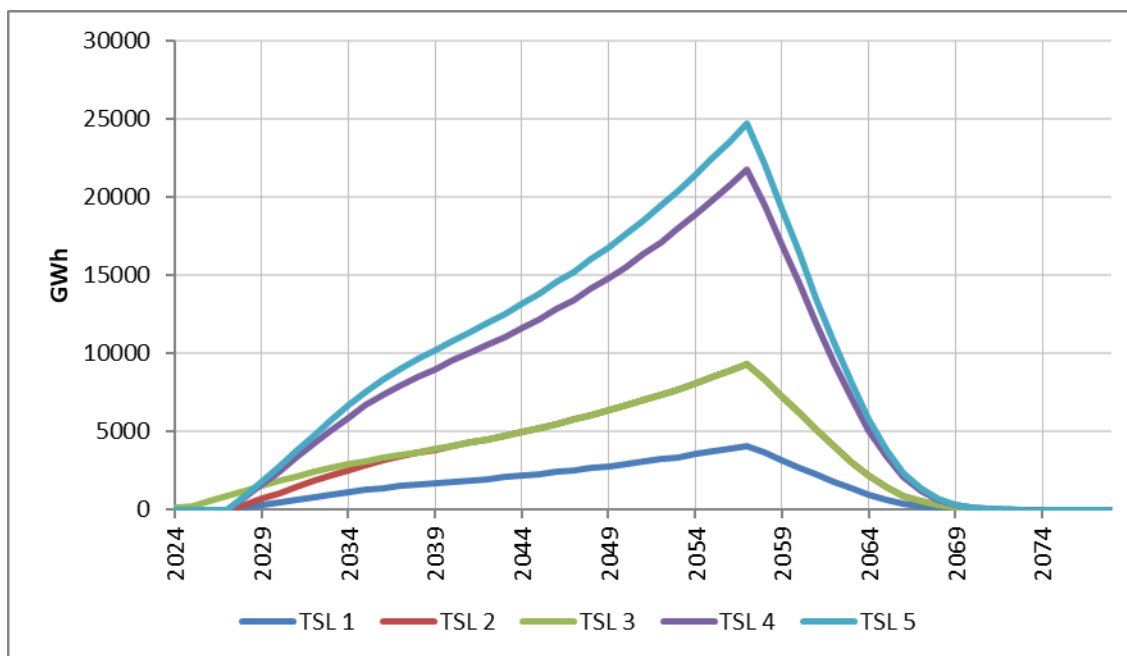


Figure 15.2.6 Air Cleaners: Total Generation Reduction

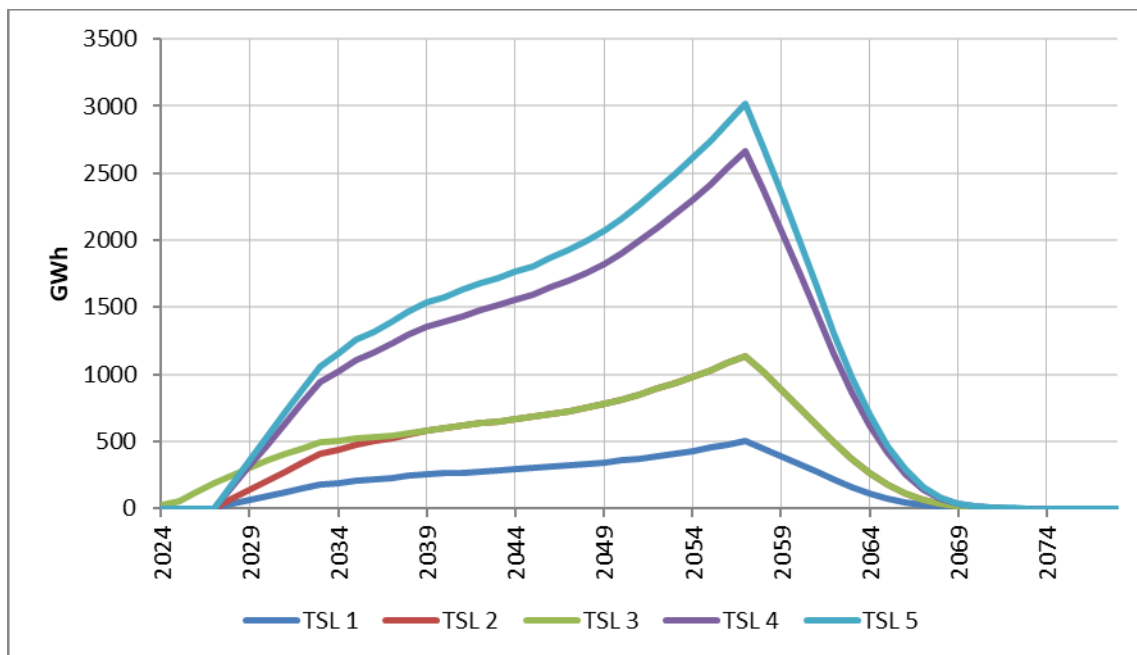


Figure 15.2.7 Air Cleaners: Coal Generation Reduction

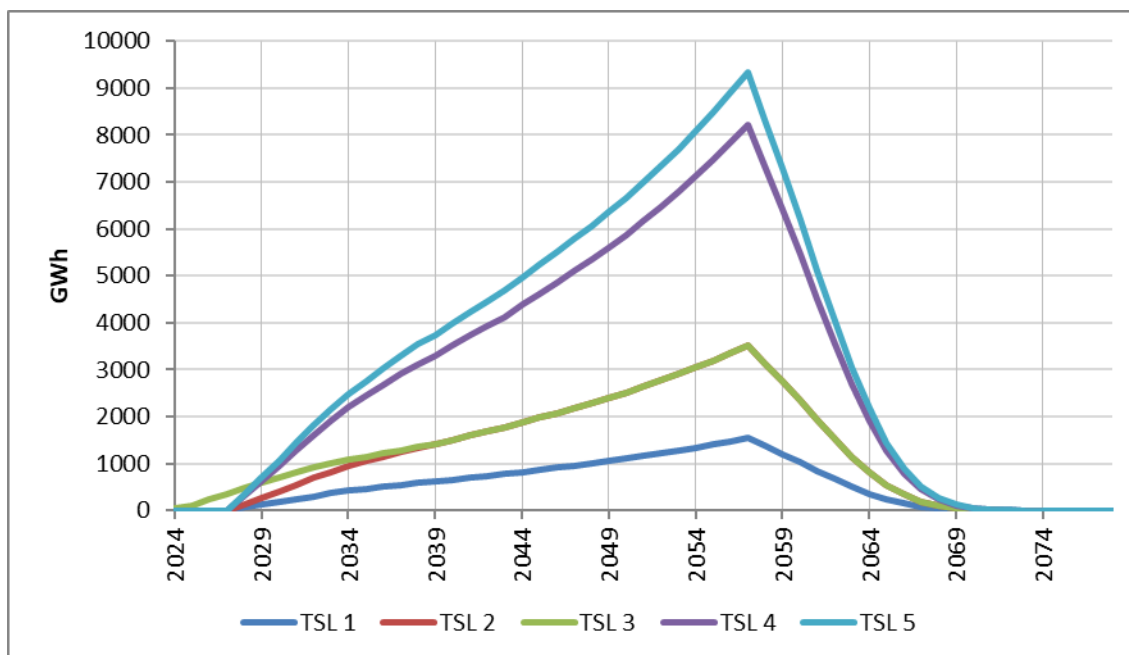


Figure 15.2.8 Air Cleaners: Gas Combined Cycle Generation Reduction

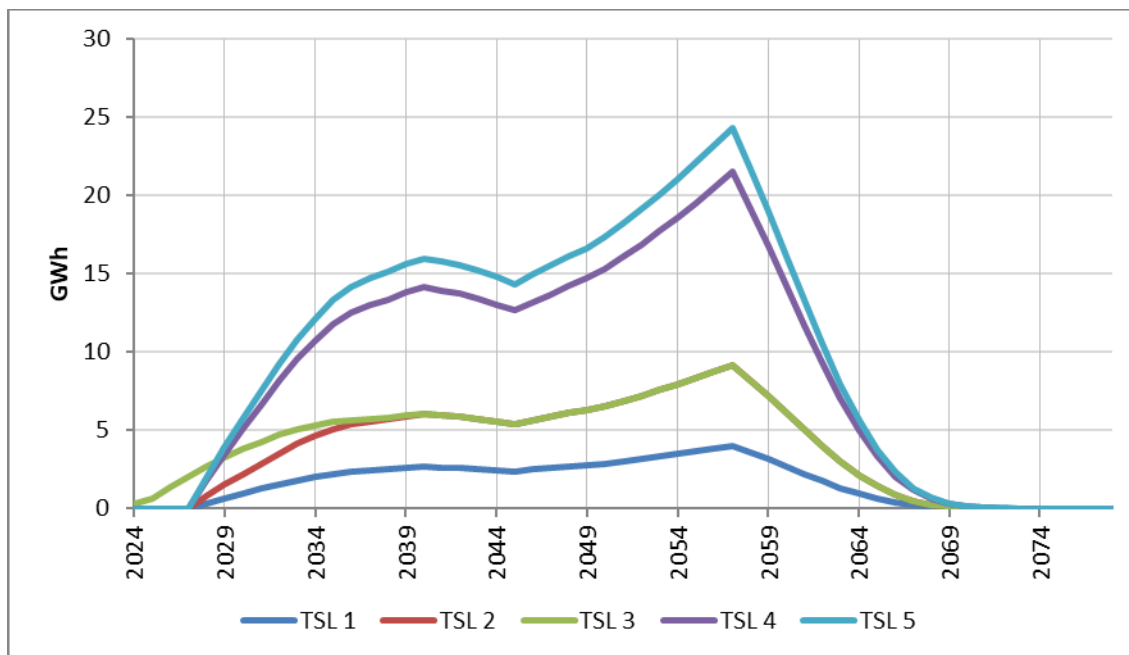


Figure 15.2.9 Air Cleaners: Oil Generation Reduction

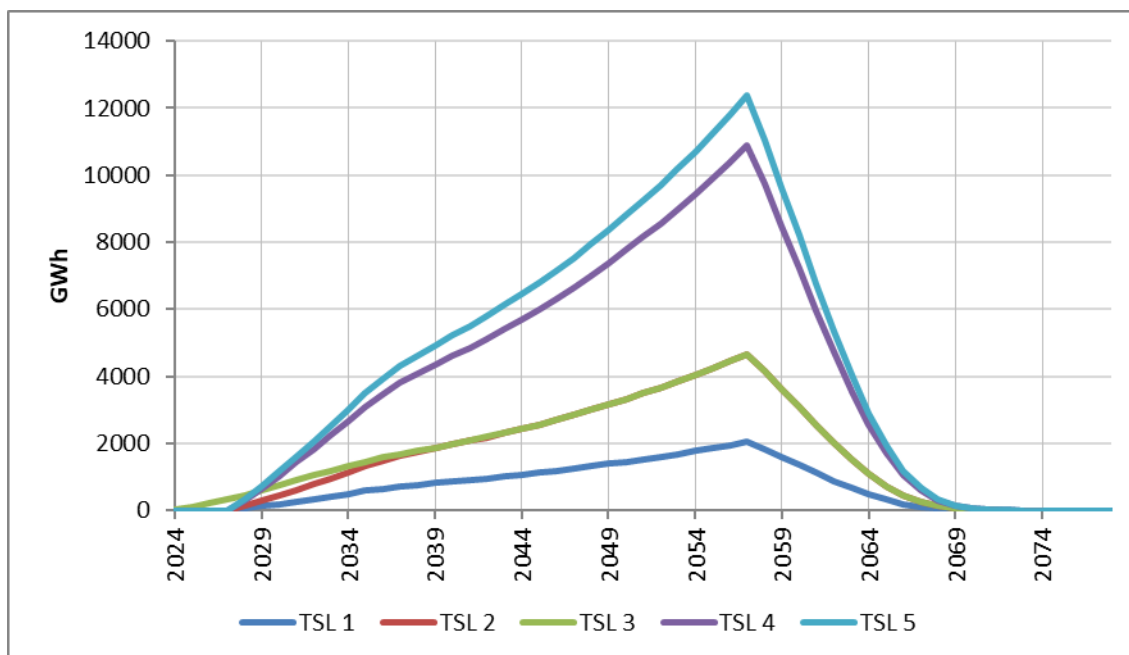


Figure 15.2.10 Air Cleaners: Renewables Generation Reduction

15.2.2.3 Results Summary

Table 15.2.1 presents a summary of the utility impact results for air cleaners.

Table 15.2.1 Air Cleaners: Summary of Electric Utility Impact Results

	TSL				
	1	2	3	4	5
Installed Capacity Reduction (MW)					
2024	-	-	44.4	-	-
2025	-	-	91.6	-	-
2030	176	402	694	932	1,047
2035	498	1,136	1,242	2,648	2,992
2040	721	1,645	1,649	3,844	4,355
Electricity Generation Reduction (GWh)					
2024	-	-	122	-	-
2025	-	-	249	-	-
2030	465	1,060	1,832	2,459	2,764
2035	1,252	2,855	3,123	6,657	7,523
2040	1,785	4,073	4,085	9,521	10,790

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CHAPTER 16. EMPLOYMENT IMPACT ANALYSIS

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CHAPTER 16. EMPLOYMENT IMPACT ANALYSIS

16.1 INTRODUCTION

DOE's employment impact analysis for air cleaners is designed to estimate indirect national job creation or elimination resulting from possible standards, due to reallocation of the associated expenditures for purchasing and operating air cleaners. Job increases or decreases reported in this chapter are separate from the direct manufacturing sector employment impacts reported in the manufacturer impact analysis (Chapter 12), and reflect the employment impact of efficiency standards on all other sectors of the economy.

16.2 ASSUMPTIONS

DOE expects energy conservation standards to decrease energy consumption, and therefore to reduce energy expenditures. The savings in energy expenditures may be spent on new investment or not at all (*i.e.*, they may remain "saved"). The standards may increase the purchase price of products, including the retail price plus sales tax, and increase installation costs.

Using the ImSET input/output econometric model of the U.S. economy, this analysis estimated the short-term effect of these expenditure impacts on net economic output and employment. DOE intends this analysis to quantify the indirect employment impacts of these expenditure changes. It evaluated direct employment impacts at manufacturers' facilities in the manufacturer impact analysis (see Chapter 12).

DOE notes that ImSET is not a general equilibrium forecasting model, and understands the uncertainties involved in projecting employment impacts, especially changes in the later years of the analysis.¹ Because ImSET does not incorporate price changes, the employment effects predicted by ImSET would over-estimate the magnitude of actual job impacts over the long run for this rule. Since input/output models do not allow prices to bring markets into equilibrium, they are best used for short-run analysis. DOE therefore includes a qualitative discussion of how labor markets are likely to respond in the longer term. In future rulemakings, DOE may consider the use of other modeling approaches for examining long run employment impacts.

16.3 METHODOLOGY

The Department based its analysis on an input/output model of the U.S. economy that estimates the effects of standards on major sectors of the economy related to buildings and the net impact of standards on jobs. The Pacific Northwest National Laboratory developed the model, ImSET 4 (Impact of Sector Energy Technologies)² as a successor to ImBuild³, a special-purpose version of the IMPLAN⁴ national input/output model. ImSET estimates the employment and income effects of building energy technologies. In comparison with simple economic multiplier approaches, ImSET allows for more complete and automated analysis of the economic impacts of energy-efficiency investments in buildings.

In an input/output model, the level of employment in an economy is determined by the relationships between different sectors of the economy and the spending flows among them. Different sectors have different levels of labor intensity, thus changes in the level of spending (*e.g.*, due to the effects of an efficiency standard) in one sector of the economy will affect flows in other sectors, which affects the overall national level of employment.

ImSET uses a 187-sector model of the national economy to predict the economic effects of residential and commercial buildings technologies. ImSET collects estimates of initial investments, energy savings, and economic activity associated with spending the savings resulting from standards (*e.g.*, changes in final demand in personal consumption, business investment and spending, and government spending). It provides overall estimates of the change in national output for each input-output sector. The model applies estimates of employment and wage income per dollar of economic output for each sector and calculates impacts on national employment.

Energy-efficiency technology primarily affects the U.S. economy along three spending pathways. First, general investment funds are diverted to sectors that manufacture, install, and maintain energy-efficient products. The increased cost of products leads to higher employment in the product manufacturing sectors and lower employment in other economic sectors. Second, commercial firm and residential spending are redirected from utilities toward firms that supply production inputs. Third, utility sector investment funds are released for use in other sectors of the economy. When consumers use less energy, utilities experience relative reductions in demand which leads to reductions in utility sector investment and employment.

DOE also notes that the employment impacts estimated with ImSET for the entire economy differ from the employment impacts in the air cleaner manufacturing sector estimated in Chapter 12 using the Government Regulatory Impact Model (GRIM). The methodologies used and the sectors analyzed in the ImSET and GRIM models are different.

16.4 SHORT-TERM RESULTS

The results in this section refer to impacts of air cleaner standards relative to the base case. DOE disaggregated the impact of standards on employment into three component effects: increased capital investment costs, decreased energy costs, and changes in operations and maintenance costs. DOE presents the summary impact.

Conceptually, one can consider the impact of the rule in its first year on three aggregate sectors, the air cleaner manufacturing sector, the energy generation sector, and the general consumer goods sector (as mentioned above ImSET's calculations are made at a much more disaggregate level). By raising energy efficiency, the rule generally increases the purchase price of air cleaners; this increase in expenditures causes an increase in employment in this sector. At the same time, the improvements in energy efficiency reduce consumer expenditures on energy, freeing up this money to be spent in other sectors. The reduction in energy demand causes a reduction in employment in that sector. Finally, based on the net impact of increased expenditures on air cleaners and reduced expenditures on energy, consumer expenditures on everything else are either positively or negatively affected, increasing or reducing jobs in that

sector accordingly. The model also captures any indirect jobs created or lost by changes in consumption due to changes in employment (*e.g.*, as more workers are hired they consume more goods, which generates more employment; the converse is true for workers laid off).

Table 16.4.1 present the modeled net employment impact from the rule in the compliance year (2024 and 2026 for TSL 3, 2028 for all other TSLs), in terms of thousands of jobs. Virtually all air cleaners are imported. The net employment impact estimate is sensitive to assumptions regarding the return to the U.S. economy of money spent on imported products. The two scenarios bounding the ranges presented in Table 16.4.1 represent situations in which none of the money spent on imported air cleaners returns to the U.S. economy and all of the money spent on imported air cleaners returns to the U.S. economy. The U.S. trade deficit in recent years suggests that between 50% and 75% of the money spent on imported products is likely to return, with employment impacts falling within the ranges presented below.

Table 16.4.1 Net National Short-term Change in Employment (1000s of Jobs)

Trial Standard Level	2024	2026	2028	2029	2031	2033
TSL 1	--	--	0.1	--	--	0.5
TSL 2	--	--	0.1 to 0.2	--	--	0.7 to 0.9
TSL 3	0.1	0.3 to 0.4	--	0.8 to 0.9	1.1 to 1.2	--
TSL 4	--	--	0.3 to 1.0	--	--	2.1 to 2.9
TSL 5	--	--	0.4 to 1.4	--	--	2.5 to 3.8

For context, the Congressional Budget Office projects that during this timeframe, the unemployment rate will be approximately 4.1%, close to “full employment.”⁵ When an economy is at full employment any effects on net employment are likely to be transitory as workers change jobs, rather than enter or exit longer-term employment.

16.5 LONG-TERM RESULTS

Over the long term DOE expects the energy savings to consumers to increasingly dominate the increase in product costs, resulting in increased aggregate savings to consumers. As a result, DOE expects demand for electricity to decline over time and demand for other goods to increase. Since the electricity generation sector is relatively capital intensive compared to the consumer goods sector, the net effect will be an increase in labor demand. In equilibrium, this should lead to upward pressure on wages and a shift in employment away from electricity generation towards consumer goods. Note that in long-run equilibrium there is no net effect on total employment since wages adjust to bring the labor market into equilibrium. Nonetheless, even to the extent that markets are slow to adjust, DOE anticipates that net labor market impacts will in general be negligible over time due to the small magnitude of the short-term effects presented in Table 16.4.1. The ImSET model projections, assuming no price or wage effects until 2029 and 2031 for TSL 3 and until 2033 for all other TSLs, are also included in Table 16.4.1.

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CHAPTER 17. REGULATORY IMPACT ANALYSIS

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CHAPTER 17. REGULATORY IMPACT ANALYSIS

17.1 INTRODUCTION

The Administrator of the Office of Information and Regulatory Affairs (OIRA) in the OMB has determined that the regulatory action in this document is a significant regulatory action under section (3)(f)(1) of Executive Order (E.O.) 12866. Regulatory Planning and Review. 58 FR 51735 (October 4, 1993). For such actions, E.O. 12866 requires Federal agencies to provide “an assessment, including the underlying analysis, of costs and benefits of potentially effective and reasonably feasible alternatives to the planned regulation, identified by the agencies or the public (including improving the current regulation and reasonably viable non-regulatory actions), and an explanation why the planned regulatory action is preferable to the identified potential alternatives.” 58 FR 51735, 51741.

To conduct the analysis of non-regulatory alternatives, DOE used an integrated National Impact Analysis (NIA)-RIA model built on a modified^a version of the NIA model discussed in chapter 10. DOE identified five non-regulatory policy alternatives that possibly could provide incentives for the same energy efficiency levels as the ones in the selected trial standard level (TSL) for the air cleaners that are the subject of this rulemaking. The non-regulatory policy alternatives are listed in Table 17.1.1, which also includes the “no new regulatory action” alternative. DOE evaluated each alternative in terms of its ability to achieve significant energy savings at a reasonable cost, and compared the effectiveness of each to the effectiveness of the selected standards for the three product classes of air cleaners covered by this RIA.

Table 17.1.1 Non-Regulatory Alternatives to National Standards

No New Regulatory Action
Consumer Rebates
Consumer Tax Credits
Manufacturer Tax Credits
Labeling
Bulk Government Purchases

Sections 17.2 and 17.3 discuss the analysis of five selected policies listed in Table 17.1.1 (excluding the alternative of “No New Regulatory Action”). Section 17.4 presents the results of the policy alternatives.

^a For this RIA, DOE developed an alternative NIA model where shipments in the standards case do not account for the effects that the price-elasticity of demand could have on those shipments. DOE believes that the national benefits from standards calculated this way are more comparable to the benefits from the alternative policies. In addition, DOE used shipment weighted average values across sectors (residential and commercial) from its former NIA model as inputs to the alternative NIA model.

17.2 NON-REGULATORY POLICIES

This section describes the method DOE used to analyze the energy savings and cost effectiveness of the non-regulatory policy alternatives for air cleaners. This section also describes the assumptions underlying the analysis.

17.2.1 Methodology

DOE used its integrated NIA-RIA spreadsheet model to calculate the national energy savings (NES) and net present value (NPV) associated with each non-regulatory policy alternative. Chapter 10 of this technical support document (TSD) describes the NIA spreadsheet model. Appendix 17A discusses the NIA-RIA integrated model approach.

DOE quantified the effect of each alternative on the purchase of equipment that meets the efficiency levels corresponding to the selected TSL. After establishing the quantitative assumptions underlying each alternative, DOE appropriately revised inputs to the NIA-RIA spreadsheet model. The primary model inputs revised were market shares of equipment meeting the target efficiency levels set for the selected TSL. The shipments of equipment for any given year reflect a shipment distribution across efficiency levels. DOE assumed that new energy efficiency standards would affect 100 percent of the shipments of products that did not meet the TSL target levels in the no-new-standards case, whereas the non-regulatory policies would affect a smaller percentage of those shipments. DOE made certain assumptions about the percentage of shipments affected by each alternative policy. DOE used those percentages to calculate the shipment-weighted average energy consumption and costs of air cleaners attributable to each policy alternative.

Increasing the efficiency of a product often increases its average installed cost. However, operating costs generally decrease because energy consumption declines. DOE therefore calculated an NPV for each non-regulatory alternative in the same way it did for the selected standards. In some policy scenarios, increases in total installed cost are mitigated by government rebates or tax credits. Because government expenditures on tax credits and rebates would be covered to a significant extent by income taxes paid by consumers in the aggregate, DOE did not include rebates or tax credits as a consumer benefit when calculating national NPV. DOE's analysis also excluded any administrative costs for the non-regulatory policies; including such costs would decrease the NPVs slightly.

The following are key measures for evaluating the impact of each alternative.

- National Energy Savings (NES), given in quadrillion Btus (quads), describes the cumulative national energy saved over the lifetime of equipment purchased during the analysis period starting in the effective date of the policy (2024-2057).
- Net Present Value (NPV), represents the value of net monetary savings in 2022, expressed in 2021\$, from equipment purchased during the analysis period starting in the effective date of the policy (2024-2057). DOE calculated the NPV as the difference between the present values of installed equipment cost and operating expenditures in the no-new-standards case and the present values of those costs in each policy case. DOE calculated operating expenses (including energy costs) for the life of the product.

17.2.2 Assumptions Regarding Non-Regulatory Policies

The effects of non-regulatory policies are by nature uncertain because they depend on program implementation, marketing efforts, and on consumers' response to a program. Because the projected effects depend on assumptions regarding the rate of consumer participation, they are subject to more uncertainty than are the impacts of mandatory standards, which DOE assumes will be met with full compliance. To increase the robustness of the analysis, DOE periodically conducts a literature review regarding each non-regulatory policy and consults with recognized experts to gather information on similar incentive programs that have been implemented in the United States. By studying experiences with the various types of programs, DOE sought to make credible assumptions regarding potential market impacts. Section 17.3 presents the sources DOE relied on in developing assumptions about each alternative policy and reports DOE's conclusions as they affected the assumptions that underlie the modeling of each alternative policy.

Each non-regulatory policy that DOE considered would improve the average efficiency of new air cleaners relative to their no-new-standards case efficiency scenario (which involves no new regulatory action). The analysis considered that each alternative policy would induce consumers to purchase units having the same technology as required by standards (the target level), according to the minimum energy efficiency set for each TSL. As opposed to the standards case, however, the policy cases may not lead to 100 percent market penetration of units that meet the target level.

Table 17.2.1 shows the minimum energy efficiency of the air cleaner product classes at each proposed TSL. In this chapter DOE evaluates the benefits from non-regulatory policy alternatives only for the selected TSL (TSL 3).

Table 17.2.1 Minimum Energy Efficiency by TSL (CADR/Watt)

	TSL 1	TSL 2	TSL 3*	TSL 4	TSL 5
$10 \leq \text{PM}_{2.5} \text{ CADR} < 100$	1.69	1.89	1.69, 1.89	1.89	5.40
$100 \leq \text{PM}_{2.5} \text{ CADR} < 150$	1.90	2.39	1.90, 2.39	2.39	12.75
$\text{PM}_{2.5} \text{ CADR} \geq 150$	2.01	2.91	2.01, 2.91	6.55	7.41

* Efficiencies shown for TSL 3 refer to standards starting in 2024 and 2026, respectively.

DOE assumed that the effects of non-regulatory policies would last from the effective date of standards through the end of the analysis period, which is 2057.

17.2.3 Policy Interactions

DOE calculated the effects of each non-regulatory policy separately from those of the other policies. In practice, some policies are most effective when implemented in combination, such as labeling implemented with consumer rebates or tax credits. However, DOE attempted to make conservative assumptions to avoid double-counting policy impacts. The resulting policy impacts are therefore not additive, and the combined effect of several or all policies cannot be inferred from summing their results.

Section 17.4 presents graphs that show the market penetration estimated under each non-regulatory policy for air cleaners.

17.3 NON-REGULATORY POLICY ASSUMPTIONS

The following subsections describe DOE's analysis of the impacts of the five non-regulatory policy alternatives to the standards selected for air cleaners. (Because the alternative of "No New Regulatory Action" has no energy or economic impacts, essentially representing the NIA no-new-standards case, DOE did not perform any additional analysis for that alternative.) DOE developed estimates of the market penetration of more efficient products both with and without each of the non-regulatory policy alternatives.

17.3.1 No New Regulatory Action

The case in which no new regulatory action is taken with regard to the energy efficiency of air cleaners constitutes the no-new-standards case, as described in chapter 10, National Impact Analysis. The no-new-standards case provides the basis of comparison for all other policies. By definition, no new regulatory action yields zero NES and an NPV of zero dollars.

17.3.2 Consumer Rebates

DOE considered the scenario in which the Federal government would provide financial incentives in the form of rebates to consumers for purchasing energy-efficient equipment. This policy provides a consumer rebate for purchasing air cleaners that operate at the same efficiency levels as stipulated in the selected TSL.

17.3.2.1 Methodology

DOE based its evaluation methodology for consumer rebates on a comprehensive study of California's potential for achieving energy efficiency. The study, performed by XENERGY, Inc.,^b summarized experiences with various utility rebate programs.¹ XENERGY's analytical method utilized graphs, or penetration curves, that estimate the market penetration of a technology based on its benefit/cost (B/C) ratio. DOE consulted with experts and reviewed other methods of estimating the effect of consumer rebate programs on the market penetration of efficient technologies. The other methods, developed after the referenced XENERGY report was published,^{2, 3, 4, 5, 6, 7, 8} used different approaches: other economic parameters (e.g., payback period), expert surveys, or model calibration based on specific utility program data rather than multi-utility data. Some models in use by energy efficiency program evaluation experts were so client-specific that generic relationships between economic parameters and consumer response could not be established.^{5, 6} DOE decided that the most appropriate available method for this RIA was the XENERGY approach of penetration curves based on B/C ratio, which incorporates lifetime operating cost savings.

XENERGY's model estimates market impacts induced by financial incentives based on the premise that two types of information diffusion drive the adoption of new technologies.

^b XENERGY is now owned by KEMA, Inc. (www.kema.com)

Internal sources of information encourage consumers to purchase new equipment primarily through word-of-mouth from early adopters. *External sources* affect consumer purchase decisions through marketing efforts and information from outside the consumer group. Appendix 17A contains additional details on internal and external information diffusion.

XENERGY's model equation accounts for the influences of both internal and external sources of information by superimposing the two components. Combining the two mechanisms for information diffusion, XENERGY's model generates a set of penetration (or implementation) curves for a policy measure. XENERGY calibrated the curves based on participation data from utility rebate programs. The curves illustrate the increased penetration (i.e., increased market share) of efficient equipment driven by consumer response to changes in B/C ratio induced by rebate programs. The penetration curves depict various diffusion patterns based on perceived market barriers (from no-barriers to extremely-high-barriers) to consumer purchase of high-efficiency equipment. DOE adjusted the XENERGY former penetration curves based on expert advice founded on more recent utility program experience.^{5, 6, 9}

DOE modeled the effects of a consumer rebate policy for air cleaners by determining the increase in market penetration of equipment meeting the target level relative to their market penetration in the no-new-standards case. It used the interpolation method presented in Blum et al (2011)¹⁰ to create customized penetration curves based on relationships between actual no-new-standards case market penetrations and actual B/C ratios. To inform its estimate of B/C ratios provided by a rebate program DOE reviewed rebate programs offered towards ENERGY STAR qualifying air cleaners. Based on that review, and considering the value of the rebates offered, DOE assumed that any rebate would cover, in full, the upfront costs for a new air cleaner. Based on this assumption, DOE calibrated the customized penetration curves it developed for each product class covered by this RIA so they can best reflect the market barrier levels that consumer rebates for air cleaners would face. Section 17.3.2.2 shows the resulting interpolated curves used in the analysis.

17.3.2.2 Analysis

DOE estimated the effect of increasing the B/C ratio of air cleaners via a rebate that would pay all of the increased installed cost of units that meet the target efficiency levels compared to units meeting the baseline efficiency level.^c DOE then rebate values for each product class covered by this RIA which it applied in the calculation of the B/C ratio of air cleaners under the effect of consumer rebates. (Appendix 17A presents the rebate amounts that DOE used.) Given that the selected TSL has two compliance years, 2024 and 2026, DOE estimated, for each product class, one rebate amount for the 2024-2025 period, and another rebate amount for the 2026-2057 period.

DOE first calculated the B/C ratio of an air cleaner without a rebate using the difference in total installed costs (C) and lifetime operating cost savings^d (B) between a unit meeting the target level and a baseline unit. It then calculated the B/C ratio given a rebate for the unit

^c The baseline technology is defined in the engineering analysis, chapter 5, as the technology that represents the basic characteristics of air cleaners. A baseline unit typically is one that just meets current Federal energy conservation standards and provides basic consumer utility.

^d The cash flow of the operating cost savings is discounted to the purchase year using a 7 percent discount rate.

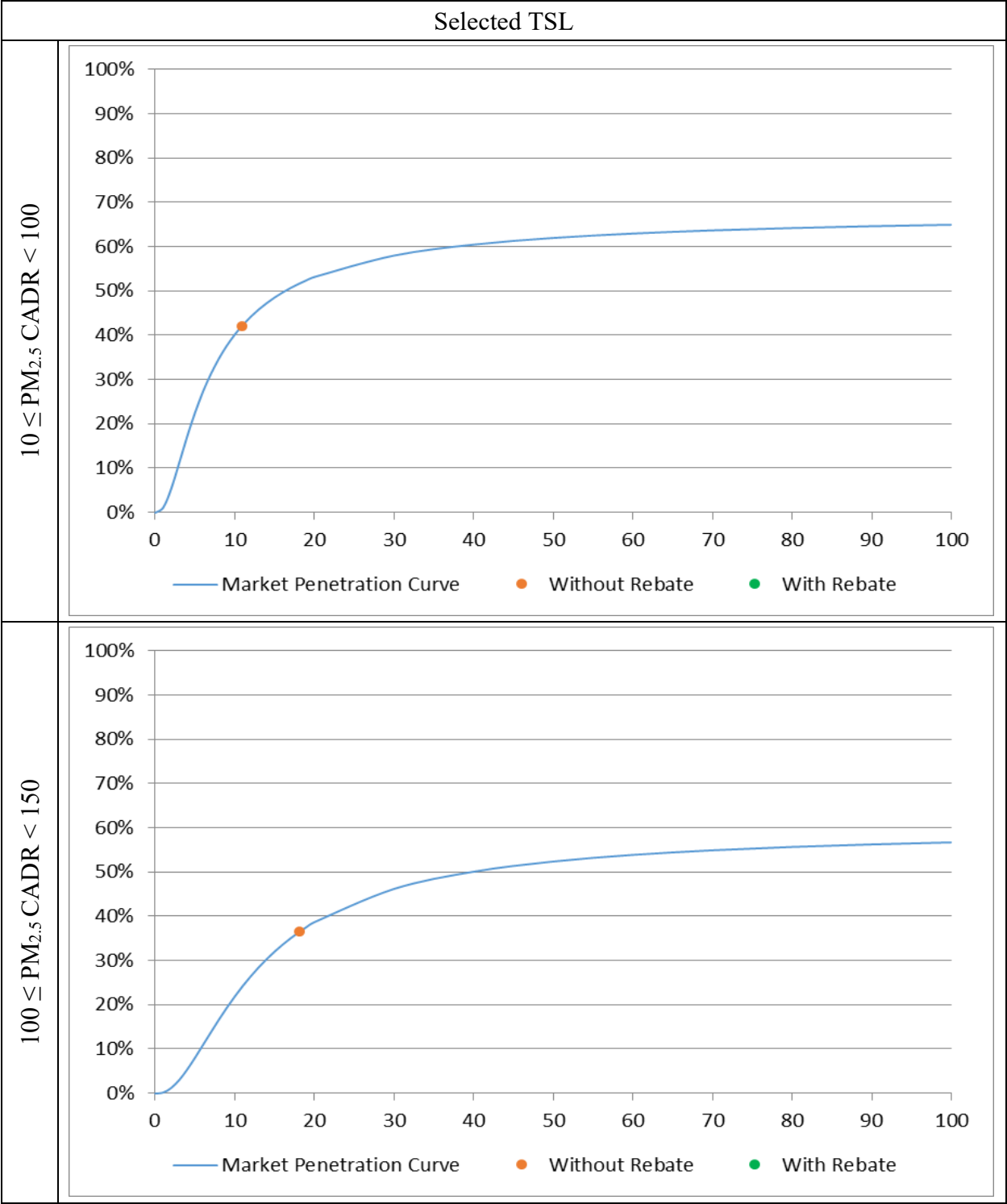
meeting the target efficiency level. Because the rebate reduced the incremental cost, the unit receiving the rebate had a larger B/C ratio. Table 17.3.1 shows the effect of consumer rebates for the selected TSL on the B/C ratio of air cleaners shipped in the first year of the analysis period.

Table 17.3.1 Benefit/Cost Ratios Without and With Rebates

	TSL 3	
	2024	2026
$10 \leq \text{PM}_{2.5} \text{ CADR} < 100$		
Net-benefits Without Rebate	17.2	17.26
Rebate Amount (2021\$)	1.74	3.66
Net-benefits with Rebate*	18.94	18.95
Estimated Market Barriers	Moderate	High
$100 \leq \text{PM}_{2.5} \text{ CADR} < 150$		
Net-benefits Without Rebate	38.04	28.17
Rebate Amount (2021\$)	2.22	4.46
Net-benefits with Rebate*	40.26	106.21
Estimated Market Barriers	Moderate-High	High
$\text{PM}_{2.5} \text{ CADR} \geq 150$		
Net-benefits Without Rebate	103.90	104.22
Rebate Amount (2021\$)	2.03	3.41
Net-benefits with Rebate*	105.93	106.21
Estimated Market Barriers	High	High

* Since rebates would cover in full the incremental installed cost, the net-benefits with rebate correspond to the net-benefits when the market penetration of efficient air cleaners are the highest possible share estimated for this equipment, given the market barriers estimated for the equipment class.

DOE used the B/C ratio along with the customized penetration curves shown in Figure 17.3.1 to estimate the percentage of consumers who would purchase air cleaners that meet the target levels both with and without a rebate incentive. Table 17.3.1 indicates the estimated levels of market barriers corresponding to the penetration curves DOE calculated to represent the market behavior for air cleaners at the selected TSL.



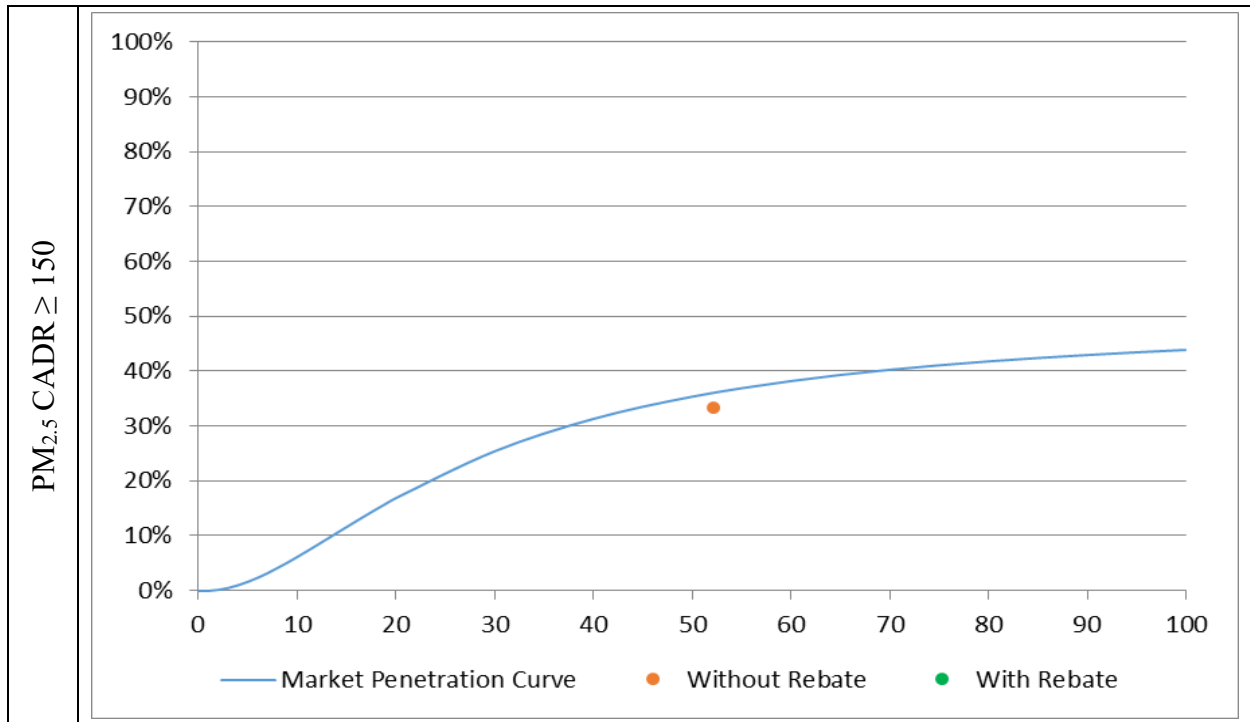


Figure 17.3.1 Market Penetration Curves for Air Cleaners^e

DOE next estimated the percent increase represented by the change in penetration rate shown on the corresponding penetration curve. It then added this percent increase to the market share of units that meet the target level in the no-new-standards case to obtain the market share of units that meet the target level in the rebate policy case.

Table 17.3.2 summarizes DOE's assumptions for air cleaners regarding the market penetration of products that meet the target levels at each TSL given a consumer rebate.

^e Because the B/C ratio with rebates is infinite for all product classes (see Table 17.3.1), the data points that refer to the market penetration with rebates are not shown in the charts.

Table 17.3.2 Market Penetrations in Respective Compliance Years Attributable to Consumer Rebates

	TSL 3	
	2024	2026
10 ≤ PM_{2.5} CADR < 100		
Base-Case Market Share	42.1%	19.1%
Policy Case Market Share	67.7%	67.7%
Increased Market Share	25.6%	48.6%
100 ≤ PM_{2.5} CADR < 150		
Base-Case Market Share	36.6%	28.1%
Policy Case Market Share	60.1%	60.1%
Increased Market Share	23.6%	32.0%
PM_{2.5} CADR ≥ 150		
Base-Case Market Share	33.3%	37.7%
Policy Case Market Share	50.0%	50.0%
Increased Market Share	16.7%	12.3%

DOE used the resulting annual increases in market shares as inputs to represent the rebate policy case scenario in its NIA-RIA model. Appendix 17A shows the annual market share increases due to this policy for the whole forecast period. Section 17.4 presents the resulting market penetration trends for the policy case of consumer rebates for air cleaners.

17.3.3 Consumer Tax Credits

DOE estimated the effects of tax credits on consumer purchases based on its previous analysis of consumer participation in tax credits. DOE supported its approach using data from Oregon State’s tax credit program for energy-efficient appliances. DOE also incorporated previous research that disaggregated the effect of rebates and tax credits into a *direct price effect*, which derives from the savings in purchase price, and an *announcement effect*, which is independent of the amount of the incentive.^{11, 12} The announcement effect derives from the credibility that a technology receives from being included in an incentive program, as well as changes in product marketing and modifications in markup and pricing. DOE assumed that the rebate and consumer tax credit policies would encompass both direct price effects and announcement effects, and that half the increase in market penetration associated with either policy would be due to the direct price effect and half to the announcement effect.

In estimating the effects of a tax credit on purchases of consumer products that meet new efficiency standards, DOE assumed the amount of the tax credit would be the same as the corresponding rebate amount discussed above.

DOE estimated that fewer consumers would participate in a tax credit program than would take advantage of a rebate. Research has shown that the delay required for a consumer to receive a tax credit, plus the added time and cost in preparing the tax return, make a tax credit incentive less effective than a rebate received at the time of purchase. Based on previous analyses, DOE assumed that only 60 percent of the consumers who would take advantage of a rebate would take advantage of a tax credit.

In preparing its assumptions to estimate the effects of tax credits on consumer purchases of air cleaners, DOE also reviewed other tax credit programs that have been offered at both the Federal and State levels for energy-efficient appliances.

The Energy Policy Act of 2005 (EPACT 2005) included Federal tax credits for consumers who purchase certain energy-efficient products.¹⁴ Those tax credits were in effect in 2006 and 2007, expired in 2008, were reinstated for 2009–2010 by the American Recovery and Reinvestment Act of 2009 (ARRA), extended by Congress for 2011 with some modifications, and expired at the end of 2011.^{15,16} The American Taxpayer Relief Act of 2012 extended, with some modifications, residential tax credits for air conditioners, heat pumps, furnaces, and water heaters placed in service between January 1, 2012 and December 31, 2013.¹⁷ DOE reviewed Internal Revenue Service data on the numbers of taxpayers who claimed the tax credits during tax years 2006 and 2007. DOE also reviewed data from an earlier Federal energy conservation tax credit program in place in the 1980s. However, DOE did not find data specific enough to air cleaners to warrant adjusting its analysis method for the Consumer Tax Credits policy case.^f Appendix 17A contains more information on Federal consumer tax credits.

DOE also reviewed its previous analysis of Oregon’s tax credits for clothes washers to provide support for its assumptions.¹⁸ In that previous analysis, DOE compared the market shares of ultra-high efficiency (UHE) residential clothes washers in Oregon, which offered both State tax credits and utility rebates, with those in Washington State, which offered only utility rebates during the same period. Based on this analysis, DOE estimated that in Oregon the impact of tax credits was 62 percent of the impact of rebates for UHE clothes washers having equivalent efficiency. This finding supports its original assumption that participation in a tax credit program would be about 60 percent of participation in a rebate program. Additional discussion of State tax credits for Oregon and other states is in appendix 17A.

DOE applied the assumed 60 percent participation described above to the increase in penetration rates estimated for the rebate policy to estimate penetration rates attributable to consumer tax credits. In doing so, DOE incorporated the assumptions for consumer response to financial incentives from the customized penetration curves it developed for air cleaners (See Figure 17.3.1).

Table 17.3.3 summarizes DOE’s assumptions for air cleaners regarding the market penetration of products that meet the target levels at the selected TSL given a consumer tax credit.

^f The Inflation Reduction Act (2022) provides for tax credits for the purchase of some energy-efficient products, but there are no data yet to examine their impact.

Table 17.3.3 Market Penetrations in Respective Compliance Years Attributable to Consumer Tax Credits

	TSL 3	
	2024	2026
$10 \leq \text{PM}_{2.5} \text{ CADR} < 100$		
Base-Case Market Share	42.1%	19.1%
Policy Case Market Share	57.5%	48.3%
Increased Market Share	15.4%	29.2%
$100 \leq \text{PM}_{2.5} \text{ CADR} < 150$		
Base-Case Market Share	36.6%	28.1%
Policy Case Market Share	50.7%	47.3%
Increased Market Share	14.1%	19.2%
$\text{PM}_{2.5} \text{ CADR} \geq 150$		
Base-Case Market Share	33.3%	37.7%
Policy Case Market Share	43.3%	45.1%
Increased Market Share	10.0%	7.4%

The increased market shares attributable to consumer tax credits shown in Table 17.3.3 were used as inputs in the NIA-RIA model. Appendix 17A shows the annual market share increases due to this policy for the whole forecast period. Section 17.4 presents the resulting market penetration trends for the policy case of consumer tax credits for air cleaners that meet the efficiency level for the selected TSL. Because the change in market penetration for consumer tax credits is proportional to the change in market penetration DOE calculated for consumer rebates, they follow similar trends over the analysis period.

17.3.4 Manufacturer Tax Credits

To analyze the potential effects of a policy that offers tax credits to manufacturers that produce air cleaners that meet the target efficiency levels at TSL 3, DOE assumed that a manufacturer tax credit would lower the consumer's purchase cost by an amount equivalent to that provided by the consumer rebates or tax credits described above. DOE further assumed that manufacturers would pass on some of their reduced costs to consumers, causing a direct price effect. DOE assumed that no announcement effect would occur, because the program would not be visible to consumers.⁸ Because the direct price effect is approximately equivalent to the announcement effect,¹¹ DOE estimated that a manufacturer tax credit would induce half the number of consumers assumed to take advantage of a consumer tax credit to purchase more efficient products. Thus, the assumed participation rate is equal to 30 percent of the number of consumers who would participate in a rebate program.

DOE attempted to investigate manufacturer response to the Energy Efficient Appliance Credits for manufacturers mandated by EPACT 2005.¹⁴ Those manufacturer tax credits have been in effect for dishwashers, clothes washers and refrigerators produced beginning in 2009.

⁸ Note that this is a conservative assumption, since it is possible that manufacturers or utility/agency efficiency programs might promote the models for which manufacturers increase production due to the tax credits, which in turn might induce some announcement effect. However, DOE found no data on such programs on which to base an estimate of the magnitude of this possible announcement effect on consumer behavior.

DOE was unable to locate data from the Internal Revenue Service or other sources on manufacturer response to the Federal credits. Appendix 17A presents details on Federal manufacturer tax credits.

DOE applied the assumption of 30 percent participation to the increase in penetration rates predicted for the rebate policy to estimate the effects of a manufacturer tax credit policy. In doing so, DOE incorporated the assumptions for consumer response to financial incentives from the customized penetration curves it developed for air cleaners. (See Figure 17.3.1).

Table 17.3.4 summarizes DOE’s assumptions for air cleaners regarding the market penetration of products that meet the target levels at the selected TSL given a manufacturer tax credit.

Table 17.3.4 Market Penetrations in Respective Compliance Years Attributable to Manufacturer Tax Credits

	TSL 3	
	2024	2026
$10 \leq \text{PM}_{2.5} \text{ CADR} < 100$		
Base-Case Market Share	42.1%	19.1%
Policy Case Market Share	49.8%	33.7%
Increased Market Share	7.7%	14.6%
$100 \leq \text{PM}_{2.5} \text{ CADR} < 150$		
Base-Case Market Share	36.6%	28.1%
Policy Case Market Share	43.6%	37.7%
Increased Market Share	7.1%	9.6%
$\text{PM}_{2.5} \text{ CADR} \geq 150$		
Base-Case Market Share	33.3%	37.7%
Policy Case Market Share	38.3%	41.4%
Increased Market Share	5.0%	3.7%

The increased market shares attributable to a manufacturer tax credit shown in Table 17.3.4 were used as inputs in the NIA-RIA model. Appendix 17A shows the annual market share increases due to this policy for the whole forecast period. Section 17.4 presents the resulting market penetration trends for the policy case of manufacturer tax credits for air cleaners. Because the change in market penetration for manufacturer tax credits is proportional to the change in market penetration DOE calculated for consumer rebates, they follow similar trends over the analysis period.

17.3.5 Voluntary Labeling

DOE assumed that a voluntary labeling program, similar to the ENERGY STAR program conducted by the Environmental Protection Agency (EPA) and DOE, in conjunction with industry partners, would lead to an increase in purchase of air cleaners with the efficiency levels set for TSL 3. The ENERGY STAR program specifies the minimum energy efficiencies that various products must have to receive the ENERGY STAR label. ENERGY STAR encourages consumers to purchase efficient products via marketing that promotes consumer label recognition, various incentive programs that adopt the ENERGY STAR specifications, and

manufacturers' promotion of their qualifying appliances. ENERGY STAR projects market penetration of compliant appliances and estimates the percentage of sales of compliant appliances that are attributable to the ENERGY STAR program.

Researchers have analyzed the ENERGY STAR program's effects on sales of several consumer products. Program efforts generally involve a combination of information dissemination and utility or agency rebates. The analyses have been based on State-specific data on percentages of shipments of various appliances that meet ENERGY STAR specifications. The analyses generally have concluded that the market penetration of ENERGY STAR-qualifying appliances is higher in regions or States where ancillary promotional programs have been active.^{19, 20, 21}

DOE believes that informational incentive programs – like ENERGY STAR, or any other labeling program sponsored by industry or other organizations – are likely to reduce the market barriers to more efficient products over time. During the rebate analysis, when assessing the B/C ratio and market penetration in the no-new-standards case for air cleaners, DOE observed that the level of market barriers for more efficient air cleaners are in the range of moderate barriers to a high level of market barriers. DOE estimates that a voluntary labeling program could reduce these barriers to lower levels over 10 years. Table 17.3.5 presents the levels of market barriers DOE estimated for air cleaners in the no-new-standards case and in the labeling policy case (10 years later). DOE followed the methodology presented by Blum et al (2011)¹⁰ to evaluate the effects that such a reduction in market barriers would have on the market penetration of efficient air cleaners.^h The methodology relies on interpolated market penetration curves to calculate – given a B/C ratio – how the market penetration of more efficient units increases as the market barrier level to those units decreases.

Table 17.3.5 Market Barriers Changes Attributable to Voluntarily Labeling (TSL 3)

	No-new-standards case	Voluntary Labeling
$10 \leq \text{PM}_{2.5} \text{ CADR} < 100$	Moderate	No
$100 \leq \text{PM}_{2.5} \text{ CADR} < 150$	Moderate-High	Low
$\text{PM}_{2.5} \text{ CADR} \geq 150$	High	Moderate

Table 17.3.6 summarizes DOE's assumptions for air cleaners regarding the market penetration of products that meet the target levels at the selected TSL given a labeling program. Table 17.3.7 expands on Table 17.3.6 to include, for the selected TSL, DOE's assumptions regarding the market penetration of units in selected years.

^h For the calculation of B/C ratios DOE discounted the cash flow of the operating cost savings to the purchase year using a 7 percent discount rate.

Table 17.3.6 Market Penetrations in Respective Compliance Years Attributable to Voluntary Labeling

	TSL 3	
	2024	2026
10 ≤ PM_{2.5} CADR < 100		
Base-Case Market Share	42.1%	19.1%
Policy Case Market Share	57.3%	70.9%
Increased Market Share	15.2%	51.8%
100 ≤ PM_{2.5} CADR < 150		
Base-Case Market Share	36.6%	28.1%
Policy Case Market Share	48.0%	58.6%
Increased Market Share	11.5%	30.5%
PM_{2.5} CADR ≥ 150		
Base-Case Market Share	33.3%	37.7%
Policy Case Market Share	46.0%	57.5%
Increased Market Share	12.7%	19.8%

Table 17.3.7 Market Penetrations in Selected Years Attributable to Voluntary Labeling for TSL 3

	2024	2026	2033	2057
10 ≤ PM_{2.5} CADR < 100				
Base-Case Market Share	42.1%	19.1%	19.1%	19.1%
Policy Case Market Share	57.3%	70.9%	89.2%	89.2%
Increased Market Share	15.2%	51.8%	70.1%	70.1%
100 ≤ PM_{2.5} CADR < 150				
Base-Case Market Share	36.6%	28.1%	28.1%	28.1%
Policy Case Market Share	48.0%	58.6%	72.7%	73.8%
Increased Market Share	11.5%	30.5%	44.6%	45.6%
PM_{2.5} CADR ≥ 150				
Base-Case Market Share	33.3%	37.7%	37.7%	37.7%
Policy Case Market Share	46.0%	57.5%	64.9%	65.5%
Increased Market Share	12.7%	19.8%	27.2%	27.8%

The increased market shares attributable to labeling shown in Table 17.3.6 were used as inputs in the NIA-RIA model. Appendix 17A shows the annual market share increases due to this policy for the whole forecast period. Section 17.4 presents the resulting market penetration trends for the policy case of labeling for air cleaners that meet the efficiency level for the selected TSL.

17.3.6 Bulk Government Purchases

Bulk government purchases can lead to Federal, State, and local governments purchasing large quantities of products that meet a certain, target efficiency level. Combining the market demands of multiple public sectors can provide a market signal to manufacturers and vendors

that some of their largest customers seek products that meet an efficiency target at favorable prices. Such a program also can induce “market pull,” whereby manufacturers and vendors would achieve economies of scale for high efficiency products.

Most of the previous bulk government purchase (procurement) initiatives at the Federal, State, and municipal levels have not tracked data on numbers of purchases or degree of compliance with procurement specifications. In many cases, procurement programs are decentralized, being part of larger State or regional initiatives. DOE based its assumptions regarding the effects of this policy on studies the Federal Energy Management Program (FEMP) performed regarding the savings potential of its procurement specifications for appliances and other products. FEMP, however, does not track purchasing data, because of the complex range of purchasing systems, large number of vendors, and so on. States, counties, and municipalities have demonstrated increasing interest and activity in “green purchasing.” Although many of the programs target office equipment, the growing infrastructure for developing and applying efficient purchasing specifications indicates that bulk government purchase programs are feasible.^{22, 23}

DOE assumed that government agencies would administer bulk purchasing programs for air cleaners. At the federal level, this type of program could be similar to the current FEMP procurement guidelines for air cleaners, which refer to the ENERGY STAR requirements for air cleaners.ⁱ DOE reviewed its own previous research on the potential for market transformation through bulk government purchases. Its major study analyzed several scenarios based on the assumption that 20 percent of Federal equipment purchases in 2000 already incorporated energy efficiency requirements based on FEMP guidelines. One scenario in the DOE report showed energy efficient purchasing ramping up during 10 years from 20 percent to 80 percent of all Federal purchases.²⁴ Based on this study, DOE estimated that a bulk government purchase program instituted within a 10-year period would result in at least 80 percent of government-purchased air cleaners meeting the target efficiency level.

DOE assumed that bulk government purchases would affect the subset of housing units and commercial buildings for which government agencies purchased or influenced the purchase of air cleaners. According to the 2009 Residential Energy Consumption Survey (RECS 2009), about 4.3 percent of all U.S. households are housing units in public housing authority.²⁵ DOE therefore estimated that 4.3 percent of the U.S. housing units constitute the market to which this policy would apply. DOE used the 2018 Commercial Buildings Energy Consumption Survey (CBECS 2018)²⁶ to calculate the share of floor space in commercial buildings that corresponds to federal government-owned buildings. The calculation resulted in 1.8 percent. DOE assumed that percentage as the share of the market to which this policy would apply.

DOE estimated that starting in 2024, each year of a bulk government purchase policy would result in an increasing percent of shipments of government-purchased units beyond the no-new-standards case that would meet the target efficiency level. DOE estimated that within 10 years (by 2033) bulk government purchasing programs would result in 80 percent^j of the market

ⁱ <https://www.energy.gov/eere/femp/search-energy-efficient-products>

^j The 80 percent target to be achieved within 10 years may not be reached, as it is constrained by the market share below the target level in the no-new-standards case scenario.

for air cleaners used in publicly owned housing and commercial buildings meeting the target level. DOE modeled the bulk government purchase program assuming that the market share for air cleaners achieved in 2033 would be at least maintained throughout the rest of the forecast period.

Table 17.3.8 summarizes DOE's assumptions for air cleaners regarding the market penetration of products that meet the target levels at the selected TSL given bulk government purchases.

Table 17.3.8 Market Penetrations in Respective Compliance Years Attributable to Bulk Government Purchases

	TSL 3	
	2024	2026
$10 \leq \text{PM}_{2.5} \text{ CADR} < 100$		
Base-Case Market Share	42.1%	19.1%
Policy Case Market Share	42.2%	20.1%
Increased Market Share	0.1%	1.0%
$100 \leq \text{PM}_{2.5} \text{ CADR} < 150$		
Base-Case Market Share	36.6%	28.1%
Policy Case Market Share	36.7%	28.7%
Increased Market Share	0.1%	0.6%
$\text{PM}_{2.5} \text{ CADR} \geq 150$		
Base-Case Market Share	33.3%	37.7%
Policy Case Market Share	33.4%	38.0%
Increased Market Share	0.1%	0.3%

The increased market shares attributable to bulk government purchases shown in Table 17.3.8 were used as inputs in the NIA-RIA model. Appendix 17A shows the annual market share increases due to this policy for the whole forecast period. Section 17.4 presents the resulting market penetration trends for the policy case of bulk government purchases for air cleaners. Market penetrations increase over the first 10 years of the forecast period, and steady for the rest of the analysis period.

17.4 IMPACTS OF NON-REGULATORY ALTERNATIVES

Figure 17.4.1 through Figure 17.4.3 show the effects of each non-regulatory policy alternative on the market penetration of more efficient air cleaners. Relative to the no-new-standards case, the alternative policy cases increase the market shares that meet the target level.

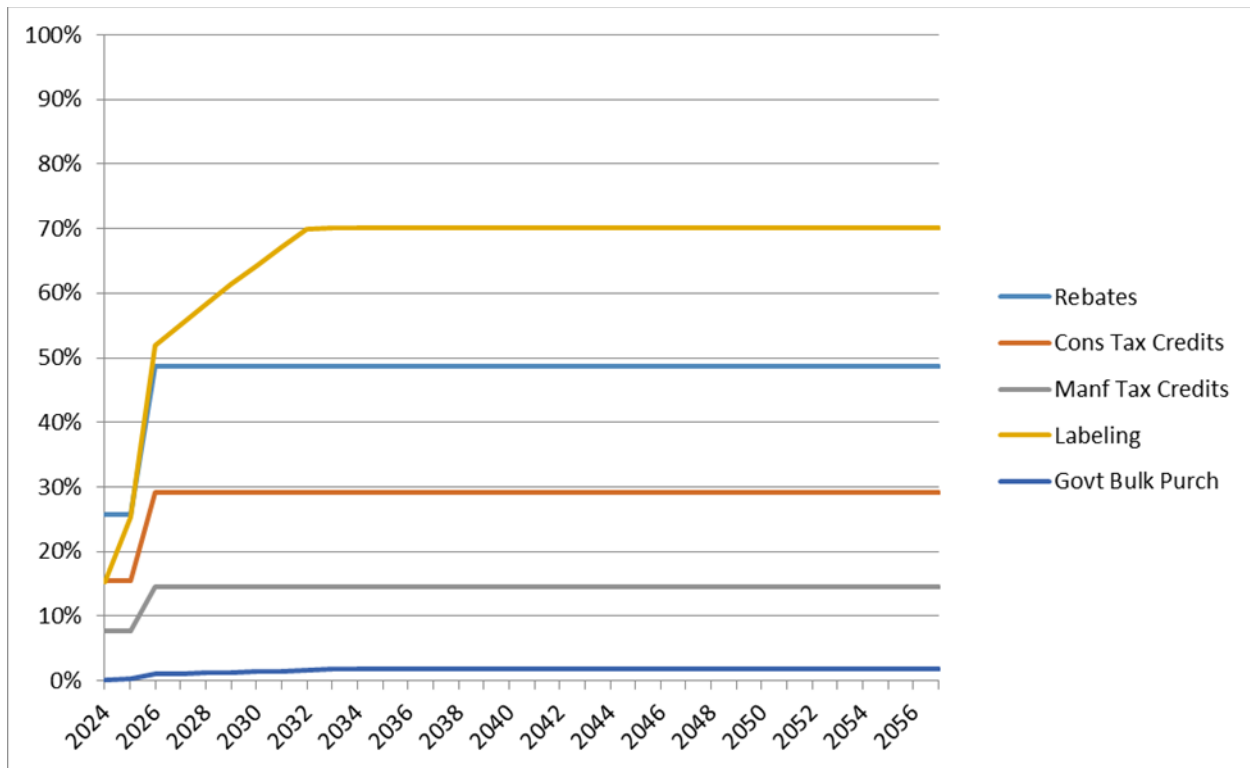


Figure 17.4.1 Increase in Market Penetration of Efficient $10 \leq \text{PM}_{2.5} \text{ CADR} < 100$ (TSL 3)

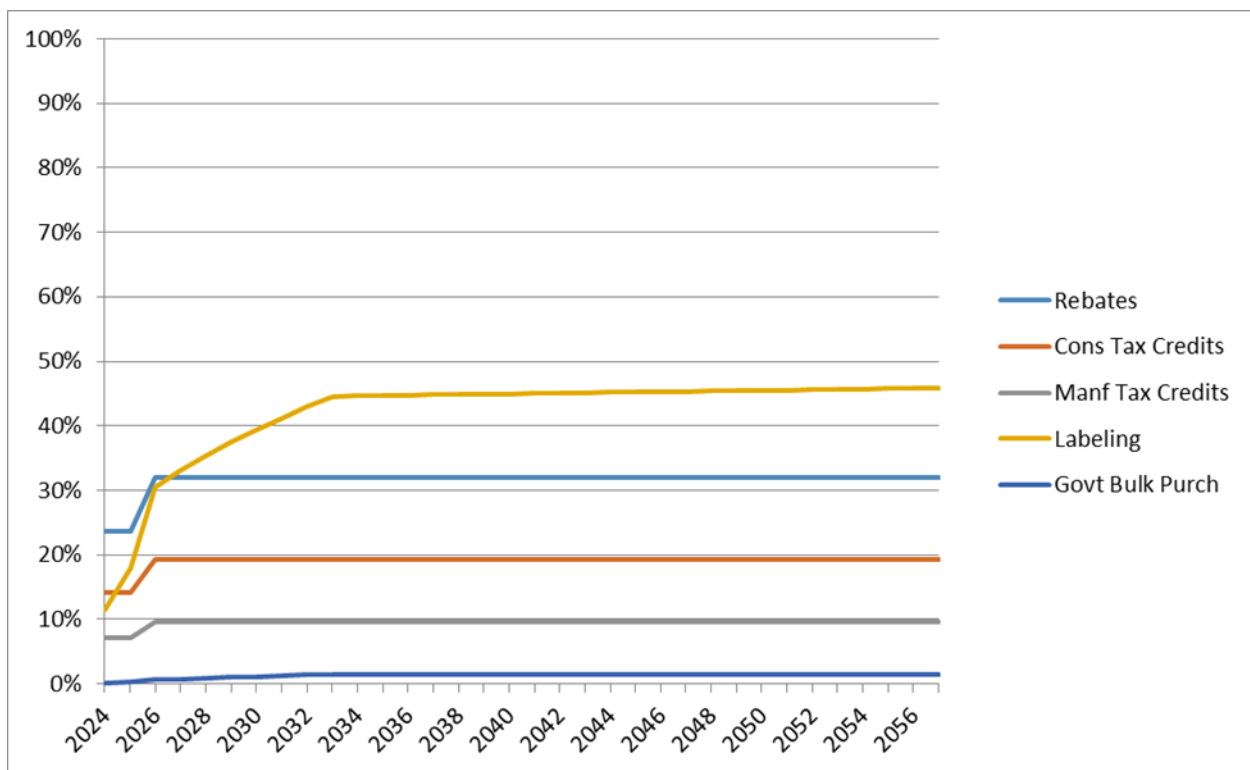


Figure 17.4.2 Increase in Market Penetration of Efficient $100 \leq \text{PM}_{2.5} \text{ CADR} < 150$ (TSL 3)

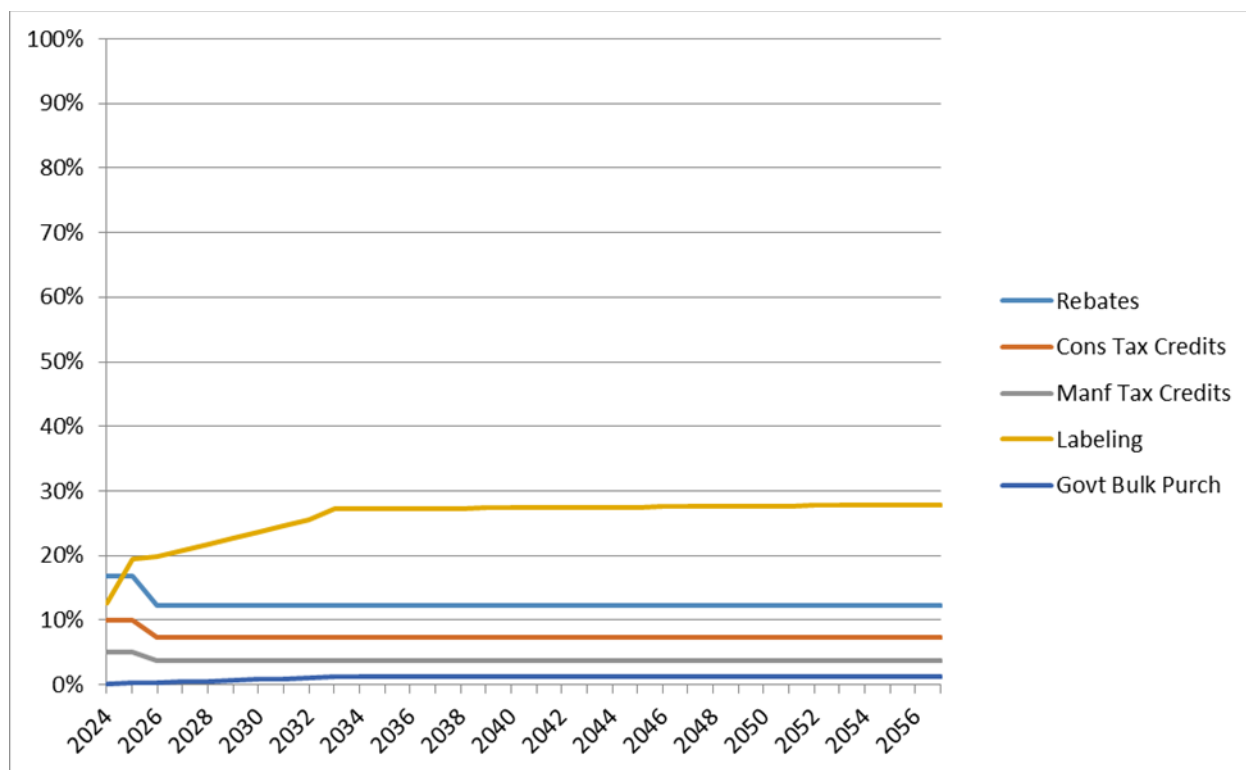


Figure 17.4.3 Increase in Market Penetration of Efficient PM_{2.5} CADR ≥ 150 (TSL 3)

Table 17.4.1 shows the national energy savings and net present value for the five non-regulatory policy alternatives analyzed in detail for air cleaners. The target level for each policy corresponds to the same efficient technology selected for standards at TSL 3. The case in which no regulatory action is taken with regard to air cleaners constitutes the no-new-standards case (or "No New Regulatory Action" scenario), in which NES and NPV are zero by definition. For comparison, the tables include the impacts of the selected standards calculated as described in footnote 'a'. Energy savings are given in quadrillion British thermal units (quads) of primary energy savings.^k The NPVs shown in Table 17.4.1 are based on two discount rates, 7 percent and 3 percent. Under both discount rates, the proposed standards carry a considerably higher NPV than any non-regulatory alternative.

The policy with the highest projected cumulative energy savings is labeling, followed by consumer rebates. However, savings from these two alternative policies would lead, respectively, to only 41 percent and 29 percent of savings from standards. Savings from manufacturer tax credits and consumer tax credits range from 8.5 to 17.0 percent of the savings from standards calculated as described in footnote 'a'. Bulk government purchasing has the lowest cumulative energy savings. Overall, the energy saving benefits from the alternative policies, range from 0.7

^k For the alternative policies whose market penetration depends on B/C ratio, the energy savings in Table 17.4.1 correspond to the case where the cash flow of the operating cost savings was discounted to the purchase year using a 7 percent discount rate.

percent to 40.8 percent of the benefits from the selected standards calculated as described in footnote ‘a’.

Table 17.4.1 Impacts of Non-Regulatory Policy Alternatives (TSL 3)

Policy Alternative	Energy Savings* <i>quads</i>		Net Present Value* <i>million 2021\$</i>	
			7% Disc Rate	3% Disc Rate
Consumer Rebates	0.502	28.9%***	2,028	4,601
Consumer Tax Credits	0.296	17.0%	1,217	2,761
Manufacturer Tax Credits	0.148	8.5%	608	1,380
Labeling	0.709	40.8%	2,602	6,154
Bulk Government Purchases	0.012	0.7%	45.0	107
Selected Standards	1.736	100%	6,138	14,567

* For products shipped 2024-2057.

** Calculated as described in footnote ‘a’.

*** The percentages show how the energy savings from each policy alternative compare to the (primary) energy savings from the selected standards (represented in the table as 100%), when the latter are calculated as described in footnote ‘a’.

17.5 Application of a Labeling Rule Under 42 U.S.C. 6294

42 U.S.C. 6295(l)(1) grants DOE authority to prescribe an energy conservation standard when it determines, among other things, that “the application of a labeling rule under 42 U.S.C. 6294 to such type (or class) is not likely to be sufficient to induce manufacturers to produce, and consumers and other persons to purchase, covered products of such type (or class) which achieve the maximum energy efficiency which is technologically feasible and economically justified.”

A literature review found only one study of the effectiveness of the EnergyGuide labels that are authorized by 42 U.S.C. 6294.¹ This study, which focused on water heaters, found that the degree to which the EnergyGuide label guided cost-efficient decisions depends importantly on the discount rate assumed. Using individual discount rates separately elicited in the study, it found that the EnergyGuide label came very close to guiding cost-efficient decisions, on average. However, using a uniform five percent discount rate—which was much lower than the average elicited rate—the EnergyGuide label resulted in choices that placed too little weight on energy operating cost savings, resulting in a one-third undervaluation of energy efficiency. (A five percent discount rate approximately corresponds to the average consumer discount rate used in DOE’s LCC analysis.)

¹Richard G. Newell and Juha V. Siikamäki. Nudging Energy Efficiency Behavior: The Role of Information Labels. NBER Working Paper No. 19224. July 2013. (available at : https://www.nber.org/system/files/working_papers/w19224/w19224.pdf) (last accessed Mar. 13, 2023)

A water heater is a significant consumer of energy, and it is plausible that EnergyGuide labels for products with much lower energy costs, such as air cleaners, would have a much lower effect.

Other research has similarly shown a limit to the effectiveness of mandatory energy labels.^m In a comparison of different energy labeling program designs, a categorical label appears to be a more effective design.ⁿ The ENERGY STAR labeling program is a type of categorical label. As such, the estimated impacts in 17.3.5 of a labeling program, which are based on the ENERGY STAR program, are likely to be an upper bound on the impacts of a labeling rule under 42 U.S.C. 6294 and would not achieve the same level of energy savings as an energy conservation standard.

Based on the above evidence, DOE concludes that the application of a labeling rule under 42 U.S.C. 6294 to air cleaners is not likely to be sufficient to induce manufacturers to produce, and consumers and other persons to purchase, air cleaners that achieve the maximum energy efficiency which is technologically feasible and economically justified.

^m For example:

Waechter, S., et al. “Desired and Undesired Effects of Energy Labels—An Eye-Tracking Study,” PLoS One. 2015; 10(7). (available at: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4521693/>) (last accessed Mar. 13, 2023)

Wang, B. et al. “Effect of energy efficiency labels on household appliance choice in China: Sustainable consumption or irrational intertemporal choice?” Resources, Conservation and Recycling, June 2021, 169. (Available at: <https://www.sciencedirect.com/science/article/pii/S0921344921000653>) (last accessed Mar. 13, 2023)

ⁿ Thorne, J. and Egan, C. “The EnergyGuide Label: Evaluation and Recommendations for an Improved Design,” Proceedings of the 2002 Summer Study on Energy Efficiency in Buildings. 2002. (available at: https://www.aceee.org/files/proceedings/2002/data/papers/SS02_Panel8_Paper29.pdf) (last accessed Mar. 13, 2023)

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**APPENDIX 8A. USER INSTRUCTIONS FOR THE LIFE-CYCLE COST ANALYSIS
SPREADSHEET**

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APPENDIX 8A. USER INSTRUCTIONS FOR THE LIFE-CYCLE COST ANALYSIS SPREADSHEET

8A.1 USER INSTRUCTIONS

The results obtained in this analysis can be examined and reproduced using the Microsoft Excel spreadsheets available on DOE's Home Appliances Rulemaking website: http://www1.eere.energy.gov/buildings/appliance_standards/standards.aspx?productid=52. From that page, follow the links to the Notice of Public Rulemaking.

8A.2 STARTUP

DOE's spreadsheet enables users to perform Life-Cycle Cost (LCC) and Payback Period (PBP) analysis for each product class.^a Within the spreadsheet, a separate page, or worksheet, exists for each product class.

To use the spreadsheet, DOE assumes that the user has access to a personal computer with a hardware configuration capable of running Windows 2010. The LCC spreadsheet requires Microsoft Excel 2010 or later installed under the Windows operating system. Because certain variables inside the spreadsheets are defined as distributions, the user's computer requires a copy of Crystal Ball (a commercially available add-on program).

8A.3 DESCRIPTION OF LIFE-CYCLE COST WORKSHEETS

For all of the product classes, DOE created one spreadsheet containing a collection of worksheets. Each worksheet represents a conceptual component within the LCC calculation. To facilitate navigability and identify how worksheets are related, each worksheet contains an area on the left side of the worksheet page showing variables imported to and exported from the current worksheet. Each LCC spreadsheet contains the following worksheets:

Overall Summary	The <i>Overall Summary</i> worksheet contains LCC and PBP simulation results for each design option and product class with residential and commercial sectors combined.
Summary	The <i>Summary</i> worksheet contains LCC and PBP simulation results for each design option, product class, and sector (residential and commercial).
LCC & Payback	The <i>LCC&PB</i> worksheet shows LCC calculation results for different efficiency levels for Residential Energy Consumption Survey (<i>RECS</i>) households and Commercial Buildings Energy Consumption Survey (<i>CBECS</i>) buildings. ^{1, 2}

^a The spreadsheet (AirCleaner_DFR_LCC_Publication.xlsm) can be found on the www.regulations.gov website.

Rebuttable Payback	The <i>Rebuttable Payback</i> worksheet contains the total and incremental manufacturer costs, retail prices, installation costs, energy use calculations, and simple payback period calculations for each efficiency level.
Equipment Cost	This worksheet calculates retail price values used as inputs in the LCC calculations in the <i>Summary</i> worksheet.
RECS Households	The <i>RECS Households</i> worksheet contains the <i>RECS 2020</i> household data for each product class.
CBECS Buildings	The <i>CBECS Buildings</i> worksheet contains the <i>CBECS 2018</i> building data for each product class.
Energy Use	This worksheet contains the power and operating hour parameters used in the energy use calculations for air cleaners.
Air Quality Data	This worksheet contains the Air Quality Index by county published by the EPA and used for estimating the operating hours for certain <i>RECS 2020</i> households.
Base Case Eff Dist	The <i>Base Case EFF Dist</i> worksheet determines the efficiency of the no-new-standards case unit.
Energy Prices	The <i>Energy Prices</i> worksheet calculates energy prices, depending on a number of variables.
Energy Price Trends	The <i>Energy Price Trends</i> worksheet shows the future electricity price trends of the different user sectors.
Discount Rate	The <i>Discount Rate</i> worksheet contains the distributions of discount rates for residential and commercial sector.
Lifetime	The <i>Lifetime</i> worksheet presents the average lifetime, in years, for all product classes, the Weibull parameters used for the survival function, and a graph of the Weibull retirement function for air cleaners.

8A.4 BASIC INSTRUCTIONS FOR OPERATING THE LIFE-CYCLE COST SPREADSHEETS

Basic instructions for operating the LCC spreadsheet are numerated below:

1. Once the LCC spreadsheet has been downloaded, open the file using Excel. Click “Enable Macro” when prompted and then click on the tab for the *Summary* worksheet.

2. Use Excel's View/Zoom commands at the top menu bar to change the size of the display to fit your monitor.
3. The user can change the parameters listed under USER INPUT on the *Summary* worksheet. There are five drop-down boxes and one command button. The default parameters are:
 - a. Energy Price Trend: Defaults to "AEO 2022 - Reference." To change the input, use the drop-down menu and select the desired trend (Reference, High, or Low).
 - b. # of Trials: Defaults to "10,000." To change the value, use the drop-down menu and select the desired number of trials (1,000, 2,000, 3,000, 5,000, or 10,000).
 - c. Analysis Group: Defaults to "National." To analyze a subgroup, use the drop-down menu and select the desired subgroup.
 - d. Learning Curve: Defaults to "Yes."
 - e. Compliance Year: Defaults to "2028"
4. To run the Crystal Ball simulation, click the "run" button (you must re-run after changing any parameters). The spreadsheet will then be minimized. You can monitor the progress of the simulation by watching the count of iterations in the left bottom corner of the screen. When the simulation is finished, the spreadsheet will re-open.
5. Additional information can be found in the *Forecast Cells*, *Summary* and *Overall Summary* worksheets.

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APPENDIX 8B. UNCERTAINTY AND VARIABILITY IN THE LIFE-CYCLE COST AND PAYBACK PERIOD ANALYSIS

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APPENDIX 8B. UNCERTAINTY AND VARIABILITY IN THE LIFE-CYCLE COST AND PAYBACK PERIOD ANALYSIS

8B.1 INTRODUCTION

This appendix discusses uncertainty and variability and describes how the U.S. Department of Energy (DOE) incorporated these into the life-cycle cost (LCC) and payback period (PBP) analysis in this technical support document (TSD) for the air cleaners energy conservation standards (ECS) rulemaking. The two key approaches are (1) to use distributions to capture uncertainties and variations in input variables when such distributions are reasonably well defined, and (2) to use scenarios that capture the bounds of uncertainty when the bounds are less well defined.

8B.2 UNCERTAINTY AND VARIABILITY

DOE develops mathematical models to analyze the impacts of proposed energy conservation standards. The models generate outputs (*e.g.*, the LCC impact of proposed standards) based on inputs that are often uncertain, variable, or both.

Variability means that the quantity of interest takes on different values at different times or under different conditions. Variability may be caused by many factors. For example, the hours of use of a lamp depend on environmental factors (*e.g.*, diurnal variations in light) and behavioral factors (*e.g.*, the schedules and preferences of the inhabitants of a house). Manufacturing irregularities can also cause variability. For example, 10 lamps of the same model may each have slightly different power consumptions. DOE attempts to account for major sources of variability in its analyses.

Uncertainty has many sources. Variability may lead to uncertainty in model inputs, because analysts frequently must estimate the values of interest based on samples of a variable quantity (for example, the hours of use of lighting in a home). Measurement uncertainty is another source of uncertainty, which may result from instrumental uncertainties (resulting, for example, from drift, bias, and precision of resolution) and human factors (*e.g.*, variations in experimental setup, errors in instrument readings or recordings). Uncertainty can also arise when there is limited data available to estimate a particular parameter. DOE attempts to address the major sources of uncertainties in its analyses.

8B.2.1 Approaches to Uncertainty and Variability

This section describes two approaches to address uncertainty and variability in numerical modeling that in practice are often used in tandem, as they are in this rulemaking: (1) probability analysis and (2) scenario analysis.

Probability analysis considers the probability that a variable has a given value over its range of possible values. For quantities with variability (*e.g.*, electricity rates in different

households), data from surveys or other forms of measurement can be used to generate a frequency distribution of numerical values to estimate the probability that the variable takes a given value. By sampling values from the resulting distribution, it is possible to quantify the impact of known variability in a particular variable on the outcome of the analysis. In this analysis, DOE used probability distributions to estimate air cleaner lifetime, discount rates, and other variables.

Unlike probability analysis, which considers the impact of known variability, scenario analysis estimates the sensitivity of an analysis to sources of uncertainty and variability whose probability distribution is not well known. Certain model inputs are modified to take a number of different values, and models are re-analyzed, in a set of different model scenarios. Because only selected inputs are changed in each scenario, the variability in the results for each scenario helps to quantify the impact of uncertainty in the input parameters. Whereas it is relatively simple to perform scenario analyses for a range of scenarios, scenario analyses provide no information regarding the likelihood of any given scenario's actually occurring.

Scenario and probability analysis provide some indication of the robustness of the policy given the uncertainties and variability. A policy is robust when the impacts are acceptable over a wide range of possible conditions.

8B.3 PROBABILITY ANALYSIS AND THE USE OF MONTE CARLO SIMULATION IN THE LCC AND PBP ANALYSIS

To quantify the uncertainty and variability that exist in inputs to the LCC and PBP analyses, DOE used Monte Carlo simulation and probability distributions to conduct probability analyses.

Simulation refers to any analytical method meant to imitate a real-life system, especially when other analyses are too mathematically complex or too difficult to reproduce. Without the aid of simulation, a model will only reveal a single outcome, generally the most likely or average scenario. Probabilistic risk analysis uses both a spreadsheet model and simulation to automatically analyze the effect of varying inputs on the outputs of a modeled system. One type of simulation is Monte Carlo simulation, which repeatedly generates random values for uncertain variables, drawn from a probability distribution, to simulate a model.

For each uncertain variable, the range of possible values is controlled by a probability distribution. The type of distribution selected is based on the conditions surrounding that variable. Probability distribution types include normal, triangular, uniform, and Weibull distributions, as well as custom distributions where needed. Example plots of these distributions are shown in Figure 8B.3.1.

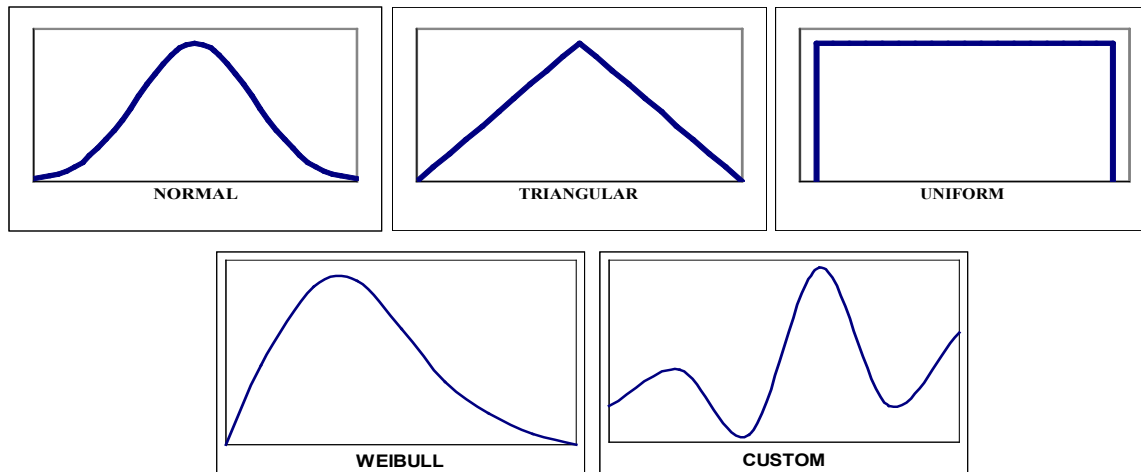


Figure 8B.3.1 Normal, Triangular, Uniform, Weibull, and Custom Probability Distributions

During a simulation, multiple scenarios of a model are calculated by repeatedly sampling values from the probability distributions for the uncertain variables and using those values for that input. Monte Carlo simulations can consist of as many trials as desired, with larger numbers of trials yielding more accurate average results. During a single trial, the simulation randomly selects a value from the defined possibilities (the range and shape of the probability distribution) for each uncertain variable and then recalculates the result for that trial.

DOE conducted probability analyses using Microsoft Excel spreadsheets combined with Crystal Ball, a commercially available add-in software. Crystal Ball simulations can consist of as many trials (or scenarios) as desired—hundreds or even thousands. To calculate the LCC and PBP for air cleaners, DOE performed 10,000 Monte Carlo simulations for each variable. During a single trial, Crystal Ball randomly selected a value from the defined possibilities (the range and shape of the probability distribution) for each uncertain variable and then recalculated the spreadsheet.

APPENDIX 8C. LIFETIME DISTRIBUTIONS

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APPENDIX 8C. LIFETIME DISTRIBUTIONS

8C.1 INTRODUCTION

For each product class, DOE characterized the product lifetime using a Weibull probability distribution that ranged from the minimum to maximum lifetime estimates described in chapter 8, Life Cycle Cost and Payback Period Analyses. The Weibull distribution is recommended for applications to lifetime data because it can be shaped to match low, average, and high values while still allowing some probability of exceeding the high value.¹

8C.1.1 Product Lifetime

For air cleaners, DOE developed a distribution of lifetimes from which specific values are assigned to the appliances in the samples. DOE ensured that the average lifetime estimate of 9 years aligned with those lifetime estimates suggested by ENERGY STAR² and by CA IOUs³ (who cited EPA and various State Technical Reference Manuals). NEEA⁴ also cited an estimated lifetime of 9 years.

8C.2 ESTIMATION OF SURVIVAL FUNCTION

The Residential Energy Consumption Survey (RECS)⁵ records the presence of various appliances in each household, and places the age of each appliance into bins comprising several years. However, the air cleaner ownership information is not available currently in RECS. DOE estimated the percentage of appliances of a given age still in operation using a survival function, which DOE assumed has the form of a cumulative Weibull distribution, provides an average and a median appliance lifetime.

The Weibull distribution is a probability distribution commonly used to measure failure rates.^a Its form is similar to an exponential distribution, which models a fixed failure rate, except that a Weibull distribution allows for a failure rate that changes over time in a particular fashion. The cumulative Weibull distribution takes the form:

$$P(x) = e^{-\left(\frac{x-\theta}{\alpha}\right)^\beta} \text{ for } x > \theta \text{ and} \\ P(x) = 1 \text{ for } x \leq \theta$$

Where:

$P(x)$ =	probability that the appliance is still in use at age x ,
x =	appliance age,
α =	scale parameter, which would be the decay length in an exponential distribution,
β =	shape parameter, which determines the way in which the failure rate changes through time, and

^a For reference on the Weibull distribution, see sections 1.3.6.6.8 and 8.4.1.3 of the *NIST/SEMATECH e-Handbook of Statistical Methods*, www.itl.nist.gov/div898/handbook/.

$\theta =$ delay parameter, which allows for a delay before any failures occur.

When $\beta = 1$, the failure rate is constant over time, giving the distribution the form of a cumulative exponential distribution. In the case of appliances, β commonly is greater than 1, reflecting an increasing failure rate as appliances age.

Table 8C.2.1 shows the lifetime parameters for air cleaners.

Table 8C.2.1 Lifetime Parameters

Product Class	Calculated Values			Weibull Parameters	
	Minimum (years)	Average (years)	Maximum (years)	Alpha (scale)	Beta (shape)
Air Cleaner	2.0	9.0	18.0	7.89	2.50

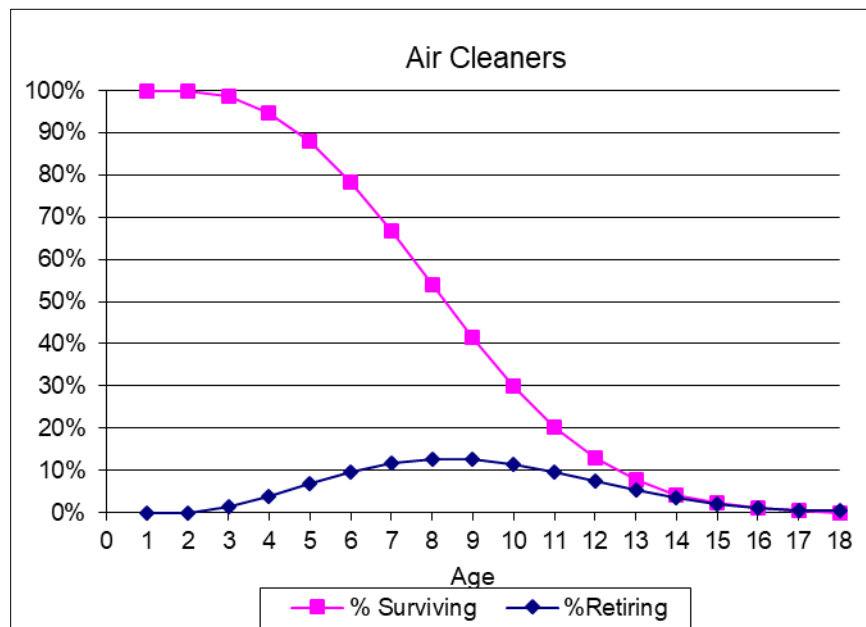


Figure 8C.2.1 Air Cleaners: Lifetime Function

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APPENDIX 8D. DISTRIBUTIONS USED FOR RESIDENTIAL DISCOUNT RATES

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APPENDIX 8D. DISTRIBUTIONS USED FOR RESIDENTIAL DISCOUNT RATES

8D.1 INTRODUCTION

The Department of Energy (DOE) derived consumer discount rates for the life-cycle cost (LCC) analysis using data on interest or return rates for various types of debt and equity to calculate a real effective discount rate for each household in the Federal Reserve Board's *Survey of Consumer Finances (SCF)* in 1995, 1998, 2001, 2004, 2007, 2010, 2013, 2016, and 2019.¹ To account for variation among households in rates for each of the types, DOE sampled a rate for each household in its building sample from a distribution of discount rates for each of six income groups. This appendix describes the distributions used.

8D.1.1 Distribution of Rates for Equity Classes

Figure 8D.1.1 through Figure 8D.1.6 show the distribution of real interest rates for different types of equity. Data for equity classes are not available from the Federal Reserve Board's *SCF*, so DOE derived data for these classes from national-level historical data (1992-2021). The rates for stocks are the annual returns on the Standard and Poor's 500 for 1992-2021.² The interest rates associated with AAA corporate bonds were collected from Moody's time-series data for 1992-2021.³ Rates on Certificates of Deposit (CDs) accounts came from Cost of Savings Index (COSI) data covering 1992-2021.^{4,a} The interest rates associated with state and local bonds (20-bond municipal bonds) were collected from Federal Reserve Board economic data time-series for 1992-2016 and Bartel Associates for 2017-2021.^{10,11,b} The interest rates associated with treasury bills (30-Year treasury constant maturity rate) were collected from Federal Reserve Board economic data time-series for 1992-2021.¹² Rates for money market accounts are based on three-month money market account rates reported by Organization for Economic Cooperation and Development (OECD) from 1992-2021.¹³ Rates for savings accounts are assumed to be half the average real money market rate. Rates for mutual funds are a weighted average of the stock rates and the bond rates.^c The 30-year average nominal interest rates are shown in Table 8D.1.1. DOE adjusted the nominal rates to real rates using the annual inflation rate in each year (see Figure 8D.1.7). In addition, DOE adjusted the nominal rates to real effective rates by accounting for the fact that interest on such equity types is taxable. The capital gains marginal tax rate varies for each household based on income as shown in chapter 8 (the impact of this is not shown in Figure 8D.1.1 through Figure 8D.1.6, which are only adjusted for inflation).

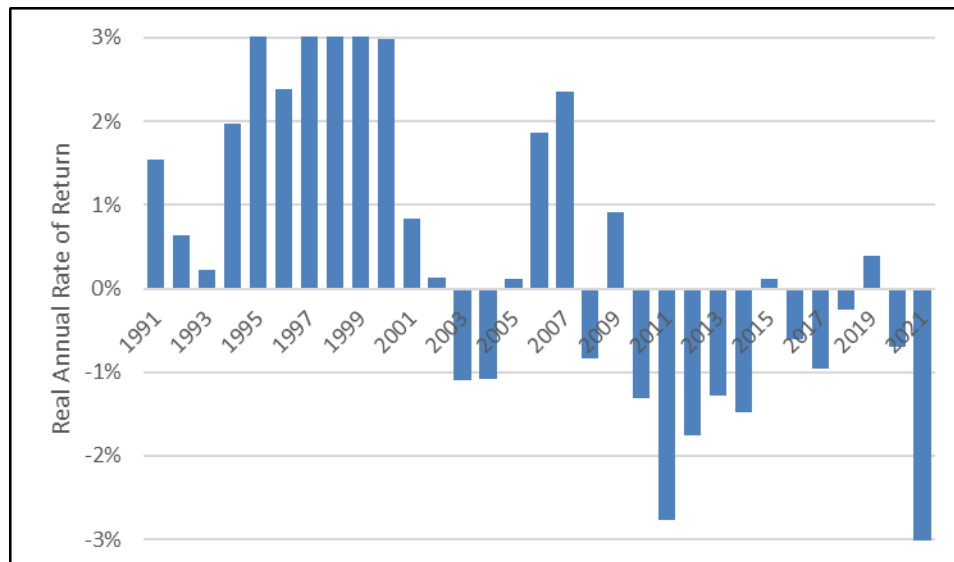
^a The Wells COSI is based on the interest rates that the depository subsidiaries of Wells Fargo & Company pay to individuals on certificates of deposit (CDs), also known as personal time deposits. Wells Fargo COSI started in November 2009.^{5,6} From July 2007 to October 2009 the index was known as Wachovia COSI⁷ and from January 1984 to July 2007 the index was known as GDW (or World Savings) COSI.^{8,9}

^b This index was discontinued in 2016. To calculate the 2017 and after values, DOE used data collected by Bartel Associates.

^c SCF reports what type of mutual funds the household has (e.g. stock mutual fund, savings bond mutual fund, etc.). For mutual funds with a mixture of stocks and bonds, the mutual fund interest rate is a weighted average of the stock rates (two-thirds weight) and the savings bond rates (one-third weight).

Table 8D.1.1 30-Year Average Nominal Interest Rates for Household Equity Type

Type of Equity	30 Year Average Nominal Rate (%)
Savings accounts	2.49
Money market accounts	2.65
Certificate of deposit	2.92
Treasury Bills (T-bills)	4.62
State/Local bonds	4.57
AAA Corporate Bonds	5.48
Stocks (S&P 500)	11.98
Mutual funds	9.52

**Figure 8D.1.1 Distribution of Annual Rate of Money Market Accounts**

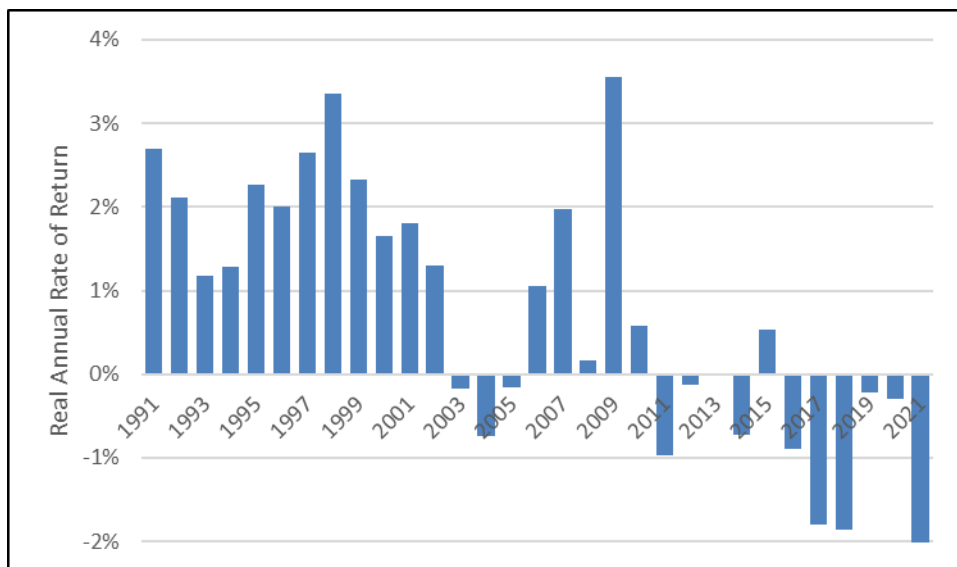


Figure 8D.1.2 Distribution of Annual Rate of Return on CDs

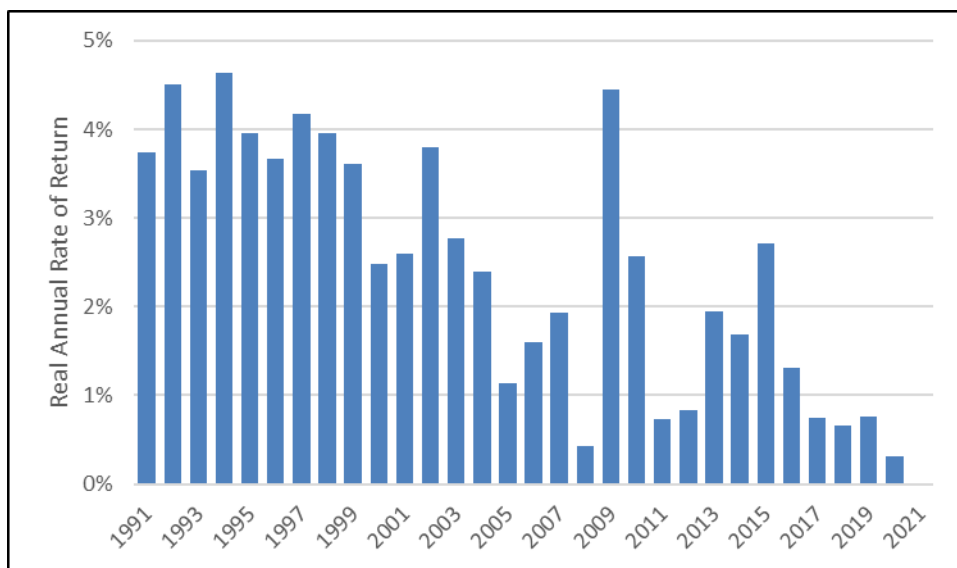


Figure 8D.1.3 Distribution of Annual Rate of Return on Savings Bonds (30 Year Treasury Bills)

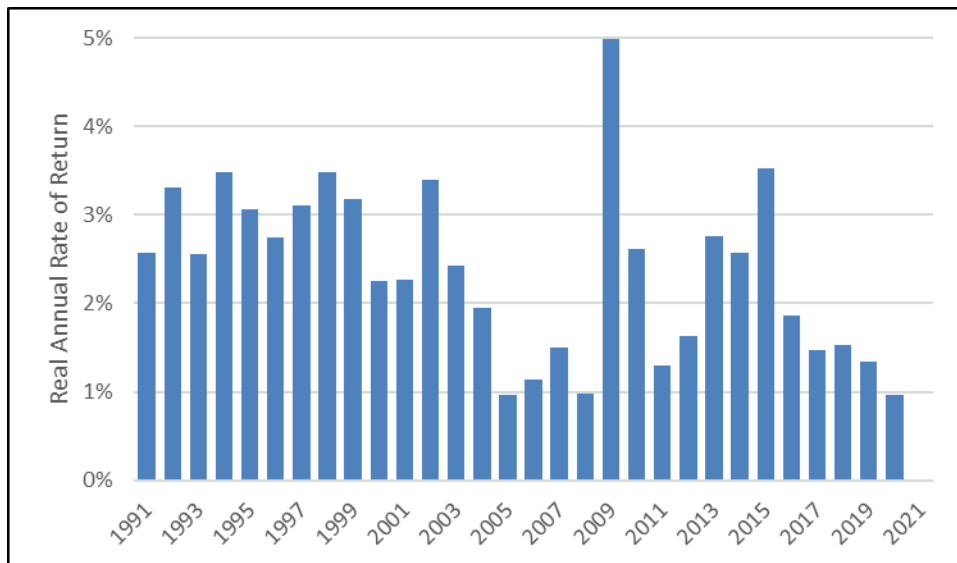


Figure 8D.1.4 Distribution of Annual Rate of State and Local Bonds

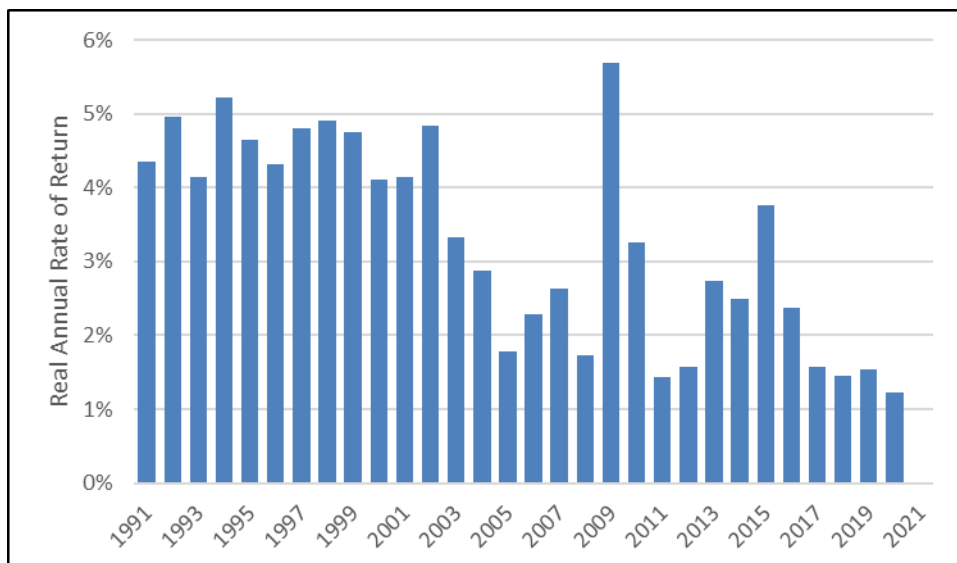


Figure 8D.1.5 Distribution of Annual Rate of Return on Corporate AAA Bonds

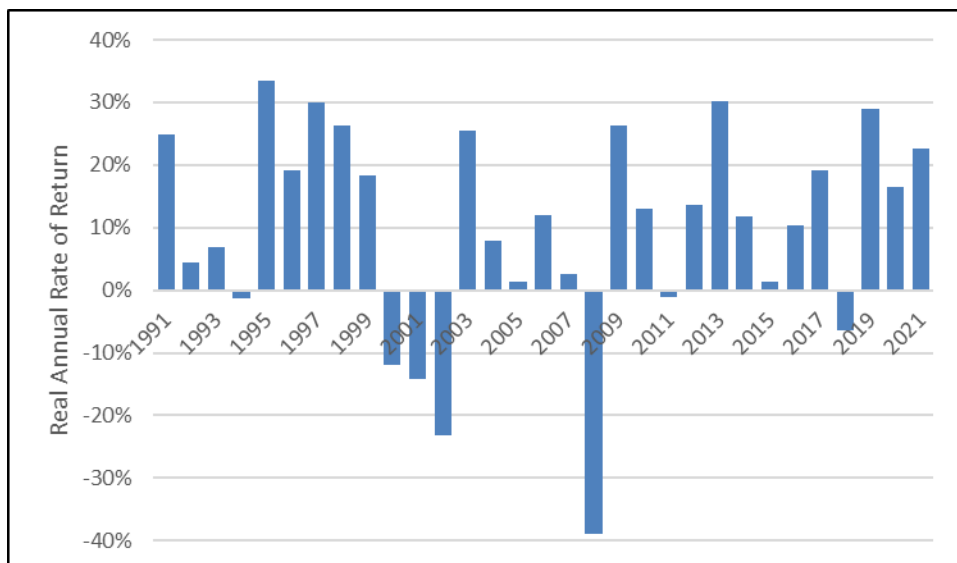


Figure 8D.1.6 Distribution of Annual Rate of Return on S&P 500

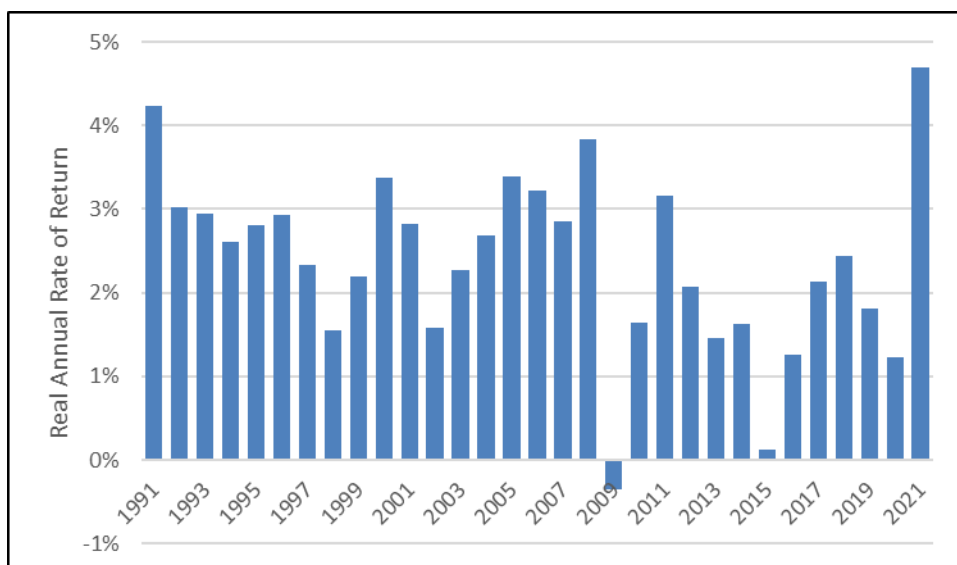


Figure 8D.1.7 Annual Consumer Price Index (CPI) Rate

8D.2 DISTRIBUTION OF REAL EFFECTIVE DISCOUNT RATES BY INCOME GROUP

Real effective discount rates were calculated for each household of the SCF using the method described in chapter 8. Interest rates for asset types were as described in 8D.1.1. The data source for the interest rates for mortgages, home equity loans, credit cards, installment loans, other residence loans, and other lines of credit is the Federal Reserve Board's *SCF* in 1995, 1998, 2001, 2004, 2007, 2010, 2013, 2016, and 2019. DOE adjusted the nominal rates to real rates using the annual inflation rate in each year.

Using the appropriate *SCF* data for each year, DOE adjusted the nominal mortgage interest rate and the nominal home equity loan interest rate for each relevant household in the *SCF* for mortgage tax deduction and inflation. In cases where the effective interest rate is equal to or below the inflation rate (resulting in a negative real interest rate), DOE set the real effective interest rate to zero. Figure 8D.2.1 provides a graphical representation of the real effective discount rate distributions by income group, while Table 8D.2.1 provides the full distributions as used in the LCC analysis.

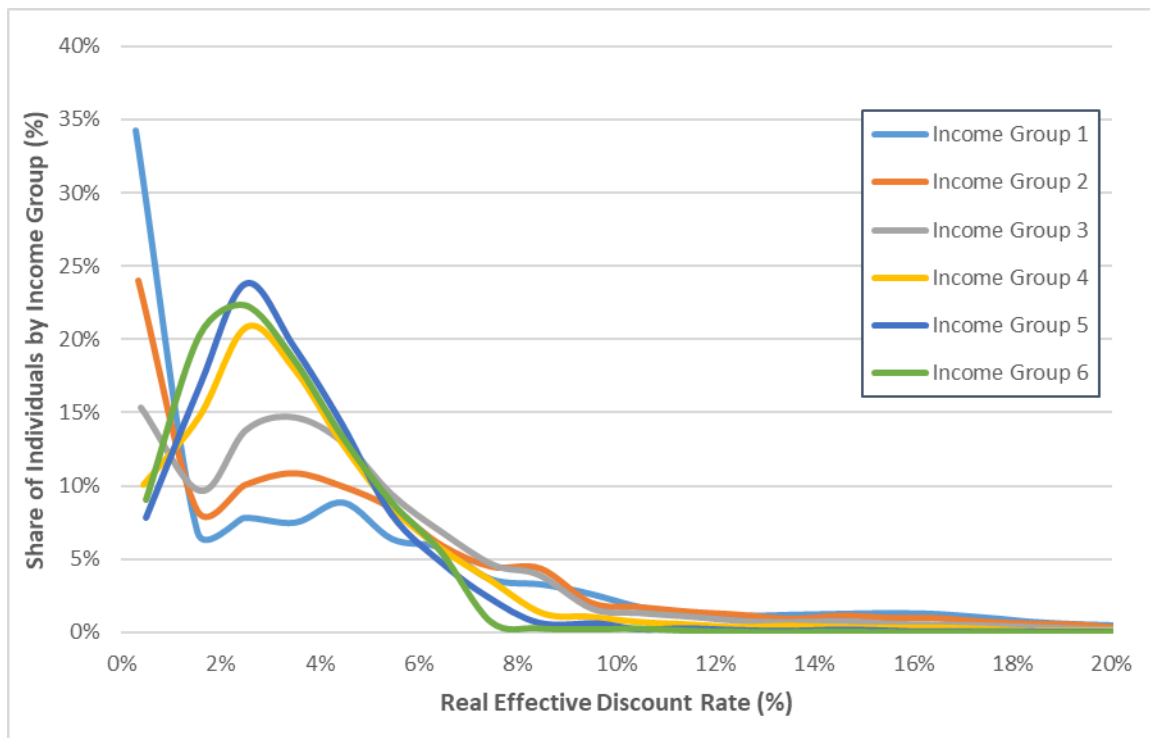


Figure 8D.2.1 Distribution of Real Discount Rates by Income Group

Table 8D.2.1 Distribution of Real Discount Rates by Income Group

DR Bin (%)	Income Group 1 (0-19.9 percentile)		Income Group 2 (20-39.9 percentile)		Income Group 3 (40-59.9 percentile)		Income Group 4 (60-79.9 percentile)		Income Group 5 (80-89.9 percentile)		Income Group 6 (90-100 percentile)	
	Rate %	Weight %	Rate %	Weight %	Rate %	Weight %	Rate %	Weight %	Rate %	Weight %	Rate %	Weight %
0-1	0.23	34.42	0.29	24.30	0.34	15.54	0.40	10.28	0.46	7.85	0.50	9.20
1-2	1.54	6.76	1.53	8.26	1.58	9.63	1.57	14.70	1.57	16.87	1.58	20.30
2-3	2.46	7.62	2.51	10.01	2.52	13.80	2.52	20.73	2.51	23.71	2.50	22.09
3-4	3.51	7.55	3.49	10.77	3.49	14.57	3.49	17.87	3.47	19.54	3.47	18.86
4-5	4.48	8.80	4.47	9.95	4.48	12.88	4.47	12.82	4.45	14.15	4.48	13.18
5-6	5.47	6.38	5.46	8.47	5.46	9.39	5.46	8.41	5.46	8.07	5.46	8.92
6-7	6.47	5.67	6.47	5.93	6.46	6.80	6.45	5.70	6.50	4.66	6.45	5.57
7-8	7.46	3.61	7.47	4.46	7.51	4.59	7.44	3.54	7.40	2.42	7.47	0.73
8-9	8.52	3.23	8.47	4.36	8.41	3.91	8.49	1.30	8.43	0.66	8.41	0.29
9-10	9.46	2.63	9.49	2.01	9.49	1.60	9.46	1.05	9.62	0.62	9.63	0.22
10-11	10.50	1.66	10.46	1.70	10.44	1.33	10.43	0.69	10.44	0.22	10.36	0.25
11-12	11.48	1.16	11.53	1.39	11.52	1.03	11.54	0.51	11.41	0.28	11.54	0.14
12-13	12.52	1.14	12.47	1.19	12.54	0.72	12.45	0.33	12.48	0.16	12.39	0.06
13-14	13.54	1.13	13.52	0.91	13.50	0.69	13.48	0.45	13.43	0.11	13.52	0.02
14-15	14.52	1.23	14.56	1.13	14.60	0.74	14.50	0.34	14.53	0.19	14.45	0.06
15-16	15.55	1.29	15.55	0.97	15.53	0.56	15.48	0.31	15.43	0.13	15.64	0.02
16-17	16.49	1.22	16.40	0.96	16.46	0.51	16.43	0.30	16.17	0.06	16.40	0.01
17-18	17.58	0.95	17.51	0.71	17.51	0.43	17.47	0.21	17.53	0.06	17.93	0.03
18-19	18.42	0.70	18.47	0.56	18.41	0.34	18.37	0.10	18.47	0.06	18.50	0.01
19-20	19.45	0.51	19.40	0.50	19.45	0.22	19.61	0.09	19.40	0.05	19.17	0.01
20-21	20.56	0.44	20.42	0.26	20.38	0.18	20.46	0.09	20.47	0.04	20.13	0.02
21-22	21.43	0.54	21.43	0.34	21.34	0.16	21.48	0.07	21.38	0.06	0.00	0.00
22-23	22.51	0.39	22.48	0.23	22.58	0.08	22.72	0.03	0.00	0.00	0.00	0.00
23-24	23.41	0.17	23.52	0.13	23.40	0.10	23.44	0.02	0.00	0.00	23.89	0.03
24-25	24.61	0.18	24.47	0.10	24.56	0.04	24.08	0.01	0.00	0.00	0.00	0.00
25-26	25.35	0.16	25.40	0.10	25.47	0.06	25.33	0.03	25.79	0.00	0.00	0.00
26-27	26.52	0.13	26.47	0.03	26.50	0.05	0.00	0.00	0.00	0.00	0.00	0.00
27-28	27.49	0.07	27.41	0.02	27.41	0.03	27.27	0.03	27.14	0.00	0.00	0.00
28-29	28.14	0.09	28.29	0.05	28.38	0.01	0.00	0.00	0.00	0.00	0.00	0.00
29-30	29.87	0.01	29.37	0.01	29.31	0.00	0.00	0.00	0.00	0.00	0.00	0.00
>30	68.17	0.14	125.34	0.19	135.28	0.02	53.82	0.00	0.00	0.00	0.00	0.00
Total	4.71	100.00	4.95	100.00	4.51	100.00	3.80	100.00	3.44	100.00	3.19	100.00

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APPENDIX 8E. DISTRIBUTIONS USED FOR COMMERCIAL DISCOUNT RATES

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APPENDIX 8E. DISTRIBUTIONS USED FOR COMMERCIAL DISCOUNT RATES

8E.1 DISTRIBUTIONS USED FOR COMMERCIAL/INDUSTRIAL DISCOUNT RATES

Table 8E.1.1 Education Sector Discount Rate Distribution

Bin	Bin Range	Rates	Weight (% of companies)	# of Companies
1	<0%			
2	0-1%			
3	1-2%			
4	2-3%			
5	3-4%			
6	4-5%			
7	5-6%	5.33%	16.9%	141
8	6-7%	6.62%	38.3%	320
9	7-8%	7.43%	12.1%	101
10	8-9%	8.40%	19.9%	166
11	9-10%	9.37%	12.9%	108
12	10-11%			
13	11-12%			
14	12-13%			
15	≥13%			
Weighted Average		7.21%		

Table 8E.1.2 Food Sales Sector Discount Rate Distribution

Bin	Bin Range	Bin Average Discount Rate	Weight (% of companies)	# of Companies
1	<0%			
2	0-1%			
3	1-2%			
4	2-3%			
5	3-4%	3.79%	2.8%	25
6	4-5%	4.63%	45.6%	409
7	5-6%	5.60%	24.1%	216
8	6-7%	6.28%	12.8%	115
9	7-8%	7.61%	3.7%	33
10	8-9%	8.75%	5.6%	50
11	9-10%	9.52%	3.7%	33
12	10-11%	10.32%	1.7%	15
13	11-12%			
14	12-13%			
15	≥13%			
Weighted Average		5.66%		

Table 8E.1.3 Food Service Sector Discount Rate Distribution

Bin	Bin Range	Bin Average Discount Rate	Weight (% of companies)	# of Companies
1	<0%			
2	0-1%			
3	1-2%			
4	2-3%			
5	3-4%			
6	4-5%	4.91%	13.8%	264
7	5-6%	5.63%	25.5%	488
8	6-7%	6.56%	35.4%	677
9	7-8%	7.24%	17.4%	332
10	8-9%			
11	9-10%	9.78%	4.1%	79
12	10-11%			
13	11-12%	11.10%	3.7%	70
14	12-13%			
15	≥13%			
Weighted Average		6.51%		

Table 8E.1.4 Health Care Sector Discount Rate Distribution

Bin	Bin Range	Bin Average Discount Rate	Weight (% of companies)	# of Companies
1	<0%			
2	0-1%			
3	1-2%			
4	2-3%			
5	3-4%			
6	4-5%			
7	5-6%	5.59%	29.9%	1710
8	6-7%	6.47%	25.0%	1428
9	7-8%	7.40%	21.4%	1222
10	8-9%	8.42%	18.5%	1056
11	9-10%	9.11%	5.3%	304
12	10-11%			
13	11-12%			
14	12-13%			
15	≥13%			
Weighted Average		6.90%		

Table 8E.1.5 Lodging Sector Discount Rate Distribution

Bin	Bin Range	Bin Average Discount Rate	Weight (% of companies)	# of Companies
1	<0%			
2	0-1%			
3	1-2%			
4	2-3%			
5	3-4%			
6	4-5%	4.78%	23.1%	389
7	5-6%	5.48%	16.3%	274
8	6-7%	6.47%	22.8%	385
9	7-8%	7.28%	24.7%	416
10	8-9%	8.36%	5.3%	89
11	9-10%	9.98%	3.9%	66
12	10-11%			
13	11-12%	11.30%	3.9%	66
14	12-13%			
15	≥13%			
Weighted Average		6.55%		

Table 8E.1.6 Mercantile Discount Rate Distribution

Bin	Bin Range	Bin Average Discount Rate	Weight (% of companies)	# of Companies
1	<0%			
2	0-1%			
3	1-2%			
4	2-3%			
5	3-4%			
6	4-5%	4.74%	0.9%	50
7	5-6%	5.58%	16.5%	941
8	6-7%	6.50%	34.4%	1969
9	7-8%	7.43%	32.9%	1884
10	8-9%	8.18%	9.4%	536
11	9-10%	9.44%	4.0%	231
12	10-11%	10.28%	1.6%	91
13	11-12%	11.50%	0.3%	16
14	12-13%			
15	≥13%			
Weighted Average		6.99%		

Table 8E.1.7 Office Sector Discount Rate Distribution

Bin	Bin Range	Bin Average Discount Rate	Weight (% of companies)	# of Companies
1	<0%			
2	0-1%			
3	1-2%			
4	2-3%			
5	3-4%	3.77%	6.1%	2902
6	4-5%	4.58%	16.3%	7771
7	5-6%	5.49%	20.5%	9772
8	6-7%	6.44%	14.0%	6653
9	7-8%	7.43%	10.0%	4779
10	8-9%	8.60%	15.8%	7517
11	9-10%	9.36%	9.5%	4530
12	10-11%	10.38%	3.7%	1745
13	11-12%	11.35%	1.6%	776
14	12-13%	12.81%	1.8%	838
15	≥13%	14.36%	0.7%	342
Weighted Average		6.89%		

Table 8E.1.8 Public Assembly Sector Discount Rate Distribution

Bin	Bin Range	Bin Average Discount Rate	Weight (% of companies)	# of Companies
1	<0%			
2	0-1%			
3	1-2%			
4	2-3%			
5	3-4%			
6	4-5%	4.98%	1.9%	73
7	5-6%	5.70%	7.4%	285
8	6-7%	6.51%	38.5%	1487
9	7-8%	7.44%	26.7%	1031
10	8-9%	8.50%	13.6%	525
11	9-10%	9.21%	12.0%	465
12	10-11%			
13	11-12%			
14	12-13%			
15	≥13%			
Weighted Average		7.27%		

Table 8E.1.9 Service Sector Discount Rate Distribution

Bin	Bin Range	Bin Average Discount Rate	Weight (% of companies)	# of Companies
1	<0%			
2	0-1%			
3	1-2%			
4	2-3%			
5	3-4%	3.85%	5.1%	818
6	4-5%	4.44%	13.3%	2133
7	5-6%	5.55%	31.6%	5078
8	6-7%	6.41%	22.7%	3645
9	7-8%	7.54%	12.2%	1968
10	8-9%	8.57%	9.6%	1549
11	9-10%	9.16%	4.3%	698
12	10-11%	10.24%	1.1%	179
13	11-12%			
14	12-13%			
15	≥13%			
Weighted Average		6.25%		

Table 8E.1.10 All Commercial Sectors Discount Rate Distribution

Bin	Bin Range	Bin Average Discount Rate	Weight (% of companies)	# of Companies
1	<0%			
2	0-1%			
3	1-2%			
4	2-3%			
5	3-4%	3.79%	4.4%	3745
6	4-5%	4.57%	13.2%	11168
7	5-6%	5.53%	22.4%	18962
8	6-7%	6.46%	19.7%	16680
9	7-8%	7.44%	14.0%	11787
10	8-9%	8.55%	13.6%	11488
11	9-10%	9.33%	7.7%	6514
12	10-11%	10.37%	2.4%	2030
13	11-12%	11.33%	1.1%	928
14	12-13%	12.81%	1.0%	838
15	≥13%	14.36%	0.4%	342
Weighted Average		6.77%		

Table 8E.1.11 Industrial Sectors Discount Rate Distribution

Bin	Bin Range	Bin Average Discount Rate	Weight (% of companies)	# of Companies
1	<0%			
2	0-1%			
3	1-2%	1.61%	0.0%	13
4	2-3%	2.62%	0.1%	59
5	3-4%	3.66%	1.5%	1257
6	4-5%	4.61%	6.6%	5350
7	5-6%	5.55%	18.6%	15185
8	6-7%	6.47%	20.3%	16529
9	7-8%	7.51%	15.7%	12759
10	8-9%	8.50%	22.6%	18455
11	9-10%	9.44%	9.9%	8101
12	10-11%	10.45%	3.7%	2980
13	11-12%	11.54%	0.5%	386
14	12-13%	12.52%	0.3%	285
15	≥13%	13.05%	0.1%	121
Weighted Average		7.25%		

Table 8E.1.12 Agriculture Sector Discount Rate Distribution

Bin	Bin Range	Bin Average Discount Rate	Weight (% of companies)	# of Companies
1	<0%			
2	0-1%			
3	1-2%			
4	2-3%			
5	3-4%			
6	4-5%			
7	5-6%			
8	6-7%	6.68%	67.6%	207
9	7-8%	7.37%	10.1%	31
10	8-9%	8.45%	22.2%	68
11	9-10%			
12	10-11%			
13	11-12%			
14	12-13%			
15	≥13%			
Weighted Average		7.15%		

Table 8E.1.13 R.E.I.T./Property Management Sector Discount Rate Distribution

Bin	Bin Range	Bin Average Discount Rate	Weight (% of companies)	# of Companies
1	<0%			
2	0-1%			
3	1-2%			
4	2-3%			
5	3-4%			
6	4-5%	4.90%	10.1%	466
7	5-6%	5.48%	18.0%	833
8	6-7%	6.34%	41.3%	1913
9	7-8%	7.47%	13.2%	609
10	8-9%	8.48%	10.8%	502
11	9-10%	9.43%	6.7%	308
12	10-11%			
13	11-12%			
14	12-13%			
15	≥13%			
Weighted Average		6.62%		

Table 8E.1.14 Investor-Owned Utility Sector Discount Rate Distribution

Bin	Bin Range	Bin Average Discount Rate	Weight (% of companies)	# of Companies
1	<0%			
2	0-1%			
3	1-2%	1.61%	0.6%	13
4	2-3%	2.50%	0.7%	16
5	3-4%	3.66%	49.2%	1064
6	4-5%	4.31%	38.5%	832
7	5-6%	5.42%	4.2%	91
8	6-7%	6.47%	3.8%	83
9	7-8%	7.36%	2.9%	63
10	8-9%			
11	9-10%			
12	10-11%			
13	11-12%			
14	12-13%			
15	≥13%			
Weighted Average		4.18%		

Table 8E.1.15 State/Local Government Discount Rate Distribution

Bin	Bin Range	Bin Average Discount Rate	Weight (% of years)	# of Quarters
1	<0%	-1.9%	3.0%	4
2	0-1%	0.9%	2.3%	3
3	1-2%	1.6%	23.3%	31
4	2-3%	2.5%	25.6%	34
5	3-4%	3.5%	35.3%	47
6	4-5%	4.2%	10.5%	14
7	5-6%			
8	6-7%			
9	7-8%			
10	8-9%			
11	9-10%			
12	10-11%			
13	11-12%			
14	12-13%			
15	>13%			
Weighted Average		2.67%		

Table 8E.1.16 Federal Government Discount Rate Distribution

Bin	Bin Range	Bin Average Discount Rate	Weight (% of months)	# of Months
1	<0%	-0.7%	11.0%	44
2	0-1%	0.5%	22.3%	89
3	1-2%	1.6%	15.5%	62
4	2-3%	2.5%	18.0%	72
5	3-4%	3.5%	18.0%	72
6	4-5%	4.3%	12.0%	48
7	5-6%	5.8%	3.0%	12
8	6-7%			
9	7-8%			
10	8-9%			
11	9-10%			
12	10-11%			
13	11-12%			
14	12-13%			
15	>13%			
Weighted Average		2.06%		

8E.2 ASSIGNMENT OF DETAILED DATA TO AGGREGATE SECTORS FOR DISCOUNT RATE ANALYSIS

Table 8E.2.1 Detailed Industries Assigned to Each Aggregate CBECS PBA Sector

Aggregate Sector for CBECS Mapping	Detailed Sector Names as Provided in Damodaran Online Data Sets (1998-2018)
Education	Education; Educational Services
Food Sales	Food Wholesalers; Grocery; Retail (Grocery and Food); Retail/Wholesale Food
Food Service	Restaurant; Restaurant/Dining
Health Care	Healthcare Facilities; Healthcare Information; Healthcare Services; Healthcare Support Services; Healthcare Information and Technology; Hospitals/Healthcare Facilities; Medical Services
Lodging	Hotel/Gaming
Mercantile	Drugstore; Retail (Automotive); Retail (Building Supply); Retail (Distributors); Retail (General); Retail (Hardlines); Retail (Softlines); Retail (Special Lines); Retail Automotive; Retail Building Supply; Retail Store
Office	Advertising; Bank; Bank (Canadian); Bank (Midwest); Bank (Money Center); Banks (Regional); Broadcasting; Brokerage & Investment Banking; Business & Consumer Services; Cable TV; Computer Services; Computer Software; Computer Software/Svcs; Diversified; Diversified Co.; E-Commerce; Human Resources; Insurance (General); Insurance (Life); Insurance (Prop/Cas.); Internet; Investment Co.; Investment Co.(Foreign); Investment Companies; Investments & Asset Management; Property Management; Public/Private Equity; R.E.I.T.; Real Estate (Development); Real Estate (General/Diversified); Real Estate (Operations & Services); Reinsurance; Retail (Internet); Retail (Online); Securities Brokerage; Software (Entertainment); Software (Internet); Software (System & Application); Telecom. Utility; Thrift
Public Assembly	Entertainment; Recreation
Service	Financial Svcs.; Financial Svcs. (Div.); Financial Svcs. (Non-bank & Insurance); Foreign Telecom.; Funeral Services; Industrial Services; Information Services; Internet software and services; IT Services; Office Equip/Supplies; Office Equipment & Services; Oilfield Svcs/Equip.; Pharmacy Services; Telecom. Services
All Commercial	All detailed sectors included in: Education, Food Sales, Food Service, Health Care, Mercantile, Office, Public Assembly, Service
Industrial	Aerospace/Defense; Air Transport; Aluminum; Apparel; Auto & Truck; Auto Parts; Auto Parts (OEM); Auto Parts (Replacement); Automotive; Beverage; Beverage (Alcoholic); Beverage (Soft); Biotechnology; Building Materials; Cement & Aggregates; Chemical (Basic); Chemical (Diversified); Chemical (Specialty); Coal; Coal & Related Energy; Computers/Peripherals; Construction; Construction Supplies; Copper; Drug; Drugs (Biotechnology); Drugs (Pharmaceutical); Electric Util. (Central); Electric Utility (East); Electric Utility (West); Electrical Equipment; Electronics; Electronics (Consumer & Office); Electronics (General); Engineering; Engineering & Const; Engineering/Construction; Entertainment Tech; Environmental; Environmental & Waste Services; Food Processing; Foreign Electronics; Furn/Home Furnishings; Gold/Silver Mining; Green & Renewable Energy; Healthcare Equipment; Healthcare Products; Heavy Construction; Heavy Truck & Equip; Heavy Truck/Equip Makers; Home Appliance; Homebuilding; Household Products; Machinery; Manuf. Housing/RV; Maritime; Med Supp Invasive; Med Supp Non-Invasive; Medical Supplies; Metal Fabricating; Metals & Mining; Metals & Mining (Div.); Natural Gas (Div.); Natural Gas Utility; Newspaper; Oil/Gas (Integrated); Oil/Gas (Production and Exploration); Oil/Gas Distribution; Packaging & Container; Paper/Forest Products; Petroleum (Integrated); Petroleum (Producing); Pharma & Drugs; Pipeline MLPs; Power; Precious Metals; Precision Instrument; Publishing; Publishing & Newspapers; Railroad; Rubber& Tires; Semiconductor Equip; Shipbuilding & Marine; Shoe; Steel; Steel (General); Steel (Integrated); Telecom (Wireless); Telecom. Equipment; Textile; Tire & Rubber; Tobacco; Toiletries/Cosmetics; Transportation; Transportation (Railroads); Trucking; Utility (Foreign); Utility (General); Utility (Water); Water Utility; Wireless Networking
Agriculture	Farming/Agriculture
Utilities	Natural Gas Utility; Utility (Foreign); Utility (General); Utility (Water); Water Utility
R.E.I.T. / Property	Property Management; R.E.I.T.; Real Estate (Development); Real Estate (General/Diversified); Real Estate (Operations & Services)

**APPENDIX 10A. USER INSTRUCTIONS FOR NATIONAL IMPACT ANALYSIS
SPREADSHEET MODEL**

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APPENDIX 10A. USER INSTRUCTIONS FOR NATIONAL IMPACT ANALYSIS SPREADSHEET MODEL

10A.1 USER INSTRUCTIONS

The results obtained in this analysis can be examined and reproduced using the Microsoft Excel® spreadsheets accessible on the Internet from the Department of Energy's (DOE's) air cleaners rulemaking <http://energy.gov/eere/buildings/appliance-and-equipment-standards-program>. From that page, follow the links to the final rule phase of the rulemaking and then to the analytical tools.

10A.2 STARTUP

The national impact analysis (NIA) spreadsheets enable the user to perform an NIA for air cleaners. In order to use the spreadsheet tool, a user will need a computer with at least 1GB of memory and an installation of Microsoft Excel 2010 or later.

10A.3 DESCRIPTIONS OF NATIONAL IMPACT ANALYSIS WORKSHEETS

The NIA spreadsheet performs calculations to project the change in national energy use and net present value of financial impacts produced by potential energy conservation standards for air cleaners. The energy use and associated costs for a given standard level are determined by calculating (1) shipments under that standard and (2) energy use and costs for all air cleaners shipped under that standard. The differences between conditions under a given standards case and the no-new-standards case can be compared and the overall energy savings and net present values determined. The NIA spreadsheet worksheets are described below.

Worksheet	Description
Input and Summary	This worksheet enables the user to stipulate inputs under "User Inputs." It also contains a summary table, "National Impact Summary," for the selected efficiency level. The sheet provides the incremental total installed prices associated with the standard levels being evaluated for air cleaners. This sheet also contains efficiency-weighted average energy use and the total installed product price for air cleaners under the no-new-standards case and standards cases.
CADR 10-99	This worksheet calculates the energy consumption costs and savings for the CADR 10-99 product class
CADR 100-149	This worksheet calculates the energy consumption costs and savings for the CADR 100-149 product class
CADR 150+	This worksheet calculates the energy consumption costs and savings for the CADR 150+ product class
Shipments	This worksheet contains historical data regarding sales of air cleaners
Electricity Prices	This worksheet contains projected average and marginal residential and commercial electricity prices for the three

Worksheet	Description
	economic growth scenarios in <i>AEO 2022</i> .
Price Indices	This worksheet includes multipliers to adjust the manufacturer's cost throughout the analysis period based on learning curves.
Energy Factors	The worksheet contains the site-to-power plant and full-fuel-cycle conversion factors used to calculate the primary and full-fuel-cycle energy savings, respectively.
Lifetime	This worksheet contains the lifetime and retirement function for air cleaners.

10A.4 BASIC INSTRUCTIONS

Basic instructions for operating the NIA spreadsheets are as follows.

1. After downloading the NIA spreadsheet file from DOE's website, use Microsoft Excel to open it. Click "Enable Macro" when prompted. At the bottom of the workbook, click on the tab for the worksheet *Input and Summary*.
2. Use Excel's "View/Zoom" command in the top menu bar to change the size of the display so that it fits your monitor.
3. Use the graphical interface in the worksheet to choose parameters or enter data. You can change the default choices for some of the inputs. To change a default input, select the desired value from the drop-down choices by the input box or enter the number manually where appropriate. The input parameters are as follows:
 - A. Discount Rate. To change the value, which is set to 3 percent, click on the drop-down and change the value to 7 percent.
 - B. Trial Standard Level (TSL). To change the level, click on the drop-down arrow to select TSL 1, 2, 3, 4, and 5.
 - C. Price Elasticity. To change the applicable elasticity, use the drop-down arrow to select whether or not price elasticity should be applied.
 - D. Economic Growth. To change the growth scenario, use the drop-down arrow to select the desired *AEO* growth case (Reference, Low, or High).
 - E. Price Trend. To change the value, use the drop-down arrow to select the desired learning-curve level (Default, High Price Decline, or Low Price Decline).

After you have selected the desired parameters, the spreadsheet will automatically generate updated NIA results based on the new inputs. Results are reported in the "National Impact Summary" table to the right of the "User Inputs" box.

APPENDIX 10B. FULL-FUEL-CYCLE ANALYSIS

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APPENDIX 10B. FULL-FUEL-CYCLE ANALYSIS

10B.1 INTRODUCTION

This appendix summarizes the methods the U.S. Department of Energy (DOE) used to calculate the estimated full-fuel-cycle (FFC) energy savings from potential energy conservation standards. The FFC measure includes point-of-use (site) energy; the energy losses associated with generation, transmission, and distribution of electricity; and the energy consumed in extracting, processing, and transporting or distributing primary fuels. DOE's method of analysis previously encompassed only site energy and the energy lost through generation, transmission, and distribution of electricity. In 2011 DOE announced its intention, based on recommendations from the National Academy of Sciences, to use FFC measures of energy use and emissions when analyzing proposed energy conservation standards.¹ This appendix summarizes the methods DOE used to incorporate impacts of the full fuel cycle into the analysis.

In the national energy savings calculation, DOE estimates the site, primary and full-fuel-cycle (FFC) energy consumption for each standard level, for each year in the analysis period. DOE defines these quantities as follows:

- Site energy consumption is the physical quantity of fossil fuels or electricity consumed at the site where the end-use service is provided.^a The site energy consumption is used to calculate the energy cost input to the net present value (NPV) calculation.
- Primary energy consumption is defined by converting the site fuel use from physical units, for example cubic feet for natural gas, or kWh for electricity, to common energy units (million Btu or MMBtu). For electricity the conversion factor is a marginal heat rate that incorporates losses in generation, transmission and distribution, and depends on the sector, end use and year.
- The full-fuel-cycle (FFC) energy use is equal to the primary energy use plus the energy consumed "upstream" of the site in the extraction, processing and distribution of fuels. The FFC energy use was calculated by applying a fuel-specific FFC energy multiplier to the primary energy use.

For electricity from the grid, site energy is measured in terawatt-hours (TWh). The primary energy of a unit of grid electricity is equal to the heat content of the fuels used to generate that electricity, including transmission and distribution losses.^b DOE typically measures the primary energy associated with the power sector in quads (quadrillion Btu). Both primary fuels and electricity are used in upstream activities. The treatment of electricity in full-fuel-cycle analysis must distinguish between electricity generated by fossil fuels and electricity generated from renewable sources (wind, solar, and hydro). For the former, the upstream fuel cycle relates

^a For fossil fuels, this is the site of combustion of the fuel.

^b For electricity sources like nuclear energy and renewable energy, the primary energy is calculated using the EIA convention as described below.

to the fuel consumed at the power plant. There is no upstream component for the latter, because no fuel *per se* is used.

10B.2 SITE-TO-PRIMARY ENERGY FACTORS

DOE uses heat rates to convert site electricity savings in TWh to primary energy savings in quads. The heat rates are developed as a function of the sector, end-use and year of the analysis period. For this analysis DOE uses output of the DOE/Energy Information Administration (EIA)'s National Energy Modeling System (NEMS).² EIA uses the NEMS model to produce the *Annual Energy Outlook (AEO)*. DOE's approach uses the most recently available edition, in this case *AEO 2022*.³ The *AEO* publication includes a reference case and a series of side cases incorporating different economic and policy scenarios. DOE calculates marginal heat rates as the ratio of the change in fuel consumption to the change in generation for each fossil fuel type, where the change is defined as the difference between the reference case and the side case. DOE calculates a marginal heat rate for each of the principal fuel types: coal, natural gas and oil. DOE uses the EIA convention of assigning a heat rate of 10.5 Btu/Wh to nuclear power and 9.5 Btu/Wh to electricity from renewable sources.

DOE multiplied the fuel share weights for sector and end-use, described in appendix 15A of this TSD, by the fuel specific marginal heat rates, and summed over all fuel types, to define a heat rate for each sector/end-use. This step incorporates the transmission and distribution losses. In equation form:

$$h(u,y) = (1 + TDLoss) * \sum_{r,f} g(r,f,y) H(f,y)$$

Where:

$TDLoss$ = the fraction of total generation that is lost in transmission and distribution, equal to 0.07037

u = an index representing the sector/end-use (e.g. commercial cooling)

y = the analysis year

f = the fuel type

$H(f,y)$ = the fuel-specific heat rate

$g(r,f,y)$ = the fraction of generation provided by fuel type f for end-use u in year y

$h(u,y)$ = the end-use specific marginal heat rate

The sector/end-use specific heat rates are shown in Table 10B.2.1. These heat rates convert site electricity to primary energy in quads; *i.e.*, the units used in the table are quads per TWh.

Table 10B.2.1 Electric Power Heat Rates (MMBtu/MWh) by Sector used for Air Cleaners

	2025	2030	2035	2040	2045	2050+
Residential	9.590	9.390	9.340	9.284	9.226	9.226
Commercial	9.477	9.269	9.226	9.177	9.125	9.127

10B.3 FFC METHODOLOGY

The methods used to calculate FFC energy use are summarized here. The mathematical approach to determining FCC is discussed in Coughlin (2012).⁴ Details related to the modeling of the fuel production chain are presented in Coughlin (2013).⁵

When all energy quantities are normalized to the same units, FFC energy use can be represented as the product of the primary energy use and an FFC multiplier. Mathematically the FFC multiplier is a function of a set of parameters that represent the energy intensity and material losses at each stage of energy production. Those parameters depend only on physical data, so the calculations require no assumptions about prices or other economic factors. Although the parameter values may differ by geographic region, this analysis utilizes national averages.

The fuel cycle parameters are defined as follows.

- a_x is the quantity of fuel x burned per unit of electricity produced for grid electricity. The calculation of a_x includes a factor to account for losses incurred through the transmission and distribution systems.
- b_y is the amount of grid electricity used in producing fuel y , in MWh per physical unit of fuel y .
- c_{xy} is the amount of fuel x consumed in producing one unit of fuel y .
- q_x is the heat content of fuel x (MBtu/physical unit).

All the parameters are calculated as functions of an annual time step; hence, when evaluating the effects of potential new standards, a time series of annual values is used to estimate the FFC energy and emissions savings in each year of the analysis period and cumulatively.

The FFC multiplier is denoted μ (mu). A separate multiplier is calculated for each fuel used on site. Also calculated is a multiplier for electricity that reflects the fuel mix used in its generation. The multipliers are dimensionless numbers applied to primary energy savings to obtain the FFC energy savings. The upstream component of the energy savings is proportional to $(\mu-1)$. The fuel type is denoted by a subscript on the multiplier μ .

The method for performing the full-fuel-cycle analysis utilizes data and projections published in the *AEO 2022*. Table 10B.3.1 summarizes the data used as inputs to the calculation of various parameters. The column titled "AEO Table" gives the name of the table that provided the reference data.

Table 10B.3.1 Dependence of FFC Parameters on AEO Inputs

Parameter(s)	Fuel(s)	AEO Table	Variables
q_x	All	Conversion factors	MMBtu per physical unit
a_x	All	Electricity supply, disposition, prices, and emissions Energy consumption by sector and source	Generation by fuel type Electric energy consumption by the power sector
b_c, c_{nc}, c_{pc}	Coal	Coal production by region and type	Coal production by type and sulfur content
b_p, c_{np}, c_{pp}	Petroleum	Refining industry energy consumption Liquid fuels supply and disposition International liquids supply and disposition Oil and gas supply	Refining-only energy use Crude supply by source Crude oil imports Domestic crude oil production
c_{nn}	Natural gas	Oil and gas supply Natural gas supply, disposition, and prices	U.S. dry gas production Pipeline, lease, and plant fuel
z_x	All	Electricity supply, disposition, prices, and emissions	Power sector emissions

The *AEO 2022* does not provide all the information needed to estimate total energy use in the fuel production chain. Coughlin (2013) describes the additional data sources needed to complete the analysis. The time dependence in the FFC multipliers, however, arises exclusively from variables taken from the *AEO*.

10B.4 ENERGY MULTIPLIERS FOR THE FULL FUEL CYCLE

FFC energy multipliers for selected years are presented in Table 10B.4.1. The 2050 value was held constant for the analysis period beyond 2050, which is the last year in the *AEO 2022* projection. The multiplier for electricity reflects the shares of various primary fuels in total electricity generation throughout the forecast period.

Table 10B.4.1 Energy Multipliers for the Full Fuel Cycle (Based on AEO 2022)

	2025	2030	2035	2040	2045	2050+
Electricity (grid)	1.044	1.041	1.039	1.039	1.039	1.039

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APPENDIX 10C. NATIONAL NET PRESENT VALUE OF CONSUMER BENEFITS USING ALTERNATIVE PRODUCT PRICE FORECASTS

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APPENDIX 10C. NATIONAL NET PRESENT VALUE OF CONSUMER BENEFITS USING ALTERNATIVE PRODUCT PRICE FORECASTS

10C.1 INTRODUCTION

The net present value (NPV) results presented in chapter 10 of this Direct Final Rule are based on future price projection for air cleaners derived from a combined producer price index (PPI) series of historical data for “small electric household appliances, except fans” from 1982 to 2015 and the more aggregated small electrical appliance manufacturing from 2015 to 2021. DOE also investigated the impact of different product price forecasts on the consumer NPV for various standard levels considered in this analysis. The two price sensitivity scenarios DOE considered for air cleaners are based on the same combined PPI series used in the default case but covering different periods of time to estimate a low price decline scenario and a high price decline scenario respectively.

10C.2 PRICE SCENARIOS

For the price sensitivity analysis, DOE used the same experience curve approach as the default case to forecast their future price. Given the historical trend of the combined PPI series for air cleaners, the high price decline scenario is based on the combined PPI series from 2014 to 2021 and the low price decline scenario is based on the same combined PPI series from 2009 to 2014 (Figure 10C.2.1).

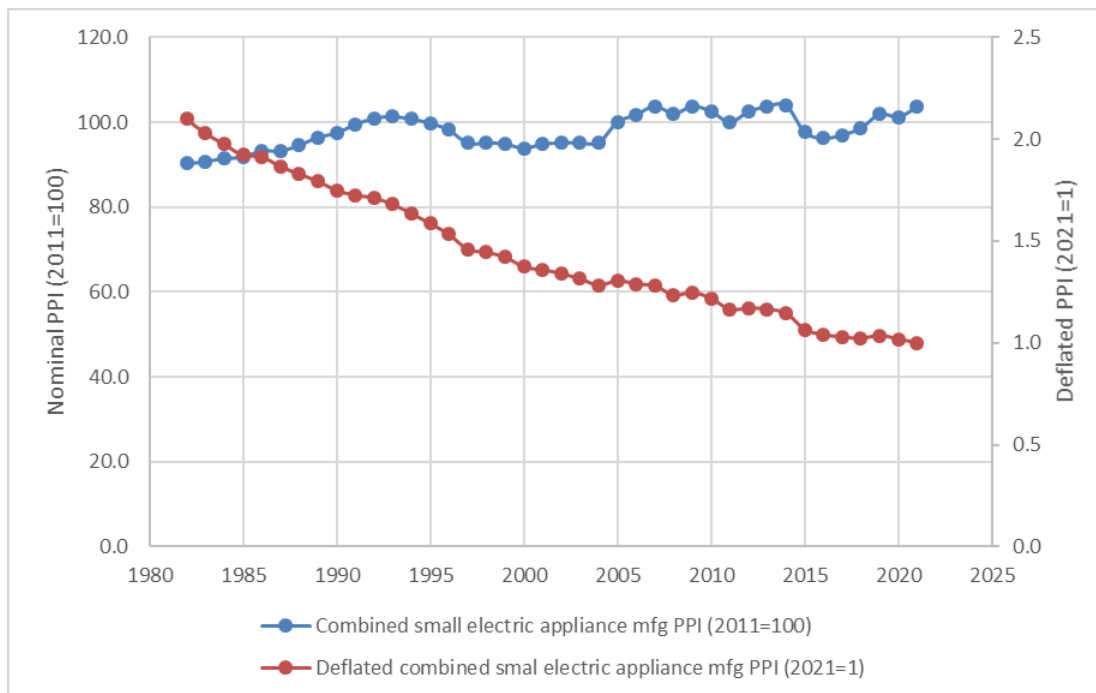


Figure 10C.2.1 Nominal and Deflated Combined PPI Series of Small Electric Household Appliances from 1982 to 2021

10C.2.1 High Price Decline Scenario

In the experience curve method, the real cost of production is related to the cumulative production, or experience, with a manufactured product. DOE modeled the experience curve by fitting the inflation-adjusted PPI series to the corresponding cumulative shipments, a proxy of cumulative production, of air cleaners. The percentage reduction in cost that occurs with each doubling of cumulative production is known as the learning rate.

To estimate an experience rate parameter, a least-squares power-law fit was performed on the unified price index versus cumulative shipments. The form of the power-law equation is:

$$P_t = P_o X_t^{(-b)},$$

where the two parameters, b (the learning rate parameter) and P_o (the price or cost of the first unit of production), are obtained by fitting the model to the data. DOE notes that the cumulative shipments on the right hand side of the equation can have a dependence on price, so there is an issue with simultaneity where the independent variable is not truly independent. DOE's use of a simple least squares fit is equivalent to an assumption of no significant first price elasticity effects in the cumulative shipments variable.

Figure 10C.2.2 presents the fit of experience curve for air cleaners under high price decline scenario.

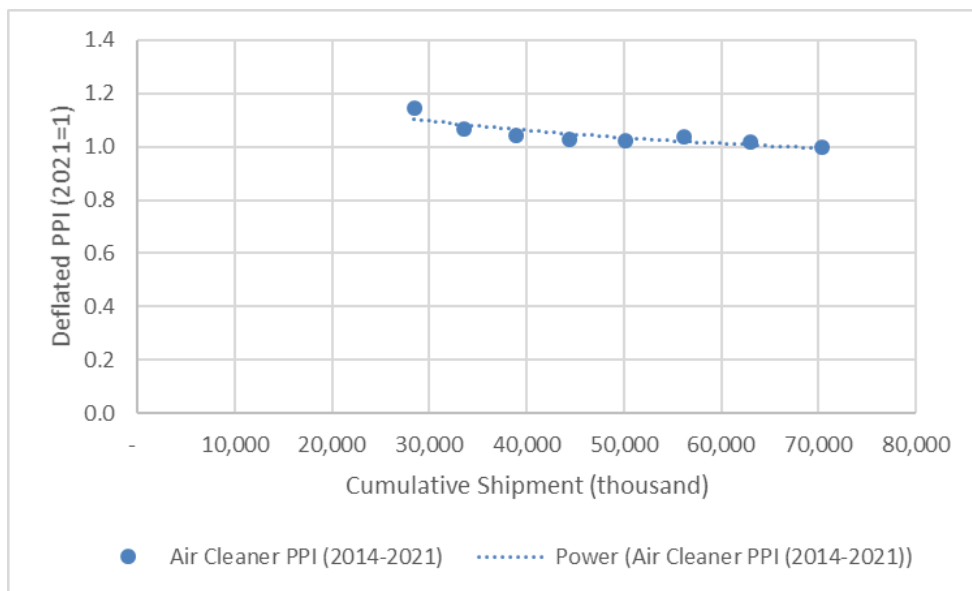


Figure 10C.2.2 High Price Decline Scenario: Relative Price versus Cumulative Shipments of Air Cleaners from 2014 to 2021

For the high price decline scenario for air cleaners, the parameter values obtained are:

$$P_o = 3.60^{+3.846}_{-1.860} \text{ (95\% confidence), and}$$

$$b = 0.115 \pm 0.068 \text{ (95\% confidence)}$$

The estimated experience rate for the high price decline scenario (defined as the fractional reduction in price expected from each doubling of cumulative production) is $7.7^{+4.2}_{-4.4}\%$ (95% confidence).

10C.2.2 Low Price Decline Scenario

For the low price decline scenario, DOE took a similar approach as the high price decline scenario that was based on the combined PPI series between 2009 and 2014. Figure 10C.2.3 presents the fit of experience curve for air cleaners under low price decline scenario.

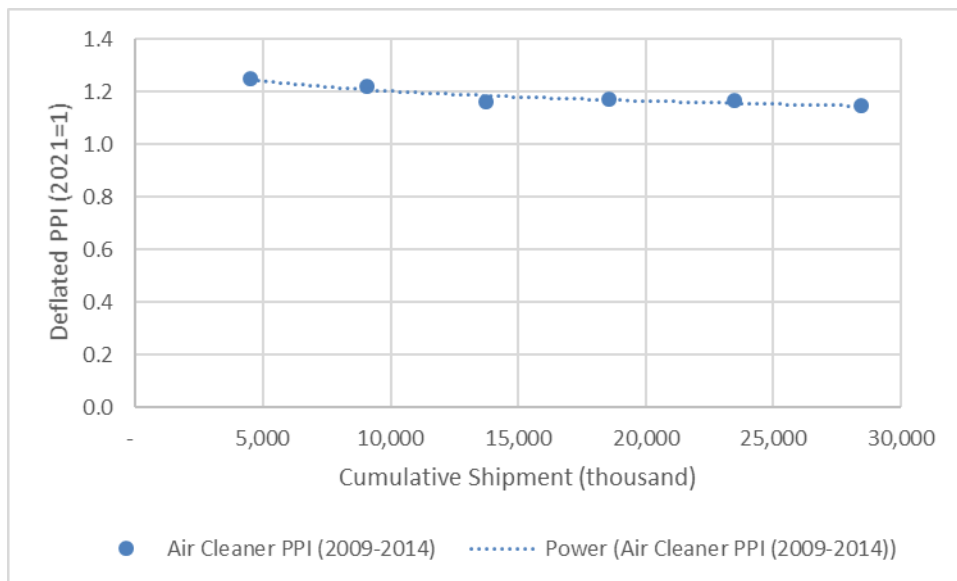


Figure 10C.2.3 Low Price Decline Scenario: Relative Price versus Cumulative Shipments of Air Cleaners from 2009 to 2014

For the low price decline scenario for air cleaners, the parameter values obtained are:

$$P_0 = 1.83^{+0.372}_{-0.310} \text{ (95\% confidence), and}$$

$$b = 0.046 \pm 0.019 \text{ (95\% confidence)}$$

The estimated experience rate for the low price decline scenario (defined as the fractional reduction in price expected from each doubling of cumulative production) is $3.1^{+1.3}_{-1.3}\%$ (95% confidence).

10C.3 SUMMARY

Table 10C.3.1 shows the summary of the learning rates and average annual price decline rates for air cleaners in each scenario. Figure 10C.3.1 shows the resulting price trends throughout the analysis period, normalized to 2021 (2021 = 1).

Table 10C.3.1 Price Trend Scenarios for Air Cleaners

Scenario	Price Trend	Learning Rate %	Average Annual Price Decline Rate %
Default	Experience curve estimation using the combined PPI of small electric household appliances (2009 to 2021)	5.98	0.51
Low Price Decline	Experience curve estimation using the combined PPI of small electric household appliances (2009 to 2014)	3.12	0.26
High Price Decline	Experience curve estimation using the combined PPI of small electric household appliances (2014 to 2021)	7.69	0.66

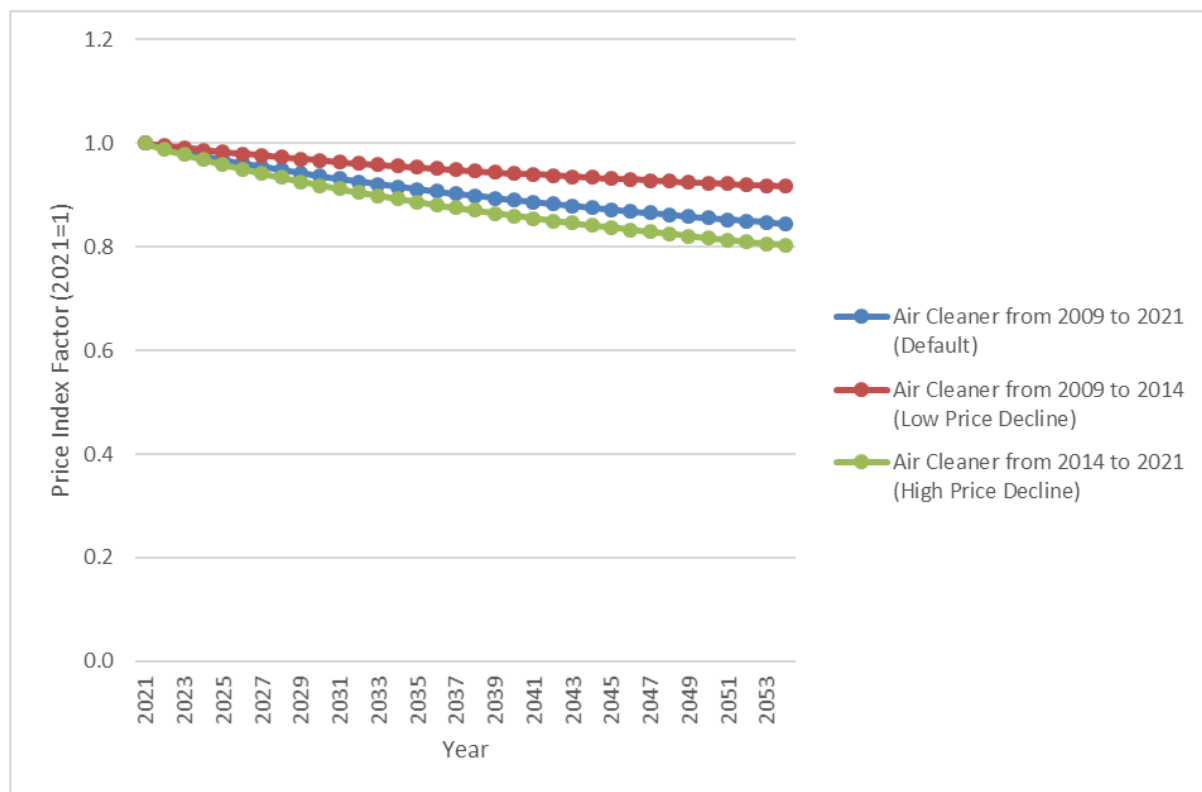


Figure 10C.3.1 Air Cleaner Price Factor Indexes for the Default Case and Sensitivity Cases

10C.4 AIR CLEANER NPV RESULTS USING ALTERNATIVE LEARNING RATES

Table 10C.4.1 Air Cleaners: Net Present Value of Consumer Impacts Under Alternative Product Price Forecasts (billion 2021\$)

Price Trend	TSL	Discount Rate	
		3%	7%
Default	1	5.44	2.18
	2	12.81	5.11
	3	13.69	5.76
	4	(8.39)	(3.40)
	5	(4.50)	(1.92)
Low Price Decline	1	5.43	2.17
	2	12.78	5.09
	3	13.66	5.75
	4	(8.40)	(3.41)
	5	(4.58)	(1.95)
High Price Decline	1	5.45	2.18
	2	12.83	5.11
	3	13.71	5.77
	4	(8.37)	(3.40)
	5	(4.45)	(1.90)

Parentheses indicate negative values.

APPENDIX 10D. NATIONAL ENERGY SAVINGS AND NET PRESENT VALUE USING ALTERNATIVE GROWTH SCENARIOS

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APPENDIX 10D. NATIONAL ENERGY SAVINGS AND NET PRESENT VALUE USING ALTERNATIVE GROWTH SCENARIOS

10D.1 INTRODUCTION

This appendix presents national energy savings (NES) and net present value (NPV) results using inputs from alternative economic growth scenarios. The scenarios use the energy price and housing starts forecasts in the High Economic Growth case and the Low Economic Growth case from EIA's *Annual Energy Outlook 2022* (AEO 2022).¹

In order to model the effect that different economic growth scenarios may have on new purchasing behavior and therefore national impacts, DOE applied an adjustment to the growth rate of new demand. In the reference scenario, DOE assumed that new demand would grow at an annual rate of 4.87 percent in both the residential and commercial sectors (see chapter 9 of this TSD). The equation below shows how annual growth is adjusted in economic indicators in different scenarios.

$$G_{scen}(y) = (4.87\%) \left(\frac{S_{scen}(y) - S_{scen}(y-1)}{S_{ref}(y) - S_{ref}(y-1)} \right)$$

Where:

$G_{scen}(y)$ = The new demand growth rate in year y of an AEO scenario (Reference, Low Growth, or High Growth), and

$S_{scen}(y)$ = The housing stock (for residential growth), or commercial floorspace (for commercial growth) in year y in a scenario

Figure 10D.1.1 shows the resulting shipments projections in each of these growth scenarios. Figure 10D.1.2 and Figure 10D.1.3 show residential and commercial electricity prices trend under the different economic growth scenarios, respectively.

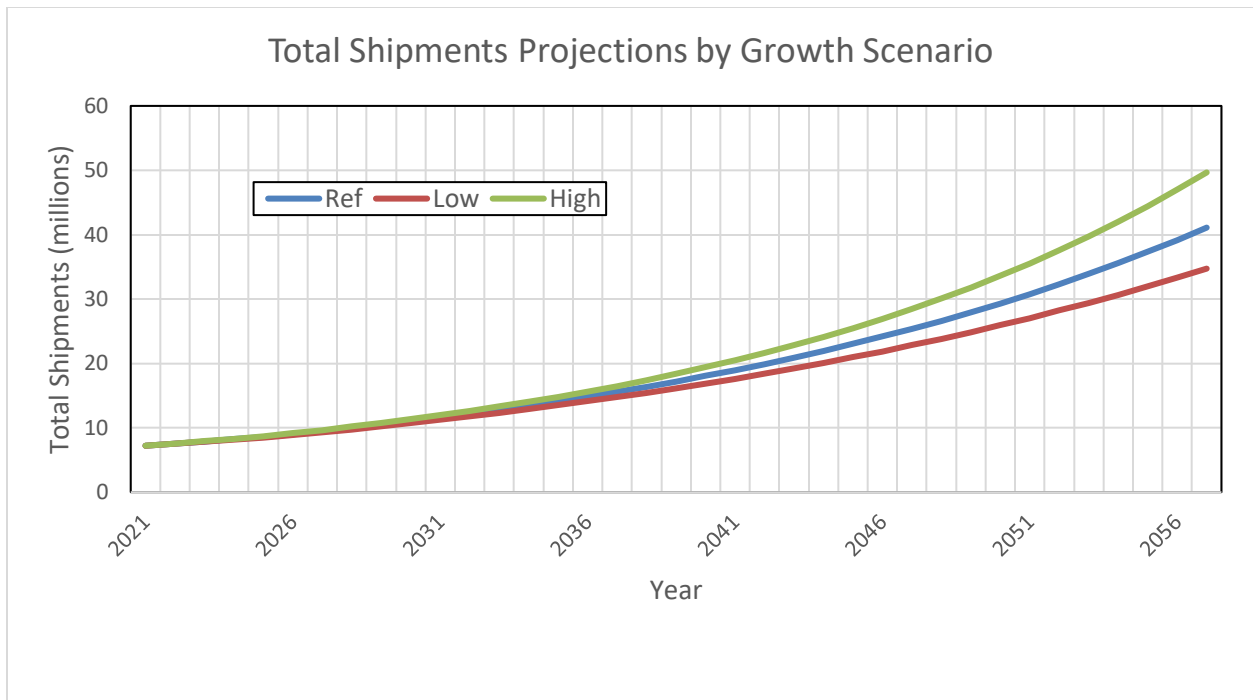


Figure 10D.1.1 Total Shipments Projections under AEO 2022 Economic Growth Scenarios

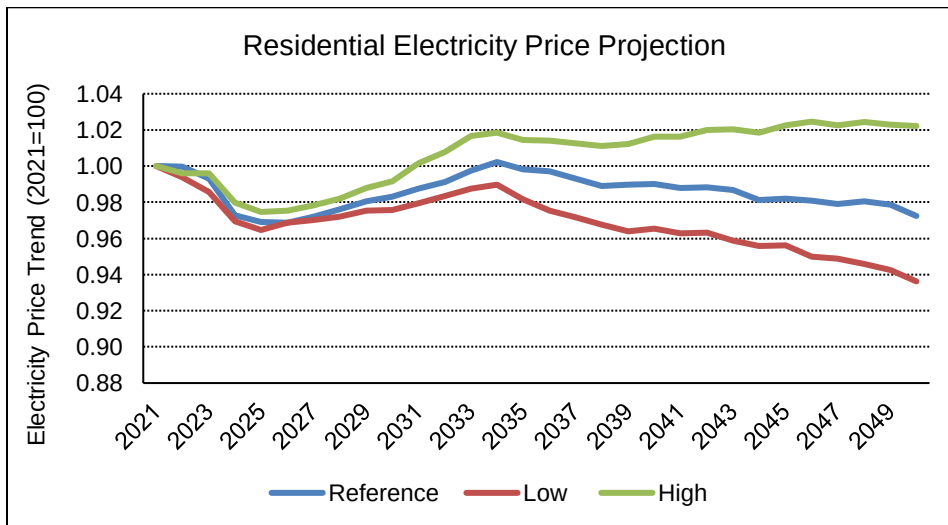


Figure 10D.1.2 Average Residential Electricity Price Projection under Alternative AEO 2022 Economic Growth Scenarios

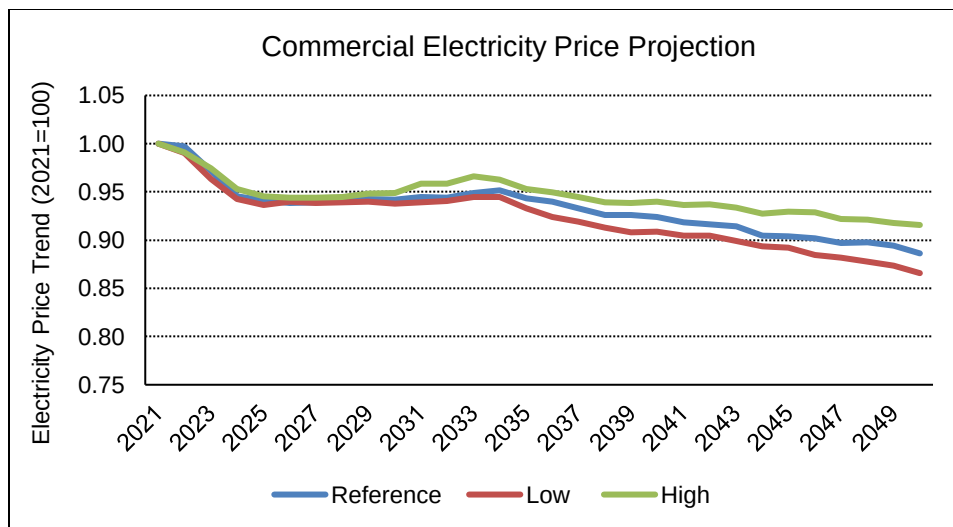


Figure 10D.1.3 Average Commercial Electricity Price Forecasts under Alternative AEO 2022 Economic Growth Scenarios

10D.2 NIA RESULTS BY ECONOMIC GROWTH SCENARIO

Table 10D.2.1 Cumulative Full-Fuel Cycle Energy Savings in Quads, by Economic Growth Scenario

TSL	Reference	Low Growth	High Growth
	<i>Quads</i>	<i>Quads</i>	<i>Quads</i>
1	0.76	0.69	0.85
2	1.73	1.57	1.94
3	1.80	1.64	2.01
4	4.05	3.66	4.53
5	4.59	4.15	5.14

Table 10D.2.2 Cumulative Net Present Value of Consumer Benefits, by Economic Growth Scenario

TSL	Reference	Low Growth	High Growth
<i>3% Discount Rate, billion \$2021</i>			
1	5.44	4.89	6.15
2	12.81	11.51	14.47
3	13.69	12.37	15.36
4	(8.39)	(8.07)	(8.65)
5	(4.50)	(4.58)	(4.25)
<i>7% Discount Rate, billion \$2021</i>			
1	2.18	1.99	2.41
2	5.11	4.66	5.66
3	5.76	5.31	6.32
4	(3.40)	(3.30)	(3.49)
5	(1.92)	(1.95)	(1.85)

Parentheses indicate negative values.

REFERENCES

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<http://www.eia.gov/forecasts/aeo/>

APPENDIX 12A. GOVERNMENT REGULATORY IMPACT MODEL OVERVIEW

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APPENDIX 12A. GOVERNMENT REGULATORY IMPACT MODEL OVERVIEW

12A.1 INTRODUCTION AND PURPOSE

The purpose of the Government Regulatory Impact Model (GRIM) is to help quantify the impacts of energy conservation standards on manufacturers in aggregate. The basic mode of analysis is to estimate the change in the value of the industry, or industry net present value (INPV), following new and/or amended energy conservation standard, as represented by trial standard levels.

Industry net present value is defined, for the purpose of this analysis, as the discounted sum of industry free cash flows plus a discounted terminal value. The model calculates the actual cash flows by year and then determines the present value of those cash flows both without an energy conservation standard (*i.e.*, the no-standards case) and under different trial standard levels (*i.e.*, the standards cases).

Outputs from the model consist of summary financial metrics, graphs of major variables, and access to the complete cash flow calculation.

12A.2 MODEL DESCRIPTION

The basic structure of the GRIM is a standard annual cash flow analysis that uses financial parameters, shipments from the national impact analysis, and manufacturing production costs as inputs and accepts a set of regulatory conditions as changes in costs and investments. The cash flow analysis is separated into two major blocks: an industry income statement and an industry cash flow statement. The income calculation determines net operating profit after taxes. The cash flow calculation converts net operating profit after taxes into an annual cash flow by including investment and non-cash items. Below are definitions of listed items on the output sheet (“No STDs Case DCF” tab) of the GRIM. Please refer to Figure 12A.2.1.

Industry Income Statement

- (1) **Revenues:** The GRIM presents annual revenues for the industry. Revenues are calculated by multiplying unit sales at each efficiency level by the associated manufacturer sales price. Annual revenues are the sum of revenues from all efficiency levels in a given year.
- (2) **Total Shipments:** Total annual shipments for the industry were obtained from the National Impact Analysis. Total shipments are the sum of shipments for all efficiency levels in a given year. Shipments by TSL, product class, and efficiency level can be found in the “Shipments” tab of the GRIM.
- (3) **MPC:** The manufacturer production cost (MPC).
- (4) **Overhead:** The portion of MPC that accounts for production facility overhead, including utilities, maintenance, property tax, and insurance. The annual overhead cost is the sum of the overhead component of MPC for all units shipped in a year.
- (5) **Standard SG&A:** Selling, general, and administrative (SG&A) expenses are calculated by multiplying revenue by the SG&A percentage found on the “Financials” tab of the GRIM.

- (6) **R&D:** Research and development (R&D) expenses are calculated by multiplying revenue by the R&D value found on the “Financials” tab of the GRIM.
- (7) **Product Conversion Costs:** Product conversion costs are one-time investments in research, development, testing, marketing, and other costs focused on making equipment designs comply with new and/or amended energy conservation standards. The GRIM allocates these costs over the period between the standard’s announcement year (*i.e.*, publication of a final rule) and the compliance year. Product conversion cost details can be found in the “Conversion Costs” tab of the GRIM.
- (8) **Annual Testing:** Annual costs for testing new model offerings that have not been previously certified to DOE.
- (9) **Stranded Assets:** In the compliance year of the standard, the GRIM can include a one-time write-off of assets that become obsolete or non-performing due to new and/or amended standards. Stranded asset details can be found in the “Conversion Costs” tab of the GRIM.
- (10) **Earnings Before Interest and Taxes (EBIT):** Includes profits before deductions for interest paid and taxes.
- (11) **Taxes:** Industry tax expenses calculated by multiplying EBIT by the tax rate contained in “Financials” tab of the GRIM.
- (12) **Net Operating Profits After Taxes (NOPAT):** Computed by subtracting manufacturer production costs (Materials + Labor + Overhead + Depreciation), SG&A, R&D, Product Conversion Costs, and Taxes from Revenues.

Industry Cash Flow Statement

- (1) **NOPAT:** This is a repeat of NOPAT in the Industry Income Statement.
- (2) **Depreciation:** Industry depreciation is added back into the Statement of Cash Flows because it is a non-cash expense.
- (3) **Loss on Disposal of Stranded Assets:** This is a repeat of Stranded Assets in the Industry Income Statement. This is added back into the Statement of Cash Flows because it is a non-cash expense.
- (4) **Change in Working Capital:** Change in cash tied up in accounts receivable, inventory, and other cash investments necessary to support operations is calculated by multiplying working capital (as a percentage of revenues) by the change in annual revenues. The Working Capital percentage can be found on the “Financials” tab of the GRIM.
- (5) **Cash Flow from Operations:** Calculated by taking NOPAT, adding back the non-cash items Depreciation and Loss on Disposal of Stranded Assets, and subtracting the Change in Working Capital.

- (6) **Ordinary Capital Expenditures:** Ordinary investments in property, plant, and equipment to maintain and replace existing production assets, computed as a percentage of revenues based on the value on the “Financials” tab of the GRIM.
- (7) **Capital Conversion Costs:** Capital conversion costs are one-time investments in property, plant, and equipment to adapt or change existing production facilities so that new equipment designs can be fabricated and assembled under the new regulation. The GRIM allocates these costs over the period between the standard’s announcement and compliance dates. Capital conversion cost details can be found in the “Conversion Costs” tab of the GRIM.
- (8) **Free Cash Flow:** Annual cash flow from operations and investments; computed by subtracting Ordinary Capital Expenditures and Capital Conversion Costs from Cash Flows from Operations.
- (9) **Free Cash Flow:** This is a repeat of Free Cash Flow from the Industry Cash Flow Statement.
- (10) **Terminal Value:** Estimate of the continuing value of the industry after the analysis period. Computed by growing the Free Cash Flow at a constant rate in perpetuity. The terminal growth rate can be found in the “Financials” tab of the GRIM.
- (11) **Present Value Factor:** Factor used to calculate an estimate of the present value of an amount to be received in the future that is calculated using the industry’s Weighted Average Cost of Capital, found on the “Financials” tab of the GRIM.
- (12) **Discounted Cash Flow:** Free Cash Flows multiplied by the Present Value Factor. For the final year of the analysis, the discounted cash flow includes the discounted Terminal Value.
- (13) **Industry Net Present Value (INPV):** The sum of Discounted Cash Flows from the reference year to the terminal year of the GRIM analysis.

Standard Case DCF

Navigation

Industry Income Statement (in 2021\$ millions)	Ancmt Yr						
	2022	2023	2024	2025	2026	2027	2028
Revenues	\$ 534.7	\$ 656.6	\$ 678.5	\$ 595.9	\$ 624.3	\$ 651.8	\$ 682.2
Total Shipments (million units)	5.602	7.909	8.195	6.354	6.645	6.987	7.357
- MPC	\$ 368.8	\$ 452.8	\$ 468.3	\$ 409.7	\$ 431.3	\$ 450.3	\$ 471.3
- SG&A	\$ 96.2	\$ 118.2	\$ 122.1	\$ 107.3	\$ 112.4	\$ 117.3	\$ 122.8
- R&D	\$ 9.1	\$ 11.2	\$ 11.5	\$ 10.1	\$ 10.6	\$ 11.1	\$ 11.6
- Product Conversion Costs	\$ -	\$ 17.2	\$ 1.3	\$ 1.7	\$ -	\$ -	\$ -
- Annual Testing	\$ -	\$ -	\$ -	\$ -	\$ 1.8	\$ 1.8	\$ 1.8
- Stranded Assets	\$ -	\$ -	\$ 2.7	\$ -	\$ 3.0	\$ -	\$ -
Earnings Before Interest and Taxes (EBIT)	\$ 60.6	\$ 57.3	\$ 72.5	\$ 67.1	\$ 65.2	\$ 71.3	\$ 74.7
Per Unit EBIT (\$/unit)	\$ 10.82	\$ 7.24	\$ 8.85	\$ 10.56	\$ 10.08	\$ 10.46	\$ 10.40
EBIT/Revenues (%)	11.3%	8.7%	10.7%	11.3%	10.4%	10.9%	11.0%
- Taxes	\$ 16.6	\$ 15.7	\$ 19.9	\$ 18.4	\$ 17.9	\$ 19.5	\$ 20.5
Net Operating Profit after Taxes (NOPAT)	\$ 44.0	\$ 41.6	\$ 52.7	\$ 48.7	\$ 47.3	\$ 51.8	\$ 54.2
Industry Cash Flow Statement							
NOPAT	\$ 44.0	\$ 41.6	\$ 52.7	\$ 48.7	\$ 47.3	\$ 51.8	\$ 54.2
+ Depreciation	\$ 18.7	\$ 23.0	\$ 23.7	\$ 20.9	\$ 21.8	\$ 22.8	\$ 23.9
+ Loss on Disposal of Stranded Assets	\$ -	\$ -	\$ 2.7	\$ -	\$ 3.0	\$ -	\$ -
- Change in Working Capital	\$ -	\$ 3.9	\$ 3.6	\$ (13.7)	\$ 4.7	\$ 4.6	\$ 5.0
Cash Flows from Operations	\$ 62.7	\$ 60.7	\$ 75.5	\$ 83.3	\$ 67.5	\$ 70.0	\$ 73.1
- Ordinary Capital Expenditures	\$ 21.4	\$ 26.3	\$ 27.1	\$ 23.8	\$ 25.0	\$ 26.1	\$ 27.3
- Capital Conversion Costs	\$ -	\$ 13.6	\$ 5.3	\$ 6.8	\$ -	\$ -	\$ -
Free Cash Flow	\$ 41.3	\$ 20.8	\$ 43.1	\$ 52.7	\$ 42.5	\$ 43.9	\$ 45.8
Discounted Industry Cash Flow							
Free Cash Flow	\$ 41.3	\$ 20.8	\$ 43.1	\$ 52.7	\$ 42.5	\$ 43.9	\$ 45.8
Terminal Value	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
Present Value Factor	0.000	1.000	0.938	0.880	0.826	0.774	0.726
Discounted Cash Flow	\$ -	\$ 20.8	\$ 40.4	\$ 46.4	\$ 35.1	\$ 34.0	\$ 33.3
INPV at TSL 1 \$ 1,293.5							
Net PPE	\$ 56.7	\$ 73.6	\$ 79.5	\$ 89.3	\$ 89.4	\$ 92.6	\$ 96.1
Net PPE as % of Sales	10.6%	11.2%	11.7%	15.0%	14.3%	14.2%	14.1%
Net Working Capital	\$ 88.8	\$ 109.0	\$ 112.6	\$ 98.9	\$ 103.6	\$ 108.2	\$ 113.2
Return on Invested Capital (ROIC)	30.25%	22.77%	27.41%	25.89%	24.53%	25.77%	25.91%
Weighted Average Cost of Capital (WACC)	6.60%	6.60%	6.60%	6.60%	6.60%	6.60%	6.60%
Return on Sales (EBIT/Sales)	11.33%	8.72%	10.69%	11.26%	10.44%	10.94%	10.95%

Figure 12A.2.1 Detailed Income Statement and Cash Flow Statement Example

APPENDIX 13A. EMISSIONS ANALYSIS METHODOLOGY

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APPENDIX 13A. EMISSIONS ANALYSIS METHODOLOGY

13A.1 INTRODUCTION

The emissions analysis consists of two components. The first component estimates the effect of potential energy conservation standards on power sector and site combustion emissions of carbon dioxide (CO₂), nitrogen oxides (NO_x), sulfur dioxide (SO₂) and mercury (Hg). The second component estimates the impacts of a potential standard on emissions of two additional greenhouse gases, methane (CH₄) and nitrous oxide (N₂O), as well as the reductions to emissions of all species due to “upstream” activities in the fuel production chain. These upstream activities comprise extraction, processing, and transporting fuels to the site of combustion. The associated emissions are referred to as upstream emissions. Together, these emissions account for the full-fuel-cycle (FFC), in accordance with DOE’s FFC Statement of Policy. 76 FR 51282 (Aug. 18, 2011).

The analysis of power sector emissions uses marginal emissions intensity factors calculated by DOE. DOE’s methodology is based on results published with the most recent edition of the *Annual Energy Outlook (AEO)* which is published by the Energy Information Agency (EIA). For this analysis DOE used *AEO 2022*.¹ DOE developed end-use specific emissions intensity coefficients, in units of mass of pollutant per kWh of site (grid) electricity, for each pollutant. The methodology is based on the more general approach used for all the utility sector impacts calculations, which is described in appendix 15A of this TSD and in the report “Utility Sector Impacts of Reduced Electricity Demand” (Coughlin, 2014; Coughlin, 2019).^{2,3} This appendix describes the methodology used to estimate the upstream emissions factors, and presents the values used for all emissions factors.

13A.2 POWER SECTOR AND SITE EMISSIONS FACTORS

Power sector marginal emissions factors are calculated by looking at the difference, over the full analysis period, in fuel consumption and emissions across a variety of cases published with the AEO. The analysis produces a set of emissions intensity factors that quantify the reduction in emissions of a given pollutant per unit reduction of fuel used in (grid) electricity generation for each of the primary fossil fuel types (coal, natural gas and oil). These factors are combined with estimates of the fraction of generation allocated to each fuel type, also calculated from *AEO 2022* data, for each sector and end-use. The result is a set of end-use specific marginal emissions intensity factors, summarized in the tables below. Total emissions reductions are estimated by multiplying the intensity factors times the energy savings calculated in the national impact analysis (chapter 10). Power sector emissions factors are presented in Table 13A.4.2 through Table 13A.4.7.

Site combustion of fossil fuels in buildings (for example in water-heating, space-heating or cooking applications) also produces emissions of CO₂ and other pollutants. To quantify the reduction in these emissions from a considered standard level, DOE used emissions intensity factors from Environmental Protection Agency (EPA) publications.⁴ These factors, presented in Table 13A.4.1, are constant in time. The EPA defines SO₂ emissions in terms of a formula that depends on the sulfur content of the fuel. The typical use of petroleum-based fuels in buildings if

for heating, and a typical sulfur content for heating oils is a few hundred parts-per-million (ppm). The value provided in Table 13A.4.1 corresponds to a sulfur content of approximately 100 ppm.

13A.3 UPSTREAM FACTORS

The FFC upstream emissions are estimated based on the methodology developed by Coughlin (2013).⁵ The upstream emissions include both emissions from fuel combustion during extraction, processing and transportation of fuel, and “fugitive” emissions (direct leakage to the atmosphere) of CH₄ and CO₂.

The FFC accounting approach is described briefly in appendix 10B and in Coughlin (2013).⁵ When demand for a particular fuel is reduced, there is a corresponding reduction in the upstream activities associated with production of that fuel (mining, refining etc.) These upstream activities also consume energy and therefore produce combustion emissions. The FFC accounting estimates the total consumption of electricity, natural gas and petroleum-based fuels in these upstream activities. The relevant combustion emissions factors are then applied to this fuel use to determine the total upstream emissions intensities from combustion, per unit of fuel delivered to the consumer.

In addition to combustion emissions, extraction and processing of fossil fuels also produces fugitive emissions of CO₂ and CH₄. Fugitive emissions of CO₂ are small relative to combustion emissions, comprising about 2-3 percent of total CO₂ emissions for natural gas and 1-2 percent for petroleum fuels. In contrast, the fugitive emissions of methane from fossil fuel production are relatively large compared to combustion emissions of CH₄. Hence, fugitive emissions make up over 99 percent of total methane emissions for natural gas, about 95 percent for coal, and 93 percent for petroleum fuels.

Fugitive emissions factors for CO₂ and methane from coal mining and natural gas production were estimated based on a review of recent studies compiled by Burnham (2011).⁶ This review includes estimates of the difference between fugitive emissions factors for conventional production of natural vs. unconventional (shale or tight gas). These estimates rely in turn on data gathered by EPA under new GHG reporting requirements for the petroleum and natural gas industries.^{7,8} The value for methane, if it were translated to a leakage rate, would be equivalent to 1.3%. Actual leakage rates of methane at various stages of the production process are highly variable and the subject of ongoing research. In a comprehensive review of the literature, Brandt et al. (2014)⁸ find that, while regional studies with very high emissions rates may not be representative of typical natural gas systems, it is also true that official inventories have most likely underestimated methane emissions. As more data are made available, DOE will continue to update these estimated emissions factors.

Upstream emissions factors account for both fugitive emissions and combustion emissions in extraction, processing, and transport of primary fuels. For ease of application in its analysis, DOE developed all of the emissions factors using site (point of use) energy savings in the denominator. Table 13A.4.1 presents the electricity upstream emissions factors for selected years. The caps that apply to power sector NO_x emissions do not apply to upstream combustion sources, so some components of the upstream fuel cycle (particularly off-road mobile engines) can contribute significantly to the upstream NO_x emissions factors.

13A.4 DATA TABLES

Summary tables of all the emissions factor data used by DOE for rules using *AEO 2022* are presented in the tables below. Table 13A.4.1 provides combustion emissions factors for fuels commonly used in buildings. Table 13A.4.2 to Table 13A.4.7 present the marginal power sector emissions factors as a function of sector and end use for a selected set of years. Table 13A.4.8 to Table 13A.4.10 provide the upstream emissions factors for all pollutants, for site electricity, natural gas and petroleum fuels. In all cases, the emissions factors are defined relative to the site electricity supplied from the grid and site use of the fuel.

Table 13A.4.1 Site Combustion Emissions Factors

Species	Natural Gas g/mcf	Distillate Oil g/bbl
CH ₄	1.03E+00	1.33E+01
CO ₂	5.47E+04	4.46E+05
N ₂ O	1.03E-01	8.65E+00
NO _x	4.36E+01	3.62E+02
SO ₂	2.73E-01	2.20E+02

Table 13A.4.2 Power Sector Emissions Factors for CO₂ (Million Short Tons (MMsT)/Quad of Site Electricity Use)

	2025	2030	2035	2040	2045	2050
Residential Sector						
Clothes Dryers	477	417	369	341	324	313
Cooking	472	412	365	337	321	310
Freezers	486	424	376	347	330	318
Lighting	486	424	375	346	328	316
Refrigeration	485	424	376	347	329	318
Space Cooling	457	400	358	333	318	308
Space Heating	492	429	379	349	331	319
Water Heating	478	417	369	340	324	312
Other Uses	477	416	368	340	324	312
Commercial Sector						
Cooking	440	383	340	316	302	292
Lighting	449	391	347	322	307	297
Office Equipment (Non-Pc)	426	370	330	307	294	285
Office Equipment (Pc)	426	370	330	307	294	285
Refrigeration	473	412	365	337	321	310
Space Cooling	449	393	352	328	314	305
Space Heating	495	432	381	351	333	321
Ventilation	473	413	366	338	321	310
Water Heating	439	382	339	315	301	291
Other Uses	432	376	334	311	297	288
Industrial Sector						
All Uses	432	376	334	311	297	288

Table 13A.4.3 Power Sector Emissions Factors for CH₄ (Million Short Tons (MMst)/Quad of Site Electricity Use)

	2025	2030	2035	2040	2045	2050
Residential Sector						
Clothes Dryers	0.0374	0.0328	0.0282	0.0250	0.0227	0.0214
Cooking	0.0365	0.0320	0.0276	0.0244	0.0222	0.0210
Freezers	0.0385	0.0338	0.0291	0.0258	0.0234	0.0221
Lighting	0.0389	0.0342	0.0294	0.0260	0.0236	0.0223
Refrigeration	0.0384	0.0337	0.0290	0.0257	0.0234	0.0221
Space Cooling	0.0330	0.0289	0.0249	0.0221	0.0201	0.0190
Space Heating	0.0398	0.0350	0.0301	0.0267	0.0242	0.0228
Water Heating	0.0377	0.0331	0.0284	0.0252	0.0229	0.0216
Other Uses	0.0373	0.0327	0.0282	0.0250	0.0227	0.0214
Commercial Sector						
Cooking	0.0317	0.0278	0.0239	0.0212	0.0193	0.0182
Lighting	0.0330	0.0289	0.0249	0.0221	0.0201	0.0190
Office Equipment (Non-Pc)	0.0294	0.0257	0.0221	0.0196	0.0179	0.0169
Office Equipment (Pc)	0.0294	0.0257	0.0221	0.0196	0.0179	0.0169
Refrigeration	0.0366	0.0322	0.0277	0.0245	0.0223	0.0210
Space Cooling	0.0317	0.0278	0.0239	0.0213	0.0193	0.0183
Space Heating	0.0403	0.0354	0.0305	0.0270	0.0245	0.0231
Ventilation	0.0368	0.0323	0.0278	0.0246	0.0224	0.0211
Water Heating	0.0316	0.0277	0.0238	0.0211	0.0192	0.0181
Other Uses	0.0304	0.0266	0.0229	0.0203	0.0185	0.0175
Industrial Sector						
All Uses	0.0304	0.0266	0.0229	0.0203	0.0185	0.0175

Table 13A.4.4 Power Sector Emissions Factors for Hg (Short Tons (sT)/Quad of Site Electricity Use)

	2025	2030	2035	2040	2045	2050
Residential Sector						
Clothes Dryers	1.207	1.067	0.994	0.955	0.886	0.939
Cooking	1.175	1.038	0.967	0.929	0.862	0.913
Freezers	1.249	1.104	1.029	0.989	0.918	0.973
Lighting	1.267	1.120	1.044	1.004	0.932	0.988
Refrigeration	1.246	1.101	1.026	0.987	0.916	0.970
Space Cooling	1.033	0.910	0.846	0.812	0.752	0.796
Space Heating	1.304	1.153	1.075	1.034	0.960	1.018
Water Heating	1.220	1.078	1.005	0.965	0.896	0.950
Other Uses	1.205	1.065	0.992	0.953	0.885	0.937
Commercial Sector						
Cooking	0.996	0.878	0.816	0.782	0.725	0.767
Lighting	1.045	0.922	0.857	0.822	0.762	0.807
Office Equipment (Non-Pc)	0.907	0.798	0.740	0.709	0.656	0.694
Office Equipment (Pc)	0.907	0.798	0.740	0.709	0.656	0.694
Refrigeration	1.181	1.043	0.971	0.933	0.866	0.917
Space Cooling	0.984	0.867	0.805	0.772	0.715	0.756
Space Heating	1.322	1.169	1.090	1.049	0.974	1.033
Ventilation	1.186	1.048	0.976	0.938	0.870	0.922
Water Heating	0.992	0.874	0.812	0.779	0.721	0.763
Other Uses	0.945	0.832	0.773	0.741	0.686	0.725
Industrial Sector						
All Uses	0.945	0.832	0.773	0.741	0.686	0.725

Table 13A.4.5 Power Sector Emissions Factors for N₂O (Million Short Tons (MMsT)/Quad of Site Electricity Use)

	2025	2030	2035	2040	2045	2050
Residential Sector						
Clothes Dryers	0.00528	0.00463	0.00398	0.00351	0.00317	0.00299
Cooking	0.00515	0.00452	0.00388	0.00343	0.00310	0.00292
Freezers	0.00544	0.00478	0.00410	0.00362	0.00327	0.00309
Lighting	0.00550	0.00484	0.00415	0.00366	0.00331	0.00312
Refrigeration	0.00543	0.00477	0.00410	0.00362	0.00327	0.00308
Space Cooling	0.00464	0.00406	0.00349	0.00308	0.00278	0.00263
Space Heating	0.00564	0.00496	0.00425	0.00376	0.00339	0.00320
Water Heating	0.00532	0.00467	0.00401	0.00354	0.00320	0.00301
Other Uses	0.00527	0.00463	0.00397	0.00350	0.00317	0.00298
Commercial Sector						
Cooking	0.00445	0.00390	0.00334	0.00295	0.00267	0.00251
Lighting	0.00464	0.00407	0.00349	0.00308	0.00278	0.00262
Office Equipment (Non-Pc)	0.00411	0.00359	0.00308	0.00272	0.00246	0.00232
Office Equipment (Pc)	0.00411	0.00359	0.00308	0.00272	0.00246	0.00232
Refrigeration	0.00517	0.00454	0.00390	0.00344	0.00311	0.00293
Space Cooling	0.00445	0.00390	0.00334	0.00296	0.00267	0.00252
Space Heating	0.00572	0.00502	0.00431	0.00381	0.00344	0.00324
Ventilation	0.00519	0.00456	0.00391	0.00345	0.00312	0.00294
Water Heating	0.00443	0.00388	0.00332	0.00294	0.00265	0.00250
Other Uses	0.00426	0.00372	0.00319	0.00282	0.00255	0.00240
Industrial Sector						
All Uses	0.00426	0.00372	0.00319	0.00282	0.00255	0.00240

Table 13A.4.6 Power Sector Emissions Factors for NO_x (Million Short Tons (MMsT)/Quad of Site Electricity Use)

	2025	2030	2035	2040	2045	2050
Residential Sector						
Clothes Dryers	0.186	0.221	0.186	0.167	0.139	0.136
Cooking	0.183	0.218	0.183	0.164	0.137	0.134
Freezers	0.190	0.227	0.190	0.171	0.142	0.139
Lighting	0.191	0.227	0.190	0.171	0.142	0.139
Refrigeration	0.190	0.226	0.190	0.170	0.142	0.139
Space Cooling	0.173	0.206	0.174	0.157	0.131	0.129
Space Heating	0.194	0.231	0.194	0.173	0.144	0.141
Water Heating	0.186	0.222	0.186	0.167	0.139	0.136
Other Uses	0.185	0.221	0.185	0.166	0.139	0.136
Commercial Sector						
Cooking	0.166	0.196	0.165	0.149	0.124	0.122
Lighting	0.170	0.202	0.170	0.153	0.128	0.126
Office Equipment (Non-Pc)	0.158	0.187	0.157	0.142	0.119	0.117
Office Equipment (Pc)	0.158	0.187	0.157	0.142	0.119	0.117
Refrigeration	0.183	0.218	0.183	0.164	0.137	0.134
Space Cooling	0.169	0.201	0.170	0.153	0.128	0.126
Space Heating	0.196	0.233	0.195	0.175	0.145	0.142
Ventilation	0.184	0.218	0.183	0.165	0.137	0.135
Water Heating	0.165	0.196	0.165	0.148	0.124	0.122
Other Uses	0.161	0.191	0.161	0.145	0.121	0.120
Industrial Sector						
All Uses	0.161	0.191	0.161	0.145	0.121	0.120

Table 13A.4.7 Power Sector Emissions Factors for SO₂ (Million Short Tons (MMsT)/Quad of Site Electricity Use)

	2025	2030	2035	2040	2045	2050
Residential Sector						
Clothes Dryers	0.269	0.217	0.169	0.147	0.135	0.137
Cooking	0.262	0.211	0.165	0.144	0.132	0.134
Freezers	0.278	0.224	0.175	0.153	0.140	0.142
Lighting	0.281	0.226	0.177	0.154	0.142	0.144
Refrigeration	0.278	0.224	0.175	0.152	0.140	0.142
Space Cooling	0.236	0.189	0.147	0.128	0.117	0.118
Space Heating	0.288	0.233	0.182	0.159	0.146	0.148
Water Heating	0.271	0.218	0.170	0.149	0.136	0.139
Other Uses	0.268	0.216	0.169	0.147	0.135	0.137
Commercial Sector						
Cooking	0.224	0.179	0.140	0.122	0.111	0.113
Lighting	0.234	0.188	0.147	0.128	0.117	0.118
Office Equipment (Non-Pc)	0.205	0.164	0.128	0.111	0.101	0.103
Office Equipment (Pc)	0.205	0.164	0.128	0.111	0.101	0.103
Refrigeration	0.263	0.212	0.165	0.144	0.132	0.134
Space Cooling	0.226	0.181	0.141	0.122	0.111	0.113
Space Heating	0.292	0.236	0.184	0.161	0.148	0.150
Ventilation	0.264	0.213	0.166	0.145	0.133	0.135
Water Heating	0.223	0.179	0.139	0.121	0.111	0.112
Other Uses	0.213	0.171	0.133	0.116	0.105	0.107
Industrial Sector						
All Uses	0.213	0.171	0.133	0.116	0.105	0.107

Table 13A.4.8 Electricity Upstream Emissions Factors

Species	Unit	2025	2030	2035	2040	2045	2050+
CO ₂	kg/MWh	27.1	24.8	23.3	22.8	22.7	22.6
CH ₄	g/MWh	2233.3	2072.0	1959.8	1937.3	1957.8	1955.2
Hg	g/MWh	5.4E-06	4.7E-06	3.9E-06	3.3E-06	2.9E-06	2.6E-06
N ₂ O	g/MWh	0.152	0.136	0.121	0.110	0.102	0.098
NO _x	g/MWh	363.0	334.7	317.0	311.7	312.1	312.3
SO ₂	g/MWh	2.4	2.1	1.8	1.6	1.4	1.3

Table 13A.4.9 Natural Gas Upstream Emissions Factors

Species	Unit	2025	2030	2035	2040	2045	2050+
CO ₂	kg/MMcf	7.1	7.1	7.2	7.2	7.1	7.2
CH ₄	g/MMcf	691.1	692.9	694.2	694.2	692.8	693.7
Hg	g/MMcf	0.0E+00	0.0E+00	0.0E+00	0.0E+00	0.0E+00	0.0E+00
N ₂ O	g/MMcf	0.011	0.011	0.011	0.011	0.011	0.011
NO _x	g/MMcf	100.3	100.5	101.5	101.7	101.2	102.2
SO ₂	g/MMcf	0.0	0.0	0.0	0.0	0.0	0.0

Table 13A.4.10 Petroleum Fuels Upstream Emission Factors

Species	Unit	2025	2030	2035	2040	2045	2050+
CO ₂	kg/bbl	69.7	69.8	70.3	71.6	71.8	72.1
CH ₄	g/bbl	950.3	944.3	943.6	960.6	963.1	965.8
Hg	g/bbl	4.7E-06	4.7E-06	4.4E-06	4.1E-06	3.8E-06	3.7E-06
N ₂ O	g/bbl	0.582	0.587	0.596	0.605	0.604	0.605
NO _x	g/bbl	762.3	770.8	785.5	799.3	799.2	802.3
SO ₂	g/bbl	13.8	13.9	14.0	14.2	14.2	14.2

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APPENDIX 14A. SOCIAL COST OF GREENHOUSE GAS VALUES, 2020-2070

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APPENDIX 14A. SOCIAL COST OF GREENHOUSE GAS VALUES, 2020-2070

14A.1 VALUES FOR SOCIAL COST OF GREENHOUSE GASES

The values in this appendix are taken from the model input files supporting the “Revised 2023 and Later Model Year Light-Duty Vehicle GHG Emissions Standards: Regulatory Impact Analysis” published by EPA in December 2021.^{1,a} These values are themselves based on the 2020-2050 values in “Technical Support Document: Social Cost of Carbon, Methane, and Nitrous Oxide; Interim Estimates under Executive Order 13990”, published by the Interagency Working Group on Social Cost of Greenhouse Gases in February 2021.² To derive values for 2051-2070, EPA extrapolated based on methods, assumptions, and parameters identical to the 2020-2050 estimates published by the Interagency Working Group. The values in the EPA files are in 2018\$. DOE converted these to 2020\$ using the GDP deflator.^b Due to a lack of available SC-CO₂ estimates for emissions years beyond 2070, DOE did not monetize the climate benefits of GHG emissions reductions occurring after 2070.

Table 14A.1.1 Interim Social Cost of CO₂ Values Based on 2021 Interagency Update and 2021 EPA Light-Duty Vehicle Regulatory Impact Analysis, 2020–2070 (converted to 2020\$ per Metric Ton of CO₂)*

Discount Rate and Statistics				
Emissions Year	5%, Average	3%, Average	2.5%, Average	3%, 95th
2020	14	51	76	151
2021	15	52	77	155
2022	15	53	79	158
2023	16	54	80	162
2024	16	55	81	165
2025	17	56	83	169
2026	17	57	84	172
2027	18	58	85	176
2028	18	59	87	179
2029	19	60	88	183
2030	19	62	89	186
2031	20	63	91	190
2032	20	64	92	194
2033	21	65	93	198
2034	22	66	95	201
2035	22	67	96	205
2036	23	68	97	209

^a Model files available at: www3.epa.gov/otaq/ld/EPA-CCEMS-PostProcessingTool-Project-FRM.zip (last accessed January 18, 2022).

^b For 2020-2050, there are slight differences from the IWG report in a few cases that are likely due to the GDP deflator used.

Discount Rate and Statistics				
Emissions Year	5%, Average	3%, Average	2.5%, Average	3%, 95th
2037	23	70	99	213
2038	24	71	100	217
2039	25	72	101	220
2040	25	73	103	224
2041	26	74	104	228
2042	26	75	105	231
2043	27	76	107	235
2044	28	78	108	238
2045	28	79	109	242
2046	29	80	111	245
2047	30	81	112	249
2048	30	82	113	252
2049	31	83	115	256
2050	32	84	116	259
2051	32	85	118	260
2052	33	86	119	261
2053	34	87	120	262
2054	34	88	121	263
2055	35	89	122	265
2056	35	90	123	267
2057	36	91	124	269
2058	37	92	125	271
2059	37	92	127	273
2060	38	93	128	275
2061	39	95	129	280
2062	40	96	131	285
2063	41	98	132	290
2064	42	99	134	295
2065	44	100	135	300
2066	45	102	137	305
2067	46	103	138	311
2068	47	105	140	316
2069	48	106	141	321
2070	49	108	143	326

* Values are rounded off to the nearest dollar.

Table 14A.1.2 Interim Social Cost of CH₄ Values from 2021 Interagency Update and Interagency Update, 2021 EPA Light-Duty Vehicle Regulatory Impact Analysis, 2020–2070 (converted to 2020\$ per Metric Ton of CH₄)*

Discount Rate and Statistics				
Emissions Year	5%, Average	3%, Average	2.5%, Average	3%, 95th
2020	663	1,480	1,946	3,893
2021	691	1,527	2,002	4,021
2022	718	1,574	2,057	4,149
2023	745	1,620	2,112	4,277
2024	772	1,667	2,167	4,405
2025	799	1,714	2,223	4,533
2026	826	1,761	2,278	4,661
2027	853	1,807	2,333	4,789
2028	880	1,854	2,388	4,917
2029	908	1,901	2,444	5,045
2030	935	1,948	2,499	5,173
2031	969	2,003	2,563	5,326
2032	1,003	2,058	2,626	5,479
2033	1,038	2,113	2,690	5,632
2034	1,072	2,168	2,754	5,786
2035	1,106	2,224	2,817	5,939
2036	1,140	2,279	2,881	6,092
2037	1,175	2,334	2,945	6,245
2038	1,209	2,389	3,008	6,399
2039	1,243	2,444	3,072	6,552
2040	1,277	2,500	3,136	6,705
2041	1,315	2,555	3,199	6,849
2042	1,352	2,611	3,261	6,993
2043	1,389	2,667	3,324	7,138
2044	1,427	2,722	3,387	7,282
2045	1,464	2,778	3,450	7,426
2046	1,502	2,834	3,512	7,570
2047	1,539	2,890	3,575	7,714
2048	1,576	2,945	3,638	7,859
2049	1,614	3,001	3,701	8,003
2050	1,651	3,057	3,763	8,147
2051	1,680	3,096	3,807	8,193
2052	1,703	3,128	3,841	8,228
2053	1,726	3,159	3,874	8,263
2054	1,749	3,190	3,908	8,297
2055	1,772	3,221	3,942	8,332

Discount Rate and Statistics				
Emissions Year	5%, Average	3%, Average	2.5%, Average	3%, 95th
2056	1,797	3,256	3,979	8,373
2057	1,823	3,291	4,017	8,415
2058	1,848	3,326	4,055	8,456
2059	1,873	3,360	4,092	8,497
2060	1,899	3,395	4,130	8,539
2061	2,021	3,548	4,296	9,067
2062	2,143	3,702	4,462	9,594
2063	2,264	3,856	4,628	10,122
2064	2,386	4,009	4,794	10,650
2065	2,508	4,163	4,960	11,177
2066	2,632	4,325	5,141	11,758
2067	2,757	4,488	5,323	12,338
2068	2,881	4,651	5,504	12,919
2069	3,006	4,814	5,686	13,499
2070	3,130	4,976	5,867	14,079

* Values are rounded off to the nearest dollar.

Table 14A.1.3 Interim Social Cost of N₂O Values from 2021 Interagency Update and 2021 EPA Light-Duty Vehicle Regulatory Impact Analysis, 2020–2070 (converted to 2020\$ per Metric Ton of N₂O)*

Discount Rate and Statistics				
Emissions Year	5%, Average	3%, Average	2.5%, Average	3%, 95th
2020	5,760	18,342	27,037	48,090
2021	5,961	18,777	27,592	49,293
2022	6,162	19,213	28,147	50,497
2023	6,363	19,649	28,702	51,700
2024	6,565	20,084	29,257	52,904
2025	6,766	20,520	29,811	54,108
2026	6,967	20,955	30,366	55,311
2027	7,168	21,391	30,921	56,515
2028	7,370	21,827	31,476	57,718
2029	7,571	22,262	32,031	58,922
2030	7,772	22,698	32,585	60,125
2031	8,019	23,188	33,195	61,480
2032	8,266	23,678	33,804	62,834
2033	8,513	24,168	34,413	64,189
2034	8,760	24,659	35,023	65,543
2035	9,007	25,149	35,632	66,898

Discount Rate and Statistics				
Emissions Year	5%, Average	3%, Average	2.5%, Average	3%, 95th
2036	9,253	25,639	36,241	68,252
2037	9,500	26,129	36,850	69,606
2038	9,747	26,619	37,460	70,961
2039	9,994	27,110	38,069	72,315
2040	10,241	27,600	38,678	73,670
2041	10,530	28,127	39,320	75,089
2042	10,819	28,655	39,962	76,508
2043	11,109	29,183	40,604	77,928
2044	11,398	29,710	41,246	79,347
2045	11,687	30,238	41,888	80,766
2046	11,976	30,765	42,530	82,186
2047	12,265	31,293	43,172	83,605
2048	12,555	31,820	43,814	85,024
2049	12,844	32,348	44,456	86,443
2050	13,133	32,875	45,098	87,863
2051	13,479	33,426	45,727	88,606
2052	13,798	33,954	46,354	89,984
2053	14,118	34,483	46,981	91,362
2054	14,438	35,011	47,609	92,739
2055	14,758	35,539	48,236	94,117
2056	15,091	36,092	48,890	95,463
2057	15,425	36,644	49,544	96,808
2058	15,758	37,196	50,199	98,154
2059	16,091	37,748	50,853	99,499
2060	16,424	38,300	51,507	100,845
2061	17,077	39,165	52,485	103,794
2062	17,730	40,030	53,463	106,743
2063	18,382	40,895	54,441	109,692
2064	19,035	41,760	55,419	112,641
2065	19,687	42,625	56,397	115,590
2066	20,354	43,515	57,403	118,657
2067	21,020	44,404	58,409	121,725
2068	21,686	45,293	59,416	124,793
2069	22,352	46,183	60,422	127,860
2070	23,018	47,072	61,428	130,928

* Values are rounded off to the nearest dollar.

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**APPENDIX 14B. BENEFIT-PER-TON VALUES FOR NO_x AND SO₂ EMISSIONS
FROM ELECTRICITY GENERATION**

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APPENDIX 14B. BENEFIT-PER-TON VALUES FOR NO_x AND SO₂ EMISSIONS FROM ELECTRICITY GENERATION

14B.1 INTRODUCTION

This appendix describes the analytical methodology DOE uses to incorporate regional and end use sector variability in NO_x and SO₂ valuations into the emissions monetization. The regional values assigned to these emissions are based on benefit-per-ton estimates published by EPA for a variety of sectors, including electricity generation. EPA provides high and low estimates of benefit-per-ton of NO_x and SO₂ emissions reductions in forty regions of the continental USA. DOE combined these data with regional information on electricity consumption and emissions to define weighted-average national values for NO_x and SO₂ as a function of sector.

DOE's methodology uses results associated with the most recent edition of the *Annual Energy Outlook (AEO)* published by the Energy Information Agency (EIA). For this analysis DOE used the Reference case from *AEO2022*.¹ The *AEO* data are used to define two sets of factors that enter into the calculation: the distribution of sectoral electricity consumption by region, and the magnitude of NO_x and SO₂ emissions in each region.

14B.2 METHODOLOGY

14B.2.1 EPA Data

In 2022 EPA published an updated Technical Support Document (TSD) describing an approach for estimating the average avoided human health impacts and monetized benefits related to emissions of PM_{2.5} and ozone precursors including NO_x and SO₂ from 21 sectors.^a The EPA TSD includes estimates of the present value of the benefits of NO_x and SO₂ emissions reductions (*benefit-per-ton* estimates or BPT) for 2025, 2030, 2035 and 2040. For NO_x, EPA provides values for PM_{2.5} –related benefits and for ozone-related benefits. Because the pollutants associated with NO_x as PM_{2.5} and SO₂ emissions persist in the atmosphere over a period of years, reductions in any given year will have benefits in subsequent years. These future benefits are discounted and summed to provide a single value for the reduction of one ton of emissions in the emissions year.

For Electricity generating units, EPA estimated a benefit per-ton for each of the 48 contiguous continental states. Some states are aggregated into larger regions (CT-RI, DE-NJ, ID-OR-WA, ME-MA-NH-VT, and ND-SD), resulting in separate BPT estimates for forty regions. BPT values for NO_x and SO₂ as precursors to PM_{2.5} include high and low impact scenarios; BPT

^a U.S. Environmental Protection Agency. *Estimating the Benefit per Ton of Reducing Directly-Emitted PM_{2.5}, PM_{2.5} Precursors and Ozone Precursors from 21 Sectors*. January 2022.
https://www.epa.gov/system/files/documents/2021-10/source-apportionment-tsd-oct-2021_0.pdf

values for NO_x as a precursor to ozone include short and long-term impacts. For all data two rates of discounting (3% and 7%) are provided.

DOE used linear interpolation to define values for the years between 2025 and 2030, 2030 and 2035, and 2035 and 2040; for years beyond 2040 the value is held constant. DOE defined the total value of NO_x emissions reductions as the sum of the BPT value for PM_{2.5} plus one half of the BPT value for ozone; the factor of one half accounts for the fact that ozone is primarily produced during the May-September period, so approximately half of NO_x emissions will produce ozone emissions.

14B.2.2 AEO Data

For this calculation DOE used the total annual emissions of NO_x and SO₂ for each of the AEO's 25 Electricity Market Module (EMM) regions,² and data tables published with the NEMS code package.^b The latter are used to map EPA regions to EMM regions, and to calculate the contribution of each utility customer sector (residential, commercial and industrial) to total pollutant emissions in each EMM region. The data are then combined to create time series of BPT values for each end use sector.

14B.2.3 Equations and Results

Consistent with its treatment of other utility and environmental impacts, DOE defines a times series of national average estimates of NO_x and SO₂ values.

The same methodology is applied to each pollutant type and EPA scenario (low-7%, low-3%, etc.). The notation is:

- y is the analysis year,
- m is a label for the EMM region,
- z is a label for the EOA region,
- $w(z,m)$ is a matrix that maps EPA regions to EMM regions; it is defined as the fraction of total electricity sales within m to region z ; $\sum_z w(z,m) = 1$ for all m ,
- $p(z,y)$ is the BPT estimate in EPA region z and year y ,
- s is a label for the customer sector (commercial, residential, industrial)
- $v(s, m)$ is the weight of sector s in EMM region m , defined as the fraction of total electricity sales within m to sector s ; $\sum_s v(s,m) = 1$ for all m ,

^b The NEMS package can be downloaded at https://www.eia.gov/outlooks/aoe/info_nems_archive.php. Once installed, the file path to the data files is aeo2021\reference\input\emm_db.zip. The data files are EMMCNTRL_RDB.xlsx and LDSMSTR_RDB.xlsx.

- $M(m,y)$ is total pollutant emissions in EMM region m and year y .

The calculation proceeds in four steps:

1. Pollutant emissions are allocated to sector:

$$M1(m, s, y) = M(m, y) * v(s, m)$$

2. Sectoral pollutant emissions are mapped from EMM regions to EPA regions:

$$M2(z, s, y) = \sum_m M1(m, s, y) * w(z, m)$$

3. A weight is defined for EPA region z and sector s , based on pollutant emissions:

$$u(z, s, y) = M2(z, s, y) / [\sum_z M2(z, s, y)]$$

4. The regional weights are used to define a national average BPT value for each sector:

$$P(s, y) = \sum_z u(z, s, y) * p(z, y)$$

The results of this calculation are provided in Table 14B.3.1 for NO_x and in Table 14B.3.2 for SO₂. DOE's prices are not significantly different than the EPA estimate of the US average. Although the EPA prices are held constant after 2040, the DOE prices may vary slightly in the period 2040-2050 due to the projected changes in regional emissions.

Table 14B.3.1 NO_x Benefit-per-ton Values by Sector (2016\$/ Short Ton)

Sector	High, 3% Discount Rate				High, 7% Discount Rate			
	2025	2030	2040	2050	2025	2030	2040	2050
Commercial	59,241	66,019	82,131	82,876	53,063	59,110	73,642	74,313
Industrial	58,828	65,753	80,552	81,237	52,694	58,888	72,242	72,859
Residential	59,102	65,922	81,783	82,420	52,939	59,018	73,327	73,901
Sector	Low, 3% Discount Rate				Low, 7% Discount Rate			
	2025	2030	2040	2050	2025	2030	2040	2050
Commercial	59,116	65,718	81,447	82,181	52,955	58,840	73,030	73,691
Industrial	58,685	65,389	79,743	80,418	52,572	58,561	71,522	72,129
Residential	58,985	65,632	81,129	81,756	52,839	58,758	72,743	73,308

Table 14B.3.2 SO₂ Benefit-per-ton Values by Sector (2016\$/ Short Ton)

Sector	High, 3% Discount Rate				High, 7% Discount Rate			
	2025	2030	2040	2050	2050	2030	2040	2050
Commercial	81,598	92,092	115,167	116,405	73,416	82,827	103,617	104,725
Industrial	81,144	91,307	113,106	114,293	73,001	82,120	101,782	102,844
Residential	81,160	91,610	114,227	115,430	73,023	82,398	102,776	103,852
Sector	Low, 3% Discount Rate				Low, 7% Discount Rate			
	2025	2030	2040	2050	2025	2030	2040	2050
Commercial	80,231	88,263	106,241	107,326	72,148	79,360	95,597	96,573
Industrial	79,821	87,548	104,421	105,453	71,775	78,712	93,950	94,878
Residential	79,820	87,850	105,429	106,477	71,780	78,983	94,859	95,802

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APPENDIX 15A. UTILITY IMPACT ANALYSIS METHODOLOGY

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APPENDIX 15A. UTILITY IMPACT ANALYSIS METHODOLOGY

15A.1 INTRODUCTION

In the utility impact analysis, the U.S. Department of Energy (DOE) analyzes the changes in electric installed capacity and power generation that result for each trial standard level (TSL). These changes are estimated by multiplying the site savings of electricity by a set of *impact factors* which measure the corresponding change in generation by fuel type, installed capacity, and power sector emissions. This Appendix describes the methods that DOE used to calculate these impact factors. The methodology is more fully described in Coughlin (2014; 2019).^{1,2}

DOE's analysis uses output of the DOE/Energy Information Administration (EIA)'s most recent *Annual Energy Outlook (AEO)*.³ The *AEO* includes a reference case and a set of side cases that implement a variety of economic and policy scenarios. In 2015 EIA announced the adoption of a two-year release cycle for the *AEO*, alternating between a full set of scenarios and a shorter edition containing only five scenarios. DOE has adapted its calculation methodology to be independent of the type of scenarios available with each *AEO* publication.

15A.2 METHODOLOGY

Marginal reductions in electricity demand lead to marginal reductions in power sector generation, emissions, and installed capacity. Generally, DOE quantifies these reductions using marginal impact factors, which are time series defining the change in some power sector quantity that results from a unit change in site electricity demand. Because load shapes affect the mix of generation types on the margin, these impact factors depend on end-use and sector.

DOE's approach examines a series of *AEO* side cases to estimate the relationship between changes to power sector generation (TWh) by fuel type and changes to other supply-side power sector variables, including fuel consumption (quads) by fuel type, and installed capacity (GW) by fuel and technology type. DOE also calculates changes to power sector emissions; the methodology for computing these impacts is described in appendix 13A.

DOE uses load shape information from the NEMS code to relate marginal generation reductions by fuel type to marginal demand reductions by sector and end use. Because *AEO* side cases with electricity demand reductions are not always available, DOE defines the relationship between sector/end-use and generation fuel type using Reference case data. Specifically, DOE defines, for each sector and end-use, fuel-share weights equal to the percentage of each MWh used to serve that end-use load that is provided by each generation fuel type.

The load shape data provide an hourly profile defining total consumption of electricity for each sector/end-use. For each load DOE allocates consumption to one of 3 periods: on-peak, shoulder, and off-peak. These categories are used in the utility sector to correlate end-use consumption with supply types. On-peak hours are defined as 12pm to 5pm Monday through Saturday, June through September. Off-peak hours are 9pm to 6am daily and all day Sunday. All other hours are allocated to the shoulder period. This leads to a set of weights $w(p,u,y)$ where:

y = the analysis year

u = an index representing the sector/end-use (e.g. commercial cooling)

p = the time-of-day period

$w(p,u,y)$ = the fraction of load u that is served in period p

By definition the sum of $w(p,u,y)$ over periods p is equal to one. On the supply-side, DOE allocates generation by each fuel type to one of the time-of-day periods. The allocation is based on the following rules:

- 1.1. The data are normalized so that total annual generation equals total annual consumption by sector and end-use;
- 1.2. The demand-side data are summed over sector/end-use to define a total demand for generation in each time-of-day period;
- 1.3. All petroleum-based generation is allocated to peak periods;
- 1.4. Base-load generation (nuclear and coal) is assumed to be equally likely to be on in all hours; hence, it is allocated to each period in proportion to the number of hours in that period;
- 1.5. Any unmet peak period demand is allocated to natural gas;
- 1.6. The remaining generation of all types is allocated to the remaining periods proportionally.

This leads to a second set of weights $z(p,f,y)$ where:

f = the fuel type

$z(p,f,y)$ = the fraction of load in period p that is served by fuel f

These weights are used to allocate a MWh of demand reduction for a given end-use to each fuel type. In defining the fuel-share weights for demand reductions, DOE makes one adjustment to the factors calculated from the Reference case data. An examination of all available *AEO* scenarios shows that both generation and installed capacity for nuclear power are unchanged across the projection period. This implies that the use of nuclear power is not affected by small changes in the supply/demand balance; hence, DOE assumes that the factor $z(p,f,y)$ is zero for nuclear power. The values of $z(p,f,y)$ for the other fuels are renormalized so that the sum of $z(p,f,y)$ across the remaining fuel types is equal to one.

DOE defines the generation fuel share weights $g(u,f,y)$ as the product

$$g(u,f,y) = \sum_p w(p,u,y) z(p,f,y).$$

Eq. 15A.1

For the sector/end-use defined by u , the product of the total annual site electricity savings times the factor $g(u,f,y)$ defines the marginal generation reductions by fuel type. These marginal generation reductions can be related to marginal fuel use reductions (see appendix 10.B of this TSD) and to the marginal emissions reductions (see appendix 13A of this TSD). They are also related to the marginal installed capacity reductions through the capacity factor.

DOE uses a capacity factor to relate reductions in generation by fuel type to reductions in installed capacity by technology type. The capacity factor is defined as the magnitude of change in capacity given a unit change in generation. The technology types are coal, natural gas combined-cycle (NGCC), oil and gas steam (OGS), combustion turbine-diesel (CTD), and renewable sources. For NGCC the capacity factor is defined as the ratio of NGCC capacity to natural gas generation. DOE combines CTD and OGS DOE into a single *peak* capacity type, with capacity factor equal to the ratio of the sum of CTD plus OGS capacity to oil-fired generation. Each fuel type is then related to a unique capacity type. While marginal capacity factors can be calculated from *AEO* data, this approach produces results that are dominated by computational noise. Hence, DOE uses data for the *AEO Reference Case* to calculate grid-average capacity factors for each year of the analysis period, defined as $c(f,y)$. The capacity change for fuel/technology type f induced by a unit reduction in demand for sector/end-use u is given by the product $g(u,f,y) * c(f,y)$.

15A.3 MODEL RESULTS

Representative values of the impact factors for fuel share by fuel type, and capacity by technology type are provided in the tables below. The tables show the factors for two years, 2025 and 2050. The marginal heat rates are presented in appendix 10B and emissions factors are presented in in appendix 13A.

15A.3.1 Electricity Generation

Table 15A.3.1 and Table 15A.3.2 show the distribution across fuel types of a unit reduction in electricity demand by sector and end-use, referred to above as fuel-share weights. The fuel types are coal, natural gas, petroleum, and renewables. The values for cooling are representative of peaking loads, while the values for refrigeration are representative of flat loads. The data are shown for 2025 and 2050.

Table 15A.3.1 Fuel-Share Weights by Sector and End-Use (Values for 2025)

	Coal	Natural Gas	Oil	Renewables
Residential Sector				
Clothes Dryers	25.3%	38.9%	0.2%	35.6%
Cooking	24.6%	39.3%	0.2%	35.9%
Freezers	26.2%	38.5%	0.2%	35.1%
Lighting	26.6%	37.9%	0.1%	35.5%
Refrigeration	26.1%	38.5%	0.2%	35.2%
Space Cooling	21.7%	42.4%	0.6%	35.3%
Space Heating	27.3%	37.3%	0.0%	35.4%
Water Heating	25.6%	38.5%	0.1%	35.8%
Other Uses	25.3%	38.9%	0.2%	35.7%
Commercial Sector				
Cooking	20.9%	41.6%	0.3%	37.3%
Lighting	21.9%	41.0%	0.3%	36.9%
Office Equipment (Non-Pc)	19.0%	42.9%	0.4%	37.7%
Office Equipment (Pc)	19.0%	42.9%	0.4%	37.7%
Refrigeration	24.8%	39.2%	0.2%	35.9%
Space Cooling	20.6%	43.1%	0.7%	35.5%
Space Heating	27.7%	37.1%	0.0%	35.2%
Ventilation	24.9%	39.1%	0.2%	35.9%
Water Heating	20.8%	41.6%	0.3%	37.4%
Other Uses	19.8%	42.3%	0.3%	37.5%
Industrial Sector				
All Uses	19.8%	42.3%	0.3%	37.5%

Table 15A.3.2 Fuel-Share Weights by Sector and End-Use (Values for 2050)

	Coal	Natural Gas	Oil	Renewables
Residential Sector				
Clothes Dryers	14.1%	36.2%	0.1%	49.6%
Cooking	13.7%	36.5%	0.1%	49.7%
Freezers	14.6%	36.1%	0.1%	49.2%
Lighting	14.8%	35.1%	0.0%	50.0%
Refrigeration	14.6%	36.1%	0.1%	49.3%
Space Cooling	11.9%	40.5%	0.2%	47.3%
Space Heating	15.3%	34.6%	0.0%	50.2%
Water Heating	14.2%	35.7%	0.0%	50.1%
Other Uses	14.1%	36.2%	0.1%	49.7%
Commercial Sector				
Cooking	11.5%	38.1%	0.1%	50.3%
Lighting	12.1%	37.7%	0.1%	50.1%
Office Equipment (Non-Pc)	10.4%	39.3%	0.1%	50.1%
Office Equipment (Pc)	10.4%	39.3%	0.1%	50.1%
Refrigeration	13.8%	36.4%	0.1%	49.8%
Space Cooling	11.3%	41.2%	0.3%	47.3%
Space Heating	15.5%	34.4%	0.0%	50.1%
Ventilation	13.8%	36.3%	0.1%	49.8%
Water Heating	11.4%	38.0%	0.1%	50.4%
Other Uses	10.9%	38.8%	0.1%	50.2%
Industrial Sector				
All Uses	10.9%	38.8%	0.1%	50.2%

15A.3.2 Installed Capacity

Table 15A.3.3 and Table 15A.3.4 show the total change in installed capacity (GW) per unit of site electricity demand reduction for the five principal capacity types: coal, natural gas, peaking, renewables, and nuclear. The peaking category is the sum of the two NEMS categories oil and gas steam and combustion turbine/diesel. Data are shown for 2025 and 2050.

Table 15A.3.3 Capacity Impact Factors in GW per TWh Reduced Site Electricity Demand (Values for 2025)

	Coal	Natural Gas	Oil	Renewables
Residential Sector				
Clothes Dryers	0.061	0.097	0.046	0.134
Cooking	0.060	0.098	0.052	0.135
Freezers	0.063	0.096	0.050	0.132
Lighting	0.064	0.095	0.017	0.133
Refrigeration	0.063	0.096	0.049	0.132
Space Cooling	0.052	0.106	0.174	0.133
Space Heating	0.066	0.093	0.002	0.133
Water Heating	0.062	0.096	0.028	0.135
Other Uses	0.061	0.097	0.043	0.134
Commercial Sector				
Cooking	0.051	0.104	0.078	0.140
Lighting	0.053	0.103	0.072	0.139
Office Equipment (Non-Pc)	0.046	0.107	0.106	0.142
Office Equipment (Pc)	0.046	0.107	0.106	0.142
Refrigeration	0.060	0.098	0.047	0.135
Space Cooling	0.050	0.108	0.188	0.134
Space Heating	0.067	0.093	0.000	0.132
Ventilation	0.060	0.098	0.045	0.135
Water Heating	0.050	0.104	0.074	0.140
Other Uses	0.048	0.106	0.095	0.141
Industrial Sector				
All Uses	0.048	0.106	0.095	0.141

Table 15A.3.4 Capacity Impact Factors in GW per TWh Reduced Site Electricity Demand (Values for 2050)

	Coal	Natural Gas	Oil	Renewables
Residential Sector				
Clothes Dryers	0.031	0.111	0.052	0.187
Cooking	0.030	0.112	0.059	0.187
Freezers	0.032	0.110	0.057	0.185
Lighting	0.032	0.107	0.019	0.188
Refrigeration	0.032	0.110	0.056	0.185
Space Cooling	0.026	0.124	0.197	0.178
Space Heating	0.033	0.106	0.002	0.189
Water Heating	0.031	0.109	0.032	0.188
Other Uses	0.031	0.111	0.049	0.187
Commercial Sector				
Cooking	0.025	0.117	0.088	0.189
Lighting	0.027	0.115	0.082	0.188
Office Equipment (Non-Pc)	0.023	0.120	0.120	0.189
Office Equipment (Pc)	0.023	0.120	0.120	0.189
Refrigeration	0.030	0.111	0.054	0.187
Space Cooling	0.025	0.126	0.214	0.178
Space Heating	0.034	0.105	0.000	0.189
Ventilation	0.030	0.111	0.051	0.188
Water Heating	0.025	0.116	0.084	0.190
Other Uses	0.024	0.119	0.108	0.189
Industrial Sector				
All Uses	0.024	0.119	0.108	0.189

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APPENDIX 17A. REGULATORY IMPACT ANALYSIS: SUPPORTING MATERIALS

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APPENDIX 17A. REGULATORY IMPACT ANALYSIS: SUPPORTING MATERIALS

17A.1 INTRODUCTION

This appendix contains sections discussing the following topics:

- Projections of annual market share increases for the alternative policies;
- NIA-RIA Integrated Model;
- Market penetration curves used to analyze consumer rebates and voluntary labeling, including:
 - Background material on XENERGY's approach,
 - DOE's adjustment of these curves for this analysis, and
 - The method DOE used to derive interpolated, customized curves;
- Detailed table of rebates offered for the considered product, as well as DOE's approach to estimate a market representative rebate value for this RIA; and
- Background material on Federal and State tax credits for appliances.

17A.2 MARKET SHARE ANNUAL INCREASES BY POLICY

Table 17A.2.1 through Table 17A.2.3 show the annual increases in market shares of air cleaners meeting the target efficiency levels for the selected TSL (TSL 3). DOE used these market share increases as inputs to the NIA-RIA spreadsheet model.

Table 17A.2.1 Annual Increases in Market Shares Attributable to Alternative Policy Measures for $10 \leq \text{PM}_{2.5} \text{ CADR} < 100$ (TSL 3)

Year	Consumer Rebates	Consumer Tax Credits	Manufacturer Tax Credits	Voluntary Labeling	Bulk Govt Purchases
2024	25.6%	15.4%	7.7%	15.2%	0.1%
2025	25.6%	15.4%	7.7%	25.3%	0.2%
2026	48.6%	29.2%	14.6%	51.8%	1.0%
2027	48.6%	29.2%	14.6%	55.2%	1.1%
2028	48.6%	29.2%	14.6%	58.3%	1.2%
2029	48.6%	29.2%	14.6%	61.4%	1.3%
2030	48.6%	29.2%	14.6%	64.3%	1.4%
2031	48.6%	29.2%	14.6%	67.2%	1.5%
2032	48.6%	29.2%	14.6%	69.9%	1.6%
2033	48.6%	29.2%	14.6%	70.1%	1.8%
2034	48.6%	29.2%	14.6%	70.1%	1.8%
2035	48.6%	29.2%	14.6%	70.1%	1.8%
2036	48.6%	29.2%	14.6%	70.1%	1.8%
2037	48.6%	29.2%	14.6%	70.1%	1.8%
2038	48.6%	29.2%	14.6%	70.1%	1.8%
2039	48.6%	29.2%	14.6%	70.1%	1.8%
2040	48.6%	29.2%	14.6%	70.1%	1.8%
2041	48.6%	29.2%	14.6%	70.1%	1.8%
2042	48.6%	29.2%	14.6%	70.1%	1.8%
2043	48.6%	29.2%	14.6%	70.1%	1.8%
2044	48.6%	29.2%	14.6%	70.1%	1.8%
2045	48.6%	29.2%	14.6%	70.1%	1.8%
2046	48.6%	29.2%	14.6%	70.1%	1.8%
2047	48.6%	29.2%	14.6%	70.1%	1.8%
2048	48.6%	29.2%	14.6%	70.1%	1.8%
2049	48.6%	29.2%	14.6%	70.1%	1.8%
2050	48.6%	29.2%	14.6%	70.1%	1.8%
2051	48.6%	29.2%	14.6%	70.1%	1.8%
2052	48.6%	29.2%	14.6%	70.1%	1.8%
2053	48.6%	29.2%	14.6%	70.1%	1.8%
2054	48.6%	29.2%	14.6%	70.1%	1.8%
2055	48.6%	29.2%	14.6%	70.1%	1.8%
2056	48.6%	29.2%	14.6%	70.1%	1.8%
2057	48.6%	29.2%	14.6%	70.1%	1.8%

Table 17A.2.2 Annual Increases in Market Shares Attributable to Alternative Policy Measures for $100 \leq \text{PM}_{2.5} \text{ CADR} < 150$ (TSL 3)

Year	Consumer Rebates	Consumer Tax Credits	Manufacturer Tax Credits	Voluntary Labeling	Bulk Govt Purchases
2024	23.6%	14.1%	7.1%	11.5%	0.1%
2025	23.6%	14.1%	7.1%	17.9%	0.2%
2026	32.0%	19.2%	9.6%	30.5%	0.6%
2027	32.0%	19.2%	9.6%	33.2%	0.7%
2028	32.0%	19.2%	9.6%	35.4%	0.9%
2029	32.0%	19.2%	9.6%	37.5%	1.0%
2030	32.0%	19.2%	9.6%	39.4%	1.1%
2031	32.0%	19.2%	9.6%	41.2%	1.2%
2032	32.0%	19.2%	9.6%	42.9%	1.4%
2033	32.0%	19.2%	9.6%	44.6%	1.5%
2034	32.0%	19.2%	9.6%	44.6%	1.5%
2035	32.0%	19.2%	9.6%	44.7%	1.5%
2036	32.0%	19.2%	9.6%	44.7%	1.5%
2037	32.0%	19.2%	9.6%	44.8%	1.5%
2038	32.0%	19.2%	9.6%	44.8%	1.5%
2039	32.0%	19.2%	9.6%	44.9%	1.5%
2040	32.0%	19.2%	9.6%	45.0%	1.5%
2041	32.0%	19.2%	9.6%	45.0%	1.5%
2042	32.0%	19.2%	9.6%	45.1%	1.5%
2043	32.0%	19.2%	9.6%	45.1%	1.5%
2044	32.0%	19.2%	9.6%	45.2%	1.5%
2045	32.0%	19.2%	9.6%	45.2%	1.5%
2046	32.0%	19.2%	9.6%	45.3%	1.5%
2047	32.0%	19.2%	9.6%	45.3%	1.5%
2048	32.0%	19.2%	9.6%	45.4%	1.5%
2049	32.0%	19.2%	9.6%	45.4%	1.5%
2050	32.0%	19.2%	9.6%	45.5%	1.5%
2051	32.0%	19.2%	9.6%	45.5%	1.5%
2052	32.0%	19.2%	9.6%	45.6%	1.5%
2053	32.0%	19.2%	9.6%	45.6%	1.5%
2054	32.0%	19.2%	9.6%	45.7%	1.5%
2055	32.0%	19.2%	9.6%	45.7%	1.5%
2056	32.0%	19.2%	9.6%	45.8%	1.5%
2057	32.0%	19.2%	9.6%	45.8%	1.5%

Table 17A.2.3 Annual Increases in Market Shares Attributable to Alternative Policy Measures for PM_{2.5} CADR $\geq$ 150 (TSL 3)

Year	Consumer Rebates	Consumer Tax Credits	Manufacturer Tax Credits	Voluntary Labeling	Bulk Govt Purchases
2024	16.7%	10.0%	5.0%	12.7%	0.1%
2025	16.7%	10.0%	5.0%	19.4%	0.3%
2026	12.3%	7.4%	3.7%	19.8%	0.3%
2027	12.3%	7.4%	3.7%	20.8%	0.4%
2028	12.3%	7.4%	3.7%	21.8%	0.5%
2029	12.3%	7.4%	3.7%	22.8%	0.7%
2030	12.3%	7.4%	3.7%	23.7%	0.8%
2031	12.3%	7.4%	3.7%	24.6%	0.9%
2032	12.3%	7.4%	3.7%	25.5%	1.1%
2033	12.3%	7.4%	3.7%	27.2%	1.2%
2034	12.3%	7.4%	3.7%	27.2%	1.2%
2035	12.3%	7.4%	3.7%	27.3%	1.2%
2036	12.3%	7.4%	3.7%	27.3%	1.2%
2037	12.3%	7.4%	3.7%	27.3%	1.2%
2038	12.3%	7.4%	3.7%	27.3%	1.2%
2039	12.3%	7.4%	3.7%	27.4%	1.2%
2040	12.3%	7.4%	3.7%	27.4%	1.2%
2041	12.3%	7.4%	3.7%	27.4%	1.2%
2042	12.3%	7.4%	3.7%	27.4%	1.2%
2043	12.3%	7.4%	3.7%	27.5%	1.2%
2044	12.3%	7.4%	3.7%	27.5%	1.2%
2045	12.3%	7.4%	3.7%	27.5%	1.2%
2046	12.3%	7.4%	3.7%	27.5%	1.2%
2047	12.3%	7.4%	3.7%	27.6%	1.2%
2048	12.3%	7.4%	3.7%	27.6%	1.2%
2049	12.3%	7.4%	3.7%	27.6%	1.2%
2050	12.3%	7.4%	3.7%	27.7%	1.2%
2051	12.3%	7.4%	3.7%	27.7%	1.2%
2052	12.3%	7.4%	3.7%	27.7%	1.2%
2053	12.3%	7.4%	3.7%	27.8%	1.2%
2054	12.3%	7.4%	3.7%	27.8%	1.2%
2055	12.3%	7.4%	3.7%	27.8%	1.2%
2056	12.3%	7.4%	3.7%	27.8%	1.2%
2057	12.3%	7.4%	3.7%	27.9%	1.2%

17A.3 NIA-RIA INTEGRATED MODEL

For this analysis, DOE used its integrated NIA-RIA^a model approach that the Department built on the NIA model discussed in chapter 10 and documented in appendix 10-A. The resulting integrated NIA-RIA model features both the NIA and RIA inputs, analyses and results. It has the capability to generate results, by product class and TSL, for the mandatory standards and each of the RIA policies. Separate modules estimate increases in market penetration of more efficient equipment for consumer rebates, voluntary labeling and bulk government purchases.^b The consumer rebates module calculates benefit-cost (B/C) ratios and market barriers, and generates customized market penetration curves for each product class; the voluntary labeling module relies on the market barriers calculated in the consumer rebates module to project a reduction in those barriers over the first ten years of the forecast period and estimate the market effects of such a reduction; and the bulk government purchases module scales down the market for air cleaners to housing units in public housing authority and Federal government-owned commercial buildings. A separate module summarizes the market impacts from mandatory standards, calculated under the same market conditions as the alternative policies, and all policy alternatives. An additional module produces all tables and figures presented in chapter 17 as well as the tables of market share increases for each policy reported in Section 17A.2 of this appendix.

17A.4 MARKET PENETRATION CURVES

This section first discusses the theoretical basis for the market penetration curves that DOE used to analyze the Consumer Rebates and Voluntary Labeling policies. Next it discusses the adjustments it made to the maximum penetration rates. It then refers to the method it used to develop interpolated penetration curves for air cleaners that meet the target efficiency levels at each TSL. The resulting curves for the proposed TSL are presented in chapter 17.

17A.4.1 Introduction

XENERGY, Inc.^c, developed a re-parameterized, mixed-source information diffusion model to estimate market impacts induced by financial incentives for purchasing energy efficient appliances.¹ The basic premise of the mixed-source model is that information diffusion drives the adoption of technology.

Extensive economic literature describes the diffusion of new products as technologies evolve. Some research focuses primarily on developing analytical models of diffusion patterns applicable to individual consumers or to technologies from competing firms.^{2, 3, 4} One study records researchers' attempts to investigate the factors that drive diffusion processes.⁵ Because a new product generally has its own distinct characteristics, few studies have been able to conclusively develop a universally applicable model. Some key findings, however, generally are accepted in academia and industry.

^a NIA = National Impact Analysis; RIA = Regulatory Impact Analysis

^b As mentioned in chapter 17, the increase in market penetrations for consumer tax credits and manufacturer tax credits are estimated as a fraction of the increase in market penetration of consumer rebates.

^c XENERGY is now owned by KEMA, Inc. (www.kema.com)

One accepted finding is that, regardless of their economic benefits and technological merits, new technologies are unlikely to be adopted by all potential users. For many products, a ceiling must be placed on the adoption rate. A second conclusion is that not all adopters purchase new products at the same time: some act quickly after a new product is introduced; others wait for the product to mature. Third, diffusion processes can be characterized approximately by asymmetric S-curves that depict three stages of diffusion: starting, accelerating, and decreasing (as the adoption ceiling is approached).

A so-called epidemic model of diffusion is used widely in marketing and social studies. The epidemic model assumes that (1) all consumers place identical value on the benefits of a new product, and (2) the cost of a new product is constant or declines monotonically over time. What induces a consumer to purchase a new product is information about the availability and benefits of the product. In other words, information diffusion drives consumers' adoption of a new product.³ The model incorporates information diffusion from both internal sources (spread by word of mouth from early adopters to prospective adopters) and external sources (the "announcement effect" produced by government agencies, institutions, or commercial advertising). The model incorporates both internal and external sources by combining a logistic function with an exponential function.^{4, 5}

The relative degree of influence from the internal and external sources determines the general shape of the diffusion curve for a specific product.^{4, 5} If adoption of a product is influenced primarily by external sources of information (the announcement effect), for instance, a high rate of diffusion occurs at the beginning of the process. In this scenario, external sources provide immediate information exposure to a significant number of prospective adopters. In contrast, internal sources (such as a network of prospective adopters) are relatively small in size and reach, producing a more gradual exposure to prospective adopters. Graphically speaking, information diffusion dominated by external sources is represented by a concave curve (the exponential curve in Figure 17A.4.1). If adoption of a new product is influenced most strongly by internal sources of information, the number of adopters increases gradually, forming a convex curve (the logistic curve in Figure 17A.4.1).

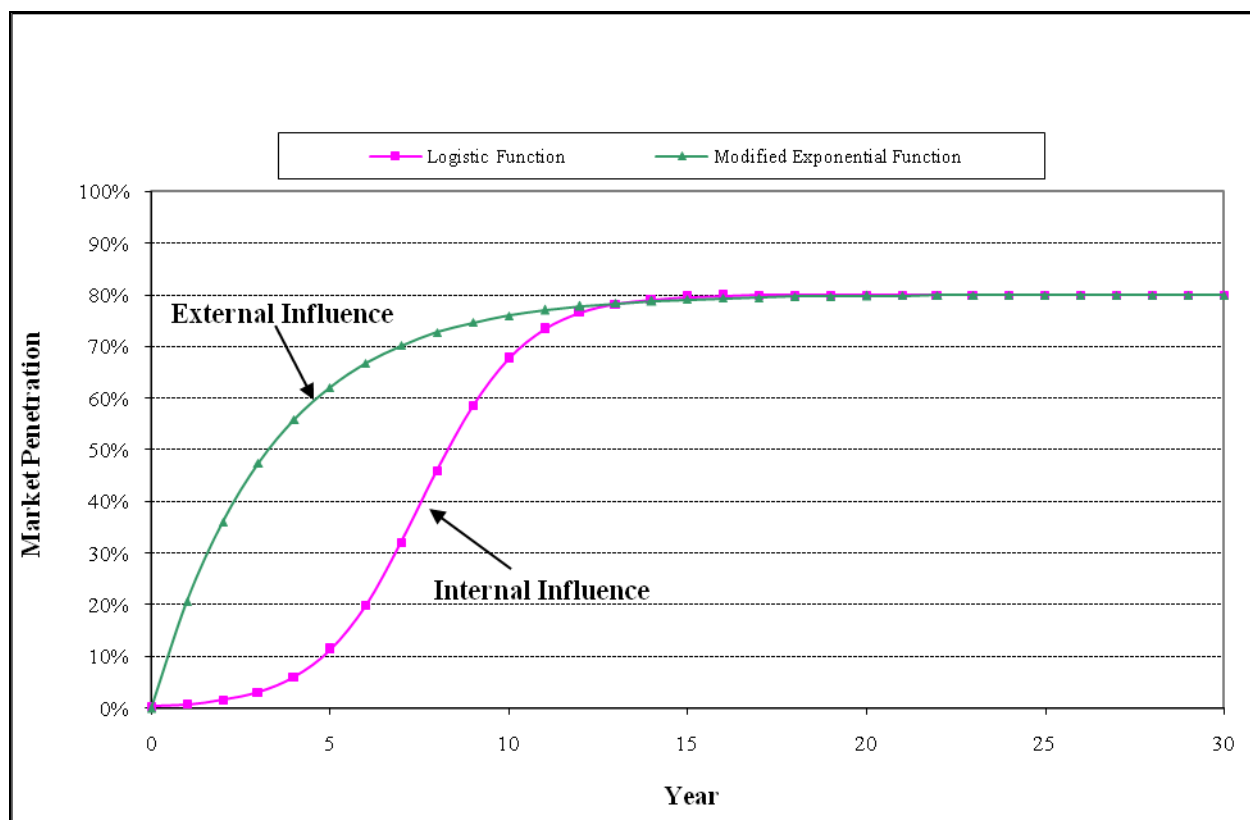


Figure 17A.4.1 S-Curves Showing Effects of External and Internal Sources on Adoption of New Technologies

17A.4.2 Adjustment of XENERGY Penetration Curves

In consultation with the primary authors of the 2002 XENERGY study who later conducted similar California studies, DOE made some adjustments to XENERGY's original implementation (penetration) curves.^{6, 7} The experiences with utility programs since the XENERGY study indicate that incentive programs have difficulty achieving penetration rates as high as 80 percent. Consumer response is limited by barriers created by consumer utility issues and other non-economic factors. DOE therefore adjusted the maximum penetration parameters for some of the curves from 80 percent to the following levels:

Moderate Barriers:	70%
High Barriers:	60%
Extremely High Barriers:	50%

The *low barriers* and *no barriers* curves (the latter used only when a product has a very high base-case-market share) remained, respectively, with 80 percent and 100 percent as their maximum penetration rates. For the interpolated penetration curves (discussed below), DOE set the *no barriers* and *extremely high barriers* curves as the upper and lower bounds, respectively, for any benefit/cost ratio points higher or lower than the curves. It set another constraint such that the policy case market share cannot be great than 100 percent, as might occur for products with high no-new-standards case market shares of the target-level technology.

17A.4.3 Interpolation of Penetration Curves

As discussed above, the XENERGY penetration (implementation) curves followed a functional form to estimate the market implementation rate caused by energy efficiency measures such as consumer rebates.^d The XENERGY report presents five reference market implementation curves that vary according to the level of market barriers to technology penetration.¹ Such curves have been used by DOE in the Regulatory Impact Analyses for rulemakings for appliance energy efficiency standards to estimate market share increases in response to rebate programs.^e They provide a framework for evaluating technology penetration, yet require matching the studied market to the curve that best represents it. This approximate matching can introduce some inaccuracy to the analysis.

Blum et al (2011, appendix A)⁸ resents an alternative approach to such evaluation: a method to estimate market implementation rates more accurately by performing interpolations of the reference curves. The referred report describes the market implementation rate function and the reference curves, the method to calibrate the function to a given market, and the limitations of the method.

DOE used the above referred method to interpolate market implementation curves, to generate customized curves that were used to estimate the effects of consumer rebates and voluntary labeling for each product class covered by this RIA. For consumer rebates, DOE derived such curves based on an algorithm that finds the market implementation curve that best fits, for the first year of the analysis period, the B/C ratio of the target efficiency level and the market penetration of equipment with that level of energy efficiency in the no-new-standards case. For the analysis of voluntary labeling, DOE departs from the market barriers level corresponding to the market implementation curve it derived for consumer rebates, to linearly decrease it over the ten initial years of the analysis period. For each year, as market barriers decline, the corresponding market implementation curve leads – for the same B/C ratio – to higher market penetrations.

^d The RIA chapter refers to these curves as *penetration curves*. This section, in references to the original source, uses the term *implementation curve*.

^e DOE has also used this method to estimate market share increases resulting from consumer tax credit and manufacturer tax credit programs, since the effects of tax credits on markets are considered in this RIA proportional to the impacts from rebates.

17A.5 CONSUMER REBATE PROGRAMS

DOE reviewed rebate programs offered towards ENERGY STAR qualifying air cleaners in January, 2023.^f Based on that review, and considering the value of the rebates offered, DOE assumed that any rebate would cover, in full, the upfront costs for a new air cleaner. Table 17A.5.1 shows the rebate amounts DOE estimated for the proposed efficiency levels of the product classes covered by this RIA. Given that the selected TSL has two compliance years, 2024 and 2026, DOE estimated, for each product class, one rebate amount for the 2024-2025 period, and another rebate amount for the 2026-2057 period.

Table 17A.5.1 Rebates Amounts for Air Cleaners*

Product Class	2024-2025	2026-2057
$10 \leq \text{PM}_{2.5} \text{ CADR} < 100$	1.74	3.66
$100 \leq \text{PM}_{2.5} \text{ CADR} < 150$	2.22	4.46
$\text{PM}_{2.5} \text{ CADR} \geq 150$	2.03	3.41

* In 2021\$.

17A.6 FEDERAL AND STATE TAX CREDITS

This section summarizes the Federal and State tax credits available to consumers who purchase energy efficient appliances. This section also describes tax credits available to manufacturers who produce certain energy efficient appliances.

17A.6.1 Federal Tax Credits for Consumers

EPACT 2005 included Federal tax credits for consumers who installed efficient air conditioners or heat pumps; gas, oil and propane furnaces and boilers; furnace fans; and/or gas, oil, or electric heat pump water heaters in new or existing homes.⁹ These tax credits were in effect in 2006 and 2007, expired in 2008, and were reinstated for 2009–2010 by the American Recovery and Reinvestment Act (ARRA).¹⁰ There was a \$1,500 cap on the credit per home, including the amount received for insulation, windows, and air and duct sealing. Congress extended this provision for 2011, with some modifications to eligibility requirements, and reductions in the cap to \$500 per home. The American Taxpayer Relief Act of 2012 extended, with some modifications, residential tax credits for air conditioners, heat pumps, furnaces, and water heaters placed in service between January 1, 2012 and December 31, 2013.¹¹ The tax credit for furnace fans was \$50 in 2011, after which it expired.

The importance of the Federal tax credits has been emphasized in research in the residential heating industry on the impacts of the relatively large credits that were available for HVAC (heating, ventilating, and air conditioning) equipment. In a survey of HVAC distributors

^f https://www.energystar.gov/rebate-finder/?scrollTo=818.1818237304688&sort_by=utility&sort_direction=asc&page_number=0&lastpage=0&zip_code_filter=&search_text=&product_clean_filter=Air+Purifiers+%28Cleaners%29&product_clean_isopen=0&product_types=Select+a+Product+Category

conducted by Vermont Energy Investment Corporation, respondents indicated that the ample credit had had a notable impact on sales of higher-efficiency heating and cooling equipment. Some distributors combined the Federal tax credits with manufacturer rebates and utility program rebates for a greater consumer incentive. However, when the amount of the Federal tax credit was reduced, smaller utility rebate incentives had not induced the same levels of equipment sales increases. The decrease in incentive size from a \$1,500 cap in 2009-2010 to a \$500 cap in 2011, during a period when the economy continued to be sluggish, resulted in a decline in total sales of residential HVAC products. Distributors stated that an incentive needed to cover 25 to 75 percent of the incremental cost of the efficient equipment to influence consumer choice. The industry publication "2011 HVAC Review and Outlook" noted a decline in sales of air conditioning units with >14 SEER in 2011 and a return in sales of units with >16 SEER to 2009 levels (after an increase in 2010). The large majority of distributors observed no impacts from the utility programs with their lower rebate amounts available in 2011. Distributors also commented on the advantages of the Federal tax credit being nationwide in contrast to utility rebate programs that target regional markets.^{12, 13}

In an effort to evaluate the potential impact of a Federal appliance tax credit program, DOE reviewed Internal Revenue Service (IRS) data on the numbers of taxpayers who claimed the tax credits during tax years 2006 and 2007. It estimated the percentage of taxpayers who filed Form 5695, *Residential Energy Credits*.¹⁴ It also estimated the percentage of taxpayers with entries under Form 5695's section 3, *Residential energy property costs*, line 3b, *qualified natural gas, propane, or oil furnace or hot water boiler*. DOE reasoned that the percentage of taxpayers with an entry on Line 3b could serve as a rough indication of the potential of taxpayer participation in a Federal tax credit program for furnaces during the initial program years. DOE found that of all residential taxpayers filing tax returns, 0.8 percent in 2006 and 0.6 percent in 2007, claimed a credit for a furnace or boiler. DOE further found that the percentages of those filing Form 5695 for any qualifying energy property expenditure (which also included installation of efficient windows, doors and roofs) were 3.1 and 3.2 percent in 2006 and 2007 respectively.

DOE also reviewed data from an earlier Federal energy conservation tax credit program in place in the 1980s. While this tax credit was available from 1979 through 1985, DOE located data for only the first three years of the program.^{15, 16, 17} For those three years - 1979, 1980, and 1981 - the percentages of taxpayers filing Form 5695 were 6.4 percent, 5.2 percent, and 4.9 percent. Given that the data from this earlier tax credit program were not disaggregated by type of energy property, this data series served only to indicate a possible trend of greater participation in the initial program year, followed by slightly smaller participation in subsequent years. However, DOE did not find detailed analysis of this program to indicate the possible reasons for such a trend. Also, this trend varies from the more stable trend shown in the EPA's 2005 energy tax credit program data for its first two program years.

As discussed in chapter 17, DOE analyzed the percentage of participation in consumer tax credit programs using its estimates of consumer participation in rebate programs that was based on benefit/cost data specific to each product class of air cleaners covered by this RIA. Hence it was difficult to compare these detailed estimates to the more general data analysis

described above from the existing Federal tax credit program, or to use the IRS data analysis in its consumer tax credit analysis.

17A.6.2 Federal Tax Credits for Manufacturers

EPACT 2005 provided Federal Energy Efficient Appliance Credits to manufacturers that produced high-efficiency refrigerators, clothes washers, and dishwashers in 2006 and 2007.¹⁸ The Emergency Economic Stabilization Act of 2008¹⁹ amended the credits and extended them through 2010. The credits were extended again to 2011 with modifications in the eligibility requirements. Manufacturer tax credits were extended again, by the American Taxpayer Relief Act of 2012, for clothes washers, refrigerators, and dishwashers manufactured between January 1, 2012 and December 31, 2013.

Manufacturers who produce these appliances receive the credits for increasing their production of qualifying appliances. These credits had several efficiency tiers in 2011. For 2012-2013, credits for the higher tiers remain but were eliminated for the lowest (least efficient) tiers for clothes washers and dishwashers. The credit amounts applied to each unit manufactured. The credit to manufacturers of qualifying clothes washers, refrigerators and dishwashers was capped at \$75 million for the period of 2008-2010. However, the most efficient refrigerator (30%) and clothes washer (2.2 MEF/4.5 wcf) models was not subject to the cap. The credit to manufacturers was capped at \$25 million for 2011, with the most efficient refrigerators (35%) and clothes washers (2.8 MEF/3.5 WCF) exempted from this cap.^g

17A.6.3 State Tax Credits

The States of Oregon and Montana have offered consumer tax credits for efficient appliances for several years, and the States of Kentucky, Michigan and Indiana began offering such credits in 2009. The Oregon Department of Energy (ODOE) has disaggregated data on taxpayer participation in credits for eligible products. (See the discussion in chapter 17, Section 17.3.3, on tax credit data for clothes washers.) Montana's Department of Revenue does not disaggregate participation data by appliance, although DOE reviewed Montana's overall participation trends and found them congruent with its analysis of Oregon's clothes washer tax credits.

Oregon's Residential Energy Tax Credit (RETC) was created in 1977. The Oregon legislature expanded the RETC program in 1997 to include residential refrigerators, clothes washers, and dishwashers, which significantly increased participation in the program. The program subsequently added credits for high-efficiency heat pump systems, air conditioners, and water heaters (2001); furnaces and boilers (2002); and duct/air sealing, fuel cells, heat recovery, and renewable energy equipment. Beginning in 2012 a Tax Credit Extension Bill (HB3672) eliminated refrigerators, clothes washers, dishwashers, air conditioners, and boilers from the RETC program, leaving credits for water heaters, furnaces, heat pumps, tankless water heaters, and heat pump water heaters.^{20, 21} The technologies recognized by the Oregon Department of

^g For more information on federal tax credits for manufacturers see the following websites:
<https://programs.dsireusa.org/system/program/detail/1273/energy-efficient-appliance-manufacturing-tax-credit>,
<https://energytaxincentives.org/>

Energy as “premium efficiency” were eligible for a tax credit of \$0.60 per kWh saved in the first year (up to \$1,500).²⁰

Montana had an Energy Conservation Tax Credit for residential measures starting in 1998.²² The tax credit covered various residential energy and water efficient products, including split system central air conditioning; package system central air conditioning; split system air source heat pumps; package system heat pumps; natural gas, propane, or oil furnaces; hot water boilers; advanced main air circulating fans; heat recovery ventilators; gas, oil, or propane water heaters; electric heat pump water heaters; low-flow showerheads and faucets; light fixtures; and controls. In 2002 the amount of the credit was increased from 5 percent of product costs (up to \$150) to 25 percent (up to \$500) per taxpayer. The credit could be used for products installed in new construction or remodeling projects. The tax credit covered only the part of the cost and materials that exceeded established standards of construction.

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